

**THEOPHILUS AND THE THEORY AND
PRACTICE OF MEDIEVAL ART**

TEOPHILVS humilis presbit' ser-
uus seruoꝝ dī. indign' nomine et
p'fessione monachi. omib; mentis
desidia animiq; uagatione utili
manuū occupatione. ⁊ delectabili nouita-
tū meditatione declinare ⁊ calcare uo-
lentib; retributione celestis premii. Legi-
m' in exordio mundane creationis homi-
nē ad imaginē ⁊ similitudinē dī conditū.
⁊ inspiratione diuini spiraculi anima-
tū. tantēq; dignitatis excellentia ce-
teris animantib; p'rogatū. ut rationi
capax diuine prudentie cōsiliū. inge-
niūq; mereret' participiū arbitriūq;
libertate donat'. soli' cōditoris sui suspi-
ceret uoluntate. ⁊ reuereret' imperiū.
Q' astu diabolico misere decept'. licet
p'p' inobedientiē culpā p'ilegiū in
mortalitatis amiserit. tam scientiē et
intelligentiē dignitatē adeo in poster-
tatis p'paginē transtulit. ut q'cūq; curā
solicitudinēq; addiderit. toti' artis
ingenuq; capacitatē quasi hereditario
iure adipisci possit. Huimodi intentio-
nē humana suscipiens sollertia. ⁊ in-
diuersis actib; suis insistent' lucris ⁊
uoluptatib; p' tēpore incrementa. tandē
ad p'destinata xpiane religionis p'dux-
tēpora. factūq; est ut qd' ad laude et
glām nominis sui cōdidit dispositio di-
uina. in ei' obsequiū cūteret' plebs
dō deuota. Quap'p' qd' ad nrām usq;
etate sollers p'dēssōꝝ transtulit p'uisio.

adipisci. Q' adepto. nemo apud se q'si
ex se. ⁊ n' aliunde accepto glēt'. s; in dño
aq. ⁊ p'que omia ⁊ sine q' nih' humilit' gra-
tulet'. nec cēssa inuidiē sacculo recon-
dat. aut tenacis armariolo cordis occul-
tet. s; omi iactantia repulsa. hylari men-
te simplicit' querentib; erogat. metuatq;
euangelica illi' negotiatoris sententiā.
q' dño suo reassignare dissimulans.
maxoꝝ feneratā omi beneficio priuat'
oris sui iudicio nequā serui p'meruit no-
tā. Quā sententiā incurrere formidans
ego indign' ⁊ penē nulli' nominis hominum
tio. qd' in gratis cēssit. quē dat omib;
affluent'. ⁊ n' impperat diuina dignatio
cunctis humilit' discere desidantib; gra-
tis offero. ⁊ ut in me benignitatē dī recog-
noscant. largiatēq; mirent' admonēo.
et ut idē si opa addiderint sibi presto
esse. p'cul dubio credant insinuo. Sicut
enī homini qd'cūq; uentū aut indebitū
cuicūq; modi ambitione at tēptare siue
rapina usurpare iniquū est. ⁊ detesta-
bile sic iure debitū ⁊ ex patre dō here-
ditariū. intēptatū negligere. aut cōtēp-
tū ducere ignauie asseribit' ac stulti-
ciē. Tu q' q'cūq; ei' fili karissime cui d's
misit in cor capū latissimū diuersarū
artiū p'scrutari. ⁊ ut exinde qd' libue-
rit colligas intellectu curāq; appone-
re. n' uilipendas preciosa ⁊ utilia que
q'si ea t' sponte aut insperato domesti-
ca terra p'duxerit. q'a stult' negotiato-
est q' thesaurū subito fossa hōmo rep-
nerit. si illū colligere ⁊ seruare ne-

THEOPHILUS AND THE THEORY AND PRACTICE OF MEDIEVAL ART

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For my parents

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Benedictine abbeys of the Rhine, Meuse, and Weser river valleys. “Germania” by Andrei Nacu. Licensed under CC BY-SA 3.0 via Wikimedia Commons. Modifications by the author. 61

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ABBREVIATIONS

- BL British Library, London
- BnF Bibliothèque nationale de France, Paris
- BR/KB Bibliothèque royale de Belgique / Koninklijke
Bibliotheek van België
- CCCM Corpus Christianorum, Continuatio Mediae-
valis. Turnhout: Brepols, 1971–.
- CCSL Corpus Christianorum, Series Latina. Turn-
hout: Brepols, 1953–.
- DDA* Theophilus. *De diversis artibus*. Translated and
edited by C. R. Dodwell. New York: Thomas
Nelson and Sons, 1961.
- HAB Herzog-August Bibliothek, Wolfenbüttel
- MGH Monumenta Germaniae Historica
- MGH SS Monumenta Germaniae Historica, Scriptores
rerum germanicarum. Munich: Monumenta
Germaniae Historica, 1984–.
- ÖNB Österreichische Nationalbibliothek, Vienna
- PL Patrologia Latina. Edited by J.-P. Migne. Paris:
Garnier, 1844–55, 1862–65. Electronic reprint,
Alexandria, Va.: Chadwyck-Healey, 1995.

INTRODUCTION

*Theophilus—humble priest, servant of
the servants of God, unworthy of the
name and profession of monk—wishes
to all, who are willing to avoid and
spurn idleness and the shiftlessness
of the mind by the useful occupa-
tion of their hands and the agreeable
contemplation of new things, the
recompense of a heavenly reward!*

THEOPHILUS,
On Diverse Arts, prologue I

Little is known about medieval artists, and few texts directly addressing the subject of art survive from the high Middle Ages. Largely functional, often religious, and lacking textual documentation, the images and objects of the Middle Ages seem to be in a world of their own, a contrast to the images and objects of the era of “art.”¹ While the Middle Ages may serve as a foil to modernity in the popular imagination, the very characteristics that make medieval art something “other”—the lack of known artists, its functionality, its sacred character—have encouraged scholars to think about art in new ways. Concepts of originality, for example, or issues of physicality, performativity, agency, and power, have become subjects of study, enriching the field of art history as whole.² While it is clear that concepts of the artist and art-making were rather different in the Middle Ages, what precisely those notions were remains largely unknown. The medieval artist is still a mysterious, shadowy figure. This book seeks to shed light on the subject and to discover the ideas, principles, and driving forces of craft and of art-making in the central Middle Ages.

This book is an argument not for an art theory of the Middle Ages, but for a distinctly medieval theory of art. It is an investigation into a system of values according to which artists operated and made objects. Previous scholars have written much about medieval aesthetic

theory and the justification of imagery from a theological and Christian point of view, and this is invaluable.³ Precious little, however, is known about the very human act of making. Despite the paucity of sources and the lack of artists' names, I argue in this book that medieval art-making was a subject of thought, of theorization, and of speculation. It was an intellectual as well as a physical activity, a blend of the ideal and the human. At its best, it was performed according to a deliberate set of principles, values, and expectations.

One of few witnesses to the theory and practice of medieval art is *On Diverse Arts*, the only complete treatise on art to survive from the central Middle Ages. Composed by a monk under the pseudonym Theophilus in the early part of the twelfth century, the treatise comprises three books that describe the arts of painting, glass, and metalwork. In clear, direct Latin prose, the author explains procedures for color mixtures and figure painting, glass dyes and window assembly, and the casting of chalices and liturgical objects. Each book begins with a prologue addressing religious themes, such as God's creation of man or Solomon's building of the temple.

Because of its rarity, the treatise has been treasured by scholars, but it is often read out of context, according to modern expectations. The very first line of the text feeds assumptions of medieval modesty. Theophilus, meaning "lover of God," begins with a declaration of his subservience: "Theophilus—humble priest, servant of the servants of God, unworthy of the name and profession of monk—wishes to all . . . the recompense of a heavenly reward!"⁴ The instructions that follow have a strong practical bent and seem neither to obey any order nor to relate to the prologues. These characteristics have led many to liken the instructional chapters to cooking or

medical recipes and to see the treatise as a compilation with no order or internal organization. Read as piously anonymous, disorganized, and practical, *On Diverse Arts* has thus been understood to confirm the assumption that medieval artists were humble, uneducated craftsmen: it has contributed to its own myth.

It was the author and poet Gotthold Lessing who brought *On Diverse Arts* to the attention of the academic community in the eighteenth century, when he cited it as evidence that the technique of oil painting was known in Germany long before the age of Jan van Eyck.⁵ Lessing's argument was intended to counter Giorgio Vasari's *Lives of the Artists*, which credited Van Eyck with the invention of oil painting, but his interest in the text went much further. As Robert Leventhal has shown, Lessing was highly interested in the gaps between ancient and modern thought, in the fissures and fractious relations between the two.⁶ Yet it was Lessing's use of *On Diverse Arts* as a technical resource that set a pattern for later interpretations. In the next hundred years, the publications of Rudolph Raspe, Sir Charles Lock Eastlake, and Mary Merrifield brought the treatise to the attention of the English-speaking world, and their studies established Theophilus within a canon of texts treating artistic technique.⁷ French editions of *On Diverse Arts* were published around the same time.⁸ In German-speaking lands, it became a key text in the history and documentation of art through the work of the Vienna School, particularly that of Albert Ilg and Julius von Schlosser, who included Theophilus in their compilations and publications of primary source materials.⁹ These studies brought together a group of manuscripts and texts that addressed issues of artistic technique and categorized them as

“recipe books.” In addition to *On Diverse Arts*, these include the *Mappae clavicula* (Little key to techniques); the *De coloribus et artibus romanorum* (On the arts and colors of the Romans), by a mysterious figure known only as “Heraclius”; and the *Lumen animae* (Light of the spirit). The *Mappae clavicula*, like *On Diverse Arts*, contains instructions for mixing pigments, making metal alloys, and dyeing glass.¹⁰ *De coloribus et artibus romanorum* also contains instructions for mixing pigments and coloring glass, as well as gilding silver, but an emphasis on chemical constitutions gives the text an alchemical cast. The *Lumen animae*, thought to derive in part from *On Diverse Arts*, is more encyclopedic.¹¹ Similarities among these texts abound, and although scholars are still sorting out how the sources relate to one another, they are often discussed as a single genre.¹² On the whole, they have been invaluable for the history of technique.¹³ Technical treatises, or treatises containing technical information, however, need not only serve this purpose. As Bernard Cerquiglini and others have shown, the functions of medieval texts could be fluid.¹⁴ One of the central tenets of this book is that cultural values are embedded in technical discourse, and that the manuscripts themselves are witnesses to how texts were read and to the values and functions they served.

Although it is a precious resource for the history of technique, *On Diverse Arts* remains little understood. It is uncertain why, or for whom, it was written, or how it was read.¹⁵ Like the idea of the medieval artist, the text may appear transparent—with its clear and concise prose, its practical instructions, and its religious utterances—but it resists easy interpretation. Self-effacing prose and practical descriptions may seem straightforward, but tropes of humility

can belie considerable complexity.¹⁶ In addition, recent studies on the so-called minor arts have brought to light how greatly such small objects, like the censers, chalices, and book illuminations addressed by Theophilus, could evoke wonder for medieval viewers.¹⁷ Instructions, too, I argue, can reveal sophisticated value systems and agendas. Theophilus’s instructions contain significant gaps, often glossing over crucial steps of a procedure; his account of bronze casting, for example, says precious little about the important step of attaching sprues, or air vents, to the mold, making it difficult to believe the instructions were intended to teach on their own. Because images or other instructional diagrams are absent from the manuscripts, the text reads more as a literary work than an instruction pamphlet. Artists are implied, but not named; processes are described in detail, but images and final objects are not. These discontinuities put pressure on our assumptions about what an instruction manual should be and about what a medieval artist might be. While studies by Jeffrey Hamburger, Lawrence Nees, Robert Hanning, and others are invaluable and have done much to clarify ideas about medieval art-making, there is still much to be done.¹⁸

This study begins with manuscript evidence: The surviving copies of *On Diverse Arts* are direct witnesses to how the treatise was read in the twelfth century. When manuscripts are read as physical artifacts, with a sensitivity to the interaction of medium and content, new insights emerge.¹⁹ I will read the text not according to modern paradigms of technical analysis, but, as much as possible, according to medieval patterns of thought. Erwin Panofsky once wrote that art theory arose only in the Renaissance, when texts shifted from being concerned with “how to do” something, as in *On Diverse Arts*, to articulating

a philosophical position and methodological strategy, as in Leon Battista Alberti's humanist tract *On Painting*.²⁰ I argue otherwise. While it may be true that, by modern standards, there are no tracts on medieval art theory per se, in the twelfth century it was not unusual for practical questions to take on philosophical import.²¹ *On Diverse Arts* was of a piece with larger learning trends of its time, and the practical insights and philosophical content it yields are intertwined. Indeed, one of the fundamental premises of this study is that questions of "how"—instructions, recipes, practicalities—are not outside of philosophy but wholly subsumed within it.

For twelfth-century readers, particularly those educated in the monastic world in which Theophilus's treatise circulated, theory and practice existed in a dialectical relationship. Hugh of St. Victor, master and teacher at the Augustinian abbey of St. Victor in Paris and one of the most influential thinkers and teachers of the period, made it a point to explain the relationship of theory and practice in the *Didascalicon*, his treatise on education and reading. According to Hugh, the theoretical is that which is speculative, while the practical is that which is active, and although the two may appear to be different, theoretical speculation and practical action are both elements of the larger field of philosophy.²² This understanding of theory and practice resolves the seeming conflict between the religious prologues and practical instructions of *On Diverse Arts*, which have often been thought to serve entirely different agendas or even to have been written separately. Once we understand theory and practice as interrelated manifestations of philosophy, we begin to see the prologues and instructions as interdependent. Through them, *On Diverse Arts* describes an ideal

mode of behavior and set of actions: Proper practice is defined by theory, and theory is defined by proper practice. Neither exists without the other. It is a rather different conception from the theoretical treatises considered by Panofsky, but it is a philosophical framework nonetheless, and in this book I explore the medieval—and Theophilan—concepts of theory and practice. When we take the medieval, and particularly twelfth-century, ideas of theory and practice seriously and read prologues and instructions together, a cogent system of art begins to emerge.

It is a major contention of this book that the person of the artist, the practice of craft, and theories of art-making are historically situated; conceptions of artists and their work were by no means consistent through the many centuries that are dubbed the Middle Ages. Paul Binski has suggested that when "God himself appears as some sort of artificer, it is as a dividers-wielding architect or *conditor* . . . not as a goldsmith: saints or monks, not God, were workers in Gold."²³ While his differentiation of media is apt, such images of God as artist, as architect, or as creator in the artistic sense were fleshed out in the later Middle Ages, and did not exist in the earlier period. The change seems to accompany a shift in the status of artists, and while the contours of that shift are yet to be drawn, it seems likely that Theophilus's text was written in the midst of this changing world.

This study will thus approach *On Diverse Arts* within its own historical context. Scholars generally agree that the treatise was written in northern Germany, near Cologne or farther to the east, near Paderborn, and that Theophilus was a Benedictine monk.²⁴ It was probably composed in the first third of the twelfth century, or around 1120; the two earliest manuscript copies

of the text date to some decades later, around the middle of the century. Both have some connection to a Benedictine monastery, and both can be localized to the region stretching from the lower Rhine to the Weser river valley. These manuscripts will form the basis of this study.

The first, now in Wolfenbüttel, is known to have come from Cologne, for it contains an ex-libris from the Benedictine monastery of St. Pantaleon, one of the wealthiest and best-known abbeys of the city.²⁵ The second, now in Vienna, has a less certain provenance but was likely written in the region around Paderborn, a bishopric within the archdiocese of Mainz and home to the Benedictine abbey of Abdinghof.²⁶ A third manuscript, younger than the other two, is of even less certain provenance but continues the pattern. Once owned by Lord Harley and now in the British Library, it dates to the late twelfth century, and although its origins are unknown, its script is characteristic of northern Germany.²⁷

The next three copies show patterns of transmission. One, now in Brussels, dates to the end of the twelfth century or the beginning of the thirteenth and was written in the region of Liège.²⁸ The following two copies date to the early thirteenth century and come from England. The presence of early copies there is not surprising, since northern Germany and the Meuse valley had close trading ties with England in the period.²⁹ Thereafter, the geographical path of transmission is more difficult to trace. Three “nearly complete” manuscripts containing all three books and most of the prologues survive from the fourteenth and fifteenth centuries; they come from Saxony, the Rhineland, and Italy.³⁰ Five more cover a similar date range and contain at least two-thirds of the book, while another eight contain excerpts of varying length.³¹

Three existing copies date to the seventeenth century, and two to the nineteenth; all were made under the auspices of book collectors. Two of the seventeenth-century manuscripts are copies of the oldest twelfth-century manuscripts and were mostly likely produced at the library of the medical doctor and book collector Bernhard Rotendorff, who owned the Vienna manuscript.³² The third was made for Henry Wanley, advisor to the great collector Lord (Edward) Harley; it is a copy of a partial manuscript now in Cambridge.³³ Charles de l’Escalopier, who owned a fifteenth-century copy of the text, had two additional copies of the manuscript made, probably in preparation for his translation of Theophilus published in 1843. One copy, completed in 1841, was ordered by J.-Marie Guichard and written by a certain M. Baker; the other is a duplicate, a less fine copy of Baker’s text, which was included in a volume of the letters of Saint Paul and a text on glass. Both manuscripts are now in Amiens.³⁴ In all, twenty-seven manuscript copies of *On Diverse Arts* survive, and they range widely in date, completion, and place of origin.

The manuscripts have received relatively little scholarly attention; only recently has there been a growing awareness of the importance of evidence contained in the extant manuscript copies of the treatise.³⁵ This book focuses on the earliest surviving copies: the two oldest, at Wolfenbüttel and Vienna; Cambridge, University Library MS Ee. 6.39; and London, BL Harley MS 3915. These manuscripts reveal possibilities for how twelfth-century readers might have understood the treatise, particularly in the Benedictine sphere of northern Europe.

Theophilus has long been identified with the early twelfth-century goldsmith Roger of Helmarshausen. This attribution is largely based

SURVIVING MANUSCRIPTS

- † complete copy
- ‡ partial copy
- * excerpt
- ◆ manuscript is not from Theophilus, but is related
- ⌘ contains *mappae clavicula*
- key text for Dodwell's translation

6

Vienna, ÖNB Cod. 2527, mid-12th c. (Cologne?) †
Wolfenbüttel, HAB Cod. Guelph Gud. lat. 2°69, mid-12th c. (St. Pantaleon, Cologne) †
Paris, BnF MS lat. 6830F, 12th c. with 15th-c. additions ◆ ⌘

1200 — ~~Brussels, BR/KB MS 10147-58, 12th/13th c. (Liège) †~~
Paris, BnF MS lat. 11212, 12th/13th c. ◆ ⌘

London, BL Egerton MS 840a, Book I, 13th c. (England) ‡ ⌘
Cambridge, UL MS Ee. 6.39, Book I, excerpts from Book III, 13th c. (England?) ‡ ⌘
London, BL Harley MS 3915, early 13th c. (Münster?, 1444) † ⌘

1300 —
London, BL Sloane MS 1754, 14th c. (England) * ⌘
Leipzig, UB MS 1157, 14th c. (Saxony) †
Oxford, Magdalene Coll. MS 173, Book I, 14th c. ‡ ⌘
Munich, BSB Clm. 444, 14th c. * ⌘
Oxford, Corpus Christi Coll. MS 125, 14th c. ◆ ⌘
London, BL Harley MS 273, 14th c. * ⌘

1400 —
Montpellier, ÉM MS lat. 277, 14th/15th c. * ⌘
Florence, BN MS Palat. 951, Book I, 14th/15th c. ‡ ⌘
Paris, BnF MS nouv. acq. lat. 1422, 15th c. (Germany, Rhineland?) †
Paris, BnF MS lat. 6741, Book I, 1431 (Jean Le Bègue, Paris) ‡ ⌘
Wrocław, BU MS IV 8°9, 15th c. *
Wolfenbüttel, HAB Guelph Helmst. 1127, Book I, half of Book II, excerpts from Book III, 15th c. ‡ ⌘
Klosterneuburg, AC Cod. 331, Book I, half of Book II, 15th c. ‡

1500 — ~~Amiens, BM MS fonds Lescalopier 46, 15th/16th c. (Italy) †~~

1600 —
London, BL Sloane MS 781, Book I, excerpts from Book III (17th c., Henry Wanley, England) ‡ ⌘
Venice, BNM MS lat. VI, 199 (3597), 17th c. †
Vienna, ÖNB Cod. 11236, 17th c. †

on an inscription in the Vienna manuscript that reads *Theophilus, qui et Rogerus* (Theophilus, who is Roger).³⁶ Roger, in turn, is known through the survival of a document that probably dates to the late twelfth century but claims a date of 1100. Here Roger is named as the artist of a *scrinium*, or box, made for the bishop Henry of Werl of Paderborn. We do not know for certain whether Theophilus was actually Roger, and it is not the purpose of this study to determine the identity of Theophilus. The identification has in fact been problematic, for it encourages assumptions that the treatise is a practical resource, written by a craftsman with little sophistication.³⁷ Theophilus may well have been Roger; he may also have been someone in Roger's circle. The point is that *On Diverse Arts* is far more learned than previous scholars have recognized. The possibility that art-making and learned writing were not mutually exclusive is a driving force of this study. In order to keep as close as possible to the historical and geographical context of *On Diverse Arts*, objects attributed to Roger and to his milieu form the corpus of this book and serve as case studies to test and build upon the principles articulated in the treatise.

The Context

The Benedictine abbeys in which Theophilus's treatise circulated were some of the wealthiest and most learned institutions of the region. They were also closely tied to imperial powers and had long traditions of art production, as well as sizeable libraries and schools. The abbey of St. Pantaleon, for example, was founded by a monk from St. Maximian, in Trier, an abbey known for its production of manuscripts and

metalwork. It was also the burial site of the empress Theophanu and the archbishop Bruno of Cologne, brother of Otto the Great. In Paderborn, the Benedictine abbey of Abdinghof was founded by the bishop Meinwerk, named bishop of Paderborn by the emperor Henry II. Liège was home to the well-established monastic school of Saint-Laurent, where Rupert of Deutz studied and later taught before becoming abbot at St. Heribert's in Deutz, just outside of Cologne. One of Rupert's students in Liège was Wibald, who would become the abbot of Stavelot and a great patron of art. Stavelot was an independent principality, and Wibald served multiple roles in the empire. He was an advisor to the emperor Conrad III and Frederick Barbarossa and served as abbot to nearby Malmedy as well as Corvey, on the Weser, an abbey founded by Charlemagne. He was even, for a short time, abbot of Montecassino. Stavelot had been reformed in the eleventh century by the abbot Poppo, who had been appointed to the position by the emperor Henry II in 1020; in the following years Poppo's reforming activities spread, and he is credited for the reforms of the abbeys of St. Maximian in Trier, Saint-Laurent in Liège, and the imperial abbey of Echternach, in present-day Luxembourg.

The abbeys were thus enmeshed in a network that was not only religious but also political and economic. As Tjamke Snijders has shown, relations among personnel could also be fickle, changing quickly in an unstable political environment.³⁸ In such an environment, allies were all the more important. Connections between abbots, ecclesiasts, and the local aristocracy made possible the easy and frequent exchange of personnel, learning, and books and certainly facilitated the exchange of art objects, relics, and probably even artists themselves. In 1107, for

example, Helmarshausen acquired the relics of Saint Modoaldus, archbishop of Trier and advisor to King Dagobert of France, from the abbey of St. Maximian in Trier.³⁹ The procession passed through Cologne and St. Pantaleon, and a cross now in Cologne was perhaps made for the occasion by Roger of Helmarshausen. Roger, too, may have traveled, as his name is remembered at a number of monasteries along the route, including St. Pantaleon and the abbey of Helmarshausen.⁴⁰

The twelfth century was indeed a century of change. As Ute-Renate Blumenthal has pointed out, the imperial support of monasteries that took place in the eleventh century in the name of reform was also the basis of precisely the conflicts that were so problematic in the investiture crisis, and which continued to play out in the twelfth century.⁴¹ The abbey of Saint-Laurent in Liège, for example, went through periods of reform, as under Poppo, but because it was never placed under the authority of a mother house, as in the Cluniac system in France, the death of one charismatic, reform-minded leader did not necessarily mean that reform continued under his successor.⁴²

These Benedictine abbeys were thus deeply embedded in the political system and often had to take sides. In the late eleventh century, for instance, Henry II of Werl-Arnsberg was made bishop of Paderborn by Emperor Henry IV, in effect usurping the See from Henry I, count of Assel, who had been installed by Henry's rival Hermann of Salm, king of the Swabians and Saxons. In a shrewd political move, Henry of Werl seems to have then commissioned a portable altar, now in Paderborn, which, according to a charter of the late twelfth century, was made by Roger of Helmarshausen.⁴³ The altar depicts Henry of Werl as a successor to Bishop Meinwerk, who had been installed by Emperor Henry

II, but had in fact caused problems for the counts of Werl in his attempts to secure independence from lay control.⁴⁴ Probably made around 1105, the altar may be seen as an attempt to legitimize the position of the bishop and solidify ties to the imperial court. It thus highlights the shifting alliances and changing political power of the early twelfth century. In this context, the production of art and trade of relics could be usefully employed as political tools.

Pressures for monastic reform persisted into the twelfth century and came increasingly from outside the Benedictine realm. While the eleventh-century reform movements of Gorze and Cluny had largely been absorbed into the preexisting monastic fabric, the Cistercians and the Carthusians gained large numbers of converts in the twelfth century, in Germany and Belgium as well as in France.⁴⁵ Even if, as Van Engen has suggested, the Benedictine houses were less threatened by this second wave of reform, the Cistercians' growing presence in the Rhine-Meuse region in the twelfth century reflects the economic and spiritual pressures of the time.⁴⁶ The Cistercians were founded at Cîteaux in 1098, and by the early twelfth century their numbers were quickly increasing. Much of their growth was due to the conversion of older monastic houses—five abbeys in the region of Liège alone were founded or transferred to the Cistercian order in the span of just sixteen years, from 1132 to 1148. The abbey of Orval was a case of such transfer. Founded in 1070 by Calabrian Benedictines and home to a house of canons regular from around 1110, it was taken over by the Cistercians in 1132, becoming the first abbey of the order in Belgium.⁴⁷

The growing presence of Cistercian monasteries was noted by writers of the time. The

author of a small book known as the *Libellus de diversis ordinibus et professionibus qui sunt in Aecclesia*, for example, described the different monastic orders active in the region of Liège in the early twelfth century, including Cistercians alongside Benedictines, discussing the habits and strengths of each and treating the different orders with an even hand. The *Libellus de diversis ordinibus* and the case of the monastery of Orval reveal a rapidly changing monastic landscape in the region of the Rhine-Meuse. Yet as Giles Constable has shown in his edition of the *Libellus*, the book also reveals a surprising tolerance and awareness, an understanding that monasticism could take different forms and have different priorities, and a remarkable willingness to agree to disagree.⁴⁸

Theophilus probably knew of the Cistercian order, its asceticism, and its critique of art; indeed, *On Diverse Arts* was written in a context where the very notion of monasticism was a subject of discussion. But while some early scholars interpreted *On Diverse Arts* as a direct response to the growing Cistercian order, and even specifically to the *Apologia* of Saint Bernard of Clairvaux, there is no evidence that this was the case.⁴⁹ It is perhaps more accurate to view the two texts as responding to similar concerns in different ways. Many of the anxieties about art-making in the period extend back into antiquity and have to do with fundamental issues of Christianity and the status of the image and the visibility of God. As Conrad Rudolph has shown, Bernard's objections to art were not only doctrinal but also economic, social, and spiritual.⁵⁰ Theophilus's treatise takes on some of the same objections to art. He answers them, however, quite differently, particularly with regard to devotional practice, liturgy, and spirituality.

On Diverse Arts may not be a direct response to Bernard of Clairvaux, but it is part of the broader conversation about the status and function of art and its production.

The overarching context of debates about art is closely related to economic changes that were occurring all over Europe, and especially in northern Europe, in the twelfth century. There were two relevant economic shifts, no doubt related: a growth in the market economy and a change in the status of the artist. It is well known that the Benedictine abbeys had long been large landowners, with enormous power and prestige in what was a primarily agrarian economy. Accordingly, many artisans and laymen were doing business mainly on behalf of the local abbey. As the twelfth century progressed, however, trade became increasingly weighted in cities, and merchants and tradesmen did business on their own.⁵¹

The city of Liège is a good example. In the tenth and eleventh centuries, Liège grew significantly. A new city wall built by the bishop Notger around the year 1000 encompassed both the old "Holy City," with the cathedral and bishops' palace, as well as a newer part of town to the south. Known as the *vicus novus*, this neighborhood was a commercial district that bordered the market, effectively expanding it. The district must have been successful, because merchants from the *vicus novus* in Liège are mentioned in a late tenth-century toll tariff of London and an eleventh-century toll tariff from Koblenz; these merchants were known, too, at the markets of Cologne. In 1107 the merchants of Liège even gained their own market prerogative, or *judicium forense*, which allowed them further independence and mobility.⁵² The nearby city of Huy rivaled Liège in economic activity. Huy

established its own city charter in 1066 and purchased a “limitation of seignorial rights” from the bishop. It had thus been home to merchants far earlier, and with its river access it became a center for trade. Clearly the region was undergoing significant economic change in the late eleventh century and into the twelfth. These developments were both a boon to the monasteries and a threat, for as the cities grew, the abbeys gradually lost their economic control.⁵³

At the same time, there is evidence that in the early part of the twelfth century, artists moved from one abbey to another, as Roger of Helmarshausen seems to have done. Few names are known, but workshop styles emerge and disperse. The characteristic V-fold style attributed to Roger seems to reach its height at the abbey of Helmarshausen in the first decades of the century, and reminiscences of that style may be detected in later objects like the shrine of Godehard at Hildesheim and in manuscripts made for the cathedral of Lund, in Norway.⁵⁴ Most evidence that survives for the names of twelfth-century artists dates from the later part of the century, even if it refers to artists from many decades earlier. This is the case with Roger, whose name and profession is known only from the late twelfth-century charter describing the altar of Henry of Werl. By the end of the twelfth century, more artists’ names survive. As Enrico Castelnuovo has aptly described, the task of studying the persona of the artist is fraught with uncertainty.⁵⁵ The artisan Godefried, of Huy, for example, was remembered in the late twelfth-century necrology of the abbey of Neufmoutier as “Godefried, goldsmith, our brother . . . was burgher of Huy, later he became a canon and our brother. He was not superseded in the art of metalwork, by any artist of his time; he made a large number

of reliquaries, shrines of saints in different lands, vases and other objects for the use of kings.”⁵⁶ The economic history and scattered evidence of artists’ names, then, seems to suggest that over the course of the twelfth century, art-making became increasingly professionalized, practiced more and more by artisans in burgher towns like Huy or cosmopolitan centers like Liège. The cultural values and the historical context that drove these developments is less well understood, as is the shift in conception of the artist. *On Diverse Arts* is a precious resource in this regard, for it seems likely to have been written precisely in this era and may even have been precipitated by these changes. It is an articulation of the values of artisanship in a shifting world.

Twelfth-century trends in learning and education also affected the kinds of texts being written and the nature of those texts. There have been many explanations for these changes in learning, for while they were partly a response to a new influx of texts from the East, they were also fostered by the needs of growing cities, with their new mercantile infrastructure, and the political upheavals and tensions of the investiture crisis. Cathedral schools trained new generations of ecclesiastical officers, and by the mid-twelfth century schools were becoming universities, as at Paris in 1160 and Montpellier in 1180.

The Rhine-Meuse region had been famous for its schools and masters since the Carolingian period. Liège in particular was home to a famous cathedral school in the eleventh century and into the beginning of the twelfth, a tradition begun under the master Notker of Liège and continued by his pupil Wazo of Liège. In the twelfth century it was also home to Reimbald of Liège, canon of the cathedral and possibly the author of the *Libellus de diversis ordinibus*. The nearby

Benedictine abbey of Saint-Laurent had its own lineage of scholars: Adelmann, a student of Fulbert of Chartres, was the teacher of Lambert of Liège, who wrote the first version of the *vita* of Saint Heribert, later to be revised by Rupert of Deutz, himself a student of the monastic school.⁵⁷ Figures like Fulbert and Wazo were renowned for their virtue, manners, and eloquence; as C. Stephen Jaeger has shown, these men were some of the great exemplars of eleventh-century charismatic teachers.⁵⁸ Rupert, too, has long been seen as continuing this tradition, though his work is interesting precisely because it links together the concern for *mores* and *virtu* with the rigorous allegories of early scholasticism. Theophilus's fine Latin may be just one piece of evidence that he was educated in this milieu; tensions within *On Diverse Arts* may also echo this moment of transition.

Seen within the monastic, literary, and pedagogical sphere of the twelfth century, *On Diverse Arts* emerges as an articulation of a particular mode of thought, an agent and expression of a twelfth-century world. Echoes and insights emerge from reading *On Diverse Arts* alongside works by other scholastic writers of the period, particularly Hugh of St. Victor and Rupert of Deutz. Both Hugh and Rupert were well known in the region as writers and as teachers—their texts circulated among local abbeys—and Rupert is thought to have been involved in art as well.⁵⁹ Situated within the discursive structures of twelfth-century thought, Theophilus's practice of writing, his mode of thinking, and his manner of describing techniques shed new light on twelfth-century values of art and art-making.

In the midst of political, economic, and religious uncertainty, Theophilus defines and defends artistic production. *On Diverse Arts*

reveals the pressures of a twelfth-century world undergoing significant cultural change, and these pressures may well have been the motive for the composition of the text. Theophilus's treatise is neither recipe book nor religious polemic; his voice is distinct, both in its mode of argument and its ideas. It is precisely by aligning theory and practice that the treatise assigns moral value to the materiality, function, and labor of sacred art. As we shall see, larger questions of the status of art are revealed in smaller questions about technique and spirituality. A re-reading of *On Diverse Arts* suggests a new mode of organizing our categories of twelfth-century art; it also suggests that while art operated primarily on a local level, it was nonetheless part of a much larger cultural discourse. Therefore even inconsistencies and oddities in *On Diverse Arts* are informative, for they reveal a changing world of artistic practice, foreshadowing larger transformations to come.

Overview of the Chapters

This book is organized around a series of manuscript readings. Evidence from these rich volumes guides a search for correspondences and patterns within the text, bringing to light the structures organizing the treatise and its major concerns: pedagogy, materials, labor, form, and vision. An exploration of these themes guides each chapter. Together, the themes shed light on artistic practice in the period.

The first chapter introduces the structure and narrative of Theophilus's text and argues against the traditional reading of *On Diverse Arts* as a "recipe book." The examination begins with the Wolfenbüttel manuscript, in which an

eleventh-century copy of Vitruvius's *On Architecture* is followed by Theophilus's *On Diverse Arts*. This copy of Theophilus's text creates a harmony with that of Vitruvius, suggesting the treatise be read within a broad context of Latin literature. Theophilus's fundamental project, I find, is to define art as a learned and spiritual discipline. I show that the treatise follows a careful structure and that its narrative echoes Augustinian thought, while Theophilus's use of prologues and his embrace of pedagogy places him in a literary tradition that builds on classical thought and may be seen as parallel to Hugh of St. Victor and John of Salisbury. *On Diverse Arts* therefore emerges as part of a scholarly endeavor that links the "moderns" to the ancients, appropriating sources from antiquity and systematically defining a philosophy of artistic practice.

In the second chapter, I suggest that, for Theophilus, art is the transformation of material and is justified by the notion of *utilitas*, or usefulness. Textual evidence links Theophilus's interest in material to a literary tradition stretching from Pliny to Isidore of Seville, while manuscript evidence suggests we read Theophilus in the context of twelfth-century texts like the *Liber Floridus*. His emphasis on metals also has great local significance, since abbeys like Stavelot and Helmarshausen were engaged in mining and the metals trade. Theophilus's treatise aims to justify a profoundly materialist practice as a religious enterprise. It was written just as European culture was experiencing an increasing split between religion and commerce, with urban mercantile economies growing rapidly and ascetic monastic communities becoming increasingly numerous and vociferous. Theophilus declares mining and materialism to be sacrosanct and suggests that art serves God because it transforms earthly

materials into useful, even sacred, objects. His argument reaches beyond traditional defenses and brings new complexity to the monastic movement's justification of wealth.

The third chapter looks at Theophilus's concept of labor and the spirituality of manual work. His instructions, usually dismissed as only technical, are shown to be carefully written, following patterns of exegetical thought and taking much from Saint Augustine. Theophilus defines skill in terms of knowledgeable and virtuous action so that art-making becomes the exercise of rationality, morality, and free will—it helps man exercise the full potential of his humanity and restore him to God. Invoking the biblical Beseleel as a typological precedent, Theophilus inverts the usual criticism of Jewish embellishment; the objects made by Beseleel pleased God because his skill was a manifestation of the work of the Holy Spirit. Most scholars have understood Theophilus's allusion to man as *imago Dei* and his invocation of Beseleel as straightforward justifications of art through biblical exemplars. I see it rather differently. Both allusions are in fact integral parts of a far more sophisticated theory of production, where art is justified by the process through which it is made. At a time when the old Benedictine agrarian and imperial patronage system was facing competition from a new mercantile economy and increasing numbers of lay artisans, Theophilus created a theory of artistic work based on sophisticated Augustinian and Benedictine spirituality. *On Diverse Arts* changes our very conception of the art object; it is no longer a finished, self-contained thing but an analogy for man's fallen status and the process of restoration.

The fourth chapter examines how such an emphasis on technique might have repercussions

for a sense of form. Beginning with the Vienna manuscript, the chapter looks at the values of variety, moderation, and decorum espoused by Theophilus and how they are made manifest in objects attributed to Roger of Helmarshausen. From this examination, we begin to see that Theophilus's reticence regarding iconography and his vague allusions to ornament tell us much about the parameters of art-making. Abstract formal qualities take on new meaning as evidence of workshop consistency, traces of good practice, and witnesses to ingenuity.

Taking each theme in turn, the book slowly builds a theory of art that changes how we understand Theophilus's treatise, medieval liturgical art, the history of art theory, and the story of artists. Objects emerge throughout as test cases and take center stage in chapter 4. In the conclusion, I bring the themes together and clarify Theophilus's broad concepts of artistic production, proposing a corresponding theory of vision. I confront the assumption that medieval art had no discourse, or that it is to be understood primarily in anthropological or socio-historical terms. Jeffrey Hamburger has recently noted that theology and visual artifacts constitute separate discourses; images do something far more than visualize theology. Rather than attempting to make the two equivalent, or make one explicate the other, Hamburger suggests we view theology and religious discourse as "less a body of doctrine than itself a variety of methods."⁶⁰ Similarly, through *On Diverse Arts*, we discern how ideas about theology, learning, and materiality echo and speak to one another in the making of an object. *On Diverse Arts* articulates an orderly structure for the discipline of art, creating an ideal and establishing the terms by which it is to be made. Material, labor, and modes of working

are all deeply theorized and self-consciously understood. Furnishings of the church, for example, are not simply embellishments or luxuries; they also carry spiritual, cosmological, and institutional significance. Through the lens of *On Diverse Arts*, the art object begins to be seen as the result of the process by which it is made.

Theophilus's justification of art goes far beyond what scholars have heretofore seen. His treatise is embedded in the exegetical and pedagogical philosophy of the time, laden with knowledge of the ancients. He proves that art is part of the realm of learning, and he shows us that medieval artistic practice was indeed self-aware, even self-serving. Understood as a part of the larger intellectual context of the twelfth century, and read as a learned treatise, *On Diverse Arts* helps us to see medieval art on its own terms.

1.

PEDAGOGY AND EXEGESIS

*Be eager and anxious to look at this
little work on the various arts, read it
through with a retentive memory, and
cherish it with a warm affection.*

THEOPHILUS,
On Diverse Arts, prologue I

One of oldest surviving copies of Theophilus's *On Diverse Arts* is now kept in the Herzog-August Bibliothek in Wolfenbüttel, Germany. Dating to the mid-twelfth century, it has a large four-line initial, two columns of text, and wide margins (plate 1). It is neatly written on fine parchment, and the surface of the page is polished and clean. Famous because it is one of the most complete copies of *On Diverse Arts*, it is also the most elegant. As is common for texts of the period, the treatise is not bound alone. Less common is the fact that it is bound with a portion of the ancient Roman treatise *On Architecture*, composed by Marcus Vitruvius Pollo in the first century A.D., probably copied about a century earlier (fig. 1). A comparison of the texts suggests distinct efforts to make the presentation of *On Diverse Arts* consistent with that of the earlier treatise. Both are transcribed in two columns, with a gutter margin matching in width, and both contain roughly the same number of lines per page—thirty-four lines in *On Architecture*, thirty-eight in *On Diverse Arts*.¹ Vitruvius's text ends a quire, and Theophilus's text begins a new one on the facing recto, its elegant, rounded, four-line black initial "T" a visual echo of the four-line black initial "D" on the opening folio of *On Architecture*. From the first folio to the last, the volume appears to be a coherent unit.

It seems likely that the two tracts were copied in the same scriptorium. A similar high-grade

M. VITRUVII DE ARCHITECTURA LIBRI.

INCIPIT LIBER I.

Dum diuina tua mens
 et nunc imperator celsus
 no potitur orbis terrarum in
 me dāq; utriusq; cunctis hostib; statim
 triumpho me totaq; tuacius gloria
 remur et genus omne subactū tuū spe
 tantū natum. populūq; romanū et
 senatus liberatus timore amplissimū
 tuū cognitionibus consiliūq; guber
 naretur. non audeb; timor occupatio
 nibus de architec tura scripta et mag
 nis cognitionibus explicata edere. me
 tuens non nāp tē tempore interpel
 lans subire tui animi offensione.
 Cum uero attendere te non solum deu
 ti communi omnium cura publice q; re
 constitutione habere. sed etia de por
 tunitate publicorū aedificiorū. ut ci
 uitas pte non solum prouincis esset
 ducba. uerū etia ut maiestas impeni
 publicorū aedificiorū congruis habere
 duc bonitatis. Non putauit pretermi
 tendū quin primū quoq; tempore de
 his rebus tibi edere. Ideo quod primū
 parenti uo deo fuerat notus. et eius
 uirtutū studiosus. Cum aut conciliū
 celestiu in sedib; immortalitatis eū de
 deauisset et imperiū parentis intua
 potestate transisset. idē studium
 meū in eius mētib; pmanens intecon

ulit. itaq; cum a concilio
 amittio. et q; concilio ad p
 tione ballistarū et scorpionū reliquo
 rumq; tormentorū refectioe fin
 presto. et tuis commodā accepti. Quicq;
 primo mihi inuisti. recognitionem
 psonis commendatione seruasti. Cu
 ergo eo beneficio esse obligatus ut ad
 extitū uir non habere inopie timo
 re. hec tibi scribere cepi qd animad
 uerti multa te edificauisse et nunc
 edificare. Reliquo quoq; tempore
 et publicorū et priuatorū edificia
 suā pampitudine rerū gestarū.
 ut posteris memorie traderentur.
 cura habiturū. Conscripsi pscriptio
 nes terminatas ut eas adtendens
 et ante facia et futura qualia sint
 opera pte posses nota habere. Nāq;
 his uoluntatibus apertū omis disci
 pline rationes

architec tura scientia plurib; discipli
 nis et uariis eruditionib; ornata. cui
 iudicio pbant oia que abegeris
 artib; pficiunt opera. canascentur
 ex fabrica et rationatione.

abrica et continuata ac tria usus me
 ditatio. que manib; pficit. Materiam unius
 cuiuscūq; generis opus ead ppositū
 deformationis. Rationatio autē
 quens fabricatas sollertie ac ratio

de archi
 tectura
 inueniēdis.

re
 oculte ac di
 ligent ipse

parchment is used for both, and while some minimal oil and dirt on the first pages of Theophilus suggest they were thumbed through, all the pages are relatively clean. Surprisingly for such a fine codex, the quires are irregular: bifolia have been cut, as revealed by stubs within the quires or between the quires. Even though the texts differ on paleographical grounds, the irregular quires may suggest a peculiar practice of assembly at a particular scriptorium. The volume is now bound in a sixteenth-century cover, but there is every reason to believe that the texts were bound together from the mid-twelfth century, when *On Diverse Arts* was transcribed.

The Wolfenbüttel manuscript is the largest and finest of any extant manuscript of *On Diverse Arts*. The copy of *On Architecture* it contains is also one of the most accurate and complete copies of that tract, and it has been used for many modern editions. Analysis of the script has led most scholars to localize the hands to the region of Cologne, if not to Cologne itself.² A thirteenth- or fourteenth-century inscription, now largely obscured by a large eighteenth-century title, supports this hypothesis. Reading “Codex mon[asterii] s[an]c[t]i pantaleonis in Colonia,” it indicates that, from at least the later Middle Ages, the manuscript was kept at the abbey of St. Pantaleon in Cologne.³ St. Pantaleon was one of the most prominent abbeys of medieval Germany, with one of the most renowned libraries. Indeed, it seems certain the book was meant for a library. There are no drawings anywhere in the manuscript, and notations are textual: a reader has underlined certain

phrases of Vitruvius and made corrections to the Latin. The consistency across the two tracts, the fine nature of the volume, and the evidence of its prior life at the abbey of St. Pantaleon change how we read *On Diverse Arts*. Like the treatise on architecture, *On Diverse Arts* was not just a haphazard collection of recipes but part of a scholarly tradition.

Cologne was renowned for its cathedral school and monasteries, and St. Pantaleon had one of the largest monastic scriptoria and libraries of the region.⁴ While the inscription localizing the Wolfenbüttel manuscript to the abbey is somewhat later, it is nearly certain that Vitruvius’s *On Architecture* was available for study at St. Pantaleon and in greater Cologne in the twelfth century. Some seventy-eight medieval manuscripts of Vitruvius survive, of which four have been localized to Cologne: a ninth-century manuscript now in London, an eleventh-century copy in Brussels, a twelfth-century copy now in the Vatican, and our eleventh-century copy in Wolfenbüttel, containing *On Diverse Arts*.⁵

The oldest known copy of Vitruvius to survive is the ninth-century Carolingian copy, now preserved in London as British Library Harley MS 2767. Like the Wolfenbüttel manuscript, this copy has been localized to Cologne on the basis of its script. It has also been localized to St. Pantaleon, due to an inscription on an otherwise blank folio that reads “Goderamnus propositus” and is placed near a drawing of a cross. Goderamnus was head of the abbey of St. Pantaleon in the 1020s and later an abbot of St. Michael’s in Hildesheim.⁶ If the script and name of Goderamnus mean that the Harley manuscript was at St. Pantaleon’s in the eleventh century, the slightly younger Wolfenbüttel copy would have been a duplicate. Curiously, however, the

FIG. 1

Incipit Vitruvius, *De architectura*, eleventh century. Wolfenbüttel, Herzog-August Bibliothek Cod. Guelph Gudianus lat. 2°69, fol. 1r.

Wolfenbüttel copy does not seem to have been copied from the London manuscript; each represents one of the earliest recensions of the text.⁷ Did Goderamnus take his copy of Vitruvius with him to Hildesheim, so that St. Pantaleon had to obtain another? It is unknown how and why St. Pantaleon would have two copies and two recensions of the same text, but it seems possible that by the twelfth century they did. What is clear, regardless, is that throughout the twelfth century Vitruvius continued to be read, copied, and studied, especially in Cologne.

On Architecture was indeed much admired in the Middle Ages. Many of the great libraries had copies of the treatise, including Fulda, Melk, Murbach, and St. Gall. There is evidence, too, that Vitruvius was read by some of the most prominent writers of the day: he is quoted by Isidore of Seville, referred to by Bede, discussed by Alcuin of York and Hrabanus Maurus, and cited by Hugh of St. Victor and William of Malmesbury.⁸ Goderamnus seems to have read the text carefully as well, for his hand has been identified in the upper margins of four folia, noting words and phrases in the text.⁹ His notes suggest that his interest in the treatise lay in its lexicon.

Markings in the Wolfenbüttel manuscript also suggest that Vitruvius's text was read carefully. Brackets and underlinings throughout call attention to particular passages, such as those in which the Roman author considered the quality of the air and water as well as the burial of the dead.¹⁰ Interest in such environmental issues is not unusual in the Middle Ages. Hucbald of St. Amand, for example, cites Vitruvius as a source of information on the nature of the eight winds.¹¹ The London Vitruvius, too, contains a diagram of the winds.¹² Medieval readers clearly valued *On Architecture* as a text on architectural theory

more than a set of technical prescriptions. As Stefan Schuler has argued, Vitruvius's treatise was esteemed by medieval readers because it defined characteristics of an architecture and an urban space that would make the best use of the surrounding natural environment and best serve communal needs.¹³ Like Hucbald of St. Amand, the reader of the Wolfenbüttel manuscript seems to have been reading the treatise for information on the harmonious coexistence of man and nature.

Parallels between the two texts extend beyond the peculiarities of the Wolfenbüttel transcription. Theophilus was well versed in Latin; as a Benedictine priest, he was probably well educated. Given his interest in treatises on the arts and the genre of didactic literature, he would have certainly been familiar with Vitruvius's text on architecture. Indeed the two treatises have striking similarities. For example, one of the most puzzling and distinctive characteristics of *On Diverse Arts* is its structure, which alternates between prologues and instructions. This is unusual in medieval texts, where a single prologue before the entire text usually suffices.¹⁴ *On Architecture* also alternates prefaces and instructions. Moreover, while Vitruvius's treatise concerns architecture, Theophilus is concerned with what is within buildings—the furnishings of a church—thus expanding the content of the ancient text and changing the focus to Christian function.

The Wolfenbüttel manuscript seems designed to show continuity between the ancient and medieval texts. The verso containing the final words of Vitruvius's tract ends with an explicit, written in fine capitals, and a prayer: "Vitruvii liber x explicit feliciter amen" (fig. 2).¹⁵ Opposite, *On Diverse Arts* begins; it is on a new page,

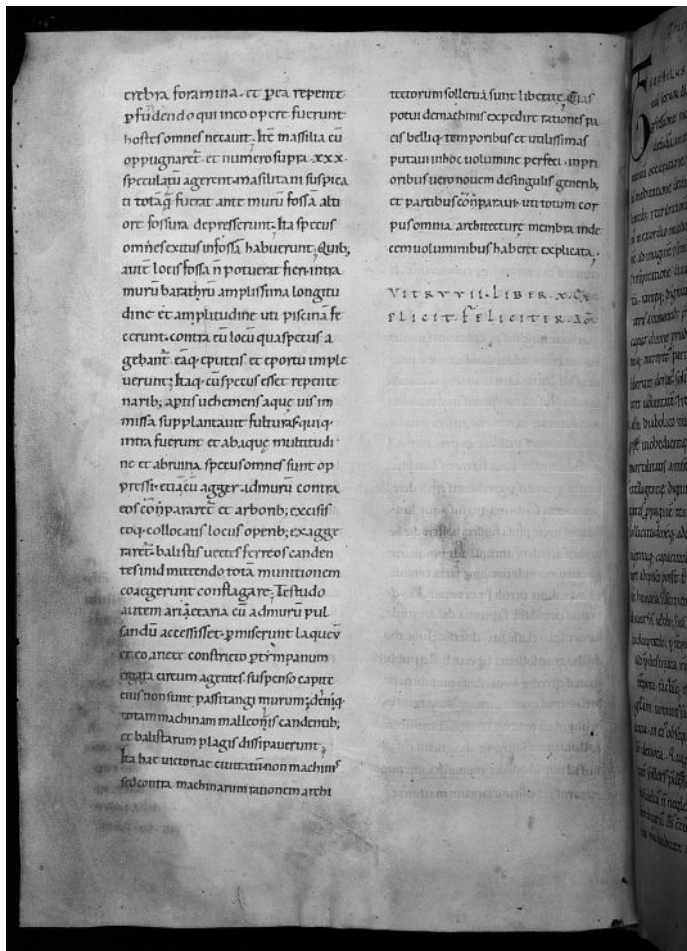


FIG. 2

Explicit Vitruvius, *De architectura*, eleventh century.
Wolfenbüttel, Herzog-August Bibliothek Cod.
Guelph Gudianus lat. 2°69, fol. 85v.

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in a new gathering, with the mark of a tab on the right column for easy navigation. Yet *On Diverse Arts* begins with its own supplication—“Theophilus Presbyter, servus servorum Dei . . .”—and here, too, the name of Theophilus is written in Roman capitals, as if to match the explicit of the preceding page. The rubrics, titles, and mise-en-page make the book flow from ancient to modern, while at the same time they present the twelfth-century treatise as a discrete text.

The goals of the two authors, and their modes of writing, run parallel. Both Vitruvius

and Theophilus use the prologues to provide an overview of the text that follows, framing each body of information according to its purpose and the principles on which it rests.¹⁶ Each author compiles in a single treatise a large body of knowledge on the practice of an art and explains the principles governing it. As Vitruvius describes it, “the several arts are composed of two things—craftsmanship and the theory of it. Of these the one, craftsmanship, is proper to those who are trained in the several arts, namely, the execution of the work; the other, namely,

theory, is shared with educated persons.”¹⁷ By explaining both abstract principles and concrete techniques, each also sought to draw broad conclusions about the nature of his art and its purpose. First, Vitruvius’s text, dedicated to the emperor Augustus, shows architecture to be part of a political sphere, while Theophilus’s treatise creates a framework for religious art that is distinctly monastic. Second, and perhaps more importantly, *On Architecture* was understood by medieval readers as a text that defined the knowledge of architecture as an essential aspect of knowledge about the world. As we will see through the course of this study, *On Diverse Arts* had a similar ambition.

It is not surprising, then, that the Wolfenbüttel volume contains none of the so-called recipe books with which *On Diverse Arts* is so often compared. Instead, it stages a smooth flow from ancient to “modern,” from Vitruvius’s descriptions of the placement and construction of pagan structures to Theophilus’s descriptions of the furnishing of the church, taking the reader step by step through painting, stained glass, and the sacred arts. It is as though *On Diverse Arts* was designed to follow the ancient treatise, completing it and Christianizing it. Because the Wolfenbüttel manuscript is a second-generation copy of *On Diverse Arts*, it is impossible to know if *On Architecture* was indeed a model for Theophilus, or if a subsequent copyist found enticing parallels. Regardless, readers of the Wolfenbüttel manuscript cannot have missed the connection between the two texts, underscored by their visual likeness.

The continuity between Vitruvius and Theophilus is an example of the practice of building upon the authority of the ancients, a practice of which twelfth-century writers were particularly fond.

Though the ancient authors were read throughout the Middle Ages, how those texts were read could vary. Schuler has shown that Carolingian readers like Godramnus were particularly interested in understanding the Latin text on its own, making notes of vocabulary and phrases.¹⁸ An eleventh-century reader of the Wolfenbüttel copy of *On Architecture* seems to have had similar motives, for he underlined phrases and words of the text. This may suggest a mode of reading that is more concerned with form, with words, than with content per se. A ninth-century catalogue from the library of the monastery of Murbach in southern Germany, meanwhile, contains the earliest known mention of a copy of *On Architecture* in a monastic library. Here the compiler treats the ancients as a separate category: Vitruvius is included alongside Sallust and Seneca under the category “Gentiles” and “Poetae gentilium.”¹⁹ In the twelfth century, however, catalogues tend to group works by subject, especially when dealing with texts on arts, sciences, and metals. The 1049 catalogue of the library of Lobbes, for example, groups medical tracts together, including Pliny among them; *florilegia*, books that brought together extracts of Latin authors in a compendium, are organized in some similar ways.²⁰ The categorizations are reflected in the contents of the manuscripts themselves: a manuscript in Avranches, for instance, includes texts of Ptolemy with Martianus Capella’s book on astronomy, while two manuscripts in Paris, both containing excerpts of texts closely related to Theophilus, bind Palladius and Varro with excerpts from the *Mappae clavicula*.²¹

Increasingly in the twelfth century, writers would compose texts explicitly modeled on ancient ones: the *Metalogicon* of twelfth-century pedagogue John of Salisbury is a defense of the study of Latin grammar and logic, for example,

and he unequivocally frames his project as part of the antique Latin tradition. He writes, “This treatise . . . is entitled *The Metalogicon*. For in it, I undertake to defend logic. According to the wont of writers, I have included various points, which each reader is at liberty to accept or reject as he sees fit: ‘Some things you will read herein are excellent, some mediocre, and several defective; But this is inevitable—as otherwise, dear Avitus, there would be no book.’ So says Martial, and I echo him.”²² John notes his admiration for contemporary writers but also aligns himself with the antique Latin authors, whose words he constantly quotes throughout his treatise. Another example of this, pointed out by Marcia Colish, is John’s *Education of Trajan*, which he authored but attributes to Plutarch, thereby creating a “new” ancient text.²³ John’s comment is fascinating because it is at once a claim to humility and imperfection and also an assertion of accomplishment. John does not decline to write for fear that it will not be perfect, nor does he place himself on a par with Martial; instead, he looks to the ancient author and his philosophy of writing as a loose model upon which he can base his own work.

The abbot Wibald of Stavelot, a rough contemporary of Theophilus, quite famously sought to compile all the known works of Cicero into one volume.²⁴ In Paris, Hugh of St. Victor urged students to read full versions of classical texts rather than short summaries. Significantly, in his treatise on education, Hugh made a point of telling his reader to pay attention to both the structures of texts and to their narration.²⁵ Classical texts, then, were admired and emulated for their overarching structure and adherence to rhetoric. Even *florilegia* attest to a desire to collect ancient texts, to interpret them, and to

use them as models.²⁶ As John of Salisbury’s comment makes abundantly clear, ancient texts were a base on which to construct one’s own work. It seems plausible to read *On Diverse Arts* as doing just this. In turn, it might be reasonable to see the entire manuscript as a microcosm for the changes in learning that occurred from the eleventh to the twelfth centuries.

Copied as a library volume, the Wolfenbüttel manuscript guides us to read *On Diverse Arts* as a piece of literature, a continuation of Vitruvius, and a medieval answer to the ancient treatise. It seems to have been written in the mold of *On Architecture* in terms of content, structure, and visual layout. The relationship between the two texts changes our understanding of *On Diverse Arts*, leading us to read it as a text with structure, narrative, and a learned sophistication. While scholars have long recognized these attributes in twelfth-century works like those of John of Salisbury, they have not historically seen them in Theophilus. Just as his contemporaries were reading the ancients and interpreting them for a Christian context, so too Theophilus reorganized received knowledge, added to it, and Christianized it, building on ancient learning to create a Christian theory of art that accorded with the *topoi* of monastic learning. As we will see, *On Diverse Arts* may be understood as an attempt to make technical and theoretical knowledge converge; like Vitruvius, the author set out to provide an orderly account of craftsmanship and to theorize the practice of art.

Narrative Structure in *On Diverse Arts*

One of the most famous passages from *On Diverse Arts* is from the third prologue. Here,

Theophilus shows how the seven gifts of the Holy Spirit may be visited upon the artisan:

I will clearly demonstrate that whatever you can learn, understand or devise is ministered to you by the grace of the seven-fold spirit.

Through the spirit of wisdom, you know that all created things proceed from God, and without Him nothing is.

Through the spirit of understanding, you have received the capacity for skill—the order, variety and measure with which to pursue your varied work.

Through the spirit of counsel, you do not bury your talent given you by God, but, by openly working and teaching in all humility, you display it faithfully to those wishing to understand.

Through the spirit of fortitude, you drive away all the torpor of sloth, and whatever you assay with energy you bring with full vigour to completion.

Through the spirit of knowledge accorded you, you are, in the abundance of your heart, the master of your skill and, with the confidence of a full mind, employ that abundance for the public good.

Through the spirit of godliness, you regulate with pious care the nature, the purpose, the time, measure and method of the work and the amount of the reward, lest the vice of avarice or cupidity steal in.

Through the spirit of the fear of the Lord, you remember that you can do nothing of yourself, you reflect that you have or intend nothing, unless accorded by God, but by believing, by acknowledging and rendering thanks, you ascribe to the divine compassion whatever you know, or are, or are able to be.²⁷

Many scholars have noted how Theophilus's recitation of the gifts seems to justify the production of the luxury arts in the monastery.²⁸ Yet they do far more. As he lists the seven gifts of the Holy Spirit, Theophilus is drawing upon a Christian interpretation of the flower of the tree of Jesse and the many talents of the Spirit of the Lord, as it is recounted in the book of Isaiah: "And there shall come forth a rod out of the root of Jesse, and a flower shall rise up out of his root. And the spirit of the Lord shall rest upon him: the spirit of wisdom, and of understanding, the spirit of counsel, and of fortitude, the spirit of knowledge, and of godliness. And he shall be filled with the spirit of the fear of the Lord."²⁹ The idea that the gifts of the Holy Spirit were a series of virtues played a significant role in twelfth-century monastic learning and spirituality. The passage points to a larger theme that structures the entire treatise.

Saint Augustine expounded on the gifts in his sermon on the Sermon on the Mount, an important exegetical tract in the Middle Ages. Linking together the Sermon on the Mount (Matthew 5:1–20), and the passage in Isaiah, Augustine aligned the gifts of the Holy Spirit with the beatitudes named by Christ in the sermon—the poor in spirit, the meek, those who mourn, those who hunger and thirst for justice, the merciful, the pure in heart, the peacemakers, and those who are persecuted for righteousness's sake—explaining the heavenly reward promised to each. Augustine's analysis of the sermon finds analogies between these eight beatitudes and the seven gifts of the Holy Spirit: humility is linked with the poor in spirit, piety with the meek, knowledge with those who mourn, fortitude with those who seek righteousness, counsel with the merciful,

intelligence with the pure in heart, and wisdom with the peacemakers.

To make this alignment work, however, Augustine has to reverse the order of gifts given in Isaiah, so that they match the increasing virtue of the beatitudes. To justify the reversal, Augustine cites a passage from Ecclesiasticus stating that “the fear of the Lord is the beginning of wisdom,” thus making the whole progression a circular one. The reversal is important, for it makes the gifts an ascent toward virtue: the climb begins with the humility of the poor in spirit and leads to the wisdom of the peacemakers.³⁰ Augustine’s analysis thus outlines a path for religious development:

Hence also the sevenfold operation of the Holy Ghost, of which Isaiah speaks, seems to me to correspond to these stages and sentences. But there is a difference of order: for there the enumeration begins with the more excellent, but here with the inferior. For there it begins with wisdom, and closes with the fear of God: but “the fear of the Lord is the beginning of wisdom.” And therefore, if we reckon as it were in a gradually ascending series, there the fear of God is first, piety second, knowledge third, fortitude fourth, counsel fifth, understanding sixth, wisdom seventh.³¹

By proceeding from one virtue to the next, learning and mastering each, the soul ascends to God.

An interpretative problem remains, however, as the eighth beatitude has no counterpart in the group of seven gifts. Augustine reconciles the discrepancy by explaining that while wisdom, the seventh beatitude, is the culmination of

the virtues, the eighth beatitude returns to the beginning:

The eighth, as it were, returns to the starting-point, because it shows and commends what is complete and perfect: therefore in the first and in the eighth the kingdom of heaven is named, “Blessed are the poor in spirit, for theirs is the kingdom of heaven”; and, “Blessed are they which are persecuted for righteousness’ sake, for theirs is the kingdom of heaven”: as it is now said, “Who shall separate us from the love of Christ? Shall tribulation, or distress, or persecution, or famine, or nakedness, or peril, or sword?” Seven in number, therefore, are the things which bring perfection: for the eighth brings into light and shows what is perfect, so that starting, as it were, from the beginning again, the others also are perfected by means of these stages.³²

In Augustine’s cycle, the eighth beatitude is perfection: the ability to see the whole. This eighth beatitude is both retrospective and instructive; after achieving wisdom and residence in the peace of the Lord, one must become humble again, recognize such peace as a gift, and help others on their own ascent.

Augustine’s explanation of the sermon was well known in the high Middle Ages. In the 1120s, Theophilus’s contemporary Rupert of Deutz, then abbot at the monastery of St. Heribert, just outside of Cologne, wrote an exegesis on the book of Matthew that drew heavily on Augustine’s analysis of the Sermon on the Mount.³³ His elegant explanation of the eighth beatitude and his clear notion of retrospection and return are particularly helpful. For him, the circular process

operates like a musical scale. Just like the seven gifts, it contains seven tones, and the eighth note, the octave, is likened to perfection. Perfection is the perfect ratio; resonant with the first, it returns to the beginning of the scale, yet, like the knowledgeable teacher, it operates at a higher frequency, one step closer to the Lord.

Since Theophilus lists the gifts in Isaiah's order, not Augustine's, the link to Augustinian thought has not previously been recognized. It has also puzzled some that Theophilus's discussion of the gifts appears in the last prologue. This is because the list of the gifts is, as it was for Augustine, a summary of his project and a declaration of his pedagogy. Using the Latin subjunctive, Theophilus introduces the section with the following: "Wherefore, dearest son, when you have adorned his House with such embellishment and with such variety of work, you should not doubt (*cuncteris*), but believe with a full faith, that your heart has been filled with the Spirit of God. And lest perchance you have misgivings, I will clearly demonstrate that whatever you can learn, understand or devise is ministered to you by the grace of the seven-fold spirit."³⁴ It is as though the student cannot see all the steps clearly until he has mastered them, and indeed if we turn back to the beginning of the treatise, we see that Theophilus expresses hope that his reader will "read [this little work] through with a retentive memory," adding in the next paragraph that he ought to "read through these things several times, and commend them to a retentive memory."³⁵ Reading through again at Theophilus's urging, we begin to see that the gifts appear not only in the last prologue but are also found throughout the text.

The gifts of the Holy Spirit appear throughout *On Diverse Arts*, structuring its narrative

and binding together the prologues and instructions. This begins with Theophilus's introduction of himself in the first line of the first prologue: "Theophilus—humble priest, servant of servants of God, unworthy of the name and profession of monk. . . ." As he continues, he addresses those "who are willing to avoid and spurn idleness and the shiftlessness of the mind by the useful occupation of their hands and the agreeable contemplation of new things."³⁶ Theophilus's expression of humility and his presentation of himself as priest and monk frames the entire treatise and shapes how it is to be read.³⁷

In his introduction to *On the Trinity*, Augustine explains that any inquiry—particularly one that pertains to the knowledge of God—must begin with faith. He writes, "The reader of this treatise on the Trinity should know beforehand that our pen is on the watch for the sophistries of those who consider it beneath their dignity to begin with faith, and who are thus led into error by their immature and perverted love of reason."³⁸ For Augustine, faith shapes reason and must precede it. Thus Theophilus's opening in the first prologue can be seen not only as a statement of humility, but also as a framing of authorial character in accordance with God. He refers to himself in terms of his profession as monk, as priest, as servant of God, and makes no statement about his artistic skill or knowledge.

Theophilus continues with a narrative about the beginning of the world and the creation of man: "In the account of the creation of the world we read that man was created by the divine breath, breathed into him," by which distinction "he was placed above the other living creatures, so that, capable of reason, he acquired participation in the wisdom and skill of the divine intelligence."³⁹ As the prologue continues, it tells the

story of the fall of man: “Wretchedly deceived by the guile of the Devil, through the sin of disobedience he lost the privilege of immortality, but, however, so far transmitted to later posterity the distinction of wisdom and intelligence, that whoever will contribute both care and concern is able to attain a capacity for all arts and skills (*artes et ingenium*), as if by hereditary right.”⁴⁰ Humility, faith, fall, and restoration thus open the prologue and set the stage. As will become clear through this study, this opening and the creation story that follows it shape the entire treatise, setting up a narrative of redemption—an ascent back toward God—that follows the gifts of the Holy Spirit.

Having introduced himself and begun the story, Theophilus turns his attention to his reader. Making himself an example, he invokes his own fear of the Lord: “Fearful of incurring this judgment, I, an unworthy and frail mortal of little consequence, freely offer to all, who wish to learn with humility, what has freely been given to me by the divine condescension, which gives to all in abundance and holds it against no man.”⁴¹ According to Augustine, humility is the first gift of the Holy Spirit, the first step of learning, and the first step on the ascent toward wisdom. “Blessedness,” he writes, “starts with humility: *Blessed are the poor in spirit.*”⁴² Those who are humble, he continues, are those “who stand in dread of punishment after this life despite the seeming blessedness of their earthly life.”⁴³ Humility and fear of the Lord are two sides of the same coin; furthermore, “fear of the Lord is the beginning of wisdom.”⁴⁴ Thus, “if we reckon as it were in a gradually ascending series, there the fear of God is first . . . [and] the fear of God corresponds to the humble.”⁴⁵ Theophilus’s opening, with its double reference to humility and fear,

thus invokes the first step of Augustine’s learning process.

In these opening statements Theophilus also refers to the gifts bestowed on him and his desire to pass them on: “I . . . freely offer to all, who wish to learn with humility, what has freely been given to me by the divine condescension, which gives to all in abundance and holds it against no man.”⁴⁶ The idea that all who proceed with humility have the potential to learn, and that knowledge is a gift given freely by God, expands on Augustine. In *On Christian Doctrine*, Augustine writes that God “has also given to each gifts suitable for the building up of his Church, that we may do what he points out as right to be done, not only without a murmur, but even with delight.”⁴⁷ Theophilus also echoes Hugh of St. Victor, who wrote that learning is the potential of all people, a gift of God that must be recognized: “This then, is that dignity of our nature which all naturally possess in equal measure, but which all do not equally understand.”⁴⁸ So when Theophilus urges his readers “to recognise God’s favour towards me and to appreciate His generosity, and I will have them know that they can be quite sure that the same things are at hand for themselves if they will add their own labour,”⁴⁹ he is drawing quite directly on Augustine and appears conversant and aware of contemporary theories of learning.

Theophilus moves on to explore the themes of piety and knowledge. In the next passage of the first prologue, he exhorts the reader to be pious and to value the material gifts that God has given: “do not despise useful and precious things, simply because your native earth has produced them for you unexpectedly. For foolish is the merchant who suddenly finds a treasure in a hole in the ground and fails to pick it up and keep it.”⁵⁰

The warning is against the prideful neglect of God's gifts: recognizing the source of all gifts in God is the first step; valuing them is the next. For Augustine, too, the second beatitude, "Blessed are the meek," is aligned with godliness, or piety, which he describes as treasuring the scripture of God and not neglecting its truth through pride. The soul, he writes, "makes itself acquainted with Sacred Scripture according to which it must show itself meek through piety, so that it may not make bold to censure what appears a stumbling block to the uninstructed and thus become intractable by obstinate argumentation."⁵¹

The final section of the prologue is a display of Theophilus's expertise, for he provides a grand summary of the techniques the text will cover: "If you will diligently examine it, you will find in it whatever kinds and blends of various colours Greece possesses: whatever Russia knows of workmanship in enamels or variety of niello: whatever Arabia adorns with repoussé or cast work, or engravings in relief: whatever gold embellishments Italy applies to various vessels or the carving of gems and ivories: whatever France esteems in her precious variety of windows: whatever skilled Germany praises in subtle work in gold, silver, copper, iron, wood and stone."⁵² Because it refers to technical information gained from many places, the passage has been interpreted as an attempt to be encyclopedic.⁵³ Yet it has also been noted that the bulk of the techniques listed here are contained not in the first book of the treatise but in the third, creating an odd sequence. Indeed, of the six countries and thirteen techniques Theophilus lists here, eleven are treated in the last book. Just one is mentioned in the first book (pigments) and one in the second (glass). This peculiarity has at times been understood either as a preference for metalwork,

confirming Theophilus's identity as a goldsmith, or as evidence for the haphazard organization of the treatise.⁵⁴ However, medieval artists are known to have worked in a variety of media. Even Roger of Helmarshausen seems to have worked in illumination; his hand has been identified in the Jeremiah initial of the Stavelot Bible.⁵⁵ Furthermore, the techniques described in this passage require the highest level of learning: engraving, carving, repoussé, niello (a black copper sulfate used to fill in engraved design). They are complex techniques that involve multiple steps and the careful manipulation of precious materials, mastered only with significant training and knowledge. Above all, then, the passage functions to create an authorial persona. Although Theophilus claims he will teach the initiate artistic techniques gathered from all over the world, references to locales hardly ever appear again in the text. Arabia, for example, is never mentioned again, nor is France or Germany. Greece makes an appearance, but in the discussion of blue glass drinking goblets rather than the book of color, where Theophilus seems to imply it will appear. Theophilus thus seems to be modeling his text after a medieval *accessus ad auctores*, describing for his reader the subjects the treatise will cover and making a claim for his authority in the process.⁵⁶

Yet Theophilus does more than display his knowledge. While the English translation reads "If you diligently examine [this little work] you will find . . .," the Latin term Theophilus uses for "find" is *invenies*, meaning to discover or to come upon, particularly with regard to interpreting a text. The word is also laden with spiritual connotations. It is used by Anselm of Canterbury, for example, to signify the approach to God, and it perhaps may be more closely translated as "coming near"

or “coming to” knowledge.⁵⁷ The choice of words here suggests that Theophilus is framing technical learning in terms of spiritual learning. The passage may be understood as recasting worldly knowledge for a Christian, monastic context. Continuing, Theophilus makes a point of explaining that his expertise and willingness to pass on his learning is done not for pride but with a desire to serve God: “He [God] knows that I have written the things collected here out of no love for human approbation nor greed for temporal gain, and that I have not appropriated anything precious or rare nor kept silent about something reserved especially for myself from malice or envy, but that, to increase the honour and glory of His name, I have ministered to the necessities of the many and had regard to their advantage.”⁵⁸ Here Theophilus echoes Augustine. For Augustine, the gift of learning is accompanied by loss. He writes that in the third beatitude Christ refers to “those who mourn.” According to Augustine, they mourn the loss of the worldly, but it is only by doing so that they come nearer to God. Their knowledge has led them to see what they must keep and what they must give up for the sake of God. They realize the cost of clinging to earthly things: “In this third step, then, wherein is knowledge, there is grief for the loss of the highest good through clinging to the lowest.”⁵⁹ The gift is one of perspective—the ability to differentiate between that which will lead to God and that which will not. Theophilus’s point of view seems slightly different from Augustine’s, but his insistence on serving God makes the same point, for he shows himself to be knowledgeable and on guard against the vanity of too much worldly learning. He urges his reader to keep a wide perspective, to learn with a sense for what is useful, good, and in the service of God.

Theophilus’s theory of learning is closely linked to the philosophy of Hugh of St. Victor, for whom all learning is under the umbrella of wisdom, and thus of God. For Hugh, humility is the awareness that God is the ultimate source of all things, and learning is only good insofar as it is directed to God. Knowledge requires a directed learning. This corresponds to Augustine’s statement that with knowledge, the soul “begins to know in what entanglements of this world it is held by reason of carnal custom and sins.”⁶⁰ Augustine and Theophilus shared this understanding of the nature of knowledge and purpose of learning. One may know many worldly things, but, as Theophilus demonstrates, learning is not an end in itself but must be used with knowledge of God to “increase the honour and glory of His name.”⁶¹

The prologue to the second book centers on the themes of work and obedience, the next three gifts in the sequence. The lessons build on, and add to, those of the first. Like a good teacher, Theophilus begins by summing up the themes of the prior prologue: “Actuated, dearest brother, by a sincere affection, I did not hesitate to suggest to you in the preceding book how much honour and advantage there is in eschewing idleness and in spurning laziness and sloth; and how sweet and delightful it is to give one’s attention to the practice of the various useful arts according to the saying of a certain author who declares: ‘To know something is praiseworthy; to be unwilling to learn anything is reprehensible.’”⁶² With this quotation from Cato’s *Proverbs*, Theophilus highlights the humility necessary for the acquisition of knowledge, since being unwilling to learn is a sin not only of ignorance but also of pride and impiety. He then progresses to the next step in the ascent toward wisdom: fortitude.

For Theophilus, this is related to strength and labor. He continues, “Nor should anyone be slow to approach him, of whom Solomon says: ‘He that increaseth knowledge increaseth labour,’ because, if he thinks seriously about it, he will be able to observe how much progress of the soul and body results thereby.”⁶³ Tellingly, the Latin *laborem* carries connotations of weariness or sorrow. Thus the term might refer, as well, to the sorrow and necessity of labor with which man was punished after the fall, making the allusion to spiritual progress all the richer. Wisdom, then, is not simply gaining knowledge; it is a labor of body and soul.

The exhortation to toil in body and soul seems to function on analogy to Augustine’s treatment of the fourth beatitude (those who hunger and thirst for righteousness) and gift (labor): “In the fourth step there is hard work. The soul puts forth tremendous effort to wrench itself from the pernicious delights which bind it.”⁶⁴ *Fortitudo*, Augustine continues, corresponds to those who hunger and thirst, “for they labor in a desire for the joy that comes from what is truly good and in an effort to stem their love for the earthly and corruptible.”⁶⁵ Labor thus refers to both the effort of wrenching oneself away from “the love of the earthly” and to the activity—looking toward joy, for example—that makes resistance possible. In this second prologue, Theophilus is building on a concept of labor central to Benedictine monasticism and outlined in the Rule of Saint Benedict.⁶⁶ He sees spiritual development as requiring fortitude and strength and as a continual resistance of “pernicious delights.” Like Augustine’s, Theophilus’s sense of labor is connected to the prevention of idleness and its related sins: “it is as clear as day,” he writes, “that whoever is abandoned to idleness

and irresponsibility also indulges in . . . things . . . which are repugnant in the sight of God.”⁶⁷ Medieval readers seem to have been attentive to these warnings; the exhortation against idleness is marked in the Harley manuscript by a pointing hand inserted in the margin.⁶⁸

Theophilus begins the second half of the prologue by describing the learning and teaching process, evoking the gift of *consilio*, or counsel: “Desiring to follow this man [Paul], I have approached the temple of holy wisdom. . . . I have filled the storehouse of my heart with a sufficiency of all those things, and without envy, have clearly set them forth for your study.”⁶⁹ Proclaiming his own learning and generosity, Theophilus encourages the reader to follow his example. Rather than couching his authority in humility, however, as he did in the first prologue, he expresses it in terms of “counsel.” It is as if the student has progressed. Theophilus describes his own education and his desire to pass on what he has learned. This process is echoed in artisanal practice and nomenclature, where, as Walter Cahn has argued, teaching was central, since a master was required to take on a student.⁷⁰ It also finds parallels in Augustine’s writing, where counsel is “just a proposition: if one wishes to be helped by a more powerful person, let him help someone who is weaker in a field wherein he himself holds the advantage,” and that “we help others to the best of our ability as we hope to be helped in our need.”⁷¹

Anticipating the instructional content to come, Theophilus closes the second prologue with a discussion of his increased competence in the glassmaker’s art: “Having applied myself to this task, I understand the nature of the glass, and I consider that this object can be obtained simply by the correct use of the glass

and its variety.”⁷² Here is the gift of *scientia*, or understanding, which according to Augustine corresponds to the sixth beatitude, “the clean in heart.”⁷³ “The sixth step,” Augustine writes, “is cleanness of heart from a good consciousness of works well done, enabling the soul to contemplate that supreme good which can be seen only by a mind that is pure and serene.”⁷⁴ Thus the second book builds upon the first: from humility, fear, piety, and knowledge come work, teaching, and understanding. Notably too, the sequence shifts—from virtues defined by a direction of mind to those constituted by the action of a person.

The prologue of the third book of *On Diverse Arts* presents the final theme of wisdom, the seventh gift, associated with the peacemakers. This prologue has received considerable scholarly attention, since its descriptions of the celestial house of David, the temple of Solomon, the tabernacle of Moses, and—now explicitly—the gifts of the Holy Spirit seem to be the most overt justification of luxury arts in the treatise.⁷⁵ Yet as Theophilus himself explains, the beauty evidenced in the temple begun under King David is not a foregone conclusion but the result of great wisdom:

David—renowned among the prophets, whom the Lord God, in His prescience, predestined before the world began, whom He “chose after his own heart” because of his simplicity and humility of mind, and placed as a Prince over His chosen people, strengthening him with a princely spirit so that he might nobly and wisely establish the rule of so great a name—David, applying himself with the full force of his mind to the love of his Creator, among other things uttered

these words: “Lord, I have loved the beauty of Thy House.”⁷⁶

For Theophilus, David exemplifies wisdom. His heart is full of love for God, and he is close to God; he is the one whom God “chose after his own heart.” His wisdom is manifest in this love, and significantly, in his love for the beauty of the Lord’s House: “It is true that a man of such authority and such great intellect may have meant by that House the habitation of the heavenly court, in which God presides over hymning choirs of angels in inestimable glory . . . or else the refuge of a devoted breast and pure heart where truly God dwells. . . . Nevertheless it is certain that he desired the embellishment of the material House of God, which is the place of prayer.”⁷⁷ David’s wisdom is also the product of his intellect, piety, and devotion to God. These can be understood in light of Augustine’s interpretation of the seventh gift: “the seventh step is wisdom itself, that is, contemplation of the truth, bringing peace to the whole man and effecting a likeness to God.”⁷⁸

Theophilus continues, writing that while David desires “ardently” to build a temple for God, he cannot. He must leave the project to his son Solomon: “For, he himself longed with a most ardent desire to become the founder of the House of God but, because of his frequent spilling of human, albeit enemy blood, he did not merit it. As a result, he entrusted almost all the needful resources in gold, silver, bronze and iron to his son Solomon.”⁷⁹ David’s awareness that he cannot achieve the building of the temple is a recognition of his own limitations, brought on by his sins. It thus confirms his wisdom, humility, and understanding: “By pious reflection [David] had discerned that God delighted in embellishment of this kind,

the execution of which he assigned to the power and guidance of the Holy Spirit, and he believed that nothing of this kind could be endeavoured without His inspiration.”⁸⁰ The recognition that all things come through the beneficence of God can be seen as what Augustine describes as *perfectio*. Just as the eighth note of a scale is the end of one octave and the beginning of another, *perfectio* is a return to the beginning: “to the number seven . . . an eighth is added, so that . . . we, as it were, return to the starting-point: on which day the Holy Spirit was sent, by whom we are led into the kingdom of heaven, and receive the inheritance, and are comforted; and are fed, and obtain mercy, and are purified, and are made peacemakers; and being thus perfect, we bear all troubles brought upon us from without for the sake of truth and righteousness.”⁸¹ Here in perfection is found the ability to look back—to recognize spiritual ascent as it is being accomplished and to seek mercy in the awareness of one’s limitations. It is to be wise and yet have humility, knowing that God is the source of all things.

Thus it is from the perspective of perfection that Theophilus lists the seven gifts, stating outright that the lessons he teaches have been learned only with the aid of the seven-fold grace of the Holy Spirit: “believe with a full faith, that your heart has been filled with the Spirit of God.”⁸² Not only does the list of the gifts function retrospectively, but it also corresponds to Augustinian thought. For just as Augustine likens the eighth beatitude to the Pentecost and the descent of the Holy Spirit, Theophilus now lists the gifts of the Holy Spirit in descending order, as in Isaiah. The reversal emphasizes both the cyclical quality of the approach to *perfectio* and implies how the lessons of the treatise begin anew when one becomes a teacher.

Appropriately, therefore, Theophilus writes as a teacher. Having reviewed the lessons of the treatise, presenting them as having been built upon one another, he congratulates his student: “Animated, dearest son, by these supporting virtues, you have . . . in some measure, shown to beholders the paradise of God, glowing with varied flowers, verdant with herbs and foliage, and cherishing with crowns of varying merit the souls of the saints.”⁸³ The third and last prologue, introducing the book devoted to the sacred arts, assumes a heart most full of the Holy Spirit. It is from this vantage point that the temple of Solomon and the gifts of the Holy Spirit are to be completed, using the wisdom gained through the process of learning and the perspective gained through arriving at *perfectio*: “Come now, my wise friend—in this life happy in the sight of God and man and happier in the life to come—by whose labour and zeal so many sacrifices are offered to God, be inspired henceforth to greater deeds of skill, and with the utmost exertion of your mind prepare to execute what is still lacking in the vessels of the House of God, without which the divine mysteries cannot continue.”⁸⁴ The invocation of Solomon and David as exemplary patrons is deeply rooted in the Middle Ages and was a common justification for the use of luxury arts in the church.⁸⁵ Abbot Suger, for instance, was compared to Solomon by Jocelin, and the abbey of Saint-Denis itself seen in some cases as a new Jerusalem.⁸⁶ Theophilus’s allusion to the Hebrew kings here is related to that tradition but builds on it substantially, for here David and Solomon are also the climax of the progressions, giving structure to the entire treatise. There are no references to the splendor of the house of God before this point. The treatise has built up to them, from the initial learning of painting and

the virtues of humility, piety, and knowledge; to the labor required for the techniques of glass and the virtues of fortitude, counsel, and understanding; and finally to the embellishment of the house of God, the virtue of wisdom, and the perspective of perfection.

The prologues of *On Diverse Arts* are organized according to a clear structure, creating a trajectory that follows the gifts of the Holy Spirit as interpreted by Saint Augustine. This is a teaching strategy that is distinctly monastic; it puts learning in the service of spiritual development. It also creates a model for the Christian life. As Augustine wrote, the beatitudes and the gifts of the Holy Spirit “perfectly shape the life of those who wish to live according to [the words of the Sermon of the Mount] . . . this sermon has been made up of all the precepts by which Christian life has vitality.”⁸⁷ Appearing only in the last prologue, Theophilus’s list of gifts functions as a retrospective summary of his project, emphasizing the progressive structure of his text. As we will see, this organization and argument align with the pedagogical developments and exegetical strategies of the period.

Redemption Through Learning

Theophilus’s use of the gifts of the Holy Spirit corresponds to a renewed interest in the subject in the twelfth century. A brief look at Esther Dech Schandorff’s bibliography of the doctrine of the Holy Spirit reveals the trend: she lists three authors on the topic for the eighth century, five for the ninth, one for the tenth, seven for the eleventh, and nineteen for the twelfth.⁸⁸ While the list may well be incomplete, it nevertheless gives a sense of the significant renewal of interest

in the workings of the Holy Spirit in the twelfth century. The list also reads like an inventory of the best-known authors of the day: Peter Abelard, Bernard of Clairvaux, Hugh of St. Victor, Peter Lombard, and Rupert of Deutz, among others.

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Significantly, too, there was a change in how the gifts are treated in these texts. Early medieval writers would be more likely to refer to the gifts of the Holy Spirit as a biblical metaphor. In his *Questions on the Old Testament*, for example, Isidore of Seville writes that Moses’s striking of the rod on the rock to bring forth water is analogous to the beating of Christ on the cross, resulting in the flow of the gifts of the Holy Spirit.⁸⁹ In the Carolingian period, the gifts were used as typological allegory in explication of Christian doctrine. In his Easter homily, for example, Haymo of Halberstadt connects the seven gifts to the presence of seven apostles at Christ’s appearance at Lake Tiberias, while Remigius of Auxerre describes the “Blessed Man” of the first Psalm as able to stand against the wicked because he is filled with the gifts of the Holy Spirit.⁹⁰ Rabanus Maurus, meanwhile, likens the gifts of the Holy Spirit to the seven lamps of the candelabra of the tabernacle.⁹¹

In the twelfth century, the gifts became a tool of sorts, a mode of thinking used to structure and describe daily life and its activities. In his *De quinque septenis seu septenariis opusculum*, Hugh of St. Victor describes and links occurrences of five sets of the number seven in the Old and New Testaments: the seven virtues, the seven beatitudes, the seven vices, the seven gifts of the Holy Spirit, and the seven precepts of the Lord’s prayer.⁹² Similarly, Rupert of Deutz, in setting guidelines for a “more intense Benedictine life,” turned to the teachings of the Sermon on the Mount, applying its schema of virtue to

daily and liturgical life. In his book on the Divine Office, for example, Rupert aligned each day of Holy Week with a different virtue and a different gift of the Holy Spirit.⁹³ In his commentary on the book of Matthew, he brings out further details. The beatitudes, he explains, are a salve for the human soul; they are virtues that operate on mankind through the death and suffering of Christ, and they are made accessible to mankind through scripture and the performance of the divine office.⁹⁴ Twelfth-century writers use the gifts to a far greater extent than their predecessors. Having served as a metaphor for the manifestation for the Holy Spirit in the early Middle Ages, the gifts are aligned in the twelfth century with the virtues and the ascent of the soul and are used in as an organizing mechanism in allegory.

Hugh and Rupert were both writing in the first third of the twelfth century, just when Theophilus is thought to have been composing his treatise. These three authors are just a small part of a profound shift in writing and teaching that was occurring in those decades. As C. Stephen Jaeger has shown, eleventh-century writers described the virtues as something a master inhabited.⁹⁵ The deeds of Willigis of Mainz, written between 1019 and 1039, describe Willigis's virtue as something he embodies and strives for internally: "Even as outwardly he climbed the seven steps, he strove to ascend internally with the seven-formed steps of grace, and in the refuge of his heart to give to God the offering of praise." This virtue, as Jaeger points out, is something that is then imparted to others: "He taught lovers of virtue to live according to moral perfection, in his acts, not in his speech, more with the language of his behavior than that of his words."⁹⁶ The contrast between this approach and

that of Theophilus is significant. Here the virtues are envisioned as a set, and they are learned through action, not by explanation. The kinds of virtues are also different. In a confirmation of his investiture in 1049, Pope Leo IX described the Archbishop Hugo of Besancon as one who "displays laudable dignity of merits, in the knowledge of virtue as in uprightness of manners, may display also beauty of ornaments in all plenitude of his high office."⁹⁷ Leo's reforming zeal here meets the splendor of an ecclesiastical office closely linked to imperial powers, and "manner" is the single, overarching virtue. Seventy-odd years later, Theophilus, Hugh of St. Victor, and Rupert of Deutz do something quite different. They list seven different discrete virtues, ordering and defining them in relation to one another; concrete, specific principles are enumerated in a predetermined system.

Contemporary objects bear out this interest in the virtues. A head reliquary from the abbey of Stavelot is an elaborate and learned example (fig. 3). Made under the abbot Wibald (r. 1130–58) and consecrated in 1145, the reliquary contains the relics of the second-century pope and saint Alexander I. The head of the reliquary is made of silver-gilt plating on a wooden core, with partially gilded, tonsured hair. Its neck sits in a golden collar, ornamented with jewels and enamels. The entire piece is set into a square base supported by four dragon feet, cast in bronze and gilded.⁹⁸ The four sides of the base are decorated with twelve enamel plaques, three to a side, alternating with a panel of jewels. On the front are enamels with Saints Severinus, Alexander, and Theodolus (fig. 4). Around the sides are more enamels with personifications of the gifts of the Holy Spirit, each identified by a vertical inscription and holding a small plaque with a

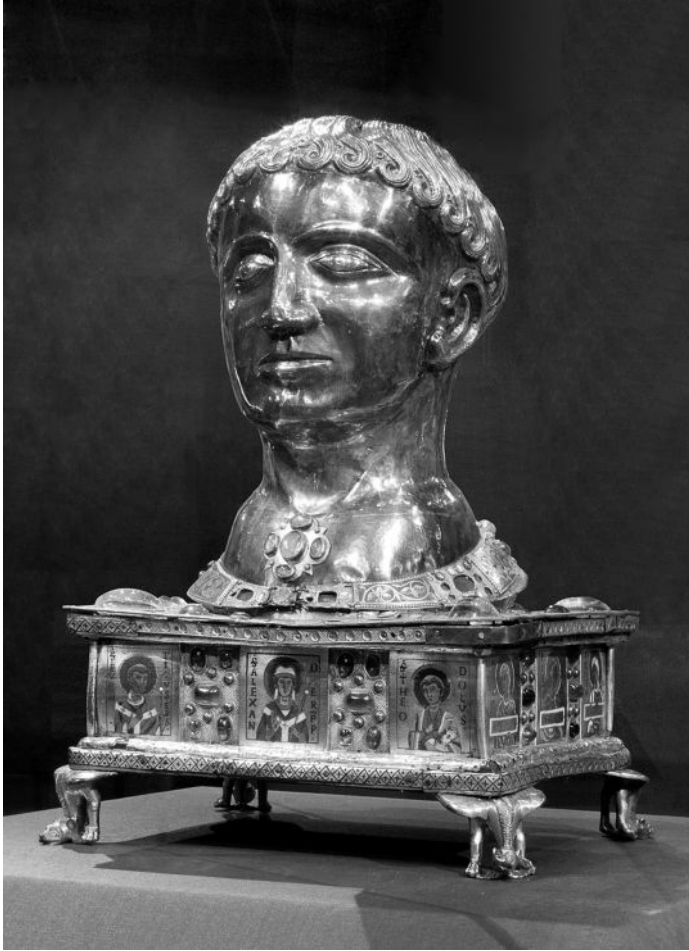


FIG. 3

Reliquary of Saint Alexander, abbey of Stavelot, 1145.
Royal Museums of Art and History, Brussels.

beatitude quoted from the book of Matthew. Moving around the reliquary in a counter-clockwise direction, the enamels begin with humility (*humilitas*), then piety (*pietas*), and knowledge (*scientia*) (fig. 5). On the back are fortitude (*fortitudo*) and counsel (*consilium*), flanking a crowned wisdom (*sapientia*) (fig. 6). Finally, completing the circuit, are intelligence (*intelligentia*), a second wisdom (*sapientia*), and perfection (*perfectio*) (fig. 7). Each figure is slightly different in details like hand gestures or the turn of the head. With the exception of the figure of

Sapientia on the back of the reliquary, which will be discussed below, the personifications appear in the ascending order given by Augustine and correlate to the beatitudes set out in his sermon. As Suzanne Wittekind has suggested, the representation of the beatitudes here makes Christ and his Sermon on the Mount present and his promise visible.⁹⁹

The two figures of Sapientia are not a duplication; they reinforce a narrative of spiritual ascent. On the central panel of the left side, she appears within the sequence, like the others, as



FIG. 4
Reliquary of Saint Alexander, abbey of Stavelot, 1145, front panel,
with enamels: Saints Severinus, Alexander, and Theodolus. Royal
Museums of Art and History, Brussels.

FIG. 5
Reliquary of Saint Alexander, abbey of Stavelot, 1145, side view
(right) with detail of enamels: *humilitas*, *pietas*, *scientia*. Royal Muse-
ums of Art and History, Brussels.



FIG. 6 (TOP)

Reliquary of Saint Alexander, abbey of Stavelot, 1145, back view with detail of enamels: *fortitudo*, *sapientia*, *consilium*. Royal Museums of Art and History, Brussels.



FIG. 7 (BOTTOM)

Reliquary of Saint Alexander, abbey of Stavelot, 1145, side view (left) with detail of enamels: *intelligentia*, *sapientia*, *perfectio*. Royal Museums of Art and History, Brussels.

a veiled figure, holding a plaque with an inscription stating “blessed are the peacemakers.” She is prominently placed in the center of the group, and her plaque is red instead of the blue or green of the other personifications. She holds her plaque with both hands covered, as if in honor of the sacred words; her still hands are a contrast to the gestures of *intelligentia* and *perfectio* on either side, who point and raise a palm. She is thus distinct and seems to hold special status. Yet it is the figure of *Sapientia* on the reverse of the reliquary base that shows wisdom to be the culmination of the series. Here, she wears a large crown and jewels around her neck, and she holds

a globe on which is inscribed “Glorious is the fruit of good labors” (*bonorum laborum gloriosus est fructus*). She is in the center of the back panel, in the same position as Saint Alexander on the opposite side; these are also the only two figures to be shown frontally, and both raise their right palm as if in prayer. In this case, Augustine’s sequence of virtues and beatitudes is not an abstract concept but a way to characterize and articulate sanctity.

The head reliquary is not the only object made under Wibald; a triptych now in New York, a portable altar in Brussels, and two now-lost retables have all been attributed to his abbacy.

Cynthia Hahn has suggested that the Alexander reliquary served as a model for initiates, and that the use of the antique-style head and surrounding virtues was a tribute to Cicero, whom Wibald admired.¹⁰⁰ The erudite theme is thus a testament to Wibald's involvement in the work, and it also shows the importance of the idea of ascending virtue in monastic communities.

Reading the prologues together, it becomes clear that the gifts of the Holy Spirit structure the treatise according to a narrative of redemption and ascent of the soul toward God. Importantly, this redemption and ascent is accomplished through the process of learning. For Theophilus, learning occurs by grace of the spirit: "I will clearly demonstrate that whatever you can learn, understand or devise is ministered to you by the grace of the seven-fold spirit."¹⁰¹ The idea of learning as a process is significant. The Latin of the text reads "quicquid discere, intelligere, vel excogitare possis artium, septiformis Spiritus gratiam tibi ministrare."¹⁰² *Excogitare*, usually translated as "devise," implies a mental activity, a thinking through, and an inventiveness. The use of the three mental activities of learning, understanding, and thinking through is noteworthy, for here Theophilus relates them specifically to art ("quicquid . . . artium"). Thus we might retranslate the entire phrase as "whatever of the arts you can learn, understand or *think through* is ministered to you by the grace of the Holy Spirit." This passage closely echoes one used by Augustine in the *Questions on the Heptateuch*, regarding the skills of the great biblical craftsman Beseleel: "dixit eum se replevisse spiritu divino sapientiae, et intellectus, et scientiae in omni opere, excogitare, et architectonari."¹⁰³ Perhaps Theophilus knew the reference, but if not, his exploitation of such intellectual terms for what had long

been considered a practical craft should catch our attention. Theophilus thus again builds on Augustine, emphasizing the link between learning, skill, and spiritual awareness.

As Jean Leclercq has famously shown, learning was considered a type of spiritual exercise, especially in the twelfth century.¹⁰⁴ Hugh of St. Victor's *Didascalicon*, for example, is dedicated to elucidating the spiritual benefits of different kinds of learning. As he explains it, the restoration of the likeness of God in man occurs through the "contemplation of truth and the practice of virtue."¹⁰⁵ Hugh explains that to learn is to advance from latent to realized potential. All men have potential to learn, but only when their potential is put into action—by the practice of virtue and the contemplation of truth—can one ascend to wisdom and, in turn, gain redemption. Learning leads to wisdom because, for Hugh, the "dignity" of man is that "the mind, imprinted with the likenesses of all things, is said to be all things and to receive its composition from all things and to contain them not as actual components, or formally, but virtually and potentially." Though all possess this potential, fulfilling it is made difficult by the temptations and confusions of earthly existence, by man's state as a fallen being. "But," Hugh writes, "we are restored through instruction, so that we may recognize our nature and learn not to seek outside ourselves what we can find within. 'The highest curative in life,' therefore, is the pursuit of wisdom."¹⁰⁶ Learning helps clear confusion. It is achieved by understanding the order of things and by recognizing that all things are due to God and that all capacities and all parts of nature are created by him. Hugh makes the sensual realm a subject of learning.¹⁰⁷

Similarly, for Theophilus, the ability to learn is latent in everyone—whoever will "contribute

care and concern,” he writes, “is able to attain a capacity for all arts and skills.”¹⁰⁸ Thus wisdom and skill are bound together. This is not dissimilar from Hugh of St. Victor’s assertion that learning and the worldly realm are pertinent to the ascent of the soul.¹⁰⁹ What is different about Theophilus’s treatise, however, is that the subject of learning he describes is the practice of art.

As we shall see, it is not just Theophilus’s prologues that cast learning, and the making of art, as intrinsic to spiritual growth. The instructions of *On Diverse Arts* are intertwined and integrated with the prologues; the pedagogical program described in the prologues is made manifest in the instructions. This is no disorderly compilation. Techniques are aligned and arranged to correspond to the progression and structure in the prologues.

For example, we might examine how the techniques progress from the simple to the complex. The sequence is clear in the first book, which begins with a simple flesh tone and proceeds to shadows, hair, draperies, and a colorful rainbow.¹¹⁰ Only the first half of the book is concerned strictly with pigments; the second half deals with compound substances like glue and gesso and their use in the assembly of complex objects like altar panels.¹¹¹ Indeed, the instructions increase exponentially in complexity. Moreover, the break between the discussion of pigments and complex objects like altars is marked by the chapter describing how to paint a rainbow. The meaning of a rainbow as a symbol of a new covenant would not have been lost on medieval Christian readers. Bruno Reudenbach has noted a parallel between the first few chapters of the first book, with its nude bodies and earthy green of paradise, and the reference to creation in the prologue.¹¹² The structure goes

even deeper. Having told of creation in the prologue, Theophilus proceeds to describe the fall of man and his potential for restoration to God. So too, the chapters begin with the nude bodies and greens of paradise; they then move to shadows, blushes, hair, drapery (suggestive of man’s need to clothe himself), and the process of aging, in what seems a reference to man’s fallen state. The sequence is only broken after the chapter on the rainbow. Just as the first prologue gives man the promise and possibility of restoration, so too in the instructions the rainbow begins a new section on church furnishings—the altars and books that will serve man’s redemption.

The second book has a similar sequence, with simpler techniques building up to skills that allow the fabrication of complicated, composite objects. Theophilus explains the process of making glass windows from the beginning; he commences with instructions on how to build a kiln and then moves deliberately and methodically through different kinds of glass, building knowledge and understanding.¹¹³ The second section of the book addresses the making and embellishment of windows, including chapters about painting on glass, molds, assembly, and the setting of gems in windows. These constitute the culmination of the prior techniques; the instructions describe, step by step, how to cut glass, paint on it “as in a coloured painting,” fire it, cast the frame of the window, and assemble it into a single piece.¹¹⁴ Finally, three chapters provide an addendum of variants on the techniques described previously, addressing “Simple Windows” (windows whose colors are assembled without interior comes, or iron rods), “How to Mend a Broken Vessel” (by applying a low-melting-point green and blue glass), and lastly, “Rings,” which can be set with various types of

glass or even gems.¹¹⁵ Just as in the first book, the techniques described in the second book build upon one another in ever more complex combinations. It begins with the simple sand of the frit and proceeds to the full-scale window, made of multicolored glass, assembled with iron, and perhaps including gems and paint.

The techniques of the third book are the most complex of the treatise, but they too follow a rough pattern from simple to complex. From the workshop and forge, tools and files, and the refining of silver, to the refining of gold and the making of a gold chalice, techniques become more complex as the chapters progress.¹¹⁶ A description of iron tools opens the book and is followed by instructions on beating out a chalice of silver. Theophilus then proceeds from the making of cast handles for the chalice to the nielloed ornament of its surface. Later, the creation of a censer of gold repoussé leads to the making of a more complicated cast censer.¹¹⁷ At the end of the book are discussions of even more complex objects, like organs and bells.¹¹⁸

Throughout the treatise, the technical instructions build upon one another; each new technique depends upon the techniques described in prior chapters. From the layering of paint and the increasingly complicated mixtures of colors to the making of the window and the painting of enamel and niello, the techniques require higher and higher levels of skill. The process of learning, and the objects themselves, progress in an additive manner. The techniques of the treatise multiply in an expanding field of knowledge, just as an object can be made of many component parts. Moreover, the selection, order, and content of the instructions roughly mirrors the narrative of the prologues. Just as the student proceeds in his spiritual learning, so

too, in a carefully gauged pedagogical program, instructions proceed from simple to complex.

The series of steps in both the prologues and the instructions are cumulative, and this guides the structure of the whole text. The idea of learning by accumulation was a major pedagogical strategy of twelfth-century monastic learning. In the *Didascalicon*, Hugh of St. Victor likens the process of understanding scripture to the building of a stone wall. For Hugh, it is important first to grasp the literal sense of words, for only then can the allegorical and spiritual meanings of scripture be built. They are as a “superstructure” upon the foundational wall of clear comprehension of the literal sense:

This then, my student, is what we propose to you. This field of your labor, well cultivated by your plough, will bear you a manifold harvest. All things were brought forth in order: move along in order yourself. Following the shadow, one comes to the body: learn the figure, and you will come to the truth. . . . Just as you see that every building lacking a foundation cannot stand firm, so also is it in learning. The foundation and principle of sacred learning, however, is history, from which, like honey from the honeycomb, the truth of allegory is extracted. As you are about to build, therefore, “lay first the foundation of history; next by pursuing ‘typical’ meaning, build up a structure in your mind to be a fortress of faith. Last of all, however, through the loveliness of morality, paint the structure over as with the most beautiful of colors.”¹¹⁹

Hugh’s argument suggests a layering of knowledge: the bricks of the Old Testament and of

knowledge of the fathers are laid and polished, and upon them is built, stone by stone, more and more complex ideas, from the historical to the typological to the moral.¹²⁰ To extend Hugh's metaphor, learning must begin with a solid foundation, with simpler ideas, so that there is ground to support the loftier ideas and finished "colors" represented by the moral sense. It is just like Theophilus's description of laying a foundation of nude color before one adds highlights or shadows.

The cumulative mode of learning evident in both Theophilus and Hugh of St. Victor is also similar to the twelfth century's analytic system of grammatical study. From the earliest times, monastic education relied heavily on the school text of Donatus, *Ars grammatica*. Written in the mid-fourth century, Donatus's text divided language into units of speech that could then be put together to form clauses and, finally, rhetorical figures.¹²¹ Such an analytic approach by no means abandoned spirituality, for the goal of the exercise was religious. According to Jean Leclercq, grammar was a foundation of monastic learning and spiritual devotion because it aided in the understanding of scripture; when combined with the compunction of desire for God, it led to wisdom.¹²² It is highly probable that Theophilus, whose Latin demonstrates a level of some sophistication, had an education based on Donatus's *Ars grammatica* or on similar works. Whatever his training, his tract is usefully compared to the structure of grammatical teaching in the period, for he organizes information according to a similar analytic system: artisanal knowledge is the sum of its parts, predicated on the understanding of its elements and of their possible arrangements. The student learns each element of technique—the mixing of pigments, the

dyeing of glass, or the purification of metal, for example—and then how to combine them so as to produce an object. Instruction in techniques is integral to the order and guiding framework of the text.

Theophilus's treatise reframes practical knowledge of art-making by organizing it according to theoretical principles. Hugh of St. Victor's *Didascalicon* does something similar with philosophy. Hugh's treatise outlines a philosophy of education that directly serves spiritual wisdom, defining the fields of learning and human activity and categorizing the arts as liberal, mechanical, or logical. The mechanical arts, manual activities that serve the necessities of life, are of special interest for Hugh; as a part of understanding of the physical world, and man's activity in it, they aid the pursuit of divine wisdom. Hugh's work defines the purpose, fields, and steps of learning, as well as the purpose, fields, and process of craft activity. By organizing knowledge according to theoretical principles and showing such activities to operate according to a theoretical system, Hugh could prove that practical knowledge could be employed in the service of philosophy.

We have seen that the separation of prologues from instructions in *On Diverse Arts* is not an indication either of multiple authors or of disorder, but that it instead serves to divide subject matter and function. Like Hugh, Theophilus uses his prologues to describe the purpose of his work, while the instructions themselves outline the procedure of an activity. This conceptual separation might be seen, roughly speaking, as analogous to antique traditions of rhetoric and grammar.¹²³ Rhetoric and grammar—eloquence and speech—are separate disciplines, but they must guide one another. As John of Salisbury

describes it, “just as eloquence, unenlightened by reason, is rash and blind, so wisdom, without the power of expression, is feeble and maimed. . . . Although he [who underestimates the importance of eloquence] may seem to attack eloquence alone, he undermines and uproots all liberal studies, assails the whole structure of philosophy, tears to shreds humanity’s social contract, and destroys the means of brotherly charity and reciprocal interchange of services.”¹²⁴ John’s defense of the trivium and quadrivium—the core subjects that constitute the seven liberal arts—relies fundamentally on the link between rhetoric and grammar, which (he argues) forms the basis of education and culture.

Rupert of Deutz’s exegeses of the bible and liturgy are notable for a similar tendency; he too seeks to reveal the underlying order of broad areas of lived experience and relate it to biblical and liturgical life. Rupert spent his early life as a teacher and writer at the abbey of St. Lawrence in Liège, but his writings eventually embroiled him in the eucharistic controversies of his day and forced him to leave Liège for Siegburg in 1113. In 1120, he returned to become abbot of the monastery of St. Heribert at Deutz, across the Rhine from Cologne, a dependency of St. Pantaleon. Rupert’s interest in an exegetical interpretation of the liturgy was expansive. In his text *On the Divine Office*, he systematically describes the significance of the material objects used in the mass as well as the actions of the clergy. The priest’s ring, for example, is treated in a section on liturgical vestments and is said to signify the “many and different gifts” distributed to the believers through the Spirit. For Rupert, it therefore “proclaims the entire faith and sensible doctrine of all the church,” analogous to the marriage of Christ to Ecclesia.¹²⁵

Analogies between the writings of Theophilus and Rupert of Deutz are particularly suggestive, since their treatises circulated in the Benedictine realm of Liège, Cologne, and the Weser river valley. John Van Engen, among others, has noted the similarity between Rupert’s justification of manual labor and Theophilus’s justification of art.¹²⁶ Scholars have found expressions of Rupert’s philosophies in surviving works of art, in connections to Wibald of Stavelot, or in the decorations of Schwarzhof, for example.¹²⁷ We have already seen that Wibald’s activity in the realm of art was significant, and the objects he commissioned bear the marks of his education and intellectual and spiritual interests. If he was directly involved in these projects, we see a model where a patron not only commissioned objects for his own abbey but served as advisor for other artistic projects. Given this local Benedictine context, where abbots, exegetes, and imperial patrons were connected, the sophisticated level of learning shown in Theophilus’s treatise should not be surprising. Indeed, it is likely the tract was intended for this audience.

Conclusions

On Diverse Arts reflects an ambition to organize large quantities of information; this ambition was characteristic of the early twelfth century and noticeable in the works of Rupert of Deutz, of Hugh of St. Victor, and of John of Salisbury, among others. Close analogies between the goals and philosophies in *On Diverse Arts* and the intellectual currents evident in the works of these writers, even in Wibald of Stavelot, suggest that Theophilus should be read within the context of a

twelfth-century intellectual sphere. More than an articulation of past expertise, more than a technical instruction manual, the treatise participates in the pedagogical trends of the period.

One might note here the famous saying of Bernard of Chartres that “we [moderns] can see farther than the ancients because we are dwarfs standing on the shoulders of giants,” quoted by John of Salisbury in the *Metalogicon*.¹²⁸ While this phrase has often been interpreted as an instance of the twelfth century’s optimism, self-awareness, and confidence in its own accomplishments, others, led by C. Stephen Jaeger, have seen it as pessimistic, a self-deprecating estimation that one cannot rise to the challenge of one’s predecessors.¹²⁹ Theophilus may be a good example of this complicated relationship with the past, for just as the Wolfenbüttel manuscript

sets the medieval author as an heir to the ancient author—and perhaps even *On Diverse Arts* as a new, modern, Christian, version of the ancient text—Theophilus’s treatise is only three books, compared to Vitruvius’s monumental ten. His intention is not to surpass the ancient model, but neither is he intimidated by it. Perhaps using Vitruvius as a model, Theophilus defines the process of art within a pedagogical structure based on Augustine’s idea of spiritual ascent. Through this treatise, art becomes something theorized, ordered, and written down. The following chapters will examine further the logic and theoretical perspective that emerges, beginning with the nature of physical matter.

41

2.

THE TRANSFORMATION OF MATTER

*Do not despise useful and precious
things, simply because your native
earth has produced them for you of
its own accord or unexpectedly. For
foolish is the merchant who sud-
denly finds a treasure in a hole in the
ground and fails to pick it up and
keep it.*

THEOPHILUS,
On Diverse Arts, prologue I

Eight manuscript copies of *On Diverse Arts* were made prior to 1300; of these, six are bound with texts loosely associated with Latin didactic texts, addressing building, agriculture, medicine, and astronomy. Six more copies of *On Diverse Arts* were made in the next century; of these, three are bound with similar texts.¹ We have seen in chapter 1 how the Wolfenbüttel manuscript is bound with Vitruvius's *On Architecture* and seems to continue the project set forth by the Latin author. A slightly later manuscript in Cambridge shows the trend even more clearly. Now known as Cambridge University Library Ms Ee. 6.39, the manuscript was probably written in England in the early thirteenth century (plate 2).² It is a small, clean volume, containing nearly all the chapters of Theophilus's first book, with its prologue, and a number of chapters from book III. *On Diverse Arts* is the third text in the volume, following Palladius's *Opus Agriculturae*, a fourth-century treatise describing the cultivation of crops and management of land, and *De viribus herbarum*, a text describing the medicinal properties of herbs that was often ascribed in the Middle Ages to the poet Macer, a first-century B.C.E. Roman, though it was probably written by Odo of Meung, an eleventh-century physician living in the Loire valley.³

The combination of texts is not by chance or later intervention. The three texts were bound together from the earliest times, for they are

listed in a contemporary table of contents on the second folio, which is followed directly by the first pages of Palladius, in the same quire.⁴ The table of contents also lists an additional five texts, now lost: *Speculum penitentis*, *Lapidarium*, *Glose super antidotarium*, *Alexander de arte predicandi*, and *Tropus magistri W. de Montibus*. These appear to have been lost since the late Middle Ages, for in the margin of the table of contents an inscription dating from the fifteenth century notes that they are “non infra” (not below).⁵ Evidence suggests that the manuscript was copied in one scriptorium in a relatively short period of time; similarities in the script suggest that the different texts are a related group. Significantly, the hand that transcribed the Palladius text also wrote the table of contents, and the layout of the volume is remarkably consistent. The number of lines on a page and prickings and rulings are similar throughout; the greatest variation is in the width of the margins, which vary due to trimmings, especially in the text of Theophilus. While they were not transcribed by the same hand, the scripts of the Theophilus and Palladius texts are very similar, with three-line initials colored in red and blue. Even the short, squat letters of the Macer text are broadly consistent with the script used in the transcription of Palladius and Theophilus.⁶ The Macer text seems to be the earliest of the three, and it presents the most differences: it is written in two columns as opposed to one and contains only red and green initials.

Though it binds together a number of disparate texts, the Cambridge manuscript was thoughtfully assembled. It contains carefully selected portions of *On Diverse Arts* that do not appear to derive directly from any surviving manuscript.⁷ With the prologue and first

book of the text making up the majority of the extract, the title was adjusted accordingly and refers to the text as one on painting: “Prologue to the first book of Theophilus monk and priest, *On Diverse Arts*, first, on colors.”⁸ There is little distinction made between the end of book I and the excerpted chapters from book III. Without capitula, incipits, or explicits, the only mark of the new section is the break in numerical chapter order. The chapters from book III do not follow the order given in other manuscripts; nevertheless, the logic of choice emerges. The extracts include two chapters on cleaning gold, silver, and gilding and the soldering of tin; five chapters on iron and the making and tempering of files; two chapters on bone carving; and one on polishing gems. With the exception of bone carving, the chapters seem to be concerned with the care of existing objects: cleaning, filing, re-soldering, and tempering. Decorative techniques such as repoussé or engraving, for example, are absent. The extracts from Theophilus are followed by forty-one chapters from the *Mappae clavicula* that run continuously from Theophilus’s text. The only marker separating them are the seven lines of hexameter that often introduce the *Mappae clavicula*.⁹ The compiler of the Theophilus extracts, then, seems to have been interested in the mixing of colors, as the title suggests, and information that might be useful for the care of objects.

Unlike other manuscripts, the Cambridge manuscript contains three very faint line drawings in its margins. Folio 6v (plate 2) shows a drawing of a woman, perhaps representing a virtue, holding a lily; folio 12r (fig. 8) contains a drawing of a lion; and folio 13v, a dragon (fig. 9). The drawings are consistent, executed with an elegant line, and seem to date to the thirteenth

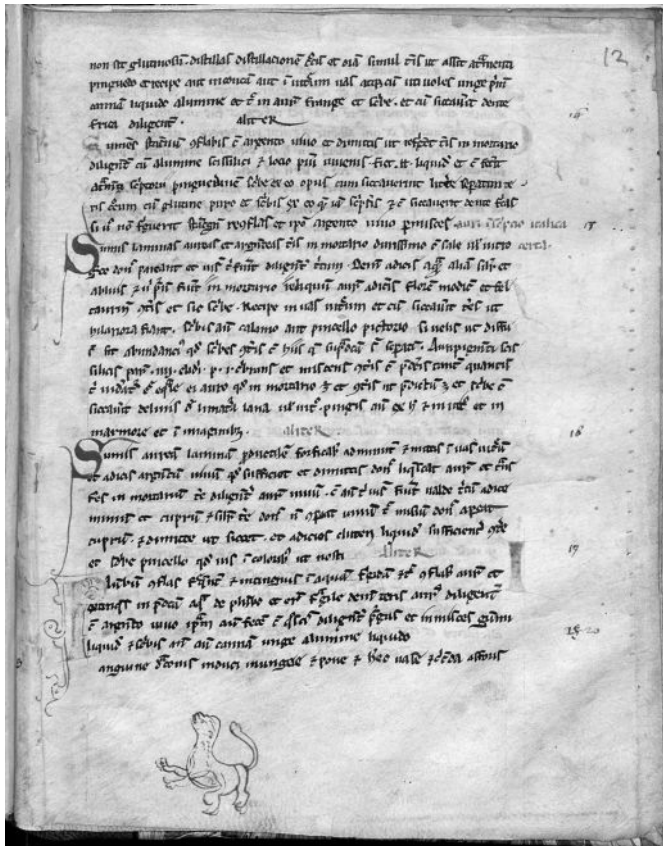


FIG. 8

Theophilus, *De diversis artibus*, early thirteenth century. Cambridge, University Library Ms Ee. 6.39, fol. 12r.

century. They are placed on the page at an off-angle, in the bottom margin, as if they are recordings for memory—for they bear no direct relation to the text itself. The parchment is dirty, and a later trimming has cropped the drawings, which may be evidence of rebinding and transformation of the text from a working pamphlet to a library book. The drawings, the specificity of the title, and the chapters chosen for transcription suggest that the text served a practical function at one time, if briefly. That it was bound with contemporary transcriptions of Palladius, Macer, and William de Montibus in the thirteenth century, however, shows that this is only a part of the story.

The book's stylistic consistency and thoughtful assembly compel us to read it as a purposeful compilation that served particular functions. Composite texts such as this one were not uncommon in the Middle Ages, and as recent scholars have shown, we may begin to see such texts' potential for carrying meanings constructed through literary patterns and structures rather than through singular intent or authorship.¹⁰

The Cambridge manuscript places Theophilus's text at the heart of a larger field of scientific knowledge. Palladius's book on agriculture treats the properties of soil, air, and water, instructing the reader on how best to employ

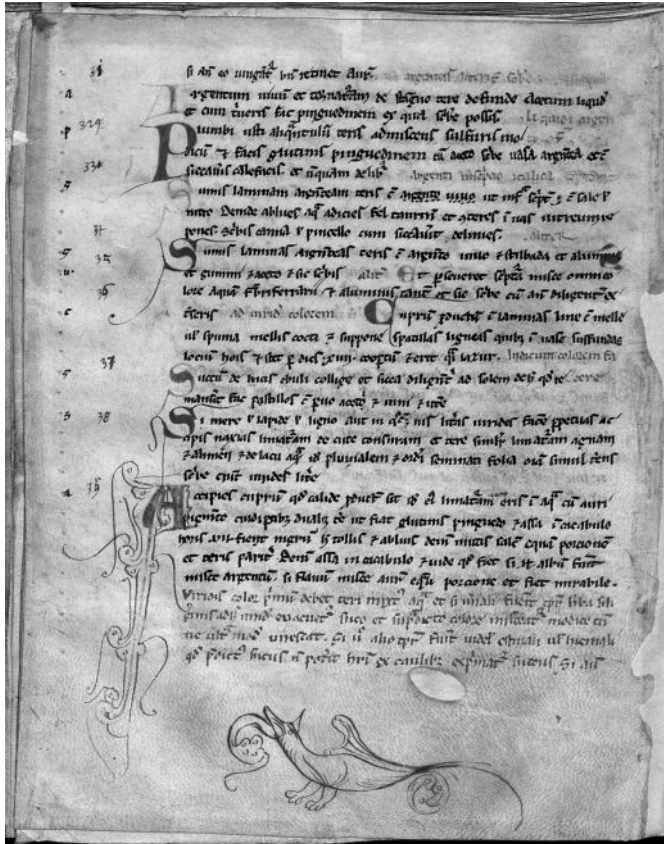


FIG. 9

Theophilus, *De diversis artibus*, early thirteenth century. Cambridge, University Library MS Ee. 6.39, fol. 13v.

these to grow plants; Macer's text categorizes and describes the medicinal properties of herbs and plants. Both were thought in the Middle Ages to be antique sources, enhancing their value as a source of knowledge.¹¹ The progression from Palladius to Macer seems to contain a logic: Palladius's text describes the earth, its soil, and the instructions for activities like sowing and reaping harvests, and Macer's goes on to explain how such plants might be used. Together, the two describe a set of natural phenomena and show a concern for the productive manipulation of the natural world.

Of the five texts no longer included in the volume, three addressed pastoral duties. The

Speculum penitentis and *Tropus magistri*, first and last in the manuscript, are works by William de Montibus, a teacher at Paris and chancellor of the school at Lincoln from 1190 until his death in 1213. The *Speculum penitentis* is a catalogue of sins and a treatise on the writing of sermons; the *Tropus magistri*, a collection of moral texts. The penultimate text, *Alexander de arte predicandi*, may be what is known as the *De artificioso modo predicandi* of Alexander of Ashby, prior of an abbey of Augustinian canons at Ashby, England, in the early thirteenth century. This text survives in only two known manuscripts, Cambridge University Library li.1.24 (fols. 332–39) and Oxford, Magdalene

College MS 168 (fols. 128v–130r). William's texts were best known in the region of East England and the Midlands; they did not circulate widely, nor for long after his death.¹² The priory of Ashby is also in this region, in modern Northamptonshire. It seems more than likely, then, that Cambridge MS Ee. 6. 39 comes from the east of England or the Midlands and that it dates to fairly early in the thirteenth century.

The other two titles included in the table of contents are concerned with the conditions of the natural world. The first of these is the *Glose super antidotarium*, perhaps a gloss on the *antidotarium* written by Nicholas of Salerno, whose mid-twelfth-century treatise on medicine and remedies was a standard work on the subject. By the later twelfth century, Mattheus Platearius wrote an analytical commentary on the *antidotarium*, shifting its emphasis from practical instructions to philosophical analysis; he describes, for example, the reason compound medicines were beneficial and defines the word *antidote* in etymological terms.¹³ Though the title of the other text, the *Lapidarium*, betrays little, it may have been a transcription of Marbode of Rennes's eleventh-century poem on the magical and natural properties of stones, also known as *De lapidibus*. Marbode's tract on stones is often found alongside Macer: seventeen extant manuscripts of *De lapidibus* also contain Macer, including a fifteenth-century manuscript now in Montpellier with excerpts from the first and third books of Theophilus.¹⁴ *De lapidibus* drew heavily on the symbolic tradition of stones and on classical sources in particular; Concetta Giliberto has suggested that these *lapidaria* were used by clerics and archbishops in their pastoral duties, either in sermons or in the explanation of themes and symbolism.¹⁵ Notable for our analysis is that

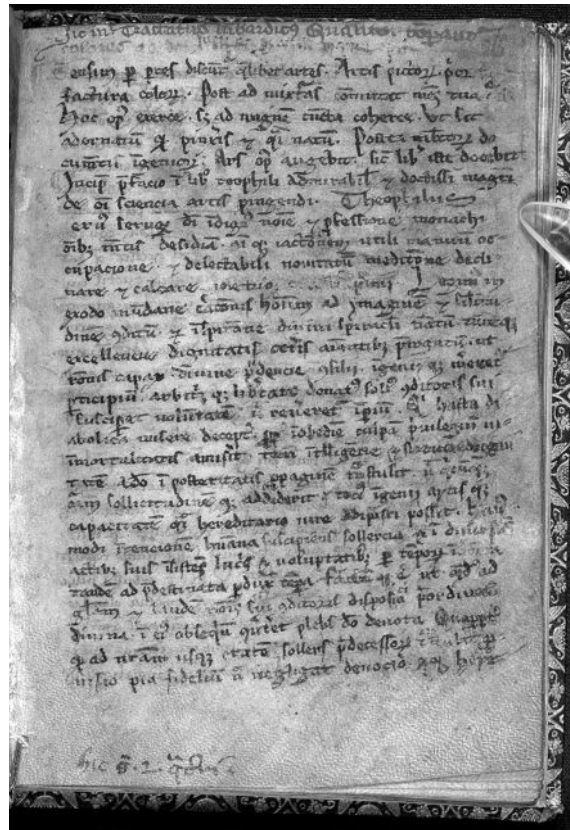


FIG. 10
Theophilus, *De diversis artibus*, thirteenth century. London, British Library Egerton MS 840a, fol. 6r.

the *De lapidibus* explains the nature of stones in a literary, poetic form. In both Marbode's poem and the *Glose super antidotarium*, the characteristics of natural elements are part of a larger philosophical conception of the world and its elements.

The Cambridge manuscript is the earliest surviving evidence that Theophilus's treatise circulated beyond northern Germany and the region of the Meuse. The only other early copy that can be localized to England is British Library

Egerton MS 840a, also made sometime in the first half of the thirteenth century (fig. 10). The origins of the Egerton manuscript are uncertain, but it seems likely that it was owned by the canons of Saint Augustine's abbey, Canterbury, from at least the fourteenth century. Before the nineteenth century, when it was transferred to London, it was bound with scientific texts, including a treatise called *De constructione et usu sphaerae et astrolabii*.¹⁶ It seems likely, then, that *On Diverse Arts* was circulating among English canons in the thirteenth century, and it was clearly read alongside other scientific and didactic texts.

The Cambridge and Egerton manuscripts are strong evidence that by the thirteenth century, scientific knowledge of material and religious writing could serve one another, as they do in *On Diverse Arts*. The treatises of Palladius and Macer served the agricultural and medicinal needs of a religious community by explaining natural phenomena and the cultivation of the earth and its treasures. The commentary on antidotes complemented these with information on medicinal compounds, and the *Lapidarium* described stones and their properties, perhaps in order that one might know how to use them in medical remedies. These works were also a reference for symbolism and thus complemented the later texts on penitence and the writing of sermons. The use of *On Diverse Arts* by an audience of canons interested in writing sermons seems a far cry from the twelfth-century Benedictine world of northern Germany, where the treatise was composed in the context of monastic craftsmanship, but there is a rather strong continuity. As we shall see, both *On Diverse Arts* and didactic literature share a concern for the properties of the natural world.

Didactic Literature in the Twelfth Century

As we saw in chapter 1, Vitruvius's *De architectura* was valued in the Middle Ages for its broad understanding of the nature of the world. For the medieval reader of Vitruvius, the architect was a "world-builder." His knowledge was of the earth, and he described an ideal city not only in formal terms but also in medical, agricultural, and astrological terms. Architecture was closely tied to the environment in which it stood, and the theories that guided architecture also guided the world around it. The properties according to which buildings and cities were built were a manifestation of a balanced cosmological system of humors, elements, and planets.¹⁷

In the early fifth century, Martianus Capella drew on Vitruvius as an authority on the balance of the planets and the harmony of the spheres and referred to the earlier author's theories of medicine for the links between disease, the circulation of air, and the location of cities.¹⁸ In the ninth century, Hucbald of St. Amand linked Vitruvius's eight primary and four secondary winds to the Pythagorean proportions of music. He also analyzed each musical interval and then proceeded to a new section entitled *Sequitur Praemissus Expositor*, in which he defined the relations of the intervals to each other and then to the set of tones in an octave. Perhaps surprisingly, he opened this section with a reference to Vitruvius.¹⁹

As Vitruvius said in the book *On Architecture*, according to natural sciences there are not more than eight winds; the principles are four, and four are subjected to these. Four truly, which are added so that there are twelve, just as there are four semi-tones to

eight tones. For in the waves of the sea and of the rivers always the first wave sounds more than the seven following.

The ninth truly is similar to the octave. Similarly you can discriminate the octave in thunder, by which Pythagoras the philosopher invented these eight consonances of tones, recited according to the harmony of heaven at the mountain of Atlantis, which is close to heaven, of whose proportions the form of the five zones of heaven he completed, of which that [zone] of yours, the world, is governed by proper mixture.

Indeed all the concords of music either to one II have twofold, or III in tri-fold, or IIII in four-fold, or V, in one and a half times, or VII in four-thirds. Finally, as was said above, out of five tones and two semi-tones an eighth whole octave is completed, which is the first tone; so that VI, XII to which all sounds of music are brought home. Whatever even the number to twelve can be divided, thus as to that same twelfth or through three and four, or four and three is divided, is music, constant in the above said proportions. For Pythagoras adjusted double in VI, one and a half as much in VIII, four thirds as much in IIII, to XII. Whence out of five simple tones, it is, with a fifth, a fourth, an octave . . .²⁰

For Hucbald, Vitruvius's winds provide a harmonious system of operation akin to the system of sounds: waves, the sounds of thunder, and the zones of heaven, "of which that [zone] of yours, the world, is governed by proper mixture" (*quarum temperie mundus iste moderatur*). By linking Vitruvius's winds to Pythagorean proportions, Hucbald drew attention to the system by

which the winds operate. He was interested not so much in the nature of the winds themselves but in the fact that their number corresponded to the proportions of tones and harmonies outlined in Pythagoras's musical system, and thus to the order that governs the world.

While the late tenth century had seen writers like Richard of Saint-Remi and Gerbert of Reims use Vitruvius alongside Boethius as an authority for the classification of philosophy and for the divisions of theory and practice,²¹ the early twelfth century saw interests shift to more systematic and scientific analyses of the natural world. In her census of Vitruvius manuscripts, Carol Krinsky describes a twelfth-century manuscript that contains nearly the entire text of Vitruvius's *De architectura* following an earlier copy of Boethius's *De arithmetica*, linking Vitruvius to a philosophy of mathematics.²² Another twelfth-century author, William of Malmesbury, was perhaps more practical, for he used Vitruvius to explain how an organ works by pushing air through the pipes at various measurements.²³ A manuscript in London, also containing the entire text of Vitruvius and dating to the mid-twelfth century, shows similar groupings of texts. It begins with Palladius's *De agricultura*, followed by Vitruvius and then Vegetius's *De re militari*.²⁴ In sum, it seems that while earlier writers like Martianus Capella and Hucbald seized upon the harmonic properties explained by Vitruvius, and tenth-century writers used it to help define theoretical issues, twelfth-century writers connected Vitruvius to agricultural and scientific subjects.

Again, Hugh of St. Victor provides insight. In his discourse on learning, Hugh classified Palladius, Vitruvius, and Virgil's *Georgics* within the realm of mechanical science as sources of world knowledge:²⁵ "Mechanical Science has had many

authors. Hesiod Ascreaeus was the first among the Greeks who applied himself to describing farming, and 'after him Democritus. A great Carthaginian likewise wrote a study of agriculture in twenty-eight volumes. Among the Romans, Cato is first with his *Concerning Agriculture*, which Marcus Terentius subsequently elaborated. Vergil too wrote his *Georgics*; then Cornelius and Julius Atticus Aemilian, or Columella, the famous orator who put together an entire corpus on this branch of knowledge.' Then there are Vitruvius, *On Architecture*, and Palladius, *On Agriculture*."²⁶ For Hugh, Vitruvius is an author of the mechanical arts, and *De architectura* is a part of didactic literature, following in the tradition of Virgil and Cato. In the greater system of Hugh's pedagogy, Vitruvius was a source of practical information relevant to his larger spiritual and pedagogical framework.

Throughout the Middle Ages, Vitruvius's text seems to have been valued for its explication of harmonies, proportional relations, and the interdependence of man-made and cosmic systems. While early medieval writers like Hucbald and Martianus Capella were concerned with numerical proportions, twelfth-century writers like William of Malmesbury and Hugh of St. Victor linked Vitruvius's sense of cosmic systems to agriculture, astronomy, and even learning itself. Agricultural treatises like Palladius's *Opus agriculturae*, which opens the Cambridge manuscript, were read similarly. Written in the fourth century, Palladius's text gives instructions for managing an estate, and, like Vitruvius, his instructions emphasize the quality of the soil and the air. The treatise follows closely in the tradition of Virgil's *Georgics*, a relationship well known in the Middle Ages.²⁷ In an eleventh-century manuscript of Palladius now

at Corpus Christi College, Cambridge, the scribe has written commentary in the margins, referring the reader to corresponding passages of the *Georgics*.²⁸

Two late twelfth- or early thirteenth-century manuscripts, one a copy of the other, both now in Paris, follow this trend. They contain not only excerpts from Theophilus's book on metalwork but also Palladius's *Opus agriculturae*. The connection was strong enough to stick; in the fifteenth century, Varro's *De re rustica*, an ancient didactic text often considered a predecessor to the *Georgics*, was added to both Parisian volumes.²⁹ *On Diverse Arts* may even have helped to shape the course of later didactic literature. Theophilus's treatise was mentioned by the sixteenth-century bibliographer Conrad Gesner in a list of authors on the history of glass that included the elder Pliny; the Florentine humanist Pietro Crinito, writing on Tiberius; and Crinito's humanist contemporary Polydore Vergil on minerals.³⁰ Josias Simmler, in his expansion of Gesner's work, noted that the Wolfenbüttel Theophilus was owned in the sixteenth century by Georgius Agricola, the author of *De re metallica*.³¹ Disappointingly, Agricola makes no mention of Theophilus in the *Bermannus*, his dialogue on metallurgy published in 1528, although he cites Vitruvius extensively.³² Regardless, Agricola's project is clear: in the first book of his treatise he makes a strong case for the importance of metallurgy and the sophisticated learning required by it, linking his modern work to the didactic tradition.³³ By intent or by the intervention of later readers, then, Theophilus's treatise thus became ever more closely linked to the ancient didactic writers.

The fact that so many manuscripts of Theophilus were bound with didactic tracts suggests

that the treatise was valued for the way in which it couched practical instruction in a theoretical framework and drew on ancient authority. We have already seen that the alternating structure of prologues and books in Vitruvius's *De architectura* seems to be repeated in *On Diverse Arts*. According to Thomas Haye, one of the defining features of didactic literature, of which *De architectura* is a famous example, is its building-block structure: these treatises are usually divided into books and chapters, each dealing with different aspects of the subject and building one upon another in hierarchies of complexity.³⁴ *On Diverse Arts* is organized similarly, with chapters treating related materials and objects progressing in a hierarchy of complexity and material preciousness. Thus the structure of the book operates as an explanation of the nature of materials, their hierarchies and relative values, and their best use. Indeed, Theophilus's interest in the nature of materials is closely related to twelfth-century trends of thought and twelfth-century readings of didactic literature. Like his contemporaries William of Malmesbury and Hugh of St. Victor, he takes great care to connect the abstract properties of material to actual functions.

From this perspective, the connection between *On Diverse Arts* and medical texts like Macer suggested by the Cambridge manuscript becomes more clear. Indeed, it is a connection that persists in later manuscripts.³⁵ Like the ancient didactic texts, Macer's treatise is written in hexameters and divided into chapters, each devoted to one plant. Allium, for example,

is called (s)cordeon in (the Greek) language. The expert physicians say that it is hot and dry in the fourth degree. Applied in a dressing or rubbed it cures the bite of

a scorpion. Harmful worms flee from the smell of pounded garlic. . . . When it is boiled with mead and mixed with vinegar and drunk it kills worms and expels them from the stomach. . . . Diocles prescribes it with centaureum for those suffering from dropsy because it dries the watery humors; he also prescribes it for those who suffer from kidney stones.³⁶

Thus Macer catalogues the natural elements and describes the properties and effects of each, drawing consciously on ancient sources.

Medical recipes of the time typically followed the pattern we see in Macer: an component like allium would be categorized, titled, or introduced according to its purpose or the ailment it was meant to cure, and each treatment would provide a list of ingredients and describe the application of those ingredients. The historian Brian Murdoch carefully distinguishes this purpose—what he calls “effecting” a cure—from those of charms or prayers, which (he states) either demand or request a cure, respectively.³⁷ A certain level of expertise was also expected on the part of the reader, as tools or details of the procedure are often lacking.

The books of artistic technique with which *On Diverse Arts* is associated follow the same patterns of instruction as the medical recipe books. In the *Mappae clavicula* and Heraclius's *De coloribus et artibus romanorum*, the recipes usually begin, or are titled with, their purpose, as in a recipe for “Making of Indigo Pigment.”³⁸ The *Mappae clavicula* reads, “Collect the juice of dwarf elderberries and dry it thoroughly in the sun. From what remains make pastilles with a little vinegar and wine, then use it.”³⁹ This recipe contains a description of the ingredients

needed and a rough description of the procedure by which they are to be mixed. Indeed, as Brian Murdoch has shown, medical, cooking, and technical recipes often overlap.⁴⁰ The *Mappae clavicula*, for instance, includes recipes for making candies of sesame or sugar alongside instructions for pigments, making little distinction between the two.⁴¹

Twelfth-century writing on medicine is best exemplified by Hildegard of Bingen's *Physica*. Written around the same time as *On Diverse Arts*, the *Physica* discusses metals and stones, describing how to use gold, silver, or pearls to aid various treatments. Here, metals are described according to their elemental properties and are used in medical recipes. Hildegard describes gold as "hot. Its nature is somewhat like that of the sun, and it is almost like the element air." If eaten as she describes, and "if the stomach is cold and full of mucus [i.e., cold and wet], it will warm it and purge it, without danger to the person."⁴² Silver, on the other hand, is cold. "The strong natural cold of silver diminishes hot, cold, and moist humors by its sharpness."⁴³ In medicine, the elements were used to bring the body into balance, and thus medical recipes focus on mixtures and properties of materials, including metals.

Medical recipes are concerned with mixing elements for a desired effect. Theophilus's chapter on flake-white follows a now-familiar pattern:

To prepare flake-white, get some sheets of lead beaten out thin, place them, dry, in a hollow piece of wood, like the copper above, and pour in some warm vinegar or urine to cover them. Then, after a month, take off the cover and remove what white there is, and again replace as at first. When you have sufficient and wish to make red lead with

it, grind this flake-white on stone without water, then put it in two or three new pots and place it over a burning fire. You have a slender curved iron rod, fitted at one end in a wooden handle and board at the top, and with this you can stir and mix this flake-white from time to time. You do this for a long time until the red lead becomes completely red.⁴⁴

His instructions begin with the purpose—making flake white—and proceed to describe the ingredients required. He then describes mixtures, whether of pigments, colored glass, even of composite metals. This pairing of purpose and the materials that serve it is the same as in the medical recipes, in Hildegard's prescription of gold, or the recipes for pigments like indigo in the *Mappae clavicula*.

Hildegard's *Physica* is guided by a philosophy of humors and elements that was a part of medieval astrology. Contemporary tetradic diagrams like the *Liber Floridus* of Lambert of St. Omer, dating from about 1120, show the vibrancy of the theory in the period. The manuscript is famous for its many diagrams and images of natural phenomena; most relevant here is a visual representation of the relationships of the planets and their links to winds, medical humors, and astrology (fig. 11).⁴⁵ Man (*homo*) is at the center, linked by the four humors to four circles, each binding a season with an element, so that the sanguine (*sanguis*) humor is connected to spring (*ver*) and the element air (*aer*), for instance, while melancholy (*melancholia*) is connected to autumn (*autumnus*) and earth (*terra*). As Faith Wallis writes, such diagrams present "a vision in which time, the cosmos, and man are harmoniously connected by the shared structure of the four

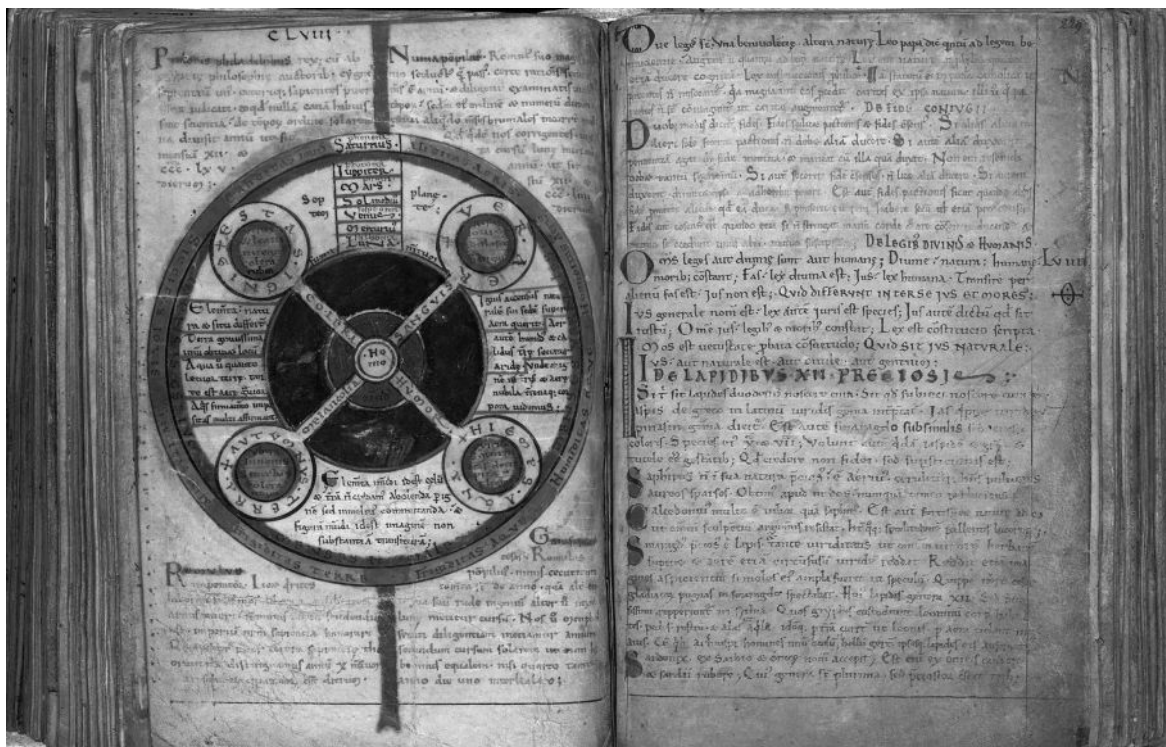


FIG. 11

Lambert of St. Omer, *Liber Floridus*, ca. 1120. Ghent, University Library MS 92 (BHSL.HS.0092), fols. 228v–229r.

elements and their qualities. In medieval didactic diagrams, these interconnections of shared pairs of qualities knit the elements, seasons, and humors into a circle.”⁴⁶ Hildegard’s medicine and Lambert’s diagram are both, like Theophilus, indebted to ideas about microcosm and macrocosm that we see in the medieval interpretations of Vitruvius. Like Theophilus, they are an example of the twelfth-century application of broad philosophical systems to concrete particulars.

On Diverse Arts was also bound with astrological texts. British Museum, Egerton MS 840a, for example, was once bound with a text on astrology and the use of the astrolabe.⁴⁷ The two

texts were separated in the nineteenth century,⁴⁸ perhaps because by then it was assumed that they belonged to two very different fields of knowledge. The logic inherent in the former binding, and the idea that the texts could all be strands of scientific learning, had not been considered.

Theophilus’s interest in *natura* may have been part of a general intellectual trend of the twelfth century. While earlier writers like Isidore of Seville were concerned with the properties of material, information gleaned from biblical sources, and Latin writers like Pliny, writers of the twelfth century were increasingly interested

in the natural world as an immediate source for study. Adelard of Bath composed his *Quaestiones naturales* in the early twelfth century, and manuscript evidence suggests that it was circulating on the continent shortly thereafter. The book answers questions ranging from why plants grow to what determines the saltiness of rivers, addressing them in terms of causes and properties and above all working under the assumption that the will of God operates according to reason. “It is indeed the will of the Creator that plants should be born from the earth,” Adelard writes, “but that will is not without reason.”⁴⁹ The extent to which Theophilus knew the Latin translations of Greek and Arabic scientific texts by figures like Adelard of Bath is unknown; his reference in the first prologue to the repoussé, cast work, and engravings of Arabia is tantalizingly vague.⁵⁰ Yet a silver reliquary box with a Kufic inscription, kept in Liège, and an eighth-century textile from Iran that was used to wrap the relics of Saint Lambert are evidence that the circulation of objects was broader than we know.⁵¹ As Eva Hoffman and others have recently shown, the transmission of knowledge and art objects from the Middle East into Europe could take many forms.⁵² In our case, the scientific bent of *On Diverse Arts* may suggest that Theophilus was aware of these new fields of learning, if only indirectly.

Seen within the context of these interpretative trends and the tradition of didactic literature, we might read *On Diverse Arts* as a contribution to twelfth-century interest in the natural world and the cosmic order. If we turn back to the text with this in mind, it becomes clear that Theophilus is presenting the materials of art in a very particular way; his treatise emerges as a set of instructions for understanding and manipulating the natural world.

Materials in *On Diverse Arts*

Modern editions of *On Diverse Arts* title its three books the *Book of Painting*, the *Book of Stained Glass*, and the *Book of Metalwork*. The division of media is a major organizational element of *On Diverse Arts*, but it is important to note that these titles do not appear in manuscripts until the seventeenth century.⁵³ The early manuscripts, now at Wolfenbüttel and Vienna, label the books by number.⁵⁴ This tendency stems from a long tradition, visible in the copy of Vitruvius’s *On Architecture* in the Wolfenbüttel manuscript.⁵⁵ In the few cases where manuscripts provide titles for the books, they are descriptive and usually attached only to excerpts, as in the case of London, Egerton 840a, with its description of Theophilus’s text as a treatise on “how to combine colors for painting.”⁵⁶ The thirteenth-century Cambridge manuscript is similar, with a rubric introducing the prologue that might imply awareness that the text is an excerpt, indicating that this is the first prologue and the first section on colors: “Prologus primum liber Theophili monachi et presbyteri de diversis artibus in primis de coloribus.”⁵⁷

A closer look at the text suggests that the categories of media within *On Diverse Arts* follow a rather different logic than might be expected, and thus our modern descriptive titles lose some accuracy. The passage addressing techniques in the first prologue provides a good starting point. As we have seen, it recounts the various techniques the book will address, but not all of these correspond directly to chapters of instructions: there are, for example, no descriptions of monumental carving of wood or stone.⁵⁸ Yet perhaps this passage needs to be read according to a medieval understanding of materiality and media. Theophilus describes

the carving and polishing of gemstones, which is a skill necessary for the metalworker, and he repeatedly introduces the arts of working wood, inasmuch as wood is described when used in the creation of precious objects: book covers, frames for shrines, the building of an organ.⁵⁹

If we consider the structure of the entire treatise, we begin to see the categorizations of media anew. *On Diverse Arts* does begin broadly with painting, moving on to glass and then to metalwork, but it does far more than that. The book on painting begins with the mixing of simple pigments, and by the end of the book it addresses altarpieces. The second book begins with the dyeing of glass but also treats glass vessels before proceeding to the complex assembly of windows. The last book addresses the making of precious objects that often involve multiple media, from enamels on a chalice to the organ, with its lead pipes and wooden chest. Thus media are broadly defined according to the class of object produced by the material, not by technique per se. This system of organization supports Bruno Reudenbach's elegant suggestion that the chapters are organized around "demonstration objects."⁶⁰ If, for example, we read the third book as treating the sacred arts and not metals alone, the appearance of disorderliness reconciles itself. As we shall see, in *On Diverse Arts* material is hierarchized according to the function of the object that it forms.

We have seen that Theophilus's techniques build upon one another in complexity. This technical order runs parallel to a hierarchy of materials, defined largely by purity and preciousness. The first book begins with the simplest of materials and pigments. The first color consists primarily of flake-white, a substance made of lead, to which is added a little vermilion. Appropriately,

it is used for the most primal of subjects, a nude body.⁶¹ The second color is that of "Green Earth," and this chapter is not so much a recipe as a description: "Green earth is a pigment, which looks like viridian mixed with black. Its nature is such that it is not ground upon stone. . . . it can be very usefully employed as a green color."⁶² The chapter seems to be lacking in detail, but as we will see, these chapters provide the basis for those that follow. Theophilus—quite literally—fills out the picture in the following sections, describing how to create the shadow color for flesh, the pink color for blushes, and the black and yellow ochre that delineates the hair of young men.⁶³ Mixtures then become more and more complex: the first shadow color is made by adding green earth, red (burnt ochre), and vermilion to the nude color, while the "First Rose" is made by adding vermilion and red lead to the flesh color.⁶⁴ The "Second Shadow Colour for Flesh" requires mixing more green and burnt ochre into the prior shadow color.⁶⁵ These mixtures are compound combinations of what came before.

These compound pigments are used for more complicated imagery. Mixtures to create shadows, blushes, and hair are layered on top of the base nude color: "If the face lacks contrast so that one highlight does not suffice," Theophilus writes in the chapter on the second highlight, "add more white to the first highlight and paint fine strokes everywhere over the first highlight."⁶⁶ The additive strategy can be seen in an image of Saint Luke from the Stavelot Bible, where the use of white highlight as something added on top of the previous color is particularly clear (plate 3).⁶⁷

The book on glass similarly progresses from simple colors to colors produced by complex mixtures and processes. It begins with a description of the kiln and frit, the core material of glass;

these are the essentials of glass production. The following chapters treat glassmaking itself. First comes the process of producing white glass and sheet glass; this, Theophilus writes, “is the kind of glass [that] is pure and clear.”⁶⁸ The importance of purity here might remind us of the nude bodies and green earth that opened the first book and the simple pigments they required, as Bruno Reudenbach has noted.⁶⁹ The next chapters describe the process for making yellow and purple glass, but, significantly, these colors are variations on basic clear glass. He begins the chapter on yellow as follows: “If you see the contents of any vessel change to a yellow colour, allow it to heat until the third hour and you will obtain a light yellow. . . . But if you wish, allow it to be heated until the sixth hour and you will have a reddish yellow.”⁷⁰ Purple glass is made similarly, by heating the glass successively until it changes from a “tawny” color, to a “flesh” color, to purple.

With much of the book devoted to glass windows, the chapters on vessels in the middle of the book might seem to be a digression.⁷¹ But here too, Theophilus’s point seems to be that glass vessels, even opaque ones, can be used as a source of color for window glass and enamels, addressed in the third book. He provides the key to this logic in chapter twelve, the first on opaque glass: “In the ancient buildings of pagans, various kinds of glass are found in the mosaic work—white, black, green, yellow, blue, red, and purple. They are not transparent but opaque like marble, and are like little square stones. From these, enamels are made in gold, silver and copper, of which we shall speak fully in their place.” He continues, expanding on their significance: “One also comes across various small vessels of the same colours, which the French—who are most skilled in this work—collect. The blue, they melt

in their kilns, adding to it a little clear and white glass, and make from it precious sheets of blue glass, which are very useful for windows. The purple and the green they also make use of in a similar way.”⁷² The next chapters, on goblets embellished with gold, opaque glass, and glazes, round out this first chapter by demonstrating the different uses of glass, but even here the stress is on modes of coloring glass. In the chapter on goblets, for instance, Theophilus describes how the metal is layered into and fused with the glass, commenting that “This gold will never come off,”⁷³ while in the chapter on earthenware vessels, he describes how a glaze is made by mixing ground colors with ground glass.⁷⁴

Theophilus’s concern for the coloring of glass runs through the book; he moves from making clear glass, to tinting glass by heating, to coloring glass with additives, to assembling a window. We might then see windows not as a set of distinct colors, but as a series of color gradations. The image of Christ from the cathedral at Strasbourg, for example, was created from glass tinted a flesh color; it was then painted with highlights and shadows, with varying levels of opacity. The variation of tints and addition of highlights and shadows is analogous to the layered pigments in the image of Saint Luke (plate 4).

The third book of *On Diverse Arts* is the longest, and it deals with the most complex objects and greatest variety of materials. It begins with the building of a workshop and furnace.⁷⁵ Next, Theophilus describes steel files and iron chisels⁷⁶ and then moves on to refining silver: “meanwhile, while [the crucibles] are drying, take the scales and weigh the gold or silver that you want to work. If the silver is not pure, refine it in this way.” Only then might one move on to dividing and casting the material and making a small chalice.⁷⁷

In the following section, Theophilus describes making a larger, cast chalice, and the process of applying niello decoration to the surface.⁷⁸ The next sections address soldering handles; gold and gilding; setting gems, pearls, and enamel; and finally creating bronze and brass alloys.⁷⁹

As in the previous books, Theophilus begins with the necessary tools and most basic materials before moving on to increasingly complex compounds and the objects they are used to create. Theophilus's concern for mixtures confirms what Paul Binski has suggested is a "particular emphasis on mixture and *varietas* as sources of aesthetic pleasure and invention" in medieval aesthetics.⁸⁰ *On Diverse Arts* also puts this theory to work, giving it a particular twist as well as a moral edge. For Theophilus, mixtures are never solely mixtures but the result of a process that is predicated on a knowledge and understanding of the hierarchy and behavior of materials. As in the other books, materials are ordered according to complexity and expense, from simple pigments to fine blue glass and then to bronze, silver, and gold. As we shall see, for the twelfth century, the process of mixture, of materiality, was also dependent on the function of the object.

Throughout the treatise, the purpose of the object determines its material, and as material increases in complexity and expense, the function of the object becomes more sacred and more central to the performance of the liturgy. Painting, for Theophilus, is ornamental and used to embellish a building: "You have embellished the ceilings or walls with varied work in different colors and have, in some measure, shown to beholders the paradise of God. . . ."⁸¹ Images painted on a surface—a wall or a panel or a book—might well encourage or focus an act of worship, but they are not essential to it. The

paintings in the apse of Saints Peter and Paul at Reichenau-Niederzell, for example, executed by the so-called Reichenau school in the 1120s, follow the architectural structure—even mimic it—with painted arches framing the apostles, yet the mural is not structurally indispensable or essential to the liturgy (plate 5).

Stained glass is more complicated; it can contain imagery, but it is also an integral part of the fabric of a church. Theophilus describes glassmaking as the "ingenious techniques [by which] a building may be embellished with a variety of colours, without excluding the light of day and the rays of the sun."⁸² Stained glass was also valued as a vehicle for the manipulation of light: as light penetrated the translucent panels, it might change color but would not fracture. The idea of penetration without damage might be seen as analogous to the miracle of the Virgin birth, and it was a property that suggested a semblance to the divine.⁸³ The last book of *On Diverse Arts* treats liturgical arts, those objects "without which the divine mysteries and service of the Offices cannot continue."⁸⁴ Crafted of precious metals, they are luminous and reflect light. Metals are a culmination of a conceptual progression from opaque painting to light-transmitting glass to the liturgical objects that seem to emit light themselves. For Theophilus, then, increasing sanctity of function is paralleled by increasingly precious and luminescent material.

The importance of material in determining function becomes clear when we compare the treatment of different kinds of vessels in *On Diverse Arts*. In the discussion of vessels in the second book, for example, Theophilus's instructions are rather vague: "To make glass vessels, prepare the glass in the above way" and "If you want to make flasks with a long neck proceed in

this way.”⁸⁵ No purpose or function is specified for these vessels. In the third book, the situation is quite different. Here Theophilus describes two metal vessels, both called *ampulla*, one made of silver and one of tin. Dodwell translates the former as a cruet, and the latter more informally as a jug, explaining that “it is hardly likely that he would make a cruet out of tin.”⁸⁶ Dodwell’s reasoning is probably sound, but if we return again to the Latin it becomes clear how Theophilus distinguishes between the two vessels. First, in “On Tin Vessels” (*de ampullis stagneis*) Theophilus uses the plural, implying a generic object without a particular purpose. The second chapter, “On the Vessel” (*de ampulla*), uses the singular and describes a specific object, made of precious silver.⁸⁷ In the following text, Theophilus notes the object’s function—it is “a cruet (*ampullam*) for pouring out the wine [of the eucharist]”⁸⁸—and that it is to be made out of silver. A similar distinction is apparent elsewhere in the chapter, for Theophilus’s discussions of lesser metals, such as iron, are not found in the context of liturgical objects, nor are they in the same section of the book. Iron is discussed as the substance from which pedestrian items such as files, rasps, and horse trappings are made.⁸⁹ Chalices, set apart by their sacred function, are listed in order of increasing preciousness of their material: first a small silver chalice, then a large one, next niello and gilding for the chalice, and last, a golden chalice.⁹⁰ Thus Theophilus articulates a hierarchy that distinguishes between the specific and the generic and defines and orders objects according to their function and materials.

The hierarchy and ordering of materials in Theophilus reveal a concern for the nature of the physical substance and what can be done with it. The pigments he describes in the first book, for

example, have distinct properties: as Theophilus explains it, “Green earth” is a soluble substance and thus doesn’t need to be ground before mixing.⁹¹ Glass, meanwhile, is rather different. Made from sand, glass was seen as a mysterious and magical medium. A story told by Pliny in his *Natural History* explains the origins of glass and its miraculous qualities, mastered by its workmen and feared by the emperor Tiberius. In the early Middle Ages, Isidore of Seville expanded on the anecdote in his *Etymologies*, emphasizing the quality and value of the medium:

It is related that, under Tiberius Caesar, an artificer invented a tempering for glass, which rendered it tender and ductile (*flexibile et ductile*). When admitted to Caesar he held out the jar to him, who, angry, threw it upon the pavement, where it bent like a brass vase. The artificer raised the jar, thrust a small hammer into the cavity, and mended the jar. This done, Caesar asked the artificer whether any other person knew this tempering for glass, and when he denied, with an oath, that any other knew of it, Caesar ordered him to be decapitated; lest, this known, gold and silver might become as clay, and the value of all metals be debased. For, in truth, did glass vessels not break they would be better than gold or silver.⁹²

The *Etymologies* was read throughout the Middle Ages, and much prized. For Isidore, and perhaps for later medieval readers as well, glass was all the more miraculous because it was made from the sand of the earth. It was treasured for its humble origins as well as its preciousness, its utility, and the curious tension between strength and fragility it exhibits as it cools and takes on

form. A writer like Theophilus would no doubt have been familiar with Isidore's work and was probably also familiar with this story about glass. Indeed, Theophilus opens the second book by describing the ash that is made from beechwood trees, emphasizing that one must keep it clean and pure: "If you should decide to make glass, first cut plenty of beechwood logs and dry them. Then burn them together in a clean pot and carefully collect the ashes, taking care not to mix in any earth or stone."⁹³

Theophilus's interest in ash and the purity of clear glass seems to reflect this fascination with glass as a material that can be transformed. This emphasis on the physical nature and potential use of material in *On Diverse Arts* sets it apart from other recipe books. Instructions in the *Mappae clavicula*, for example, do not explain the properties of material but instead give proportions, as in the recipe for green mosaic: "Take five pounds of a lump of clean glass and two ounces of lead-free copper filings, and put them in a new earthenware pot. Put fire underneath, and in the lower part of a glassworker's furnace cook them down for seven days, and after this take it out and break it up into small pieces and melt it again. It will be green colored." The recipes for making red and purple glass are similar: "Making [glass of] a reddish color: Put two ounces of white lead to a pound of glass and cook for six days," and likewise, "Making a purple color [on glass] without fire: Color thin glass pieces, mix and coat them with dragon's blood, and in this way a reddish color will result."⁹⁴

Theophilus, by contrast, is so concerned with understanding the nature of a material that he even describes the contingencies that might occur in a process. The delicate process of heating glass requires an understanding of how

glass changes properties, for example, so that Theophilus's instructions are both a warning and a solution: "Purple Glass: If you perceive that the contents of any container happen to change to a tawny colour, which is like flesh, let this glass serve for your flesh colour. Take as much of it as you want. Heat the rest for two hours, namely from the first to the third hour, and you will obtain a light purple. Heat again from the third to the sixth hour and it will be a perfect reddish purple."⁹⁵ The student, Theophilus suggests, should understand the material's properties well enough to be able to adjust the process as necessary, so that the substance is best used.

It is in the third book that Theophilus's concern for the nature and properties of materials is most clear. Four chapters are dedicated to describing four different kinds of gold, which form an introduction to the discussion of making a gold chalice. The first of these addresses the "Gold of the Land of Havilah." There are "many kinds of gold," Theophilus writes, but this one is the best, being a paradisial gold referred to in the book of Genesis.⁹⁶ The next three chapters deal with gold of varying degrees of quality from different regions of the world. The first is "Arabian Gold," which, according to Theophilus, is "very precious," and "is frequently found used in antique vases." This gold, he warns, is sometimes counterfeited by the addition of copper, which deceives "many unwary people." Such a deception can be prevented, however, by testing the gold in the fire. The second is "Spanish Gold," which is "prepared from red copper, powder of basilisk and human blood and vinegar." Theophilus then describes the method by which "the heathen, who are said to be skilled in this art," prepare the compound of basilisk, which, combined with vinegar, creates a compound that "eats through

the copper, which, thereupon, takes on the weight and colour of gold. This gold is suitable for all work.” The last is “Sand Gold” from the banks of the Rhine, which is very fine: *aurum subtilissimum*.⁹⁷ Each of these characterizations specify the gold’s source, how it can be obtained, its quality, and how it can be used. Theophilus’s interest in the material is both practical and theoretical. Practical information about obtaining gold, or identifying counterfeit gold, are placed alongside discussions of the paradisaical gold of the land of Havilah. Yet here purity trumps practicality: the gold of Havilah functions as a standard against which other types of gold is measured.

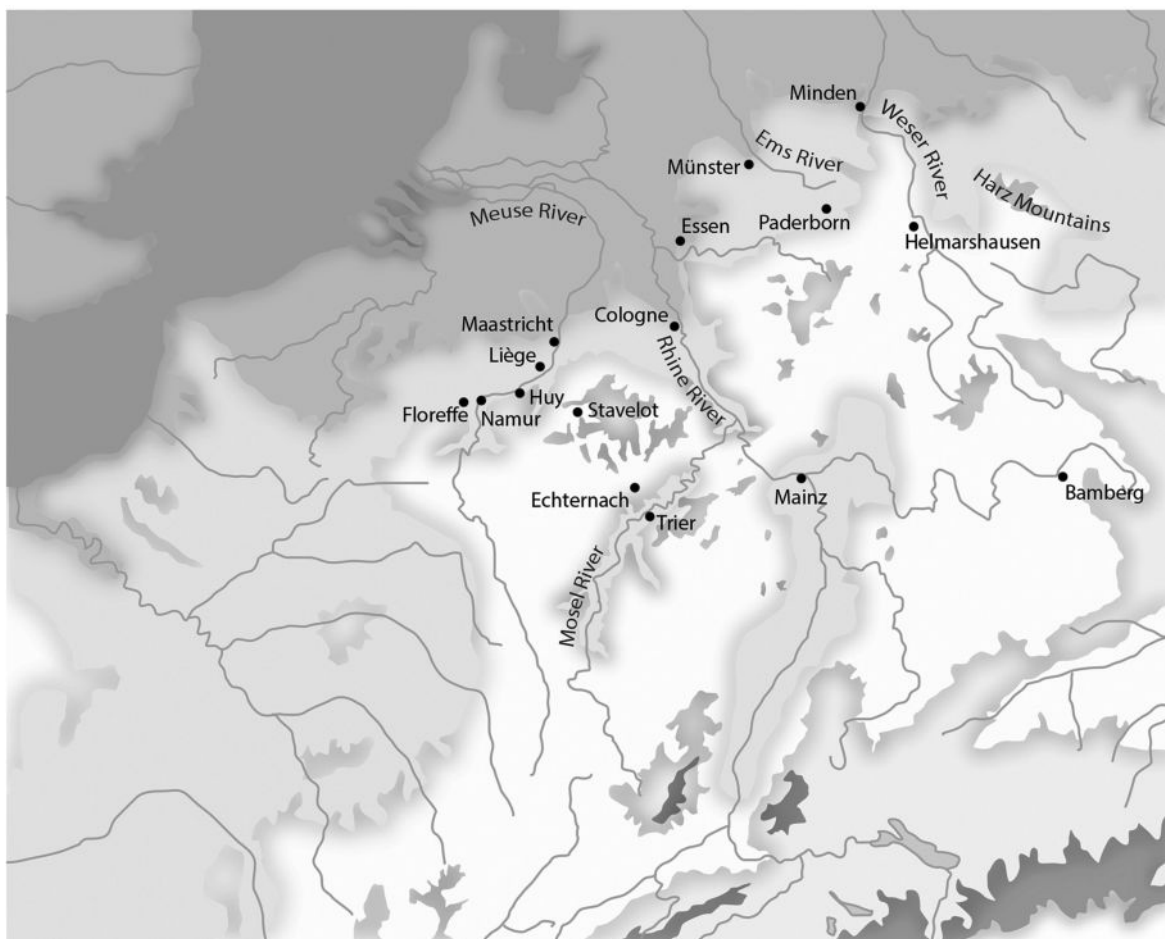
Theophilus’s concerns extend to other metals as well. He describes in detail how copper and iron are found and mined: “Copper is formed in the earth. When a vein of it is found, it is obtained with the utmost labour of digging and breaking. It is an ore of a green colour, very hard, and in its natural state mixed with lead.”⁹⁸ The description of iron is very similar: “Iron is formed in the earth as an ore. It is dug up and broken up in the same way as copper above, and melted down into pigs, then it is smelted in the iron foundry and hammered so that it becomes fit for any work. Steel is so called from Mount Calibs, where, so far as we know, it is used more than anywhere else. It is prepared in a similar way [to iron] and so made fit for work.”⁹⁹ Theophilus’s discussion of copper mining has been shown to have some inaccuracies, but what is clear is that these chapters on copper mines, iron mines, and finding gold reveal a concern for the origin of a material and its raw state; Theophilus had probably been exposed to the processes, or at least told of them.¹⁰⁰ Read thus, Theophilus’s sharp distinction between materials that are pure and those that are composites makes more

sense. Having described the nature of bronze, for example, Theophilus writes that “by adding a fifth part tin to the copper founded in this way, a metal is made from which bells are cast.”¹⁰¹

Composite materials are mixtures, a balance of elements achieved through understanding the properties of the materials from which it is made. Metals are the most complex, the most sacred, and the most precious of all materials treated by Theophilus. It is therefore not surprising that *On Diverse Arts* devotes an unusual amount of attention to their nature, their origins, and the methods of extracting them.

Acquiring Material: The Local Context

The high status accorded to metals in *On Diverse Arts*, and the deep interest in their qualities and their origins, is unusual. A closer look at the local context may explain why. Most of the religious houses of the region, and particularly those with which Roger of Helmarshausen and his works have been linked, were wealthy landowners. Helmarshausen, Corvey, Stavelot, and Echternach all played an important part in local and regional commerce, which was heavily reliant on mining and trading precious metals. Some of the imperial abbeys of the region, like Corvey and Helmarshausen, had long histories as imperial mints; their participation in the metal trade appears to have originated from this role. The abbey of Corvey was one of the earliest, having held minting rights since 883. Helmarshausen began minting coins in 997, the same year that the emperor Otto III granted the abbey rights to a market and also assigned it revenues from a toll. The earliest Helmarshausen coin to survive dates from 1050, and there is evidence that by the



Benedictine abbeys of the Rhine, Meuse, and Weser river valleys.

later twelfth century the mint was big enough to support the import and forging of silver from Scotland.¹⁰² The abbey of St. Pantaleon, meanwhile, was located in the midst of Cologne, the major urban center for the trade of metals, while abbeys like Abdinghof at Paderborn were located at the junction of routes to the mines of central Germany and the main road to Cologne.

The region was the primary one for mining and trade in metals in northern Europe. Verhulst believes that by the mid-twelfth century Cologne

had become so dominant in the metals trade that the industry, along with urban growth, came to a halt in formerly prosperous Mosan cities like Huy.¹⁰³ Cities on the Rhine-Meuse and Sambre rivers seem to have been connected by trade and shared mining interests from Roman times; these had gained significant economic power in the late tenth and early eleventh centuries under the Ottonians. The Meuse-Sambre river route provided a throughway between northern Gaul and the Rhine, from Reims to Cologne, and the

combination of mines, mints, and access to fluvial transport enriched the region's cities as well as its ecclesiastical and monastic institutions. By the fourth century, Tongeren had been made a bishopric, but in the sixth, Bishop Monulphus moved the seat of the diocese to Maastricht, a large urban center located on the Meuse and at the intersection of trading routes. By the Merovingian period, other cities on the Meuse, including Dinant, Namur, and Huy, were "moneyers," or makers of coinage. Huy and Maastricht were the largest in the region; seventh-century coins minted in Huy and Dinant have been found in England and Scandinavia.¹⁰⁴ While Adriaan Verhulst has warned that coins may not be direct evidence of trade activity, the presence of moneyers at Dinant, Huy, Namur, and Maastricht in the seventh century is certain.¹⁰⁵ The minting industry of the Meuse region grew under Charlemagne, with additional mints established at Liège, St.-Trond, and Aachen. Evidence of surviving coins suggests that the Mosan mints were some of the most productive of the empire.¹⁰⁶ Although briefly interrupted by Norman invasions in the mid-ninth century, Huy in particular was an economic force well into the Ottonian period, minting coins for a string of emperors, including Charlemagne, Louis the Pious, Lothair, Charles the Bald, Louis II (or Louis III), Louis the Child, and later, Otto I and Conrad II. Coins from these later periods can be found as far away as eastern Slovakia, northern Scandinavia, Iceland, Poland, and Russia.¹⁰⁷

Under the Ottonians, silver and copper mines opened in the Harz mountains, and ore went to Cologne to be traded. By the twelfth century, the trade network extended across Germany and Belgium and into England and northern France; tin was being imported from England, zinc from Belgium, copper and silver from Germany. The

major center for the metal trade was Cologne, and the Rhine and Meuse rivers were major thoroughways. The cities of this region, especially Cologne, Liège, Huy, and Maastricht, thus grew increasingly urban and mercantile in the course of the twelfth century, with growing populations of wealthy burghers. The abbey of regular canons at Neufmoutier, in Huy, for example, was founded by burghers of the town and earned its income from tolls on the river Meuse.¹⁰⁸ The city of Paderborn was a major trading point between the Rhine and the Weser, with the abbey of Corvey situated on the latter and Helmarshausen on the Diemel, near its junction with the Weser. The region encompassing Helmarshausen, Corvey, Paderborn, and Hildesheim was on the eastern edge of the trading corridor of the Rhine and Meuse, but it was significant for its placement on the east-west route that, from the eleventh century onward, brought silver and copper from the Harz mountains to Cologne.

Because making metal objects often required a combination of materials, artistic production was reliant on the availability of special metals. The Mosan cities, for example, were particularly powerful because they had ready access to zinc, which was required to make alloys of bronze and was found only in Belgium and Lorraine. The mines of Moresnet, in particular, were known to yield high-quality zinc, and the location of these deposits near the city of Visé, on the Meuse between Liège and Maastricht, allowed for the growth of the trade.¹⁰⁹ In addition, the Meuse region contained a number of iron deposits. Copper was mined in the Harz mountains, at Goslar, and tin came from England, especially Cornwall, where it was mined from the eleventh century onward.¹¹⁰

Zinc, more so than other metals, was difficult to extract and transport; as a result, the

convenience of the river routes made the valley of the Rhine and Meuse the center for the trade of all these metals. It is from such assets that the economic might of the region's cities grew in the tenth and eleventh centuries.¹¹¹ Traders from Huy are recorded as selling copper and tin basins and kettles in Cologne in the eleventh century, and they were in England as early as 1000.¹¹² While the production of what is now called *dinanderie*, small bronze or brass objects from the region around Dinant, dates back to the Merovingian and Carolingian periods, the industry grew quickly in the eleventh century. From 983 there is evidence that the city of Visé hosted a fair for trade, and through the tenth and eleventh centuries the presence of Mosan-made metal objects, such as cauldrons and basins, is noted at the fairs in Cologne.¹¹³ Cologne, with its proximity to the Meuse, the North Sea, and, via the Rhine and the Main, a southern and eastern route toward Frankfurt and Nuremberg, became a commercial and trading center in the eleventh century. Hildesheim, meanwhile, was at the foot of the Harz mountains and on the route to Cologne, where copper was traded and where representatives of the Mosan cities received special privileges. The river Meuse, and the proximity of the Mosan cities to its outlet in the North Sea, also gave easy access to trade with England and the acquisition of tin.¹¹⁴

Beyond the river access, the growth of trade fairs and markets encouraged urban commerce and shifted the locus of the industry from the rural areas in which the mines were located to urban centers. Representatives of Huy or Liège, for example, would buy their copper in Cologne, rather than traveling to the mines of Harz itself. Cities grew accordingly. In the tenth century, there is evidence of only one bridge crossing the

Meuse, at Maastricht. In the eleventh and twelfth centuries, bridges were built at Givet, Dinant, Namur, Huy, Liège, and Visé, all within a distance of one hundred kilometers.¹¹⁵

Trading networks linked together cities, abbeys, and mines. The Benedictine abbey of Florennes held the rights to a mine as a part of its property holding, while the Premonstrian abbey of Floreffe traded its rights to the mines of Prüm for those at Ardoisières.¹¹⁶ The abbey of Corvey was similarly endowed: the free use of the mines at Monte Eresburg was the subject of a privilege dating to 1150, given by Conrad II to Wibald of Stavelot, then custodian of Corvey. Notable here is that Corvey's privilege included not only the mining but also the refining and use of the metals for the abbey's own purposes. "Through you, of the church of Corbei," the document reads, "we grant, give over and confirm by the present writing, and it is permitted for you and your successors, apart from the objection of any person, in the same mountain, to dig all metals, which are to be found, to mine and to refine, and to freely apply for your use and that of your brothers, so that so much better the church of Corbei could serve so much the divine things as the things of the kingdom."¹¹⁷ From at least 1250, Archbishop Conrad von Hochstaden opened mines in the Berg district, east of Cologne, ostensibly to raise money for the rebuilding of Cologne Cathedral. These were not the only mines owned by the archbishop; he also owned mines at Ems, further up the Rhine, called the "Cologne Pits."¹¹⁸

The ownership of mines was a highly valued right that came under the control of imperial patrons such as Conrad II or Frederick II and could be traded for advantage. As the twelfth century progressed, burgher towns like Huy were gaining in economic prestige; meanwhile, cities

like Cologne and Paris were quickly becoming concentrations of economic power. In the western edge of the region, guilds were beginning to be founded; the earliest evidence of a guild in the region is from Valenciennes, dating to 1024, while statutes for a merchants' guild in St. Omer date to 1072–83.¹¹⁹ The guilds assured “mutual protection” while traveling abroad and also specified that members were to assist in community development.¹²⁰ It has been surmised that cities like Liège, with its powerful bishop, were slower to experience these developments, and it was not until the end of the century that guilds and communes became more common in the Meuse valley. Regardless, it seems likely that the traders who belonged to these guilds were traveling through, if not trading in, the cities of Liège and Cologne. Evidence points to a trend of increasing commercialization, and pressure on the old religious authorities, as new economic competitors rose in power.

Economic uncertainty was especially acute during the investiture controversy, and this increased with pressures for reform. By the late eleventh century, patronage was less stable, and workshops were no longer so closely associated with the court as they had been in the Ottonian and Salian periods.¹²¹ Abbeys like Corvey were in a state of decline in the early twelfth century; as charters and letters attest, their land and their inherited rights as mints were increasingly subject to dispute. The success of Wibald of Stavelot, on the other hand, is due largely to his skill as an abbey administrator and “reformer,” and in the case of Benedictine monasteries, the restoration of discipline was in many cases dependent on a restoration of financial health and regaining lost property. In a surviving letter from 1147, Herimann, the count of Winzenburg, who

calls himself the advocatus of the monastery of Corvey, commends Abbot Wibald to Pope Eugenius III for his abilities in reform and in restoring the financial assets of the abbeys of Kemnade and Fischbeck.¹²² As Herimann's letter shows, an orderly monastery meant both spiritual and financial well-being.

In this context, the high status of metals in *On Diverse Arts* is not surprising. Theophilus's interest in them may come from both a close knowledge of the monastic mining system and an understanding of the emerging economies that were threatening the old manner of production. It seems likely that a twelfth-century reader of Theophilus, a participant in the monastic system whose income was tied to the trade of metals and land, would have recognized the importance of metals and their sources in *On Diverse Arts*.

The production of art objects in locally available metals may have played a meaningful part in local economies; at the very least, it seems to have resulted from such an economy. From this perspective, material is not generic but connects a particular object to the earth that produced it, to the mines that gave up its material, and to the earthly power of the abbey that oversaw its production. Theophilus's tract may well be an argument for the value of local, monastic metal production. Indeed, this idea is found in the first prologue:

Do not despise useful and precious things, simply because your native earth has produced them for you of its own accord or unexpectedly. For foolish is the merchant who suddenly finds a treasure in a hole in the ground and fails to pick it up and keep it. . . . For however much men are accustomed to place in the first rank precious things that are

sought with much toil and acquired at great expense, and to look after them with great solicitude, yet, if meanwhile they happen to find or come across things for nothing that are comparable or better, then they keep these with a similar, even greater care.¹²³

Theophilus sets out a strategy justifying and even praising the use of material that comes from “a hole in the ground.” Perhaps his argument is a response to the reform orders that renounced material wealth, or, more likely, to the developing world of trade and commerce. He articulates a wholly different value system for material, determined not by monetary cost, or by toil, but according to purity and function. Materials are to be properly used, properly understood, and properly acquired.

Conclusions

Throughout *On Diverse Arts*, Theophilus puts extensive emphasis on the nature of a material, its properties and origins, and the manner in which it is manipulated. Given the local importance of metals mining, this interest makes it likely that the treatise functioned as a defense of precious objects and a monastic economy. It also changes how we understand twelfth-century concepts of materiality. In the earlier Middle Ages, as Herbert Kessler has shown, materials themselves were thought to “animate” the art object and bring viewers’ attention to the dual nature of Christ.¹²⁴ By the twelfth century, Abbot Suger was asserting the anagogical function of material, so that viewers, on seeing images made of luminescent glass, were brought “from the material to the immaterial.”¹²⁵ Theophilus’s ideas

about matter are related to both of these, but his concern for the process of making objects, and for the origins and natures of material, adds a new dimension.

As we have seen, the instructions of *On Diverse Arts* articulate and systematize the entire process of art-making from mining to gilding, from the most primitive materials to the most sophisticated objects. This breadth of perspective makes the manipulation of material for the production of art an activity that functions within the orderly natural system of God, serving both a monastic spiritual system and a monastic economic system. Theophilus’s justification of materiality is embedded in his local context and is in many ways the product of the economic structures and pressures experienced by twelfth-century Benedictine monasteries in the region. The organized nature of his treatise makes the learning of art a disciplined subject, and Theophilus takes the long view of production and materiality. For Theophilus, the utility of the material is dependent on the kinds of manipulations that can be performed on it, and these are operations that function within an orderly, scientific system.

It is also notable that at just this time, Hugh of St. Victor was discussing the formation of the soul as a similar kind of transformation. For Hugh, the soul is molded as a piece of clay, stamped with a seal. The substance of the clay does not change, but the form takes on the likeness of God. The arts, when well practiced, aid in the restoration to likeness because they help transform the soul, even if to varying degrees. As Hugh puts it, “This then, is what the arts are concerned with, this is what they intend, namely, to restore within us the divine likeness, a likeness which to us is a form but to God is

his nature. . . . for then there begins to shine forth again in us what has forever existed in the divine Idea or Pattern, coming and going in us but standing changeless in God.”¹²⁶ The mechanical arts thus transform what already exists into something useful. Accordingly, Hugh defined the mechanical arts as adulterine, or *moechus*, for, he claimed, they do not create from nothing, but transform that which exists.¹²⁷ Ambrose explains a similar idea in terms of a duality of flesh and spirit, explaining that there is truth only in the celestial realm; man exists in a world of images, where spirituality adheres to flesh.¹²⁸ God is immutable, but—as Theophilus and Hugh take particular care to express—earthly matter is unstable, mutable, moldable, and it is the very moldability of the soul that enables man to restore his likeness to God.

Looking again at these sources, we begin to see a rather different understanding of what materials, in essence, can be. Metal, for example, is not static, but it is instead dynamic physical matter, dug up in inchoate form from the ground, with properties aligned with herbs and humors. It is then transformed to become something useful. Material may have iconographic meaning, as in the use of gems to evoke the heavenly Jerusalem, or bronze to elicit the molten sea, but it is also multivalent, shifting and fluid as the light emitted from the shining material itself.¹²⁹ Notably, Honorius of Autun’s explanation of a church’s chandelier shows that material could function both as analogy and symbol:

Gold represents the faithful, who shine with wisdom, silver those who shine by their words, bronze, those who resound sweetly with the doctrine of heaven, iron, those who master vice; the towers of the chandelier are

those who safeguard the church with their writings; the lamps shine with their good deeds: the martyrs are gold, the virgins silver, bronze those serving near to the church, iron, those serving in wedlock. The glittering gems in the chandelier are those who glow with the virtues; the metals tempered in the fire for the ornamentation of the chandelier are chosen, as the elect, tested on the journey to heaven by tribulation, are chosen to decorate Jerusalem.¹³⁰

By the later middle ages and Renaissance, the process of casting metal would carry more explicit significance. As Michael Cole has shown, it seemed to be the taming of chaotic matter; the wonderment of that process would in turn be attributed to the artist’s person and power, in the tradition of Vulcan.¹³¹ Yet as we see in Honorius of Autun, in Theophilus, and in didactic literature, there are only inklings of this idea in the early twelfth century. For Theophilus, it is a shaping of material, a transforming of what is already there.

Like the treasure in the hole in the ground, precious matter already existed long before the lucky merchant picked it up; but to bring it into useful form requires understanding and, as we saw in the first chapter, good intention. The transformation from chaotic matter to beautiful object is a process that operates within the orderly system of the divine cosmos. Importantly, the nature of material is only the beginning. Its manipulation requires good craft, which in turn is something that must be sought, learned, and practiced. The nature of this labor and skill is the subject of the next chapter.

3.

MONASTIC LABOR AND CRAFT

*Wretchedly deceived by the guile
of the Devil, through the sin of dis-
obedience [man] lost the privilege
of immortality, but, however, so far
transmitted to later posterity the
distinction of wisdom and intelligence,
that whoever will contribute both
care and concern is able to attain a
capacity for all arts and skills, as if by
hereditary right.*

THEOPHILUS,
On Diverse Arts, prologue I

In the margins of a late twelfth-century manuscript now in London is a drawing of a small hand with an outstretched index finger, pointing toward the text (plate 6). A reading device not uncommon in medieval manuscripts, such “manicules” function like a bookmark, placing added emphasis on certain words. Here, it points to the opening words of the second prologue of *On Diverse Arts*: “Incipit prologus in librum secundum / in precedenti libello, frater karissime.”¹ The entire passage reads, “Actuated, dearest brother, by a sincere affection, I did not hesitate to suggest to you in the preceding book how much honour and advantage there is in eschewing idleness and in spurning laziness and sloth; and how sweet and delightful it is to give one’s attention to the practice of the various useful arts.”² The manicule shows us one reader’s concern for the avoidance of sin and Theophilus’s exhortation to stay away from idleness by paying “attention to the practice of the various useful arts.” This notation calling attention to the role of manual labor in mitigating temptation and poor behavior is but one of many such markings highlighting similar themes throughout the manuscript. Read together, these notations reveal one reader’s interest in the link between labor and virtue, and they illuminate a medieval concept of artistic skill in the monastic sphere.

The London manuscript is now at the British Library, where it is known as Harley MS 3915.

Nearly complete, it is one of the oldest copies of Theophilus to survive, and it was probably copied several decades later than the two oldest copies, at Wolfenbüttel and Vienna.³ It is a small handbook, the bulk of which is copied by a single hand in neat, pointed, proto-gothic lettering. It is a good and careful copy, with one column of text, ample margins, visible rulings, and fine rubrication. Theophilus's text fills the first 109 folia; the quire in which it ends is filled, without a break, by chapters of Heraclius's *De coloribus et artibus romanorum*. The rest of the book contains technical and medical treatises. After a short piece of Faventius's commentary on Vitruvius, one new hand transcribes excerpts from the *Liber de coloribus* and the *Mappae clavicula*, and a third hand a compilation known as *De unguentis*, an ancient text on ointments.

This copy of *On Diverse Arts* is unusual and bears little relation to either the grand copy now at Wolfenbüttel or the small, modest copy in Vienna. Strangely, it does not contain the first prologue, and the gatherings show that it never did. It begins with a table of contents for book I and the instructions on the art of painting. On the eighth folio, in the midst of the instructions, is a signature mark "i" in the bas-de-page. The mark numbers the gathering; with eight folia, it is the length of a standard quire, and thus was unlikely to have once contained other pages. Throughout the rest of the text, the gatherings are numbered; all but the last two are a standard length of eight folia. This system of numbered gatherings and standard quires betrays meticulous workshop scribal practice. While the absence of the prologue may be an oversight, the attention paid the rest of the volume makes it more likely there was a defect in the parent manuscript. Even though the Harley manuscript

was copied in the same region as the Vienna and Wolfenbüttel manuscripts, it represents a separate recension of the text, including chapters omitted from these two older texts but found in the manuscript now in Cambridge (discussed in chapter 2) and in a fourteenth-century manuscript now in Leipzig.⁴

Like the Wolfenbüttel and Vienna copies, the Harley manuscript was probably once part of a monastic library. On the back flyleaf, a note indicates that a certain "N." obtained the book in 1444 on the day of Saint Lambert, "in the diet of the schism between Pope Eugenium and the antipope."⁵ The name of the city is difficult to decipher. It has been suggested that it is Nuremberg, where the diet was held, and that the "N" stands for Nicholas of Cusa, who was in Nuremberg for the imperial diet.⁶ There is indeed evidence that Nicholas of Cusa was buying manuscripts and astronomical tools in Nuremberg that September—a testament to his interest in science. While it is possible that this led to an interest in Theophilus, the attribution ultimately rests on very little and seems unlikely.⁷

The inscription may instead be read as Münster, which would be a far more likely provenance, and indeed, the manuscript is written in a northern script more typical of that locale. As Tjamke Snijders has proposed, books traveled between monasteries that were already connected through personnel or relic cults, and this was probably the case with the Harley manuscript.⁸ The diocese of Münster bordered the diocese of Paderborn, and thus *On Diverse Arts* may have easily traveled there. Münster also had spiritual and political ties to Cologne and Liège. In the eleventh century, Archbishop Hermann of Cologne and Bishop Henry of Liège gave the city a gift of the relics of Saint Lambert, and this

Mosan saint is venerated there even today—his feast is a major celebration.⁹ Lambert was the patron saint of Liège; he had been the bishop of Maastricht, having helped to Christianize the pagans of the lower Meuse, but was killed in Liège. His body was kept in the city, and when the bishopric moved from Maastricht to Liège, his church became the cathedral of Saint Lambert. It seems likely that the collector N. would have been keenly aware of the feast of Saint Lambert in Münster, perhaps more than he would have been anywhere outside of Liège itself.

It is furthermore quite probable that the goings-on of the diet would have been known in Münster, for papal strife in Rome reverberated across the bishoprics of Germany. In the 1440s, for example, Pope Eugenius IV allied himself with the Dukes of Cleves in a dispute against Dietrich von Mörs, the archbishop of Cologne, and his brother Henry von Mörs, the bishop of Münster, a tussle that set papal control against episcopal authority. Monasteries suffered from these power struggles, losing property to quarrelling adversaries.¹⁰ Calls for reform led some monasteries to be emptied and their property sold. The Benedictine monastery of Liesborn, near Münster, was one such case, and perhaps N. acquired the manuscript from the library of a monastery like this one.¹¹

N. certainly bought the volume intact; quire labels run through folio 111v, and page format is consistent until folio 142v, where a new hand finishes the last pages of *De unguentis* and continues to copy an unknown text that runs until the last folio of the volume, which faces the inscription. It is not surprising that the volume was bought by a collector, for by the fifteenth century manuscripts like these had caught the interest of humanists. In Paris in 1431, for

example, Jean Le Bègue had a personal copy of *On Diverse Arts* made, which he included in a volume dedicated to the techniques and lexicon of painting. His anthology also contained excerpts from Heraclius and Saint Audemar and a glossary of painting terms;¹² Le Bègue seems to have been interested in the text as a part of the history of technique. While the interest of the collector of the Harley manuscript is impossible to ascertain, the fact that he inscribed his name and date in Latin may be evidence, at least, of humanistic tendencies.

The medieval readership of the Harley manuscript appears to have been something different, for the marks in the text note neither technique nor lexicon. In this older copy, the marks are in the prologues, not the instructions. Such a preference suggests a concern for the philosophical underpinning of the text. Further, the fact that the first manicule points out the one place in the text where Theophilus refers to the prior prologue may be evidence that this reader knew what was missing. If this is true, then the unity of the text is thus underscored, the fullness of the narrative recognized.

As we have seen, Theophilus's narrative begins in the very first prologue, where he describes the nature of man and his abilities, setting into motion the entire treatise: "In the account of the creation of the world, we read that man was created in the image and likeness of God and was animated by the divine breath, breathed into him. By the eminence of such distinction, he was placed above other living creatures, so that, capable of reason, he acquired participation in the wisdom and skill of the Divine Intelligence, and, endowed with free will, was subject only to the will of his Creator."¹³ The passage draws on the first chapter of Genesis:

“Let us make man to our image and likeness’ . . . And God created man to his own image: to the image of God he created him: male and female he created them.”¹⁴

Restoration to God

Theophilus’s narrative of creation states that man was made “in the image and likeness of God and was animated by the divine breath, breathed into him.” The story begins optimistically, but no sooner has Theophilus described man’s privileged status than he recounts its loss: “Wretchedly deceived by the guile of the Devil, through the sin of disobedience he lost the privilege of immortality.” Created man was given rationality and free will, but his will to disobey God caused his fall. His sin is as indicative of his nature as his creation. Luckily, however, some restoration to God is possible: “But, however, so far [was] transmitted to later posterity the distinction of wisdom and intelligence, that whoever will contribute both care and concern is able to attain a capacity for all arts and skills, as if by hereditary right.”¹⁵ Just as man’s fall was the product of his own doing, so too can be his restoration, for to the extent that he still possesses wisdom and intelligence, he has the potential to gain skill. As Bruno Reudenbach and others have shown, man’s rational nature is a driving force in *On Diverse Arts*.¹⁶ Importantly, however, as Theophilus shows us here, reason and free will only make it possible for man to pursue the arts; they do not guarantee aptitude. That man was made in the image and likeness of God only means that he has the rationality and free will to learn. This potentiality, which sets the stage for the development of skill, emerges

from a fundamental distinction between image and similitude that was being made by twelfth-century thinkers.¹⁷ Rupert of Deutz differentiates them as follows:

What difference there is between image and similitude, that man by sinning did not lose the image but the similitude of God, to which he can be restored through the Baptism of Christ. Man was made in the image of God, naturally, because he is rational, and to the similitude of him truly because he is driven to imitate divine goodness. . . . Therefore, as we have lost, by our volition, the similitude of God, by our volition we can nevertheless recuperate the goodness of God, and the holy spirit there always descends where we easily can reach, while it is brought to waters which are ready, requiring only will, procuring all other things freely.¹⁸

Rupert thus draws heavily on Augustine’s theology of likeness and restoration. For Augustine, only Christ is like God; man can only pursue likeness to him.¹⁹ As Rupert explains it, the *imago Dei*—a theological term expressing the idea of “man made in the image of God”—is the gift of free choice and rationality; similitude has been lost, but man’s will makes its restoration possible.²⁰ Perhaps more simply put, Rupert writes in his commentary on the Gospel of John that “by sinning man lost imitation of God’s goodness, ‘according to which he was made in the similitude of God,’ but he did not lose his reason, according to which he was made in the image of God.”²¹ The *imago Dei*, then, describes man’s potential: it is his ability to reason, his wisdom, and his intelligence. “Likeness,” or “similitude,” describes the closeness to God that was lost in the Fall. Saint

Paul, in his letter to the Corinthians, famously describes this lapsed state as a distant mirror: “We see now through a glass darkly,” he writes, “but then face to face.”²²

Other early twelfth-century thinkers also express a notion of man’s likeness to God and the importance of rationality and intelligence, but nearly all do so exclusively in terms of restoration. For Hugh of St. Victor, man’s rationality constitutes his likeness to God, and man’s motivation and potential to restore himself to God comes in large part through this rationality.²³ Hugh makes the distinction between *imago Dei* and *similitudo* clear throughout the *Didascalicon*, and especially in his opening chapter: “This, then, is that dignity of our nature which all naturally possess in equal measure, but which all do not equally understand. For the mind, stupefied by bodily sensations and enticed out of itself by sensuous forms, has forgotten what it was. . . . But we are restored through instruction, so that we may recognize our nature and learn not to seek outside ourselves what we can find within. ‘The highest curative in life,’ therefore, is the pursuit of wisdom: he who finds it is happy, and he who possesses it, blessed.”²⁴ For Hugh, like Rupert, the notion of man as *imago Dei* is a way of describing potentiality. Drawing on Paul, he likens it to a faded, covered image, a promise that man might recover some of the closeness to God that was damaged, or obscured, by the fall.²⁵ As it is for Augustine and for Theophilus, potential is latent.²⁶ The beginning of restoration is the recognition of one’s potential, of one’s nature as *imago Dei*. In a post-lapsarian world, this is the beginning of the pursuit of wisdom. Learning, therefore, is restoration. For Augustine, Hugh of St. Victor, Rupert of Deutz, and Theophilus, *imago Dei* and *similitudo* carry a sharp

distinction: the former is inherent in man, while the latter is what was lost in the fall. Similitude, then, is the goal of restoration.

The very language that Theophilus uses to describe man’s capacities and the possibility of restoration shows that man’s fallen state requires a struggle for restoration to likeness, a struggle for personal salvation. Theophilus’s description of the condition of fallen man is telling. He links the clauses with a carefully chosen set of conjunctions: although man “lost the privilege of immortality,” *nevertheless* knowledge was transferred, *so* that “whoever [contributes] care . . .” (*licet . . . privilegium immortalitatis amisierit, tamen scientiae . . . transtulit, ut . . . curam addiderit*).²⁷ With these, Theophilus makes learning a fundamental part of man’s potential for restoration and emphasizes the continuity of the sequence. The sequence of “although,” “nevertheless,” and “so” (*licet . . . tamen . . . ut*) sets in motion the progressive trajectory of *On Diverse Arts*. Man’s potential and abilities are set within the narrative of restoration that drives the entire treatise.

Man’s free will is a major component here. As Theophilus writes in his prologue, free will (*arbitriique libertate*) enables man to choose to revere his Lord. Together with his rationality, free will makes it possible for him to learn, to gain skill, to pursue the arts, and to restore himself to God. Theophilus seems to echo an argument of Augustine, who in both *The City of God* and *On Free Choice and the Will* explains free will as the key that makes restoration possible: “In fact, the very reason why a man is undoubtedly responsible for his own sin, when he sins, is because He whose foreknowledge cannot be deceived foresaw, not the man’s fate or fortune or what not, but that the man himself would be responsible for his

own sin. No man sins unless it is his choice to sin; and his choice not to sin, that, too, God foresaw.”²⁸ The idea of free will is fundamental to monasticism more broadly. For a monk, the very willingness to do something held significance for spiritual progress. Part of a monk’s initiation was a consideration of the choices available to him; this included whether or not to pursue the monastic life at all, and what kind of monastic life one would pursue.²⁹ The significance of those choices is a major theme of the *Little Book of the Different Orders and Professions which are in the Church*, known as the *Libellus*. This book by the early twelfth-century canon regular of St. Lambert of Liège often appeals to modern readers because it describes the various practices of the different orders of monks with a remarkable tolerance for the divergences of monastic practice. Rather than praise or condemn individual orders of monks, it places them into an elaborate allegory in which the diversity of monasticism contributes to the eventual triumph of the Church. Yet most significant for our purposes is the fact that the text presents the reader with a series of choices about the spiritual life. The diversity and tolerance in the *Libellus* confirm the importance of the choice itself. If we return to the manicule on folio 20r of the Harley manuscript, we see that when Theophilus suggests to his reader “how sweet and delightful it is to give one’s attention to the practice of the various useful arts”³⁰ he is giving just the first choice of many, like a monk setting out on the path of the religious life.

The distinction between image and similitude, and the importance of free will and rationality, is clearly present in twelfth-century spirituality.³¹ That it forms the basis for Theophilus’s theology of art-making, however, has not

been recognized, and its implications for how we understand the activities he describes is significant. Read in the context of restoration and free will, the manicule in the Harley manuscript, which points out the importance of avoiding sloth, takes on greater depth. It suggests that we should read Theophilus’s instructions as a series of possibilities and choices. Indeed, the instructions often begin with an “if,” as in the opening of the long section on the chalice: “When this [the silver] has been purified, if you want to make a chalice, divide the silver into two equal parts . . .” (*quo purificato, si calicem fabricare volveris, divide argentum aequaliter in duo . . .*).³² Theophilus here begins with the future perfect form of “to wish,” *volveris*. The *De coloribus et artibus romanorum* of Heraclius and the *Mappae clavicula*, in contrast, most often use the indicative *vis*.³³ Sometimes, too, as in Heraclius’s description of gilding procedures, the imperative is used: “Take soot and pure salt, and grind them well, and take white of egg, and distemper them with it.”³⁴ Theophilus’s use of the future perfect is consistent throughout the treatise, and the construction shows a command of the language, a stylistic sophistication. The Latin future perfect expresses potentiality and puts added emphasis on the verb, on the hypothetical.

Although rather awkward in English, a literal translation illustrates the effect of the original grammar. For example, the section on censers begins, “If truly you should have wished (*si vero . . . voveris*) to make censers of repoussé work in gold, silver, or copper, you first should have refined (*purificabis*) the metal in the above way, and according to the size that you will have wanted the upper part of the censer to be, you will have cast (*funde*) two, three or four marks in the iron moulds.”³⁵ The same applies to the cruet: “If you

should have wished (*volveris*) to make a cruet for pouring out the wine, you should have beaten (*percutere*) out silver in the same way. . . ." Theophilus introduces each object by one of these verbal formulas, and the entries are carefully structured. The use of *si vero . . . voveris* in the opening lines of the instructions emphasizes the choice at stake and the desire that will guide action. Only then are instructions given on how to proceed. With this introduction of a technique or object, the reader is implicated in the instructions, as if being asked to imagine the scenario and consider the decisions to be made in the process.

A copy of *On Diverse Arts* now kept in Leipzig is evidence that such decisions mattered to a monastic audience (plate 7).³⁶ Copied in Saxony in the early fourteenth century, the manuscript contains a recension of the text that is closely related to the Vienna and Wolfenbüttel manuscripts yet has significant variations. In a striking omission, the copyist has left out the very last line of Theophilus's third prologue, which encourages the reader to make objects: "These are they: chalices, candlesticks, censers, cruets, shrines, reliquaries for holy relics, crosses, covers for Gospel books and the rest of the things which usage necessarily demands for the ecclesiastical rites. If you wish to make these, begin in this way."³⁷ The Leipzig manuscript ends the prologue at "ecclesiastical rites."³⁸ Moreover, it contains only nine chapters excerpted from book III. Together these provide the information necessary to make items using niello. The first four chapters cover files and crucibles, while the remaining five pertain to the technique of niello and its application.³⁹ Also omitted is the reference to colored glass in book II.

The logic behind these omissions is not known. Analysis of the script and decoration

suggests that the manuscript was probably made at the Cistercian abbey of Altleitzsch, near Dresden, but was not kept there, for it does not contain the mark inscribed in most Altleitzsch books from the period. There is, however, an insertion naming a certain Rudolphi de Pritzen that may provide a clue. Pritzen is not far from Altleitzsch, and near Pritzen were the brothers of the order of St. Anthony at Lichtenberg. Perhaps Rudolphi was a member of this order. The fact that the Leipzig manuscript was severely damaged by fire may also link it to Lichtenberg, which suffered a fire in 1533, after which many manuscripts were given over to the secular landholder and thence brought to Leipzig.⁴⁰

If the manuscript was copied at Altleitzsch, that provenance might confirm the significance of the edited text. Perhaps the omission of chapters on colored glass and elaborate liturgical objects was a way of changing the text to comply with Cistercian ideals of asceticism, which discouraged unnecessary objects and lavish colors. It is well known, however, that Cistercian ideals were subject to historical and social forces, and that statutes governing asceticism were followed to widely varying degrees.⁴¹ The fact that there is a Cistercian copy of *On Diverse Arts* at all is significant and may yet change our understanding of Cistercian art as well as narratives of Cistercian-Benedictine relations. Above all, the emendations of the Leipzig manuscript underscore the idea that art objects were something chosen. Some objects were necessary for performing liturgical rites, and yet there was also a range of possibilities to accommodate variations in religious ideals and decorum.

In the case of *On Diverse Arts*, the reader's decision to restore himself to God sets the stage. As Theophilus writes, "whoever will contribute

both care and concern" (*curam sollicitudinemque*) can attain (*adipisci*) "a capacity for all arts and skills."⁴² Theophilus's use of the verb *adipisci* is notable, for it is also used by both Hugh of St. Victor and Bernard of Clairvaux to express the pursuit of heavenly reward, highlighting both the progressive nature of learning and its restorative goal. Andrew of St. Victor also uses it to describe the acquisition of wisdom: "if you wish to attain the wisdom which occurs through the worship of God. . . ."⁴³ Theophilus's use of the verb thus may carry connotations of this kind of turning to, intending toward, and approaching God. In classical Latin it is distinguished as a "deponent" verb, denoting action on, or within, the self, and thus it carries implications of intent. The danger, as Theophilus explains, is for man to neglect the potential of his nature as *imago Dei*, to let lapse the inheritance of rationality and intelligence: "Therefore, let not the pious devotion of the faithful neglect what the wise foresight of our predecessors has transmitted to our age; what God has given man as an inheritance, let man strive and work with all eagerness to attain" (*hoc homo omni aviditate amplectatur et laboret adipisci*).⁴⁴ The decision to turn to God orients both learning and action.

Choice and decision-making were central tenets of monastic culture and spirituality. For Augustine, the primary activity that served the pursuit of God was reading; his *De doctrina Christiana* sets out methods for interpretation intended to serve the understanding of scripture and the pursuit of God. By the twelfth century, a far wider range of activities was understood to aid the pursuit of God. As we saw in the *Libellus*, the rise of Cistercian orders and reform movements increased the range of choices available for the spiritual life; the text also shows how intent

shapes action, for in setting out the various possibilities for the spiritual life, it recounts the variety of activities entailed therein. Hugh of St. Victor's philosophy of learning follows a similar trend, for he expands Augustine's idea about reading and the pursuit of God to include a broad range of human activities, framing them all as a part of Christian spirituality: "Of all human acts or pursuits, governed as these are by Wisdom, the end and the intention ought to regard either the restoring of our nature's integrity, or the relieving of those weaknesses to which our present life lies subject."⁴⁵ Actions are defined by the intent they serve: for Augustine, the activity of reading scripture was the primary means by which to turn to God; by the time of Hugh of St. Victor, the pursuit of God could occur through a number of human activities, but the intent that guided them—the choice to turn to God—remained the same.

Learning, action, and intent are of a piece. As Mary Carruthers has shown, *intentio*, or intent, as used here, might best be understood as a "coloration," an "emotional attitude" that is also an act of intellectual activity, a pulling toward and "hooking" of ideas into spaces, enabling the manipulation and use of memory.⁴⁶ Intent shapes activity and is an activity itself, for it is a way of mentally processing and retrieving material. In the monastic realm, intent is the driver of activity, orienting life toward the pursuit of God. As Hugh of St. Victor describes it, the purpose of the arts, and indeed of all human activity, can be directed toward this pursuit: "this then, is what the arts are concerned with, this is what they intend, namely to restore within us the divine likeness, a likeness which to us is a form but to God is his nature. The more we are conformed to the divine nature, the more do we

possess wisdom, for then there begins to shine forth again in us what has forever existed in the divine Idea or Pattern, coming and going in us but standing changeless in God.”⁴⁷ For Hugh, this works because the arts are subsumed under philosophy, which is at once the yearning for God and the “discipline which investigates . . . the causes of all things, human and divine.”⁴⁸ Philosophy seeks its origin, seeks God in the world, as the ultimate cause; but despite the Augustinian, Neoplatonic cast, Hugh is concerned with concrete experience. He divides knowledge into four categories: theoretical, which is speculative; practical, which he sees as active and ethical; mechanical, concerned with labor; and logical or linguistic.

Theophilus’s argument for the practice of art likewise rests on the idea that when the mind turns toward God and philosophical intent, the arts can be a path to restoration. Theophilus is far more technical and precise than Hugh; he shows how such restoration to God might be possible through every step of making an object. The choice to make an object at all, and the choice of what object to make, begins a series of decisions that can be a step on a path to either virtue or vice. Good decision-making drives Theophilus’s defense of art-making, for intention—*intentio*—shapes action. Once again Theophilus is in tune with contemporary thought, for Hugh of St. Victor’s division of knowledge into theoretical, practical, mechanical, and logical aspects is similar. For Hugh, the theoretical and logical orient one’s mind, while the practical and mechanical are physical and active.⁴⁹ The result then, is that Theophilus’s treatise defines a mode of art-making activity that is a manifestation, and even a physical practice, of one’s orientation to God.

The Practice of Virtue

A second manicule is found in the second prologue of the Harley manuscript, at a point in the text where Theophilus makes an explicit warning against the perils of sin (fig. 12): *Nam luce clarius constat quia, quisquis otio studet ac levitati*.⁵⁰ “It is as clear as day, that whoever is abandoned to idleness and irresponsibility,” it reads in translation, “also indulges in empty chatter and scurrility, inquisitiveness, drinking, orgies, brawls, fighting, murder, fornication, theft, sacrilege, perjury and other things of this kind which are repugnant in the sight of God.”⁵¹ While the first manicule highlights the importance of intention, this one highlights the importance of action. It indicates a detailed list of the activities that the reader should actively avoid and thus defines how to behave (or not). As Hugh of St. Victor writes, “practical [philosophy] might be called active, likewise ethical, that is, moral, from the fact that morals consist in good action.”⁵²

As we have seen in the first chapter, the prologues and instructions of *On Diverse Arts* thematize the ascent of the soul to God through the seven gifts of the Holy Spirit, creating a narrative that runs through the entire treatise and binding together the religious invocations of the prologues with the techniques of the instructions. These gifts, and their appearance in the treatise, might be loosely understood in terms of intention. The virtues of the first prologue are primarily mental: fear of the Lord, obedience, and piety. The virtues in the second prologue, meanwhile, require a certain kind of activity: fortitude is a mode of working, counsel a concern for teaching, and understanding an active mastery of thought. The third prologue, on Wisdom, is the fulfillment of all of them. The

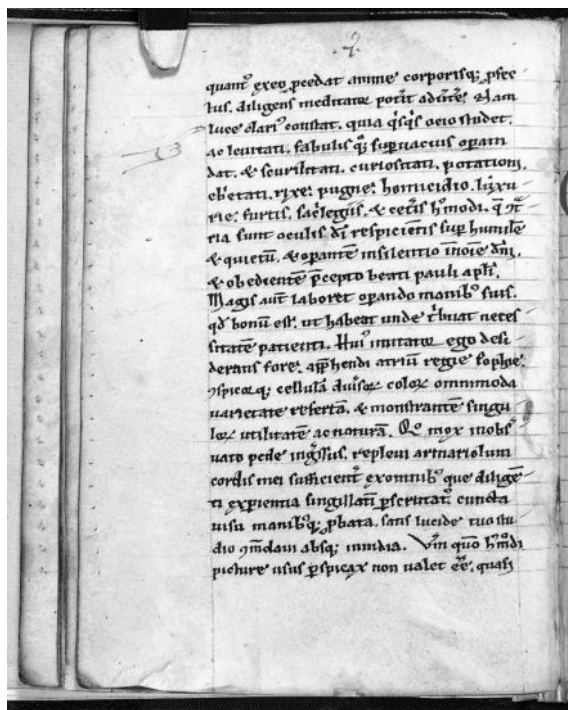


FIG. 12

Prologue II, Theophilus, *De diversis artibus*, late twelfth century. London, British Library Harley MS 3915, fol. 20v.

building of virtues one upon the next demonstrates the active nature of *intentio*. When Theophilus quotes Solomon in the second prologue and explains that “he that increaseth knowledge increaseth labour; because . . . much progress of the soul and body results thereby,” he suggests that well-intentioned, knowledgeable labor aids the progress of the soul. Just as the monk’s love for God must guide his learning, so knowledge must guide work in order for it to serve God and aid in restoration.⁵³

There has been much debate over whether or not Theophilus’s instructions are accurate, or if they even would have been helpful for the practicing artist. Once we see the importance

of choice, however, and the way in which intention and the pursuit of God shape activity, the question of accuracy becomes less important. By calling attention, in the second prologue, to the importance of avoiding sin and specifying activities that are sinful, Theophilus suggests that morality is constituted by behavior. The passage is the most explicit statement in the text of the pitfalls of bad choices. Yet as Theophilus continues, we see that the threat of sin is diminished by labor: “God is mindful of the humble and quiet man, the man working in silence in the name of the Lord.”⁵⁴ Labor and pious action defend against sloth, against laziness. As Carruthers has shown, it was seen as defense against the wandering of the mind and *curiositas*, clearing a route for intention to be carried out.⁵⁵

But this is not all. Throughout the instructions Theophilus describes not only what to do, but the manner in which to do something. He often urges diligence: when straining the color green earth, for example, or collecting sand for the making of frit, or melting glass for enamel, one must work carefully (*diligenter*).⁵⁶ The virtue of diligence is extolled thirty-five times in the text, and “diligently” is the adverb most frequently used by Theophilus by far. *Diligentia* and *diligentissime* have a similar semantic function, the former as an ablative, the latter as an amplification. The word denotes an action of care, of attentiveness. Lastly, Theophilus uses the adverb *diligentius*, such as when, in the first prologue, he describes the reward of knowledge that will accompany the careful study of his text: “If you will diligently examine it, you will find in it whatever kinds and blends of various colours Greece possesses.”⁵⁷ Care and diligence are paramount, overwhelmingly defining the proper execution of technical processes.

The word *diligentia* is not uncommon, but it carries specific connotations that Theophilus probably knew through the work of other authors. In his *De officiis*, Cicero includes diligence as that which prevents negligent or heedless action.⁵⁸ *Diligentia* is also a key term in Pliny's *Natural History*, where it indicates an accuracy or likeness in naturalistic rendering; the best artists have diligence, though as Ellen Perry notes, this did not necessarily imply artistic *gratia*, or grace, and could in fact obscure it.⁵⁹ *Diligentia* carried connotations of self-criticism, and in this respect it is related to *cura*, or care, another of Theophilus's favorite terms.⁶⁰ In the monastic context, Benedict's rule states that the true "office" where the works of God are "diligently labored" is the cloister of the monastery.⁶¹ Benedict of Nursia's rule here may be drawing upon a classical usage employed by Cicero, for whom the visual order of a home and its decorum is the manifestation of the diligence and care of the owner; the house is understood to reflect the dignity of the man who owns it.⁶² In sum, for these authors, diligence carries connotations of both assiduousness and propriety. Theophilus was no doubt educated in a tradition that included the works of Cicero, as was common in the day; his frequent use of the term *diligence* may be intended to carry on a well-established literary form.⁶³

Other adverbs and adjectives used in the text have similar connotations. Theophilus describes the preparation of color for painting glass, for example, using *diligentissime* and *cautela* (care): "Grind [the ingredients] carefully (*diligentissime*) together on the same stone with wine or urine, put them in an iron or lead vessel, and, following the drawing on the board, paint the glass with the utmost care (*cautela*)."⁶⁴ Patterns for

windows, meanwhile, must be drawn *studiose*, carefully or attentively: "this done, draw whatever figures you have chosen, first with lead or tin, and then with a red or black colour, making all the lines carefully (*studiose*) because it will be necessary when you paint the glass, to match up the shadows and highlights to [the outline on] the board."⁶⁵

Studiose is used elsewhere to denote a kind of cherishing, an extreme attentiveness. Often translated as "carefully," it is related to the word *studium*, meaning zeal or eagerness, yet it carries the additional meaning of attentiveness, like *diligenter*. It is used, for example, by the monks of Stavelot when praising the abbey of Corvey for electing Wibald abbot in 1146: "your learning has recognized well, that which ought to be cherished ardently (*studiose foveri*), held and kept diligently (*diligenter*)."⁶⁶ Theophilus uses it in a similar sense when he describes engraving patterns for die stamping: "They [the drawings on the die] are not engraved too deeply, but moderately and with care (*mediocriter ac studiose*)."⁶⁷

Related to *studiose* is *studio*, or zeal. In the third prologue, Theophilus suggests what he means by zeal: "Come now, my wise friend (*vir prudens*), in this life happy in the sight of God and man and happier in the life to come—by whose labour and zeal (*labor et studio*) so many sacrifices are offered to God, be inspired henceforth to greater deeds of skill."⁶⁸ *Studio* is used here, however, in a very specific case, for it refers to the wise man, the prudent man (*vir prudens*), who reads the text.⁶⁹ Zeal is mentioned often in the prologues, but specific actions are to be performed *studiose*, with care.

Thus a distinction is drawn between zeal and care. Zeal is reserved for the greater project, while extreme care and cherishing, even gentleness

in the detail, is made manifest in the objects themselves. A cross attributed to Roger of Helmarshausen and known as the Enger cross is a fine example.⁷⁰ The object is not complicated in its overall form, but when examined closely, more and more layers of detail emerge: the shimmering surface is filled with filigree; the wires of the filigree themselves are beaded; ends are finished with small knobs; and surfaces are bound with tiny bands (plate 8). We might interpret the cross as operating on these two levels: the artisan makes the larger, religious offering of a cross as a sign of his zeal, while the minutely careful detail is witness to the performance of attentiveness, or *studiose*. Each strand of filigree is attended to as an element to be cherished, through the endlessly repeated actions of beading, bending, and binding.

A third set of adverbs, or nouns in the ablative that function as adverbs, elaborate on the idea of *studiose* and are words of care and of moderation: *subtiliter* (subtly), *leniter* (lightly), *moderate* (with moderation), *mensurate* (measured, with moderation), and *modicum* (gently). When setting gems and pearls, Theophilus writes, “You adjust the cloisons with delicacy and care, each one in its place, and you stick them on with paste over the fire. When you have filled one piece, you solder it with the greatest care lest the delicate work and the thin gold come apart or melt.”⁷¹ Here he uses the word *confirmare* (to secure or attach) to describe how to attach the cloisons to the work, stressing the care required, and he closes the section with a warning against damaging the delicate work by carelessness. When two pieces of metal are to be bound together, as in the case of the gem settings, like those on the Enger cross, the task must be done carefully and gently, so as not to cause damage to the fine material.

The application of varnish to a painting, which may not at first appear to be delicate work, must also be performed *leniter*: “When the painting is completed and dried and the work has been carried out into the sun, carefully spread over it the sticky varnish, and when this begins to run with the heat, rub it gently with the hand.”⁷² Similarly, when making sheet glass one is to “take [the pipe with the glass on it] out at once, put it in your mouth and blow gently (*suffla modicum*); then immediately take it from your mouth and hold it near your cheek in case you accidentally draw flame into your mouth if you inhale.”⁷³ The book on glass contains a number of instances where procedures have to be carried out with speed, yet Theophilus nevertheless emphasizes that one must be cautious and take care with the procedure and warns against calamities that could occur when such precautions are not taken. Like the monastic life, it is a matter of discipline. Perhaps, then, there is a moral implication in having a light touch. The smooth sheet of glass, for example, or the clear face of Christ in the cathedral at Strasbourg, discussed in chapter 2, can thus be read as evidence of procedure and proof of the artist’s restraint and gentleness of action.

Vice is also implicit in the instructions. Often, Theophilus’s words of caution are used in conjunction with warnings against neglect, and indeed the predominant “vices” that Theophilus warns against are those of neglect and ignorance. The technique of repoussé, for example, requires a beaten gold or silver plate, which should be free of air bubbles and cracks caused by negligent casting: “When you first cast this gold or silver, examine it, by carefully scraping and scratching around it, in case there happens to be any air-bubble or crack in it. This often happens when, through the carelessness, or negligence,

or ignorance, or lack of skill of the founder, it is cast either too hot or too cold, or too quickly, or too slowly. If, when you have cast it with care and circumspection, you perceive a flaw of this kind in it, carefully dig it out, if you can, with a suitable tool.”⁷⁴ Here Theophilus warns of the problems that may occur through negligence, and how to recognize the effects of poor work. Cracks and air bubbles are caused by inattentiveness in the casting process. In contrast, good work is made by casting “with care and circumspection” (*cumque considerate et caute fuderis*). Yet even if the casting is done attentively, the risk remains of flaws in the metal, and so Theophilus also describes how to fix it by “carefully” digging out the air bubble with a “suitable tool” (*diligenter, cum ferro ad hoc apto*). Theophilus warns against negligence often, usually using a hypothetical clause: “if it happens because of negligence” (*si contigerit per negligentiam*). Such a clause begins many chapters. When applying powdered gold and silver, for example, he writes, “if, as a result of carelessness, the glue has not been properly heated.” Regarding ink, he writes, “If, as a result of carelessness, the ink is not black enough, take a piece of iron, an inch thick, put it on the fire until it is red hot and then throw it into the ink.” When casting silver, he similarly advises, “If, through some negligence, it happens that the cast silver is not sound, cast it again until it is.” When casting the large chalice, he reassures his reader: “Unless there is great negligence, all silver and gold, which is cast in this way, is invariably sound for working whatever you wish from it.” His phrasing is similar when describing how to gild the handles of the chalice (“But, if it should happen, as a result of negligence, that some blemish appears on the silver where the gold is thin and unevenly applied, reapply the gold with

the copper tool, and spread it out with the dry brush until it is even everywhere”) and when executing figures in repoussé (“Then you gently and carefully raise [the details of the figures] inside with smaller rounded tools, taking particular care that the work is not fractured or pierced. But if this should happen through ignorance or negligence, it should be soldered in this way”).⁷⁵ In all of these chapters Theophilus follows a consistent pattern: He describes the flaw that results from negligence, why it matters, and how to repair it.

Perhaps most emphatic about carefulness, restraint, and subtlety is the chapter devoted to the pinning and soldering of handles for the chalice:

Place the bowl in the fire and *carefully* cover it with coal that you put on, and blow *gently* with a long breath until the soldering melts sufficiently. Take the bowl from the fire and, when it has somewhat cooled, wash it, and if the pins are firm—well and good. But if not, treat them as before.

When they are firm, file them off inside [the bowl], *polish smoothly so that no one can see where they are*, and, refixing the handles outside, *carefully* hold them in position. Then make fine holes through the middle of the handles opposite the pins, and similarly in the same place beyond the pins, and into these you fix them with all the work *perfectly finished so that no one can see how they are attached*. After this, *carefully* scribe and engrave these handles with files and iron tools and, if you want to niello on these, proceed in this way.⁷⁶

Here, care is made manifest in the object through its proper execution and the careful, yet secure,



FIG. 13

Roger of Helmarshausen, book cover, Cod. 139, ca. 1110–20, detail: eagle. Cathedral of Trier, Domschatz.

soldering of handles. Throughout the instructions, Theophilus describes how actions are to be performed. Good labor is defined by the virtues it puts into play, be they working with care, knowing and following appropriate techniques, or understanding how to prevent and repair

problems. As Virginia Raguin has suggested, the order and care of work in Theophilus is in harmony with the ordering of the monastic day.⁷⁷

In his description of repoussé work, Theophilus describes the initial pressing out of the head of the figure. Again, diligence is required:

When this [heating] is done and the plate has cooled of itself, you again gently and carefully (*leniter et diligenter*) rub it on the underside with the rounded tool, inside the hollow of the head. Turning the plate over, you again rub the upper surface with the smooth tool and depress the ground so as to raise the relief of the head. Once more beat lightly (*mediocri*) round it with the medium hammer, and, putting it on the fire, heat it again. You do this often, carefully raising it inside and out (*diligenter elevando interius et exterius*), frequently beating and as often reheating it, until the relief is raised (*ducatur*) to a height of three or four fingers, or more or less, according to the size of the figures.⁷⁸

A book cover attributed to Roger of Helmarshausen and now in Trier is a fine example of this technique. Here the four symbols of the Evangelists are shown in repoussé, set within a frame of gems, filigree, and enamels.⁷⁹ The beak of the eagle representing John is extraordinary; it is not only pushed out from the surface of the metal but also bends back above it (fig. 13). Theophilus, in his instructions, emphasizes both the care and diligence required for the proper pushing out of the head of the figure, a repetitive task. The head of the eagle is the result of a repeated series of heating, pressing, and cooling of the metal until it is raised—in the Latin *ducatur*, literally “led” to the desired height. Similarly, the patterned wings of the eagle and the texture of feathers pressed into the metal can be interpreted in terms of this repeated, careful action.

This concept of action finds parallels in other instances of twelfth-century thought. Wibald of Stavelot, in a letter to a monk, describes the beginning of wisdom as interior knowledge and

a teacher’s role as one of stimulating and guiding. “Let our example stimulate you, imitation rouse you, concern incite you,” the letter reads. “You learn if you see him, you are instructed if you hear, you are perfected if you follow.”⁸⁰ Virtue therefore, was something to be “enacted.”⁸¹

The way in which art aids this learning process, and is evidence of it, has been studied by Ilene Forsyth in her analysis of the sculptural representation of Abbot Durand (d. 1077) made around 1100 for the cloister of Moissac. The image of the abbot in the company of apostles, set into the piers of the cloister, replaces the traditional image of Christ and invites living monks to follow the apostolic path.⁸² Cynthia Hahn has focused on such representations of saints who, because they are members of the heavenly realm and are present on earth through remembered actions, function as invitations to veneration and models of behavior.⁸³ Ienje van’t Spijker, meanwhile, shows how such veneration of external models like saints relates to contemporary discourse on the inner man. For Van’t Spijker, imitation of an exemplar’s actions is a way of bridging the exterior and interior man.⁸⁴ This method, she suggests, is inherently limited. Because man is human, he can but follow fellow humans; the quest for likeness to God must be mediated by the model of Christ and the saints. However, because the saints themselves were inaccessible, and because their actions as recounted in lives and legend were beyond the reach of most individuals, the imitation of the saints was not a wholesale mimesis. Instead, saints’ lives provided forms and patterns to be emulated; one was to understand and interiorize the virtues at stake. As Van’t Spijker points out, this was especially the case for martyrs, whose martyrdom was inimitable but nonetheless

held lessons in virtue.⁸⁵ This physical practice of virtue is an extension of what C. Stephen Jaeger has shown was the prominent mode of learning in the eleventh century, when virtue was understood as something to be learned through the physical imitation of a master; it remained prominent in Germany into the twelfth.⁸⁶ Physical activity, then, was not separated from moral activity, but integral to it.

We might read this idea of imitation of virtue in the process of art-making. With the actions so carefully defined, the procedure by which an object is made becomes a path of potential virtue, with every task along the way laden with moral significance. For Theophilus, artistic labor is a part of man's journey toward virtue or vice. Very rarely does Theophilus discuss iconography or describe what an object is to look like, and this may be why. He will allude to forms, such as Greek foliage on the niello of a chalice, or figures, animals, or flowers on a silver cruet, but imagery is usually a secondary matter.⁸⁷ His concern is instead for technique, for process. In the chapter on repoussé, for example, he describes the figures to be made, but his only comment on how they should look is that the head should be in higher relief: "Then, with the tracer, mark out the body, or bodies, of the figures, and so, by depressing and from time to time beating, you can raise them as much as you want. Take care of this, however, that the head is always in higher relief." Theophilus's concern here is for the relative height of the forms, not the forms themselves, and that attention be paid to the execution so that the appropriate effect is achieved. This is not by chance. When the text is read closely, we also notice that Theophilus refers to the activity of the artist primarily as work (*opere* or *labore*), or, as in the case of repoussé, *opere ductili*

(beaten work). It is as if objects are constituted by procedure.

Contemporary objects show that process itself can be employed as a visual motif. The portable altar made by Roger of Helmarshausen for Henry of Werl, the bishop of Paderborn, is a good example (plate 9). The entire lower beveled edge of the altar is covered with the studs used to attach the plate onto the wood beneath. The number of studs visible here far exceeds the amount necessary to attach the plate to the wood. A comparison with an earlier altar made for the countess Gertrude, now in Cleveland, is telling (plate 10). This altar, made just after 1038 (and as we shall see in the following chapter a precedent for the altar of Henry of Werl), has no nails visible on the beveled edge. It seems possible that Roger lavished special attention on the process of assembly, making traces of labor part of the decorative program. Yet elsewhere in Theophilus's instructions, traces of assembly are to be hidden: in the silver chalice, for instance, the pins securing the handles to the cup are to be filed down and "perfectly finished so that no one can see how they [e.g., the handles] are attached."⁸⁸ Some processes are to be hidden, but others are to be seen. As in the emerging head of the eagle or the disappearing pins of the chalice, process can be part of visibility. In the following chapter we will see how process as a visual form plays out, focusing particularly on the altar of Henry of Werl, the Enger cross, and the Trier book cover. For the moment, however, we will focus on only one aspect of the issue, where process is evidence of good, moral labor.

Moralized labor is not only the practice of virtue; it can also be the embodiment of virtue. As we saw in the beginning of this chapter, *imago Dei* and *similitudo* were the beginning of a longer

process of restoration following the fall. Theophilus writes that man acquired “participation in the wisdom and skill of the Divine Intelligence” by virtue of his rationality (*divinae prudentiae consilii ingeniiue mereretur participium*). According to Augustine, all beings “participate” in God to some extent:

This supreme and true God—with His Word and Holy Spirit which are one with Him—this one omnipotent God is the creator and maker of every soul and of every body. . . . He made man a rational animal, composed of soul and body. He permitted man to sin—but not with impunity—and he pursued him with his mercy. He gave men—both good and bad—their being, as He gave being to the rocks. He let men share generative life in common with the trees, and the life of the senses with the beasts of the fields, but the life of intelligence only with the angels.⁸⁹

God is the creator of all things, and thus the essence of living things—their very being-ness—comes from him. Man, being rational, has the unique ability to “participate” in God’s intelligence. As Augustine explains in *The City of God*, “To his irrational creatures he gave memory, perception and appetite, but to His rational creatures he added a mind with intelligence and skill.”⁹⁰ Because of the distance between man and God caused by the fall, however, participation is not guaranteed. This is eloquently expressed by Augustine in *De trinitate*, where it is described as a distant memory or image of the Trinity: “For although the memory of man . . . in proportion to its own small measure in this image of the Trinity, a likeness, incomparably unequal, of course, but yet a likeness of whatever kind it may be to the

Father.”⁹¹ It can, however, happen when man uses his rationality to practice intelligence and skill, because when man acts in this way, he is fulfilling his ultimate self and therefore “participating in God.” Participation may not be a permanent state for the fallen man, but if he uses his intelligence and rationality, he can participate with God at least temporarily. Rupert of Deutz draws on this idea too, for in his commentary on Benedict’s rule he quotes Paul’s first letter to the Corinthians to describe the service of the altar, saying “they who serve the altar, partake with the altar” (*cum altario participantur*).⁹² The ideal of participation links the artist to God not through his person but through his work, skill, and actions.

Monastic Labor in the Twelfth Century

In *On Diverse Arts*, artistic action is moralized, and skill is a way to participate in God. Such an emphasis changes how we might see the twelfth-century artist and his work. Theophilus invokes a model artist just once in the text, in the third prologue. Here he makes a reference to the Old Testament figure Beseleel, the builder of the ark of the covenant: “For [Solomon] had read in Exodus that the Lord had given to Moses a commandment to build a tabernacle, and had chosen by name the masters of the work, and had filled them with the spirit of wisdom and understanding and knowledge in all learning for contriving and making works in gold and silver, bronze, gems, wood and in art of every kind.”⁹³ The passage has been read as justifying the production of art in the monastery, and the allusion to Beseleel has been understood as a biblical exemplar validating the artist and affirming his special status.⁹⁴ Beseleel, however, is not mentioned by name,

nor is it he who is praised. If we read the passage within the context of Theophilus's concern with process, skill, and participation, the reference to Beseleel is seen to invoke a theory of production in which art is justified by the process of its making.

The biblical source of the reference is Exodus 31, where God names the artists who will build the tabernacle for the ark: "And the Lord spoke to Moses, saying: Behold, I have called by name Beseleel the son of Uri the son of Hur of the tribe of Juda, And I have filled him with the spirit of God, with wisdom and understanding, and knowledge in all manner of work. To devise whatsoever may be artificially made of gold, and silver, and brass, Of marble, and precious stones, and variety of wood."⁹⁵ Beseleel would have been familiar to medieval readers, for his name was invoked periodically throughout the Middle Ages. Charlemagne's *Libri Carolini* invokes the example of Beseleel as maker of the ark, filled with the spirit, while Einhard, Charlemagne's biographer and court architect, was called a new Beseleel in a poem by Walafrid of Strabo and in several of Alcuin's poems.⁹⁶ Rabanus Maurus, too, used the passage from Exodus describing Beseleel in his epitaph for Einhard, writing that like Beseleel, Einhard "was skilled in every kind of craft."⁹⁷

Walafrid of Strabo uses Beseleel in a more complex way, to explain the presence of the Holy Spirit in God's actions:

In the construction of the tabernacle the mysteries of the Trinity are shown, just as in baptism and in the transfiguration. Indeed in baptism the Father is heard: and the son is baptised, and the spirit appears in the vision of the dove. Ooliab also, who is interpreted

as my protector, the father, the person of the Father: Beseleel also, who, in the shadow of God, signifies the person of the Holy Spirit: of which it is said, and the strength of the most high will shadow over you. The strength even of the highest Holy Spirit is to be understood. So as God says: I have known the strength that comes from me. (Luke 8) I order also, it is the word, where the tabernacle is ordered to be made, what the Son is, through whom the Father in the virtue of the Holy Spirit operates all things.⁹⁸

Here we see Beseleel likened to the Holy Spirit, which is known through its workings on earth. Just as Beseleel made the objects ordered by God manifest on earth, so too the Holy Spirit makes the workings of God manifest on earth.

In the eleventh century, Peter Damian invoked Beseleel in a letter to the monks at Montecassino, in which he cites God's order to build the tabernacle as justification for the rebuilding of the church:

Indeed, if almighty God despises the efforts of those who build his churches, how is it that he commanded that a tabernacle be constructed for him with such great care in the desert? Was he speaking to Moses in a belittling or off-handed way when he said this? "I have specially chosen Bezalel, son of Hur, of the tribe of Judah. . . ." Moreover, if he ordered such meticulous care in constructing a tabernacle that he knew would soon be abandoned, how much greater is his concern in building the Church, which for the salvation of all men will stand unmoved to the end of time? The tabernacle erected in the desert and the temple constructed in the

reign of Solomon were shadows and figures of that Church that now stands so brilliantly among the Christian people.⁹⁹

This constitutes a departure from the Carolingian use of Beseleel as a personal example of skillfulness, for the letter is instead concerned primarily with the choice to build the church and expand the Christian community. The calling of the skilled Beseleel is a testament to how much God cares about the building even when he knows it will not last. Beseleel's skillful work on the temple is an example, but like the Old Testament itself, it will be overshadowed by the new works that will be made under Christianity.

Theophilus's take on Beseleel is rather different. He does not follow the Carolingian tradition of rhetorical praise and liken himself or his potential student to the Old Testament figure. Nor does he invoke the idea of Beseleel's work as a precedent to be overcome. Theophilus's concern is rather with how an artist can make manifest the workings of the Holy Spirit. The fact that God called Beseleel by name is central to the biblical passage; however, Theophilus does not mention his name. Instead, his emphasis is on God's gifts to the artist of "wisdom and understanding and knowledge in all learning for contriving and making (*excogitandum et faciendum*) works . . . of every kind." Theophilus alludes to Beseleel—again, without naming him—as evidence of the gift of skill. As the passage continues, Theophilus writes, "by pious reflection . . . [David] discerned that God delighted in embellishment of this kind, the execution of which he assigned to the power and guidance of the Holy Spirit, and he believed that nothing of this kind could be endeavoured without His inspiration."¹⁰⁰ Here Theophilus is interested in Beseleel

as an example of one who has been filled with "the spirit," following Augustinian thought.¹⁰¹ Yet while Augustine concedes that these gifts are "not yet read as the Holy Spirit," Theophilus makes no such distinction.

Rupert of Deutz's commentary on Exodus is instructive for understanding how Beseleel might function in *On Diverse Arts*. According to Rupert, Beseleel is to be interpreted as the shadow of God, because in his name the shadow of the tabernacle points to its sacred counterpart, the tabernacle of heaven.¹⁰² For Rupert, the tabernacle was considered a precursor to the embellishment of the Temple of Jerusalem by Solomon; the temple of Solomon, in turn, was a prototype and justification for the ornamentation of the Church.¹⁰³ But ultimately for Rupert what matters is Beseleel's work and intention, not his character. For God, he writes, "called them [Beseleel and Ooliab] by name and wisdom taught them both.' [So] who could doubt that all these arts of theirs are gifts of God? Hence in whatever way the arts have, for man, so much utility and are permitted to be esteemed, and the skilled artisans are to be warned, so too the talent of God in profit should be paid in profit, because it is not appropriate to the faculty of man himself, but all things of the Creator are the undertaking of tribute, and from them the tribute itself is to be extracted."¹⁰⁴ Thus for Rupert the object is significant for its function, its proper use, while Beseleel the person is esteemed because he performs the work well, using his God-given faculties.

In sum, for Theophilus and Rupert, and even to some extent for Augustine, Beseleel's labor is the operation of the Holy Spirit on earth, a physical manifestation of that presence. Perhaps this is why Theophilus does not name Beseleel outright,

for without his name, Beseleel's role as Old Testament exemplar is lessened, and the value of his good work comes to the fore.

Theophilus's reference to Beseleel sheds light on his use of Old Testament precedents more generally. Many exegetical texts of the twelfth century invoked examples from the Old Testament as prototypes; they were, as Beryl Smalley has put it, understood as a "sign."¹⁰⁵ This is well illustrated in the visual program of the Stavelot altar, made under the abbot Wibald around 1150, where images of Abel's gift of the lamb, Melchisedek's offering of wine, Abraham's sacrifice of Isaac, and Moses's raising of the brazen serpent are set as precursors to Christ's sacrifice on the cross and the celebration of the Eucharist. In each of these cases, however, it is not the person but the activity being performed that creates the parallel. The crucial point of such typologies is the Christian fulfillment of the Hebrew Bible. We might thus understand Theophilus's use of Beseleel as a similar exegetical analogy.

The use of Beseleel to demonstrate the value of artistic work not only corresponds to Theophilus's overall emphasis on procedures and skill, but it also reveals something about monastic ideas of labor in the twelfth century. Few sources directly addressing the issue survive from the twelfth century; a major resource remains the Rule of Saint Benedict, particularly its chapter on "Daily Manual Labor." Here Benedict writes, "Idleness is the enemy of the soul. Therefore, the brothers should have specified periods for manual labor as well as for holy reading. . . . They must not become distressed if local conditions or their poverty should force them to do the harvesting in themselves. When they live by the labor of their hands, as our fathers and the apostles did, then they are really monks."¹⁰⁶

Benedict's rule was certainly heeded in the twelfth century, but it is rather vague, and monastic practices varied quite widely. The relation of labor and wealth to the monastic ideal was a point of contention for monastic reformers in the twelfth century, and Benedictines were at times criticized for their wealth and restful lives. One of the most famous articulations of the debate over wealth, poverty, and work was written by Idung of Prüfening, a Benedictine monk who became a Cistercian in the twelfth century. His *Dialogus duorum monachorum*, or *Dialogue of Two Monks*, presents a Cistercian and a Cluniac debating the merits of their monastic orders.¹⁰⁷ However, since much of what we know of Cistercian labor—from Idung's writing and other sources—was dedicated to agriculture, it remains difficult to know how artistic labor was valued in the Cistercian world.

The actual status and activities of monastic labor in any house are not easy to discern, especially since the problem is known largely through critiques such as those of Idung of Prüfening. As Jacques Le Goff has argued, these written sources attest to an ambivalence about skill and leisure, for throughout the Middle Ages the need for skill and productivity in society conflicted with the desire for *otium*, or leisure. In monastic contexts, this ambivalence might be expressed in questions of participation in the world and in terms of the work itself; in questions of how, and where, the performance of the work of God could occur.¹⁰⁸ For many monks, prayer was their work, and others were left to till the fields. Yet here, too, little is known of artists, and it is difficult to know the motivations behind such questioning. John Van Engen has suggested, for example, that the growth of new orders and even the growth of market economies, rather than posing a real threat, may in

fact have been a spur that forced the Benedictines to articulate their own purposes, philosophies, and mode of life.¹⁰⁹ This certainly comes through in the *Little Book of the Orders*, in which all the monastic orders are defined and characterized but without harsh judgment. All, ultimately, are seen as intending to serve to the Lord.

The few references to monastic labor and artistic work that survive suggest that spiritual work and manual work are often melded in the monastic context. A letter from Herimann of Winzenburg to Pope Eugenius III lauds Wibald for his reform work, referring to it repeatedly as his “labor.”¹¹⁰ Isenric, a monk of St. Gall in the ninth century, for example, is remembered as having labored in humility and is praised for the works of gold and silver he created.¹¹¹

Rupert of Deutz may best exemplify the Benedictine perspective. In his discussion of the work of the divine office in the *In regulam Sancti Benedicti*, Rupert discusses manual work at length. For Rupert, manual labor is to be conducted as is necessary for the livelihood of the monastery and in accordance with the health of the monks, and all things that can be made within the monastery ought to be, to prevent the monks’ having to go out of the cloister and disengage the spirit.¹¹² Additional evidence may be taken from the commentary on Benedict’s rule by Smaragdus, a ninth-century Lotharingian commentator; twelfth-century copies of his text survive from the abbey of Saint-Laurent in Liège and from the abbey at Gembloux, and ninth- and tenth-century copies survive from Great St-Martin’s in Cologne and the abbey of Stavelot, in Belgium, the latter containing a twelfth- and thirteenth-century chapter index.¹¹³ Expanding on Benedict’s phrase “*otium* is the enemy of the soul,” Smaragdus writes,

The idle, even in body, monk will never be free in the mind from unclean thoughts. . . . If indeed the apostles were doing physical work whence they could support the life of the body, how much more, monks, it is necessary not only to produce with one’s own hands the necessities of his life but also to restore the needs of others by his work. Who with bodily strength and soundness of health stops if they are idle in work are doubly recalled to sin, because not only do they not work but also they spoil others inviting them to their imitation.¹¹⁴

Smaragdus’s comment seems to have much in common with the words of Theophilus, for here, as in *On Diverse Arts*, the actions of the body are tied to the actions of the mind. Labor not only produces the objects necessary for life, but it also serves the mind and the spirit.

Developing market economies of the twelfth century may also have had an effect on ideas of artistic labor. Indeed, it is in the late eleventh and twelfth centuries that the class of *burgus*, or artisans, emerges in the records.¹¹⁵ When Huy became an independent city of “bourgeois” in 1066, for example, it declared its main goals to be promoting commerce and credit, and defending itself.¹¹⁶

Theophilus’s treatise, on the other hand, shows artistic labor to be a performance of procedures according to ethical terms of diligence, carefulness, and attentiveness. As he writes of “avoiding sloth” in the second prologue, or carefully measuring one’s work, “lest the vice of avarice or cupidity steal in,” in the third, Theophilus makes it clear that even in artistic work one must be on guard against vices.¹¹⁷ Theophilus moralizes artistic skill so that the physical act of

art-making is the exercise of free will, rationality, and virtue. The justification of art in *On Diverse Arts* might be read as a defense of Benedictine monastic practices on the one hand and an answer to the challenge of a new mercantile class on the other.

Conclusions

Through prologues and instructions, Theophilus moralizes the act of artistic labor. Every step of a process becomes a choice, from the decision to make an object to the decisions about how to do so. Procedures, therefore, are central to his theory of art. Theophilus begins with mental orientation and shows how to apply the practice of

virtue to artistic work. In the end, the object can be understood as the trace of good work, virtue, and godliness. Taken together, the practicalities of technique define a mode of art-making that adheres to certain standards and criteria. Techniques are praised because they demonstrate values like diversity, variety, and ingenious subtlety. Material signs of skilled process, as seen in the altar of Henry of Werl or in the Enger cross, lend value to a work. Notably, Theophilus never suggests that skill resides in the manner of rendering figures or patterns, or even that it is evident in compositional complexity. In the next chapter we shall look further at this reticence and explore how good procedure can drive aesthetics.

Theophilus humilis presbyter
ius servus dei. indigni nomine et
affectione monachi omnibus mentis
desidia animique uagatione utili
manuum occupatione. et delectabili nouita
tu meditatione declinare et calcare uo
lentibus; retributione celestis premii legi
m in exordio mundane creationis homi
nis ad imaginem et similitudinem dei conditi.
et inspiratione diuini spiraculi anima
tu. tamque dignitatis excellentia ce
teris animantibus; propterea. ut rationi
capax diuine prudentie consilio inge
niique mereret participium arbitriique
libertate donatus. soli eductus sui suspi
ceret uoluntate. et reuereret imperium.
et astu diabolico misere deceptus. licet
propter inobedientie culpa priuilegium in
mortalitatis amiserit. tam scientiae et
intelligentiae dignitate adeo in posterita
tis propaginem transtulit. ut quicumque cura
solicitudinemque addiderit. totum artium
ingeniumque; capacitate quasi hereditario
iure adipisci possit. Illi uero modo intemio
ne humana suscipiens sollertia. et in
diuersis actibus suis insistentis lucris et
uoluptatibus; per tempore incrementa. tandem
ad predestinata christiane religionis produ
centia tempora. factusque est ut quod ad laudem et
gloriam nominis sui edidit dispositio di
uina. in eius obsequium conuerteret plebs
dona deuota. Quapropter quod ad nostram usque
etate sollertis predestinatio transtulit per uisio
pia fidelium non neglegat deuotio. quodque
hereditarium deus contulit homini. hoc ho
mo omni audacitate amplectatur et labore

adipsi. et adepti. nemo apud se quasi se
et se. et non aliunde accepto gloriatur. sed in domino
a quo. et per quem omnia. et sine quo nihil humiliter gra
tuletur. nec excessum inuidie sacculo recon
dat. aut tenacis armariolo cordis occul
tet. sed omni iactantia repulsa. hilariter men
te simplicitate querentibus; erogat. inquitque
euangelica illi negotiatoris sententia.
quod domino suo reassignare dissimulans.
maxime feneratorum omni beneficio priuat
oris sui iudicio nequa seruus promeruit no
tia. Quia sententia incurrere formidans
ego indignus et penitus nullius nominis hominu
m. quod in gratia excessit. quae dat omnibus
affluent. et non imperat diuina dignatio
cunctis humiliter discere desiderantibus. gra
tis offero. et ut in me benignitate dei recog
noscant. largiaturque; murentur ad monito.
et ut idem si opera addiderint sibi prelo
esse. paulo dubio credant insinuo. Sicut
enim homini quodcumque; uentum aut indebitum
cuiusque; modi ambitione at temptare siue
rapina usurpare iniquum est. et detesta
bile sic iure debitum et ex parte dei here
ditarium. interceptum negligere. aut con
tuit ducere ignauus asseribit ac stulti
cie. Tu ergo quicumque; et fili karissime cui de
us misit in cor caput latissimum diuersarum
artium perscrutari. et ut exinde quod libue
rit colligas intellectum cuiusque; appone
re. non uilipendas preciosa et utilia quaeque.
quasi ea sunt sponte aut insperato domesti
ca terra produxerit. quia stultus negotiator
est qui thesaurum subito fossa humo rep
perit. si illum colligere et seruare ne
glexerit. et si est et arbusta uilia myrra
thus. et balsama producerent. seu fontes

imponet opus tuo si libitum glutine coarctat demaps acapit colores quos impone
uolueris tenet eos diligenter oleo lino si aqua et fac mixtas multum ac uelut
mixtas sic super aqua fecit et testat siue auel aut sola uariabilis sunt coloribus
sunt libuerit. *De coloribus oleo et gummi in ligno*

xxv. **O**mnia quae coloribus oleo uel et pomis possunt in ope ligneo in his amant
rebus quae sole siccat possunt quod sciens in coloribus imponeris aliam ei suppo
ne non potes si per exerceat quod imaginibus duntaxat et adiosum nimis est
aut uolueris opus aut festinare sume gummi quod exat de arbore ceruso
siue primo et gaudens illud minutatim pone in uas fictile et aqua
abundante infunde et pone ad solem siue carbonem in hanc donec
gummi liquat et ligno rotundo diligenter gmisce. Deinde cola per pan
nam et in cere coloribus impone. Omnes colores et mixtas eorum huius gum
mi tere et pomis prius per minutum et cerosam et curram qui est clauo
omni terendi et pondo et viride hispanice non misceat suam si glutine

xxvi. **O**mnis color siue oleo siue gummi et color in ligno et debet ponere et per
tina perfecta et sicca deat ope ad solem diligenter linet illud glutine
amiciem et cum defluere apertum a calore leniter manu fabulit atque
eas sic facies et et sine donec penitus exsiccat. *De petra et fluada*

xxvii. **E**t petra in ligno qui est fluada et apud quas uocat amecola et
hoc modo operies. Tolle petram suam si minutam glutine si colo
ram et si simplicem et diligenter pollicam et in coepiet locum
quem ita pingere uolueris. Deinde cere coloribus imponendos diligen
tissime oleo lino ac ualde tenuet et hoc eos cum pincello sicque pinge
re. *De molendo anise in uis et de fundendo molendo*

xxviii. **Q**uam pferis magnetem ut licet in latere colle amperum et lina illud
minutissime mundissima pelui siue bacina sicque uariabilis illud
et pincello et oculi testum ut gehili quod de aqua tollit. De his hanc





PLATE 3

Stavelot Bible, 1097, detail: Saint
 Luke. London, British Library
 Add. MS 28107, fol. 161v.



PLATE 4

Head of Christ, glass fragment, abbey church of Saints Peter and Paul, Wissembourg, late eleventh or early twelfth century. Strasbourg, Musée de l'Oeuvre Notre-Dame.



PLATE 5

Christ in Majesty, Church
of Saints Peter and Paul,
Reichenau-Niederzell, ca.
1120–30.

ponere
 - las ollas abigne ad sole. don se nigrū incan-
 nam artibea fece purificet. Postea tolle
 folliculos ex pgameno diligenter siccos. ac
 uelicas. & infundes purū incanū. suspend
 ad sole ad sole. don omnino siccet. Cumq; sic
 oū fuit. tolle mē quatiens uolueris. & tempa
 cum vino sup carbonel & addes modicu at
 mēti sebe. Qd si nigrū pnegligentia ut
 n satis nigrū sit incanū. accipe ferrum
 gitudinē unū digiti & ponens in ignem
 sine candescē moxq; in incanū pice.

Explicit lib' primus. Incipit prologus

h pcedenti libello in libru scdm.
 te kme sine dilectionis affectu. n me pi-
 gnat tue indoli insinuare qnti honoris
 qntiq; pfect' sit. ocium declinare. q mer-
 cia desidiaq; calcare. quaq; dulce ac de-
 lectabile. diuina utilitatū exercitiis
 opam dare. iuxta uocē oratoris cuiusda
 dicentis. Scire aliqd laus est culpa est
 nil discē uelle. Nec pigritet quisquam
 eū de q salomon ait. Qui addit scien-
 tiam. addit & labore apprehendere qā



PLATE 6 (OPPOSITE)

Incipit prologue I, Theophilus, *De diversis artibus*, late twelfth century.
London, British Library Harley MS 3915, fol. 20r.

PLATE 7

Incipit Theophilus Presbyter, *De diversis artibus*, fourteenth century.
Universitätsbibliothek Leipzig, MS 1157, fol. 17v.





PLATE 8 (OPPOSITE)

Roger of Helmarshausen, Enger cross, ca. 1105. Kunstgewerbemuseum, Berlin.

PLATE 9

Roger of Helmarshausen, portable altar of the cathedral of Paderborn, made for Bishop Henry II of Werl, ca. 1105. Erzbischöfliches Diözesanmuseum, Paderborn.



PLATE 10

Altar of Countess Gertrude, possibly Lower Saxony, ca. 1045, short side, with the True Cross, Constantine, Helena, Sigismund, and Adelaide. The Cleveland Museum of Art, Gift of the John Huntington Art and Polytechnic Trust 1931.462.

seris & sabulu de sup atq; aqua
imponere lapide minore q; lig
no iunct & tenensq; i supiori
parte ipsu lignu e i te modicu
ut man tuas. & mar impulsu al
li. lapidisq; q; ligat & i feri cur
ciferet. sicq; circufiendo
mole p m. l. iiii. horas. **I**ndu
u respice & p ba digito & rur
su imitte sabulu cu aq. **C**uq;
gyrando & regyando ipsu sabulu
ceperit ebullire. & p lapide dif
fundi cu ligno geili longo &
tenui recollige sep. & i foram
reponere. ne forte auru cu sabulo
egerat & n. molat. **I**d. cu plent
molitu fuerit. eiciat & lauet.
& siccet ut sup. ponatq; sup
libra. **S**i u qe qm defuerit la
uent sordes q; fluunt ex lapide

PLATE 11

Theophilus, *De diversis artibus*,
mid-twelfth century. Vienna,
Österreichische Nationalbiblio-
thek MS 2527, fol. 71r.

In epistola prologus libri primi.

Theophilus qui est kyoeris. de iustis artibus.

Theophilus humilis pbr seruus seruorum
di. indigni nomine ypfessione monachi.
omib; mris def. dia animeq; uagatio
ne utili manu occupatione i delecta
bili nouitatū meditatione declina
re. caleare uolentib; retributionē
celestis pmi. Legim in exodo mun
dane creationis hominē ad imaginē
i similitudinē di. ad trū. i spūatione
diuini spūaculi animatū. tantēq; dig
nitatis excellentia cetis animantib;
progatū ut rationis capax diuine
prudentie cōsilii i genūq; mereret
participiū arbitriūq; libtate donat
soli creatoris sui suscipet uoluntatē
i reuereret impium. Qastu diabolico
misere decept. licet ppt inobedientie
culpā pūlegiū immortalitatis amise
rit tam scientie et intelligentie dig

Berns: Rottenдорff D
C 12. 12C. XLVII.

Medis. Caser. ad electorē.



PLATE 12 (OPPOSITE)

Incipit Theophilus, *De diversis artibus*, mid-twelfth century. Vienna, Österreichische Nationalbibliothek Cod. 2527, fol. 1r.

PLATE 13

Roger of Helmarshausen, portable altar of the cathedral of Paderborn, made for Bishop Henry II of Werl, ca. 1105, view from the side, with apostles (including Bartholomew). Erzbischöfliches Diözesanmuseum, Paderborn.





PLATE 14 (OPPOSITE)

Roger of Helmarshausen, portable altar of the cathedral of Paderborn, made for Bishop Henry II of Werl, ca. 1105, top panel, with Bishop Meinwerk (top) and Henry (bottom) and medallions of the four Evangelist symbols. Erzbischöfliches Diözesanmuseum, Paderborn.

PLATE 15

Roger of Helmarshausen, portable altar of the cathedral of Paderborn, made for Bishop Henry II of Werl, ca. 1105, underpanel with Saint Liborius. Erzbischöfliches Diözesanmuseum, Paderborn.

PLATE 16

Roger of Helmarshausen, portable altar of the cathedral of Paderborn, made for Bishop Henry II of Werl, ca. 1105, front detail: Christ, Kilian, and Liborius. Erzbischöfliches Diözesanmuseum, Paderborn.





PLATE 17

Roger of Helmarshausen, altar of Abdinghof, ca. 1115–20. Erzbischöfliches Diözesanmuseum, Paderborn.



PLATE 18

Roger of Helmarshausen, altar of Abdinghof, ca. 1115–20, side panel: scenes from the life of Saint Blaise of Sebaste. Erzbischöfliches Diözesanmuseum, Paderborn.



PLATE 19

Roger of Helmarshausen, altar of Abdinghof, ca. 1115–20, side panel: two scenes from the life of Saint Felix of Aquileia. Erzbischöfliches Diözesanmuseum, Paderborn.



PLATE 20

Roger of Helmarshausen, altar of Abdinghof, ca. 1115-20, short side:
Saint Peter performing a baptism; side panel: two scenes from the
life of Saint Felix of Aquileia. Erzbischöfliches Diözesanmuseum,
Paderborn.



PLATE 21

Roger of Helmarshausen, altar of Abdinghof, ca. 1115–20, short side: martyrdom of a saint (possibly Paul). Erzbischöfliches Diözesanmuseum, Paderborn.



PLATE 22

Roger of Helmarshausen, Enger cross, ca. 1105, reverse side, with niello panels. Kunstgewerbemuseum, Berlin.



PLATES 23–26 (CLOCKWISE FROM TOP LEFT)

Roger of Helmarshausen, Enger cross, ca. 1105, reverse side, with niello panels, details. Kunstgewerbemuseum, Berlin.



PLATE 27

Gisela brooch. Kunstgewerbemuseum, Berlin.

PLATE 28

Roger of Helmarshausen, Modoaldus cross, ca. 1107.
Museum Schnütgen, Cologne.





PLATE 29

Roger of Helmarshausen, crucifix, ca. 1110–20. Museum Angewandte Kunst, Frankfurt.

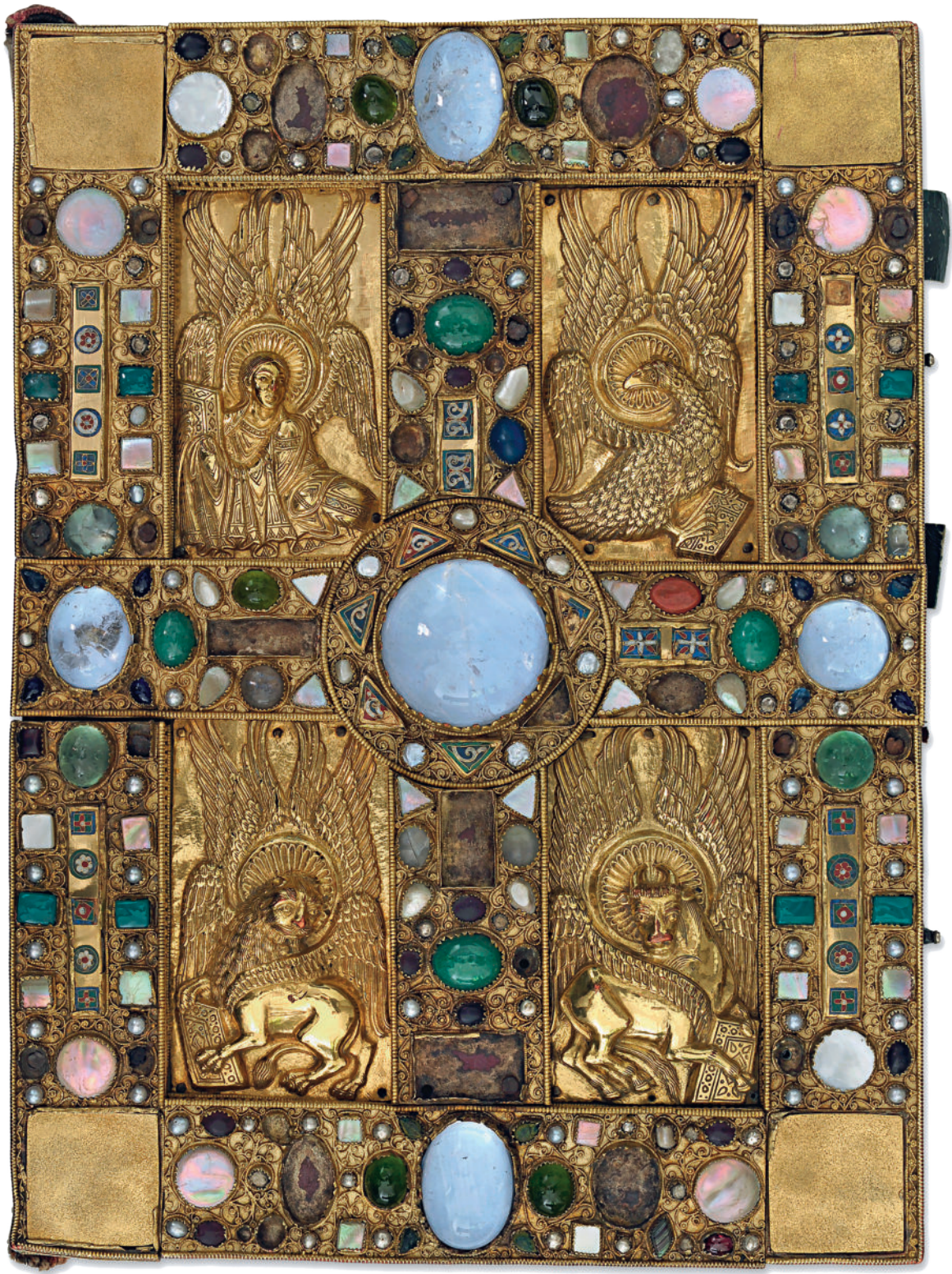


PLATE 30

Roger of Helmarshausen, book cover, Cod. 139, ca. 1110–20. Cathedral of Trier, Domschatz.

4.

THE DISPLAY OF SKILL

*When you have read through these
things several times and commended
them to a retentive memory, you
will recompense me for the labour of
instruction if every time you make
good use of my work you pray to
Almighty God to have mercy on me.*

THEOPHILUS,
On Diverse Arts, prologue I

Surprisingly, for a treatise on art making, none of the manuscripts of *On Diverse Arts* contain formal illustrations. There are no diagrams explaining procedures or drawings exemplifying techniques or objects. Yet tucked into one of the quires of the Vienna manuscript of *On Diverse Arts*, set in the middle of the third book, in the midst of the chapter on milling gold, is a small colored drawing that is worthy of attention. Found on the edge of a stub between folios 70v and 71r, the protrusion of an inserted folio, is a narrow strip of parchment upon which is drawn a set of four and a half acanthus leaves, running from the lower to the upper edge (plate 11).¹ The pattern is formed of two alternating sets of leaves: one bound by a band at its base, with four leaves curving downward, and the other bound with a circle, with four leaves turning upward.² The series repeats twice; a portion of a fifth unit is cropped at the top of the page. The designs are drawn in black ink, with one, toward the top, filled in with red ink.

The drawing appears to be roughly contemporary with the manuscript, and although its outer edge has been cropped, careful examination suggests it was drawn onto the stub after the manuscript was bound. At the very top edge of the page, the second line from the left stops just before the edge. The leaf pattern in the corner fills the space completely, with the inside line of the leaf fading out just before the

page ends. Meanwhile, there is more variation of line in the lower leaves than the upper, and the interior hatching is more roughly drawn in the upper leaves, with straighter, rather than curved, lines. The uppermost leaf is also smaller than the others, and it is crowded into its frame, rather than extending beyond it as do the others. The fullness of the lower leaves and the cramped composition of the upper suggest that the lower leaves were done first. It may also indicate that the upper leaf was more awkward to draw, and thus that the stub was already bound into the book, rather than drawn first and later cut down.

The relationship between the drawing and the holes punched for the binding is perhaps the most telling. In the second full acanthus leaf from the top, the line bends to go around the sewn binding, creating a bump in an otherwise concave line. The other binding hole may also hold a key. Placed just above the first half-leaf, this may have been the starting point for the entire drawing, for two vines perfectly frame the hole. The ink is thickest just to the right of the hole, before it curves down—forming an arc below the hole—and then up, forming a matching arc above. Indeed, throughout the drawing the variation from a thick to a thin line moves primarily vertically, in pairs of lines. It may seem that such a drawing would be awkward to execute on a stub in a bound book, but the vertical orientation of the line variations makes it possible to imagine a right-handed draftsman moving his hand up the page. Perhaps he began the pattern as a doodle, drawing just a couple of lines curving around the binding hole, and expanded it upward from there. Indeed, the half-leaf just below the binding hole is the most simply drawn of the group. Wormholes or other protrusions in manuscripts could be the starting

points for such marginalia, and this may be the case here. A right-handed writer might also have found it easier to draw in an upward direction, rather than across the bottom of the page.

While it is difficult to know how the drawing was made, and any analysis of it must remain speculative, an examination of how the drawing seems to have grown, from wormhole doodle to simple leaf to complex pattern, is a reminder that drawing happens in time, changing and developing as it goes. The forms on the page are the trace of that process, and thus this modest drawing compels viewers to pay attention to the development of formal structures and patterns. The manuscript in which this drawing is found seems to have been copied or owned by someone who was interested in just these kinds of designs. That interest in the development of form will be the focus of this chapter.

Dating to the early or mid-twelfth century, the Vienna manuscript is slightly older than the manuscript at Wolfenbüttel and can be localized to the region around Paderborn. It is bound alone and complete, containing all three books and all three prologues, and it is written entirely by one hand. The current binding of the volume dates to the seventeenth century and was probably done by its owner at the time, the humanist, doctor, and book collector Bernhard Rottendorff (d. 1671), whose name appears on the opening folio with the date 1647. The first and last folios (fols. 1r and 117v) of the volume, however, are especially dirty, and there is substantial damage to the inner corner of folio 1r, suggesting the text, at least for a time, had no outer binding (plate 12).

The manuscript is the only medieval copy of the treatise to be bound alone. Even if it is an anomaly, its existence suggests that—at least in this context of mid-twelfth century northern

Germany—the treatise was valued in its own right. Unlike the grand Wolfenbüttel manuscript, with its fine parchment and broad margins, this manuscript is quite small and scrubby, and it is written in one column rather than two. The quality of the parchment also varies, even within the same quire: some bifolia are very thin, and others are very thick. Moreover, in the Vienna manuscript three bifolia and one single folio are written on reused parchment taken from a ninth-century liturgical manuscript.³ These folia are distributed throughout the volume in the first, ninth, and eleventh quires. They have been well-scraped in preparation for the new text, as the liturgical text, which is unidentifiable, is only visible at the edge of one page. The quires, too, are fairly regular, and thus while the parchment of the Vienna manuscript seems to indicate it was not an expensive manuscript, it was nevertheless very carefully assembled. The use of liturgical parchment, meanwhile, indicates that it was copied in a religious setting.

The uncovered manuscript was probably kept in a workshop and not a library, as was the case with the large, elegant Wolfenbüttel manuscript. The first few folios are darkened with dirt and oil, suggesting the book was often consulted. A small, mysterious stain of green pigment on folios 32r and 57r may also be evidence of workshop use, for this pigment does not appear elsewhere in the volume.⁴ In addition to the drawing on the stub, there is a sketch of an eagle on folio 8r and a faint drawing of a figure in a frame on folio 52v. A small notation in the left margin of folio 18v may be precious evidence of one reader's interest in the technical aspects of the treatise. A gloved hand points to a line in chapter seventeen of book I, "The Panels of Altars and Doors and Casein Glue," which describes bathing cheese in

cold water in preparation for making glue: "Soft cheese is cut up into small pieces and washed in warm water with a pestle and mortar until the water, which you have poured on several times, comes out unclouded. Then this cheese is thinned out by hand and placed in cold water until it becomes hard. . . . It is then replaced in the mortar and carefully pounded with the pestle, and water mixed with quicklime is added until it becomes as thick as lees. With this glue the panels are fastened together."⁵ The step highlighted is indeed a crucial element of the process, for without the cold bath the cheese will not harden. This notation is unusual. No such marks are to be found in the Wolfenbüttel manuscript—its margins are broad and clean—and even in the early thirteenth-century Harley manuscript, the manicules are all inserted in the margins of the prologues.⁶ That instructions for making glue should be singled out as worthy of particular attention may support the hypothesis of a workshop context. On the whole, then, with its reused liturgical sheets, pamphlet binding, dirt, and markings, it seems likely the manuscript was made not only for a workshop, but specifically for one in a monastic setting.

The text of *On Diverse Arts* found in the Vienna manuscript derives from the same parent as the text of the Wolfenbüttel manuscript, with slight variations.⁷ Surviving copies derived from the Vienna manuscript, however, date no earlier than the seventeenth century, when it came into Rottendorff's hands. It is possible the manuscript did not circulate until then. Meanwhile, the textual variations follow a pattern and may be evidence of one twelfth-century reader's particular interests. In the third book, for example, concerning the fabrication of the smaller chalice, both the Vienna and Wolfenbüttel manuscripts

insert a clarifying *illud*, but the Wolfenbüttel manuscript alone changes the word for rub, *fricabis*, to the more generic *fabricabis*.⁸ Similarly, in the chapter on the repoussé censer, in a description of the iron molds in which to place the silver, the scribe of the Wolfenbüttel manuscript replaces the odd, but precise, construction *in infusoriis ferreis* with *in fusoris ferreis*.⁹

Perhaps the most notable variation comes at the close of the second prologue, where Theophilus describes his own process of learning the art of glassmaking. In the Wolfenbüttel manuscript and most other manuscripts, this passage is written *quod artificium, sicut visu et auditu didici; studio tuo indagare curavi*: “This art, as I have learned from what I have seen and heard, I have endeavored to unravel for your use.” The Vienna manuscript, however, does not say *visu* (seeing), but *usu* (using or experiencing): “This art, as I have learned from what I have *experienced* and heard, I have endeavored to unravel for your use.”¹⁰ These may be scribal errors, simply a case of a missing *i* and a *v* that reads as a *u*, but they may well also be evidence of how the book was read. Melanie Holcomb has noted the similar use of “as I have seen” in the sketchbooks of Villard d’Honnecourt to invoke authority, a phrase used also by eyewitnesses in the legal context of the period.¹¹ Variations are small and include scribal errors on both sides, but if we look for patterns within the variations, they support Dodwell’s conclusion that the Vienna manuscript seems to “show more feeling for the meaning of the text, even in its errors, and was probably copied by someone with a closer understanding of the subject than the scribe of G [the Wolfenbüttel manuscript].”¹² The Vienna manuscript does seem to be more technical and specific than the Wolfenbüttel manuscript. Intentional or not, the

variation of *visu/usu* changes the meaning for the reader—from a statement of witness, suggesting the author has seen the techniques performed, to a claim of expertise, suggesting the author has actually used the techniques.

The Vienna manuscript also contains a unique arrangement of prologues. Instead of preceding each chapter, all three prologues are gathered at the front of the volume, following one upon the other. All are contained in the first quire and written across bifolia, a structure that confirms they were copied together, not simply inserted or rearranged at a later time. Two folia are inserted at the end of the quire, but these do not seem to be later additions. Containing the capitula and incipit of book I, they follow the prologues without a break. It is certain, then, that the prologues were written as a set and placed at the start of the treatise from the beginning. Manuscript recensions and internal evidence suggest that the autograph version of *On Diverse Arts* had the alternating structure of the Wolfenbüttel copy, so the arrangement of prologues in the Vienna manuscript was made either by its scribe or by the scribe of its parent manuscript. Since both the Vienna and Wolfenbüttel manuscripts seem to have been copied from the same parent, the rearranging was probably done by the scribe of the Vienna manuscript.¹³

The rearrangement of prologues is not an oversight or a convenience; it would have required the scribe to read the text carefully beforehand to know where the breaks are. It is difficult to speculate on why the prologues were rearranged, but it is clear that doing so allows them to be read continuously and also allows for the instructions to be consulted without interruption. If the book were intended for a workshop, this would be a benefit. Yet it is

important to note that the prologues were not discarded, though in the interest of practicality and brevity they very well could have been. Even if the pamphlet were intended for consultation in a workshop, the prologues were considered important enough to copy and carefully rearrange. Together they would be read as a full statement about art.

The idea that the Vienna manuscript might be an artist's working copy becomes all the more enticing when we consider its most famous attribute, an inscription that seems to identify the author. Inserted between the incipit line and the beginning of the text on the first folio, the inscription functions as both a title and an attribution: "Here begins the prologue of the first book of Theophilus, who is Roger, on the diverse arts" (*Incipit prologus libri primi Theophili qui et Rogerus de diversis artibus*).¹⁴ Whether the inscription can be believed, and who this Roger might be, has been a subject of great debate among scholars. Indeed, since the Wolfenbüttel manuscript was brought to scholarly attention by Gotthold Lessing in the eighteenth century, scholars have often assumed that Theophilus was a practicing artist and labored to identify the man behind the pseudonym. Lessing, for example, dated the treatise to the ninth century and linked Theophilus to the saintly metalworker Tuotilo, of the abbey of St. Gall.¹⁵ But it was the inscription in the Vienna manuscript that made a more plausible attribution possible. In his 1874 edition of the treatise, Albert Ilg linked the "Roger" named in the Vienna manuscript to the goldsmith Roger of Helmarshausen.¹⁶

Roger of Helmarshausen is known through the chance survival of a charter in Paderborn recording an agreement that purportedly took place in 1100 between Thietmar, the abbot of

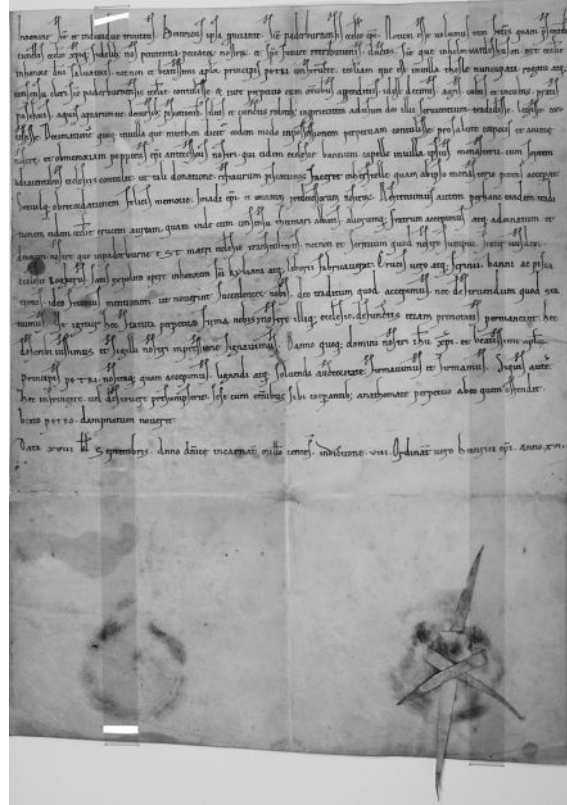


FIG. 14
Paderborn, Erzbistumsarchiv, Generalvikariat, Urkunde 11.

Helmarshausen, and Henry of Werl, the bishop of Paderborn. According to the charter, the abbey of Helmarshausen received the tithes of the village of Muthen, the parish church of Thesle, and "all things that are accessory to it," and gave to the cathedral of Paderborn a golden cross and a *scrinium*, or reliquary box, dedicated to Saints Kilian and Liborius, made by the monk Roger (fig. 14).¹⁷ A portable altar now in the Diözesanmuseum in Paderborn is probably the *scrinium* mentioned: it is inscribed with the name and image of Bishop Henry, it is adorned with images of Saints Kilian and Liborius, and it is usually dated to the first



FIG. 15

Liber vitae of Corvey, mid-twelfth century. Münster/W., Nordrhein-Westfälisches Staatsarchiv, Msc. I Nr. 133, p. 26.

quarter of the century (plate 13). Paleographical and internal evidence, however, suggest that the charter was written in the early thirteenth century. It is either a copy of an earlier document or a forgery.¹⁸ Nevertheless, this document, and the altar thus attributed to Roger of Helmarshausen, have led scholars to tentatively identify a body of works made by Roger or his followers.

Otto von Falke first outlined Roger's oeuvre, attributing to him the portable altar of Abdinghof (also now in Paderborn) and the so-called Enger cross (now in Berlin), linking the latter to the cross mentioned in the Paderborn exchange charter.¹⁹ To this oeuvre Max Creutz added the

book cover in Trier and two crosses, now in Frankfurt and Cologne. Soon a wider body of works were attributed to the larger circle of followers of Roger, including a cross in Fritzlar and an oval reliquary in Xanten.²⁰ Yet it was Eckhard Freise's study of Roger of Helmarshausen, published in 1981, that proposed a hypothetical biography. Freise found the name "Roger" in five early twelfth-century necrologies from monasteries in the Rhine-Meuse region, including the abbey of Abdinghof in Paderborn, St. Pantaleon's in Cologne, St. Mauritz in Minden, and the abbey of Echternach in modern-day Luxembourg. All include the name Roger and record a death date

of or within a few days of February 15.²¹ Such lists of the dead—whether monks, figures related to an abbey, or patrons—record the names of persons whose souls are to be remembered in prayer. One of the best-known necrologies of the time, the so-called *Liber vitae* from the abbey of Corvey, includes a list of brothers from the abbey of St. Godehard in Helmarshausen that includes a certain Roger (fig. 15).²²

From these sources, Freise deduced a hypothetical biography of the artist. Roger, he proposed, was born sometime around 1070 in the German-speaking region of the diocese of Liège, on the Meuse river in present-day Belgium, and spent the early part of his career at the royal abbey of Stavelot, in the hills nearby. There he professed, learned his trade, and eventually worked on the great Stavelot Bible, completed in 1097. Freise surmises that Roger left the royal abbey soon after, in the wake of internal strife that occurred in 1100, and went to the abbey of St. Pantaleon, in Cologne, where he wrote *On Diverse Arts*.²³ This narrative places Roger's arrival at Helmarshausen in 1107, after the relics of Modoaldus were transferred there from Trier. According to Freise, Roger probably made the *scrinium* in the following decade.

Freise's reconstruction of Roger's life and peregrinations has superseded earlier suggestions that Theophilus was Greek, was writing about Byzantine artistic techniques in Europe, was living in the Carolingian period, or was situated in southern Germany.²⁴ Yet the specific chronology of Roger's life remains rough. The arrival of relics at Helmarshausen in 1107 does not necessarily mean that Roger came at the same time. The manuscript within the Trier book cover, for example, is usually dated to around 1100, suggesting that the Helmarshausen workshop was

producing manuscripts before Roger's supposed arrival. It seems far more likely that he was there in the years prior, at least as early as 1105.

Dating and localizing portable objects is fraught with difficulty, but as Eva Hoffman has argued, their very portability may suggest new avenues of meaning.²⁵ The same goes for artists. The portability of these objects, the possibility that artists traveled throughout abbey networks, and the fact that they were remembered (even if inaccurately) for their contributions are significant. Monks probably moved about more often than is commonly thought. Roger's life would have been no exception. His movement from abbey to abbey could also explain discrepancies in the documents that record his life. In the necrology of the abbey of Echternach, in the neighboring archdiocese of Trier, an early insertion in the text names Roger as a monk of Saint Remaclus, probably a reference to the abbey of Stavelot, which Remaclus founded.²⁶ The practice of remembrance was the work of any monastery, and the shift may correspond to the point at which Roger had contact with these abbeys. He is known at Echternach as having come from Stavelot; at Paderborn he is known as a monk from Helmarshausen. Perhaps his travels and the objects he fashioned gave the abbeys reason to remember him.

The Paderborn document is evidence of a memory, forged or not. As Clemens Bayer has shown, the charter was written in the context of a larger dispute over whether the abbey of Helmarshausen would be the seat of the archdeacon of Paderborn. It was but one of a series of documents transcribed at the abbey in the late twelfth and early thirteenth centuries, each purportedly composed at an earlier date, that sought to establish rights and privileges for

Helmarshausen.²⁷ Indeed, the charter confirms the right of tithes as well as control over the pastures, lands “cultivated and uncultivated,” a meadow, and the water of the river, later referred to as “fishing rights.” John Eldevik has recently shown that tithes were one of the major modes through which social structures of aristocratic, monastic, and ecclesiastical relations were made manifest, especially in the eleventh and twelfth centuries; they were traded as signs of friendship and alliance.²⁸ Indeed, this particular document was probably written to confirm ownership of property, perhaps in response to a property dispute. According to the *Vita Meinweri*, written in 1165, Thesle, along with a number of other village churches, had been given over to the protection of Saints Kilian and Liborius at Paderborn in the eleventh century by Duke Bernhard, who was acting as proxy for a group of three sisters; it was a gift they had made for the benefit of their souls.²⁹ The Paderborn document may record a follow-up exchange, when Thesle was given to the abbey of Helmarshausen by Bishop Henry in exchange for the altar.

The Paderborn charter is a singular survival; no written records are extant to confirm or negate its claims. Nevertheless, it is clear that tithes, and documentation of them, were used to write, or re-write, memory and history, a practice that was not uncommon.³⁰ The charter was written in Helmarshausen’s interest, yet as we shall see, the *scrinium* was clearly made for Henry’s own political reasons. The discrepancy casts further doubt on the reliability of the charter’s narrative. Reliable or not, the charter offers clear evidence that by the thirteenth century the value of an art object might be enhanced by including the name of an artist. The exchange involved three parties: the bishop, the abbot, and the artisan, each of

whom made a contribution. Roger’s contribution appears to have been skill, and it does seem to have lent the object greater prestige or worth. The bishop and abbot, meanwhile, participate in the recognition of the quality of the object, confirming that it has sufficient value to seal the exchange. In this way, the memory of the agreement is made visible by a precious object; it is also ratified by its viewers, who act as witnesses willing to vouch for the fairness of the trade.

The script of the Vienna manuscript also refers to the past. It is not known exactly where the manuscript was copied, though it was probably somewhere east of the Rhine; meanwhile, the dating of the manuscript to the mid-twelfth century is usually based on the words and abbreviations it employs.³¹ The script, however, is more akin to a very particular group of manuscripts from Paderborn, transcribed in the early eleventh century around the time of Bishop Meinwerk (r. 1009–36). One of these, a fragment of a lectionary now preserved in the Stuttgart Hauptstaatsarchiv, is typical of this group (fig. 16).³² Here the rounded letters, wide spacing, and high stems are astonishingly close to the script of the Vienna *On Diverse Arts*. In both manuscripts, the lower half of the *g* is relatively large, with a squared-off lower loop that tilts up and to the right. The lower left stem of the *x* falls below the ruled line and then curves back up until it almost meets the bottom of the previous letter. The *a*’s are straight, with a belly that, although narrow, hits the stem near the top of the letter. Stems of *b*’s, *h*’s, and *d*’s are perpendicular, and the bottom stem of the *s* falls just below the ruled line of text.

The connection to eleventh-century Paderborn and the era of Bishop Meinwerk would not have been by chance. Appointed by the emperor Henry II, Meinwerk was credited with greatly

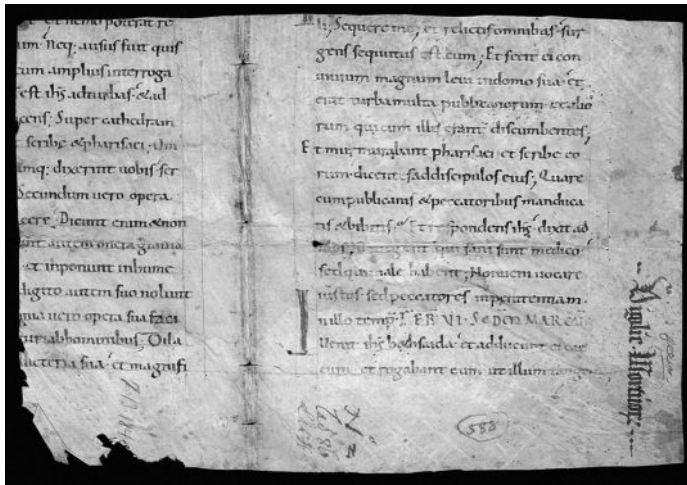


FIG. 16
Stuttgart, Hauptstaatsarchiv, J 522 B Xa 588.

increasing the wealth and stature of the bishopric of Paderborn. In 1017, Henry II gave the then-impooverished abbey of Helmarshausen to the protection of Paderborn and Bishop Meinwerk. The transfer meant that the abbey was shielded from disputes with the local aristocracy. Yet it also increased the power and control of the bishop and the emperor, who, as John Bernhardt has shown, were increasingly justifying such takeovers by claiming that bishops were responsible for their realm and for the spiritual and economic well-being of the monasteries within it.³³ Such ideas, which would cause problems later on, also served to heighten Meinwerk's reputation in certain circles. In 1031, Meinwerk founded the abbey of Abdinghof in Paderborn, and he continued to be revered there in the twelfth century. Perhaps the Vienna manuscript was copied at Abdinghof or at Helmarshausen, two abbeys with a close connection to the eleventh-century bishop.

There was, in fact, a surge of local interest in the bishop's legacy in the twelfth century, when the Vienna manuscript was copied. In 1165 the

Vita Meinwerki, composed in Paderborn, perhaps at the abbey of Abdinghof itself, recorded Meinwerk's many deeds and accomplishments. Importantly, it also glorified him by casting him as a "Courtier Bishop," a persona modelled on the ancient statesman who served the church yet was primarily loyal to the emperor.³⁴ Meinwerk's status, then, was politically loaded. His memory could be invoked as an authenticating force, a sign of loyalty and high virtue and accomplishment. Although the *Vita* probably postdates the Vienna manuscript by a couple of decades, it nevertheless exemplifies how Meinwerk was revered in Paderborn and Helmarshausen throughout the early twelfth century, particularly during and after the investiture controversies that began in 1075 and lasted until the Concordat of Worms in 1122.

The portable altar of Henry of Werl, attributed to Roger of Helmarshausen by virtue of the Paderborn exchange charter, shows Meinwerk as just such a model bishop. Now in the Diözesanmuseum in Paderborn, the altar is made of a wooden core and stands on four cast bronze

claw feet. The two long sides are gilt silver and engraved with images of the apostles, six on each side. One short end shows the Virgin with Saint John and Saint James, in repoussé and niello, while the opposite short end shows the figure of Christ, also in repoussé and set in a jeweled mandorla and flanked by Kilian and Liborius, the patron saints of the cathedral of Paderborn. The top panel consists of a green altar stone framed by a band of filigree and four nielloed plaques. Two long, narrow plaques contain a pattern of heart-shaped vines with crosses inside, and two shorter plaques have medallions of the four Evangelist symbols framing images of Meinwerk and Henry of Werl (plate 14). At the top of the panel, Meinwerk raises the chalice, while below, Henry censes the altar. Henry had been appointed bishop of Paderborn by the emperor Henry IV in 1084, replacing Henry I, count of Assel, who had been installed by Henry IV's opponent in the Great Saxon Revolt, Hermann of Salm, and recognized by Pope Gregory VII. Thus the images on the altar present Meinwerk as the predecessor and model for a bishop whose own election to the office was dubious at best.

Incorporating an image of Meinwerk on an altar is one thing, but why would a scribe have copied *On Diverse Arts* in an archaizing script associated with the bishop's time? One possibility is that it was a forgery, meant to look as though it was produced under Meinwerk. As Patrick Geary and Lawrence Nees have both shown, forged documents authenticated and authorized power structures and histories.³⁵ The archaizing script was essential to this function: it was not only a visual reference to the older history, but it could also have made the manuscript look legitimate.³⁶ It seems possible, then—if not probable—that the Vienna manuscript was written by someone

in or near Paderborn, someone trained to write in that script. Like the image of Meinwerk on the altar, the antiquating script might have been used to legitimize the text, lending it additional prestige or authority.

The inscription in the Vienna manuscript might also be interpreted as a remembrance, intended to establish authority. Dating to around the middle of the century, the manuscript is not the autograph but two generations removed. The scribe would have been copying his manuscript from a copy that had in turn been copied from the autograph. Roger of Helmarshausen, if he did exist, had lived some thirty or forty years prior. If Roger made both the altar of Henry of Werl and the altar for the abbey of Abdinghof, and if he was indeed remembered at the abbey, as the charter seems to imply, the scribe might have known of him. Thus, while the scribe's inscription could have been genuine, it could also be a false attribution, a false memory, or a constructed memory. In any case, it reveals a desire to connect the text with an actual person that may or may not have been based wholly in fact.

It is well understood that author portraits, preface pages, and even colophons in medieval manuscripts could—and often did—establish the authority of the text, asserting the expertise of the author, the veracity of the copy, or the labor of the scribe.³⁷ Authorial personas are found in medical texts as well; the figure of Trotula, author of a compendium on women's medicine, is one such example. If the Roger remembered in the Vienna manuscript is the same Roger remembered in the thirteenth-century charter of exchange, namely Roger of Helmarshausen, and if that Roger was indeed the artist of the portable altar, then the association of the text

with a known goldsmith, again, would have no doubt heightened the volume's authority and prestige. The fact that this scribe thought the author worthy of remembrance is significant, for it places value on the treatise and on knowledge about artistic work. If he furthermore intended to name Roger of Helmarshausen, then he either knew that the author was an artist or wanted to identify the author with an artist. The speculation invited by the inscription is informative. The act of naming Roger tempts readers to imagine the mysterious Roger and his accomplishments, but it remains stubbornly elusive. Roger matters, but like the pseudonym, his authority is created through his absence, as the reader is obliged to resort to preconceived notions of authors and make assumptions about Roger's expertise. Mythical figures like Trotula remind us that this may have been intentional; a mysterious identification could in fact heighten authority. Through memory and imagination, and through the construction of a fabricated memory or identity, Theophilus's role as a knowledgeable expert is strengthened.

The scribe was certainly familiar with technical terms, and the doodles and drawings suggest a reader, perhaps the scribe himself, interested in patterns of ornament and able to draw such patterns with ease. It seems quite possible that the manuscript was intended for a workshop at Paderborn. By copying the treatise, rearranging the prologues so that instructions could be easily accessed, writing in a historical script, and linking the treatise to a known artisan, the scribe created a manuscript that could be a useful and authoritative workshop resource. For this medieval annotator, the theoretical and practical aspects of the treatise were but two sides of the same coin.

Such an awareness of artistic work may have gone hand in hand with an increasing interest in the techniques of artistic production. The drawing on the stub does not obviously record something seen; however, it may be related to a pattern found on the portable altar of Henry of Werl. The underside of the object is a copper gilt panel, engraved with an image of the saint Liborius, who stands within an arch, above which is a large pedimented roof flanked by two domes (plate 15). The center of the panel is quite worn, and the wear extends to the long sides of the altar, obscuring the image. This suggests that the altar was used and that it was carried in a vertical orientation, which echoes the orientation of Liborius, as well as of Meinwerk and Henry on the top panel. The frame of this panel, however, is most relevant to our discussion. It consists of a series of semicircular frames, within which are three-leaf palmettes, with acanthus leaves that alternatively bend open and point out. All are bound with a roundel. The sequence is very close to that of the drawing in the manuscript. It is not a copy, but the combination of alternating leaf patterns, semi-circular frames, and a roundel at the base is found in both.

The pattern seen on the panel of the altar of Henry of Werl has a counterpart in a border in the sacramentary made for abbot Wibald of Stavelot between 1150 and 1158 (fig. 17). Here again, the image—now of a Crucifixion—is surrounded by a thick frame, within which is a series of semicircles, each framing a scallop-shaped palmette. The corners are filled with three-leaf palmettes, surrounded by a roundel. In the sacramentary, the palmettes have eight leaves instead of three, and the border is a broad, outlined space, rather than the double semicircle seen on the portable altar. The corner palmette,

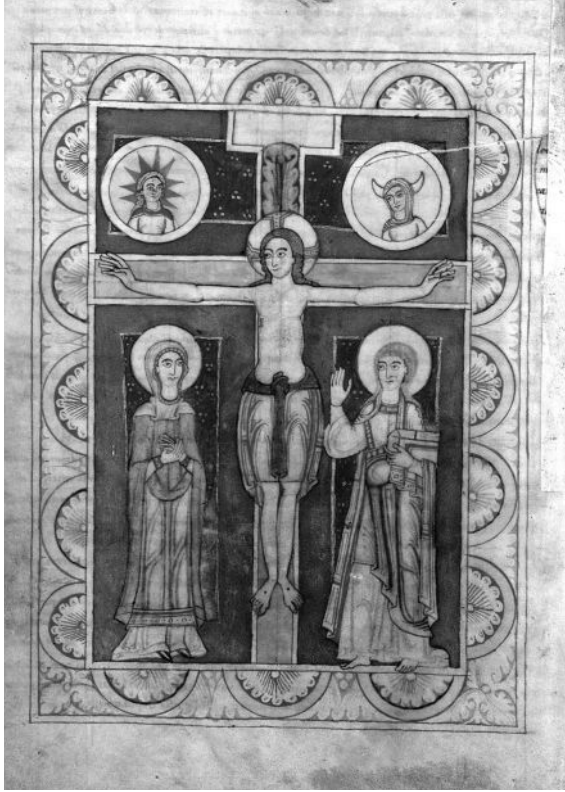


FIG. 17
Sacramentary of Wibald of Stavelot, 1150–58. Brussels, Bibliothèque royale de Belgique, MS 2034–35, fol. 25v.

meanwhile, has leaves that open outward and up, in a horizontal direction, rather than the curved leaves of the altar palmette. Freise has suggested that Roger professed at the abbey of Stavelot; a hand identified as Roger's has also been located in an image of the prophet Jeremiah on folio 161r of the Stavelot Bible (fig. 18).³⁸ More important for our purposes, however, are the curving leaves of the palmettes that form the top of the initial "H" on the Stavelot Jeremiah folio, with the roundel at the center. These seem to constitute a type of ornament, and the very fact that other patterns can be found that are like this one, but

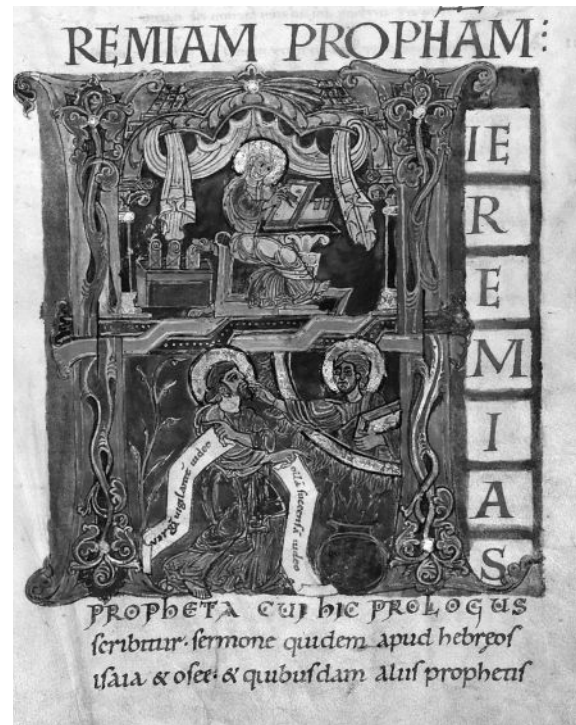


FIG. 18
Stavelot Bible, 1097, detail: Prophet Jeremiah. London, British Library Add. MS 28106, fol. 161r.

not precisely the same, suggests that together they represent an ornamental motif that could be remembered, repeated, and varied. We might see the drawing on the stub, then, as a doodle of a motif similar to one the artist had made or seen before. It records a way of working, a repetition and combination of elements.

The closest link between the pattern in the Vienna manuscript and extant objects might be found in a work produced for Theophanu, the granddaughter of Otto II and abbess at Essen (r. 1039–58). A book cover made for Theophanu is composed of a gilded repoussé plaque, into



FIG. 19 (LEFT)
Book cover of the abbess Theophanu, ca. 1050. Essen Cathedral Treasury.

FIG. 20
Book cover of the abbess Theophanu, ca. 1050, detail. Essen Cathedral Treasury.

which is set an ivory showing the Crucifixion, the Ascension, and the Nativity (fig. 19). The ivory is surrounded by a border of acanthus leaves, again divided into units, within which is a three- or five-leaf palmette. The units are divided by vines that grow out of the bottom of the leaf and rise up around it, forming a parenthesis-like space. The palmettes are all framed with a band like that in the Vienna manuscript, and the leaves vary—some growing straight up, as on the bottom-most left; others curving up and turning in, as in the second from the bottom on the left; and some growing up and then curving out, as in the fourth from the bottom on the left (fig. 20). Another ivory of the crucifixion, dating to ca. 1050 and now at the *Musées Royaux d'Art et d'Histoire* in Brussels, uses the same composition as the Theophanu ivory, but here the border units are

more distinct: the small leaves growing out of the vines at the root of the palmette are smaller, and there is no additional ornamentation or curving leaves at the top of the vine, above the palmette.³⁹ While this change is extremely small, it clarifies the structure of the ornament and the division of the space into units.

The structural resemblance of the drawing in the Vienna manuscript to patterns on the Henry altar, the Stavelot Bible, and the cross of Theophanu suggests that we may think about medieval ornament as a series of configurations, with elements that could be varied and changed. If the drawing in the manuscript was spontaneous, produced from memory, it might mean that workshops and artisans had ways of working, general schemata, upon which they relied regularly and which they could employ at

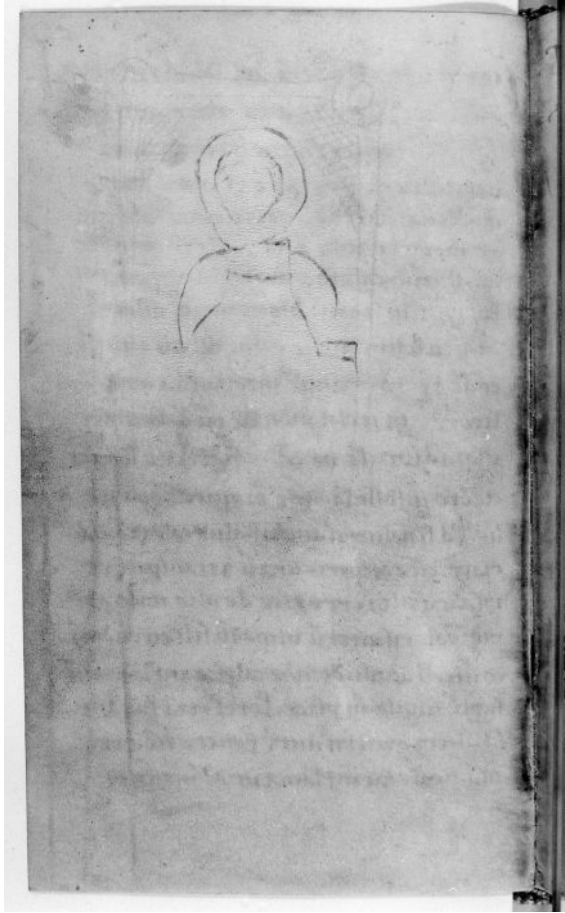


FIG. 21

Theophilus, *De diversis artibus*, mid-twelfth century. Vienna, Österreichische Nationalbibliothek Cod. 2527, fol. 52v.

will and vary according to the object. Drawing and design were dependent on visual memory. Recent research has done much to elucidate the ways in which art served memorial and meditative practices. Building on Mary Carruthers's study of memory in the twelfth century, for example, Adam Cohen has shown how images in manuscripts could function as loci for memory.⁴⁰ More important for our purposes, he has also shown how images could function as

forms upon which more complex ideas could be developed. As Cohen describes it in the example of an illustrated miscellany,

The different elements of the book were derived, in practical terms, from various textual and pictorial models, but conceptually this unique combination resulted from monastic *ductus*, the method of collating and structuring raw elements into a new whole. Whether or not the particulars of my reconstruction of that original thought process are accurate, it seems clear that Clm. 13002 was a book produced through such *ductus*. Once created, the compilation would have functioned not to delimit knowledge within the confines of the book but to stimulate further *meditatio* in the minds of the Prüfening monks using it for that purpose.⁴¹

I do not wish to suggest that the drawing in the Vienna manuscript is a locus for contemplation, or map onto this drawing the structures and functions of memory-building as described by Carruthers. Instead, I want to use the ideas of structure and complexity to think about how visual memory works, to consider patterns like that on the Vienna stub as a combination of “raw elements.” In medieval studies, memory is often discussed as a tool for learning and for establishing social networks and histories. Yet that visual memory—of patterns, structures, and modes of drawing—was also a highly utilized tool. Perhaps stocks of images and patterns and modes of drawing were memorized just like texts.

If drawing and patterns sometimes came from memory rather than pattern books, then ornament might be understood as a new iteration of a remembered arrangement of forms.

With its series of palmettes and enclosing vine, the drawing in the Vienna manuscript is based on a common compositional structure. Richard Wollheim has discussed such similar visual elements as “schemata,” or the building blocks of visual form.⁴² A pattern like that on the stub is based on a schema of palmettes and vines, with alternating roundels and bands. Importantly, schemata define a mode of working. As Wollheim describes it, schemata can be defined by material; by a technique such as punchwork, repoussé, or line; or by representative elements, such as groups of lines, thickness of line, shapes, or spacing.⁴³ The idiosyncrasies of the Vienna manuscript and the emphasis on skill in *On Diverse Arts* underscore the fact that formal elements are, quite simply, traces of making. As Melanie Holcomb has written, medieval drawings “seem to operate in a subjective mode, not a prescriptive one; that is, their subjects have been determined by a series of individual experiences, interests, and reactions rather than a focused, larger objective.”⁴⁴

A drawing found on folio 52v of the Vienna manuscript demonstrates another type of schema (fig. 21).⁴⁵ A faint outline of a semicircular frame, filled with cross-hatching, with two circular roundels in the corner, the drawing is on a blank verso that mirrors the opening chapter headings of the third book of *On Diverse Arts*. Once again, the drawing shares its structure with the underpanel of the altar of Henry of Werl. While the composition of the altar is far more complex, the drawing may have served as the reminder of a compositional type, a structure, into which can be placed more complex patterns: the motif of a haloed figure set into a curved space with anchored corners is a simple pattern that could be borrowed and reconfigured. Indeed, while

the cross-hatching motif does appear in manuscripts, it is a texturing technique seen more often in metalwork. It is particularly common in twelfth-century objects from the Mosan region and objects made at Stavelot during the abbacy of Wibald (r. 1130–58). One can find it on the reliquary of Pope Alexander, made in 1145, or the Stavelot triptych, now at the Morgan Library in New York (fig. 22). The drawing of folio 52v is not a diagram, nor is it an illustration of a technique or a drawing of a figure that might be used as reference. It seems instead to refer broadly to a framing strategy. That the pattern is more likely to be found on metalwork of the region suggests that it was recorded from memory and not from observation of a particular object. It is not a precise pattern to be copied but a building block of memory. The drawings, copied into blank spaces in the earliest surviving manuscript of Theophilus’s text, record the memory of an image. As Mary Carruthers has famously shown, the act of remembering in the Middle Ages was often achieved through visual means.⁴⁶ In her examples, the tools of remembrance are employed primarily to memorize or analyze text, but here the case is slightly different—the tools of remembrance are instead employed to remember, copy, and build visual patterns and formulas. In artistic practice, as in reading or textual analysis, the function of memory plays an important role.

The Vienna manuscript is peculiar in many ways. It is an early recension, yet it rearranges the prologues; it is copied by one hand, but in an archaizing script. It is not a large manuscript, and it incorporates reused liturgical sheets, but great care has been taken to scrape these clean and distribute them evenly throughout the volume. It seems as if it was used in a workshop, and it even contains some drawings. The



FIG. 22
Stavelot triptych, abbey of Stavelot, mid-twelfth century. The Morgan Library and Museum, New York.

manuscript seems to have been copied or owned by someone who was a practicing artist, in a monastic setting. In contrast to the philosophical bent of the Wolfenbüttel manuscript, the Vienna manuscript seems to have been made for someone who cared about how things were made. Given the drawings in the manuscript, which relate to ornamentation associated with metalsmithing; the inscription on the first folio

of the manuscript naming a monk, Roger; the unique variations of text and curiously archaizing script in the Vienna manuscript; and the survival of objects associated with Roger of Helmarshausen, it is likely that Theophilus was Roger. If he was not, and the inscription is either mistaken or refers to someone else, then it still seems likely that Theophilus was someone associated with Roger or his workshop. Whoever he was, the

naming of Roger strengthens the authority of the treatise.

Memory, Technique, and Pattern

Indeed, many of the priorities of the Vienna manuscript are expressed in *On Diverse Arts* itself. Memory, for example, is of utmost importance. At the end of the first prologue, Theophilus suggests that to master the skills and techniques of the treatise, the reader must commit the text to memory. Using the term *relegere* (to read) and *commendaveris* (to entrust), he implores his reader to “read through these things several times and [commend] them to a retentive memory.”⁴⁷ This language suggests that his intent is to provide a book that is to be read and understood, not referred to like a collection of recipes. The orderly progression of the book would no doubt have helped this endeavor.

Yet how much can one actually learn of techniques by reading the treatise? The prologue suggests that Theophilus expects understanding more than action. One might therefore surmise that the emphasis on learning, not doing, means that Theophilus is addressing viewers of art, not artists. I do not believe the two are mutually exclusive. The knowledge of how things are made, of processes, materials, and labor expended, is a benefit to the patron as much as it is a requirement for the artist. If he is to have something made, and made finely, the patron must know something of what the job requires; indeed, it was often his reputation that was on the line.⁴⁸ The intended reader might just as easily be a patron as an artist, and probably someone connected to a monastic community where art was being produced. He also, it seems, was expected

to have an understanding of all the arts, not just one.

The careful organization of the treatise would certainly help the reader—be he artist, patron or something else—remember the text. Descriptions, too, are vivid. For example, Theophilus’s description of the work furnace seems to invite the reader to picture the building process step by step: “On the left side of the sitter [whose table he has just described], near the window by the wall, you fix in the ground a board three feet long, two wide and almost two fingers thick. When it is firmly in position, make a hole in the middle, one finger across and four fingers off the ground. Also, have attached to it in front, and fixed with wooden pegs, a narrow piece of wood, four fingers wide and the same length as the larger board. In front of the latter, you should fix another board.”⁴⁹ It is not surprising that in this twelfth-century monastic world, Theophilus would be expecting readers to employ visual memory as they read through the treatise. It seems possible, too, that artists would have been expected to use visual memory in their practice, and perhaps even patrons and other educated readers would have been expected to have a highly trained visual memory. Such a visual memory might well have been employed for remembering images and procedures as well as ideas.

he previous chapter examined skill as a pious moralized labor, and suggested that the artist was expected to perform his work with care, caution and learning. Process could be a visual motif. If we also consider the possibility that readers would have been trying to commit an understanding of techniques (if not mastery of techniques) to memory, then we might begin to see how skill and process can drive formal values. The famous list of gifts in the third prologue is

perhaps the clearest on this point. Theophilus writes of the second gift—understanding—which he defines as the capacity for skill: “Through the spirit of understanding, you have received the capacity for skill—the order, variety and measure (*quo ordine qua varietate, qua mensura*) with which to pursue your varied work (*diverso operi tuo*).”⁵⁰ Skill does not exist outside of action; rather, it is the ability to work according to principles of order, variety, and measure, which are then made evident in the “varied work.”

Theophilus continues along a similar line as he describes the gift of godliness: “Through the spirit of godliness, you regulate with pious care the nature, the purpose, the time, measure and method of the work (*quid, cui, quando, quantum, vel qualiter operis*) and the amount of the reward, lest the vice of avarice or cupidity steal in.”⁵¹ In the previous chapter, we saw that work is driven by procedure and proper action. Here, we see that good skill is the ability to work within constraints—to make the appropriate object, in the expected time, with fine technique. Theophilus warns against avarice and cupidity, and one imagines that cost is implied here too. Thus the text defines a sense of decorum where order, variety, and measure are expected as well as appropriateness, suitability, and regulated execution.

These ideals come through in the instructions as well. For example, book III contains a series of sixteen chapters dedicated to the construction of the large chalice, describing its nielloed ribs, gilded handles, and elaborate foot.⁵² Toward the end of the series is a chapter titled “Embellishing the Bowl of the Chalice” that guides the reader through the last steps of the process, the decoration of the ribs without niello ornament. He writes, “You have already nielloed half of its ribs,

and now you smoothly file and scrape the ones that you left free of niello between them. Draw on them whatever designs you wish, but with some variance from the niello work, and engrave delicately with the fine graver.”⁵³ Theophilus’s instructions do not describe specific forms. Instead, his concern is for the process and the generic, technical program of the piece. In other words, he cares that the ribs on the bowl of the chalice alternate between niello and engraving, and that the designs placed on them are similarly varied. Patterns and ornamental programs are not specified, nor are particular images. The reader is given basic structures upon which to build a decorative program of technical and formal variety, and thus the object is made within an expectation of a certain “order and measure.” The notion that rib decoration might alternate, then, might be a practice of variety. Theophilus is sharply attentive to shape and proportion, as is particularly evident in his description of dividing up silver for the chalice: “If, when [the silver] is purified, you want to make a chalice, divide the silver into two equal parts and keep one half for making the foot and the paten. From the other piece you make the bowl to which you add some of the paten. . . . Later, by scraping or filing, you will recover this amount for its own portion.”⁵⁴ On the whole, then, *On Diverse Arts* reveals priorities of form and structure, of order and variety, of purpose and procedure.

The importance of variety, order, and form emerges in other chapters as well. One of the best examples is the description of the repoussé censer. The lengthy chapter begins with the ratio of the dimensions of the censer (one and one half times taller than the width) and then goes on to describe the number of towers (three rows of towers with one, four, and eight towers

on each level), the number of columns or windows to be placed on each, and a level of arches below, between which are placed the heads of lions or men to hold the chains. The structure of the object is clear; moreover, the proportions of numbers make it easy to remember. The details of decoration are more open: the level with eight towers should contain “four round [towers] corresponding to the square ones above, in which small flowers, birds, animals or little windows are made, and between them, four square ones, which are wider and in which the half figures of angels are represented, as if sitting in them with their wings.”⁵⁵ Structure is given, but the choice of how to decorate it—whether to fill the structure with flowers, birds, animals or windows, for example—is left to the discretion of the artist or demands of the patron.

In both of these chapters, Theophilus is sensitive to issues of cost and he gives many options appropriate to different budgets. We have seen this in the chapters on gold, where he describes four different varieties, from the fake gold of the Spaniards to the reddish gold of Arabia, to the sand gold of the Rhine, and most precious and best of all, the gold of Havilah.⁵⁶ Censers, too, can be made in a range of material: “If you want to make censers of repoussé in gold, silver, or copper, you first refine the metal in the above way, and according to the size that you want the upper part of the censer to be, you cast two, three or four marks in the iron moulds.”⁵⁷ Similarly, the section on the silver chalice presents options regarding size: “If you want to make a large silver chalice of four, six or ten marks, first prove and refine all your silver.”⁵⁸

In this way, *On Diverse Arts* helps the reader to remember details and forms. The artist would need only remember the censer’s four levels

and their divisions, for example, and then place towers and figures within the larger structure. Theophilus thus gives the towers and figures of the censer equal weight in his description; both are discussed primarily in relation to the greater whole. Previous chapters demonstrated how *On Diverse Arts* teaches readers to look for evidence of good process; that it also teaches them to look for structure. These priorities are not entirely abstract ideals: contemporary objects show a concern for structure and program, for technique and ornament, and for cost and consistency. In these objects, too, good form is the result of good process. Decorum drives the aesthetic of the work.

The Altar of Henry of Werl: Variety of Workmanship

The altar of Henry of Werl has long been noted as an example of the different types of techniques that Theophilus describes in his treatise.⁵⁹ Engraving brings the apostles on the sides to vivid life, while punchwork fills the background; the images of Christ, Kilian, and Liborius on the front panel are done in repoussé. Niello is found on the top panel, on the back, and in inscriptions throughout; a band of filigree borders the altar stone. Rather than seeing such techniques as an inventory of book chapters, however, we might look to see how Theophilus’s instructions define parameters and general patterns. The process of engraving, gilding, and punching, for example, is described by Theophilus as follows: “Sheets of copper . . . are engraved with delicate designs of figures, flowers, or animals, and the designs are so disposed that the grounds are small. They are then cleaned with fine sand and burnished with the



FIG. 23
Roger of Helmarshausen, portable altar of the cathedral of Paderborn, made for Bishop Henry II of Werl, ca. 1105, back detail: Virgin and saints. Erzbischöfliches Diözesanmuseum, Paderborn.

appropriate tools, and then gilded and again burnished and coloured. After this they are punched with a punch which is made in this way.”⁶⁰ The figures on the front panel of the Henry altar seem to follow Theophilus’s directives for repoussé, as well, for here the heads of the figures jut out from their bodies—they are “in highest relief”—and there is sharp undercutting below their chins (plate 16).⁶¹ The bodies of Kilian and Liborius are particularly shallow in contrast to the heads, with very little variation in depth. The fabric of their robes is evenly embossed, and their hands do not protrude beyond their heads, making them seem pressed

against the background. Christ, in the center, has been given more animation; he has an oversized head, and his knees come forward almost as far as the head. Openwork, too, is used on the back end of the altar, in the nielloed figures of the Virgin with John and James (fig. 23). As Theophilus describes it,

Thin out for yourself plates of the same copper as above, but thicker. Having drawn on them whatever designs you want, engrave as above. Then have some tools—narrow or wide according to the size of

the grounds—which are edged and sharp at one end, and thick at the other; they are called cold chisels. Placing the plate over the anvil, you cut through all the grounds with the above-mentioned tools, striking them through with a hammer. . . . In the same way one makes silver plates and silver book covers, decorated with figures, flowers, small animals and birds.⁶²

Henry's altar is indeed an excellent example of Theophilus's techniques, but there is far more to be explored. An examination of the object in light of *On Diverse Arts* can reveal what such technical variety might have meant for medieval viewers. An altar commissioned by the countess Gertrude shortly after the death of her husband, Count Liudolf of Braunschweig in 1038, may have been a precedent for the altar of Henry of Werl (fig. 24). Now in the Cleveland Museum of Art, the altar was given by Gertrude to the Collegiate Church of Dankwarderode in Braunschweig and later went to the cathedral of St. Blaise, where it became a part of the Guelph treasure. Gertrude's altar consists of an oak core, covered in gold. The top panel comprises a porphyry altar stone bordered by a band of inscription and two bands of filigree. The inscription runs in one line, its letters turning around the stone. It names Gertrude and dedicates the gift to Christ: "Gertrude offers to Christ, to live joyfully in him, this stone that glistens with gems and gold."⁶³ Underneath is a silver panel with no ornamentation and a door to an inner compartment where the relics are kept. On one long side is an image of Christ flanked by three repoussé apostles on each side. The opposite long side shows the Virgin, also flanked by three repoussé apostles on each side. On one short side is Saint Michael with the dragon, with



FIG. 24

Altar of Countess Gertrude, possibly Saxony, ca. 1045, side view, with Virgin and Apostles. The Cleveland Museum of Art, Gift of the John Huntington Art and Polytechnic Trust 1931.462.

two more angels on each side; on the opposite short side is the adoration of the True Cross, with Emperor Constantine and Helena, his mother, on either side of the cross and King Sigismund of Burgundy and Empress Adelaide flanking them.

The portable altar of Henry of Werl is similar in shape to Gertrude's altar, with beveled edges and an overhanging lid and baseboard. It, too, is a box, and it is perhaps for this reason that the charter refers to a *scrinium* and not an altar.⁶⁴ According to Michael Budde, the altar of Gertrude is one of the earliest of this type to survive, and though the type became popular in the twelfth century, the form was still relatively rare in the early part of the century, when the altar of Henry was made.⁶⁵ Yet there are significant differences. The Paderborn altar is seven and a half centimeters longer and five and a half centimeters taller than Gertrude's imperial altar.⁶⁶ It is also oriented differently. Gertrude's altar is organized horizontally, so that the long side with the image of Christ and ornate enameled arcades

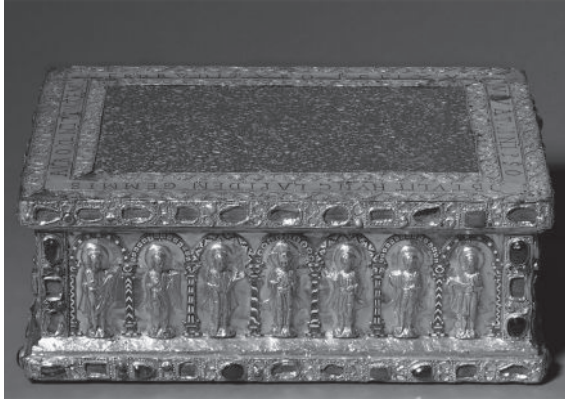


FIG. 25

Altar of Countess Gertrude, possibly Saxony, ca. 1045, long side, with Christ. The Cleveland Museum of Art, Gift of the John Huntington Art and Polytechnic Trust 1931.462.

would have been at the front, and the inscription beginning with Gertrude's name would be upright on the top panel (fig. 25). This makes Gertrude's name and the figure of Christ visible at the same time, a link repeated in the inscription itself, where Gertrude's name is placed alongside Christ's. On Henry's altar, Christ is in the mandorla on one short end, and the images of Bishop Meinwerk and Henry himself on the top panel are aligned above him, on a vertical axis.

The images on the sides of the Henry altar are arranged slightly differently, too. On the Gertrude altar, the long sides contain images of the apostles under arcades. On the front six apostles flank Christ, while on the back the remaining six flank the Virgin. On the Henry altar, five apostles remain under arcades on the long sides, but Christ has been moved to the short end, where he is shown flanked by Saints Kilian and Liborius, whose relics were kept in the altar. The Virgin, similarly, appears not on the long side but on the opposite short end, accompanied by the last

two apostles, John and Jacob. It is as if Kilian and Liborius, the former a missionary to Germany, the latter a bishop of Le Mans, have joined the apostles; flanking Christ, they provide further legitimacy for the episcopal lineage traced on top that leads ultimately to Henry of Werl.

It is highly likely that Henry would have looked to Gertrude's altar and the family lineage it depicts as a model. Henry was related to Count Liudolf, Gertrude's husband, via his grandfather, Hermann II. Hermann II was the half-brother of the empress Gisela, and Gisela, by her marriage to Bruno of Brunswick, was the mother of Liudolf. In 1016 or 1017 Gisela married Conrad II, future Holy Roman Emperor; her grandson by Conrad was the emperor Henry IV, who appointed Henry of Werl bishop of Paderborn and served as his patron until 1105. If we go back a generation further we come to Gerberga of Burgundy, the great-grandmother of Henry of Werl and the grandmother of Liudolf of Braunschweig, the former by her marriage to Hermann I of Werl and the latter by her marriage to Hermann I of Swabia. Gerberga is important here, for it is her lineage that is represented on the short end of Gertrude's altar: Empress Adelaide, who appears next to Helena on the altar and whose relics are kept within the altar, was Gerberga's paternal aunt. Henry of Werl, then, surely knew of Gertrude's altar and may have known Gertrude herself, for she did not die until 1077. But rather than present his own royal heritage on the altar, Henry shrewdly created an episcopal heritage for himself. Given the events of 1105, this is highly understandable.

Henry's altar seeks to express episcopal legitimacy, and indeed it is certain that Henry's position as bishop in the early 1100s was tenuous. In 1105, the future Emperor Henry V, having

aligned himself with the papacy to challenge the power of his father, organized a synod at Nordhausen in hopes of bringing “schismatic” bishops back into line with the papacy. Henry of Werl, Udo of Hildesheim, and Frederick of Halberstadt were a few of those bishops. Mass was celebrated at Goslar following the synod, and Henry of Werl was present, dressed in liturgical vestments and celebrating his reconciliation with the Church.⁶⁷ His creation of an altar with imagery intended to create an episcopal heritage and religious authority may well have been made in celebration of this event.

Henry emerges as a figure who was willing to align himself in whatever way was politically expedient. The inclusion of Meinwerk on his portable altar was probably a similar gamble. Henry’s grandfather, Hermann II, had fought with the Billungs against Meinwerk for control of the abbey of Helmarshausen in 1017—and lost. For Henry to align himself with Meinwerk, and commission the altar from the abbey whose independence his family had fought, is a significant turnaround. Nevertheless, Henry’s image on the altar appears closer to Christ than that of Meinwerk. The altar thus seems intended to display Henry’s capitulation alongside his ultimate victory, making him more prominent than Meinwerk in the end. Meanwhile, Henry’s act of commissioning works from Helmarshausen, and the remembrance of the exchange in a charter, may also have been a political move.

Henry’s altar may well have been modeled after Gertrude’s, but it also creates it anew, rotating the axis to create a vertical pairing of Meinwerk and Henry on the top panel while keeping the pairing of Christ and patron. It is impossible to know if viewers, especially those outside of Henry’s circle, would have made the

connection between the two altars. Yet it was not uncommon for objects to be commissioned with the intent that they be associated with other objects, particularly in political contexts like this one. Eliza Garrison has shown, for example, how the Ottonian emperors Otto III and Henry II incorporated into new objects motifs or even materials from earlier objects as a way to gain status and assert their power.⁶⁸ Building on the example of Gertrude and incorporating the image of Meinwerk, Henry probably commissioned the altar sometime after the reconciliation of 1105, when his status was particularly unstable, thereby solidifying his power and legitimacy. Even if not all viewers could make the connection, Henry’s commission of an altar that was an updated, reconfigured version of an earlier altar served his episcopal agenda well.

The ornamental programs of the two altars relate as well. All four sides of Gertrude’s altar are framed with vertical bands. Though many gems are missing, the surviving gems and empty settings suggest that each band on Gertrude’s altar consisted of an oblong gem in the center and two smaller round gems flanking it. Between each were two small pearls, and nail studs decorated each corner. Bands of gems also line the top and bottom of the long sides of Gertrude’s altar, each again separated by a pair of pearls, and with an alternating pattern of oblong and round stones.

Henry’s altar repeats much of this structure, with slight variations. Gemmed bands appear only on the two short ends, for a total of four rather than eight. They do, however, have the same structure: each band has three gems, and the side with Christ has a large, oblong carbuncle in the center. Perhaps other oblong gems could not be found, for the gems on the Virgin panel

are comparatively small. As if to make up for the lack of pearls, the spacing between the gems is wider, and the four nail holes are more prominent, framing the gems in a similar way to the pearls but at a far lower cost.

Henry's altar also reduces the number of bands and gems. Where the Gertrude altar has bands of gems framing each side panel, Henry's altar instead has niello inscriptions identifying the figures. Only one band of gems runs around the top of the piece. Even here, the number of gems has been greatly reduced; they are spaced generously and number only fourteen, compared to the thirty-four gems around the upper band of the Gertrude altar. The rest of the band is filled not with more gems or pearls, but with silver gilt quatrefoil flowers, perhaps dating to slightly later.⁶⁹ Henry's altar has just one band of filigree, while Gertrude's has two, and the precious red porphyry in the older object finds a less expensive counterpart in the green marble of the younger. These adjustments suggest an effort to greatly reduce the cost of Henry's altar while keeping the structures and ornamental patterns of Gertrude's intact. What is striking, however, is that some of the substitutions would surely have required more labor.

More importantly, the comparison alerts us to the choices that took place in the making of the object. The similarity of gem patterns, for example, might suggest that a certain decorum was expected of ornament. Theophilus's instructions bear this out. While he does not discuss portable altars, in his chapter on the ornamentation of the chalice he describes how the gems and pearls are to be placed, and it is remarkably consistent with ornamentation on contemporary altars.

When this has been done [the soldering of the handles] take a thin strip of gold and

attach it to the upper rim of the bowl, taking it round from one handle to the other. The width of this strip is the same size of the stones you want to set. Placing these in their order, you so arrange them that, first, there is a stone with four pearls set at its corners, then an enamel, next to it a stone with pearls, and then an enamel again. You arrange them like this, with the stones always standing next to the handles, and you prepare the settings and grounds of the stones, and the settings for the enamels, and solder them as above.⁷⁰

Theophilus's text here reveals some clear guidelines: the band of gems should be placed on the upper rim of the chalice, and stones and enamels should alternate, with pearls set in the corners, as in the Gertrude altar. Here and on the whole, *On Diverse Arts* provides some concrete structural guidelines for forms, but leaves space for variation.

Henry's altar reconfigures the patterns and schemata set forth in Gertrude's altar. The way that it operates within its tradition, and within its modes of decorum, is a significant part of its value. The Henry of Werl altar shows how structure can organize ornament and technique and how ornament and technique, in turn, can organize a visual program. Unlike the Gertrude altar, which has repoussé on all sides—the only variation in technique being the use of enamel on two panels—the altar of Henry of Werl changes technique according to the type of image. Christ and the dedicatory saints are in repoussé; the Virgin and primary apostles are in openwork; the others are engraved; and local bishops (Meinwerk had not yet achieved sainthood) are in niello. The program suggests a hierarchy of techniques, with repoussé used for the most sacred

of images and niello for the lesser. If Theophilus's readers "commended . . . to a retentive memory" the variety of techniques in the treatise, as they were encouraged to do, they might well have been equipped to make these kind of interpolations and to read such display of skill as a formal value. Ornamental programs could not only be determined by economic considerations, but they could also be interpreted as such. Henry was not given his bishopric but had to purchase it, and in order to do so he sold some of his inheritance to his brother. The charter, meanwhile, implies that the expense for Henry's altar was carried, at least in part, by the abbey. Given this situation, it becomes all the more likely that the altar made for Henry compensates for material wealth with a display of technical virtuosity and artistic skill. Economic realities and comparisons to other works suggest that this was not unusual.

The Abdinghof Altar: New Imagery

An altar made for the abbey of Abdinghof shows a rather different mode of technical display, with novelty taking precedence over other priorities (plate 17). Although the longstanding attribution to Roger of Helmarshausen has been disputed, it is clear that the altar was made for the abbey of Abdinghof in Paderborn in the first quarter of the twelfth century.⁷¹ Kept at the abbey until its secularization in 1803, it is now, like the altar of Henry of Werl, in the Paderborn diocesan museum. The altar was probably made between 1115 and 1120, and thus it would have been made under the auspices of Henry of Werl (d. 1127). At the time, the abbey of Abdinghof was under the close protection of the bishops of Paderborn, and the proximity of cathedral and abbey makes

it likely that a viewer of one the altars would have been aware of the existence of the other. There were also close connections between Abdinghof and Helmarshausen in this period, for Abdinghof's abbot, Hamuko (d. 1142), had come to Abdinghof from the abbey of Helmarshausen in 1115, and the altar was probably made under his rule.

The altar of Abdinghof has some similarities to the altar of Henry of Werl, but it is smaller and has no gems. Unlike the bishop's altar, with its variety of techniques and materials, the Abdinghof altar has copper gilt openwork plaques on all sides. The top of the altar has been much restored; it now comprises four gilded copper plaques with images of the dedicatory saints Paul, Felix, Peter, and Blaise framing an engraved band of silver and a modern altar stone (fig. 26).

The sides of the altar show the martyrdoms of Blaise and Felix in gilded openwork against a contrasting background. Three scenes of the martyrdom of Saint Blaise of Sebaste fill one long side, while two scenes from the life of Saint Felix of Aquileia fill the other (plates 18, 19). The short sides show images of a baptism by Saint Peter and the martyrdom of a monk opposite a scene possibly illustrating the martyrdom of Saint Paul (plates 20, 21).

The figures on the altar show clearly the V-folds of drapery that are emblematic of Roger's style. On the long side, with the martyrdom of Saint Blaise, a soldier steps forward, decapitating the saint with his two swords. His cape flutters behind him in the wind as the blindfolded head falls to the ground. A large swath of his cape falls in a series of Vs, as if to show the billowing of the fabric. Meanwhile, the narrow folds of the rest of the cape, intended to show the small flutters of fabric, are made up of a series of long,



FIG. 26

Roger of Helmarshausen, altar of Abdinghof, ca. 1115–20, top panel. Erzbischöfliches Diözesanmuseum, Paderborn.

narrow rectangles. There are similar patterns on the figure itself. His right thigh is made up of one large parabolic shape, edged on the left side with a set of V-folds to create depth around the curve of the leg; behind this leg are narrowly set rectangular forms, indicating the fold of fabric separating his legs, and, with another set of inverted Vs at the bottom, the fold of the edge of the drapery. His back leg, meanwhile, is defined by the same drapery pattern, with large nested triangles forming the space of his back thigh, and narrowly set lines outlining the edge of the figure. It is as if the artist had a schema: broad areas are made with large triangles; folds are marked by nested V's; and narrow folds are made by long, closely set rectangles.

The same method of depicting falling drapery—but far more masterfully executed—appears on what is thought to be an image of Paul on the short panel of the altar. On the right side of the panel, a figure about to sever the head of a saint wields a large sword. His pose is similar to that of the soldier on the Blaise panel: he steps forward with his right leg, his left slightly bent, with just his toe on the ground. Yet his arms are

outstretched, holding the head of the saint with his left hand and the sword with his right, his head leaning in to make the strike. As on the Abdinghof panel, the fabric falls in nested Vs, but here they are far more dynamic and subtle. As on the Blaise panel, the figure's right thigh comprises a parabola shape with smaller triangular shapes within, yet there is far more variation in the line and in the shape of the leg. The folds of drapery showing the fabric between the figure's legs are also much more convincing on the Paul panel: the fabric takes the same upward fold as it does on the Blaise panel, marked out in a T shape, but while the artist of the Blaise side has made the arms of the T perpendicular to the stem, the artist of the short panel gives the lines more curvature, making the form more dynamic and subtle. Even the head of the figure with the sword leans in further, and the curve of his left wrist is exaggerated as if to show the strength required to hold the head of the martyr in such a way. The comparison could continue: the belly of the figure in the short panel protrudes slightly outward, as if to show his weight, and folds in slightly at the curve of his back to show

movement and form. There is no such subtlety in the torso of the figure on the Blaise panel; his body is made up of a rectangular shape made by three nested lines, with too many lines coming in from his shoulder blade to convincingly indicate form or weight.

A similar comparison could be made between the other long side, with scenes of Saint Felix, and the other short side, with Peter performing a baptism. On the long panel, three soldiers stand stiffly together, each with one foot in front of the other, back heel slightly raised. The chain mail of their armor is rendered in openwork, and three horses stand in profile behind them. Two of the three martyrs have already lost their heads. Their bodies are again very schematically rendered: a large parabola forms the upper thigh of each, with V-folds falling behind to indicate curvature and depth. Upper bodies are formed with two stacked squares, the separation indicating the break at the shoulder blade. The short side with Peter baptizing an unknown figure, on the other hand, is more dynamic and more fluid in its line. Another figure on the right lunges in to behead a monk, and the drapery of his skirt, while still retaining a divided parabolic shape for the upper thigh, has a great deal more variation in its line than we saw on the long Blaise panel. The line here goes dramatically from thick, at the base of the V, to thin, and a number of closely set, similarly varied lines at the small of his back show the curving of the drapery around the form.

These variations suggest that the panels of this altar were made by at least two, if not three, artists: one for the short panel with the martyrdom of a saint who is perhaps Paul, one for the two long sides, and perhaps a third for the short side with the martyrdom of the monk. Even though the patterns are similar, the artist or

artists of the short sides are far more advanced than the one who executed the long sides. Two points are thus to be made: first, the artists must have drawn according to schemata; and second, the workshop, or patron, cared about consistency. The figures are done in the same mode, using the same patterns of drawing, across all four panels, by artists with different levels of skill. What we see here is a workshop "style," but one that is not dependent on an individual artist's hand, nor even on type of imagery, but on ways of working, on technique itself.

Stylistic differences between the panels, moreover, alert us to a particular mode of stylistic analysis. As Richard Neer has noted, style may be dependent on the recognition of certain criteria, such as V-folds of drapery, but more important is the existence of "grammatical" relationships that govern these criteria.⁷² These grammatical relationships indicate a shared habit of viewing. The altar of Henry of Werl and the altar of Abdinghof are often attributed to the workshop of Helmarshausen on the basis of the V-fold style, and we can see that in the figure of Bartholomew on the altar of Henry of Werl, for example, the drapery falls in the same pattern of V's and rectangles as on the long panel of the sword-wielding, cape-flying soldier on the panel of Saint Blaise of the Abdinghof altar. Despite these similarities, the Abdinghof altar is often noted for having more dynamic figures than the altar of Henry of Werl: they are not confined under arches, as are the seated figures on the altar of Henry; they are not all frontal; and no frames separate them. These differences have been seen as a reason to separate the two altars in date or even authorship.⁷³

Rather than signal a change of technique, however, the distinctions between the two suggest

rather differing approaches determined by the type of image. Both are made up of similar schemata of V-folds, triangles, and narrow rectangular folds, but the figures on the Abdinghof altar are actors in narrative scenes, while those on the Henry altar are intended to visually and figuratively support the operation of the church and the celebration of the mass. The different approaches to imagery seen here—the active figures on the Abdinghof altar and the still, enthroned figures on the Henry altar—reflect the function of the piece and the demands of the commission and do not necessarily indicate different artists. These kinds of variations should not hinder or confuse an understanding or dating of the works but simply demonstrate the range of modes to which workshop style could be adapted. As Willibald Sauerländer has shown in the case of sculpture at Reims Cathedral, stylistic affinities could express social ties and function as a community's identifying mark.⁷⁴ In the early twelfth-century Holy Roman Empire, in the political upheaval of the aftermath of the investiture controversy, this may have been particularly important.

The Enger Cross: Virtuosity and Ingenuity

A cross now in Berlin, probably the one mentioned in the Paderborn document and commissioned by Henry of Werl, shows some of the same patterns that we have seen in the Henry altar and the altar of Abdinghof (fig. 27). Formerly in the church of St. Dionysius at Enger, and thus known as the Enger cross, it measures just twenty-two centimeters high, half the height of earlier examples such as the cross of Abbess Mathilde and Duke Otto in Essen (fig. 28), the Lothair cross in Aachen (fig. 29), the Borghorst

cross, or the Gisela cross.⁷⁵ One side is a *crux gemmata*: gems and pearls are set on a gold field filled with filigree. Large square panels mark the ends and center of the cross; each contains large gems bordered by a string of pearls. The central panel is dominated by a large crystal, probably Carolingian, carved with a figure of an angel. Beneath it, as noted by its inscription, is a fragment of the True Cross. The reverse of the cross holds four panels, one set in each terminus, which contain anthropomorphic images of the four symbols of the Evangelists against a notched background: an angel representing Matthew, a lion for Mark, an ox for Luke, and an eagle for John. A fifth panel set in the center of the cross has an image of the Lamb of God (plate 22). Spaces between panels on the horizontal arm are filled with gold sheets and filigree. The vertical arm was probably once the same, though its panels are now lost, leaving the oak core of the cross visible.

The niello panels on the reverse side reveal much about workshop procedure (plates 23–26). Here the figures have the characteristic V-folds that have led scholars to attribute the cross to the workshop of Roger of Helmarshausen.⁷⁶ The drapery on the ox of Luke, for example, shows a series of V's just below the animal's left knee, and the hooked lines at the bottom of the drapery are very similar to those just above the left ankle of Saint Philip on the Henry of Werl altar (fig. 30). Lasko has also noted the similarity of the tiny notches in the flowers of the spandrels on the altar of Henry of Werl to the notches in the background of the Enger cross panels.⁷⁷

While it seems likely that the cross is the one made for Henry of Werl, there is significant inconsistency in the execution that sets it apart from the portable altar of Henry and



(COUNTERCLOCKWISE FROM TOP LEFT)

FIG. 27

Roger of Helmarshausen, Enger cross, ca. 1105, oblique view. Kunstgewerbemuseum, Berlin.

FIG. 28

Cross of the abbess Mathilde and Duke Otto, 973-82. Essen Cathedral Treasury.



FIG. 29

Cross of Lothair, ca. 985-91. Palatine Chapel Treasury, Aachen.



FIG. 30

Roger of Helmarshausen, portable altar of the cathedral of Paderborn, made for Bishop Henry II of Werl, ca. 1105, detail: Saint Philip. Erzbischöfliches Diözesanmuseum, Paderborn.

thus complicates the attribution. The ox, for example, has three sharp V-folds just below the left knee. The V is slightly longer on the left side but ascends straight up, the evenly spaced lines gradually decreasing in thickness as they form the stem. The drapery on the figure of Matthew's angel is rather different. It contains the same V-folds, but beneath the figure's right knee the lines are set close together, and they lean at a sharp angle. Furthermore, on this figure there is little contrast between linear pattern and blank

space: the lines are set close together and distributed evenly, leaving few open spaces to delineate the forms beneath. The ox, in contrast, has a very few lines drawn across the top of his thigh, making it stand out as a "light" contrast, and the arm, too, is defined by two halves, above and below the elbow. The two figures, then, are drawn using a similar technique, yet their forms are very different. The artist of the ox shows more of a sense of form and gives more weight to the figure, while the artist of the angel of Matthew fills the

space with an even distribution of line, arranged in falling, dynamic patterns. The disparity in the two figures probably reflects a difference in hand and skill.

A third hand may be distinguished in the eagle and lion. In the lion, the folds of drapery are composed of very tight, regular patterns. The drapery on the lower front section of the lion's figure shows a set of four V-folds, while the drapery across his body is a set of horizontal lines. The interior of his left sleeve is also a set of evenly spaced lines, here gently curved to suggest shadow. The patterns are similar on the eagle, where drapery is treated in sections: the V-folds of the lower front, the horizontal lines curving across the torso, and the V's cascading down toward the left elbow. Here we see neither the contrast of closely set lines and open spaces visible on the ox, nor the busier, dynamic pattern of the drapery on the angel. These differences again reflect individual skill: the artist of the ox is strong at depicting masses of forms, while the artist of Matthew draws with a dynamic, energetic line. The artist of the eagle and lion, meanwhile, sharply reduces each figure to its constituent parts, creating figural mass while covering the surface in decoration and line. Each artist, however, follows the same basic formula.

Individual elements bear this out. The fold of the cloth at the neck of the three animals, for example, is not identical, but in each case a sideways V at the left points toward an overlapping layer composed of two lines that create drapery folds. Hands and feet follow similar formulae, but again with varied effect. The right hands of the ox, the lion, and the angel are nearly identical; each hand is held upright, with palm outward in a presentation gesture, the smallest finger slightly bent, and a thumb muscle drawn to mirror the

backward curve of the outer palm. A line above each thumb muscle creates another fold in the flesh, but at different heights. The rendering of the angel's hand again shows a lack of clarity; the artist has added an extra crease in the hand, and the folds of the drapery behind the hand obscure its shape as it is set against the figure. The methods of drawing are consistent, but the execution is not.

Differences are visible in the backgrounds as well. The angel's halo differs from the others, as it lacks a ring of dots on its edge, and the angel is also the only figure rendered with solid black fill beneath the drapery of the raised hand. The eagle's left hand is missing, more likely a loss in the niello than part of the iconography, while its right hand, bare, holds the scroll. The background of this figure, too, has been adjusted, and a row of round notches omitted, so that the figure stands out more clearly. The backgrounds were probably filled in last and were not ruled, but done by eye. Rows behind the ox do not match from bottom left to bottom right, though the artist eventually compensated for this, adding an extra row of arches below the right wing.

As on the Abdinghof altar, the combination of surface decoration, figural depth, and pictorial organization varies greatly. The lion and eagle on the Enger cross may well have been made by the master, for they show the most sophisticated balance of movement, depth, and surface pattern of any of the nielloed figures on the reverse. Despite these differences, V-folds and background patterns remain the same across the work. Importantly, these are precisely the elements that may be taught, remembered, and repeated. It is easier to teach an artist to make drapery using V-folds than it is to teach them to reduce a figure to its essential parts, as the artist of the eagle did.



FIG. 31
Imperial cross of the Holy Roman Empire, 1024. Vienna Imperial Treasury.



FIG. 32
Osnabrück cross, late eleventh century. Osnabrück Cathedral Treasury.

Again, we see a concern for consistent form and a regularity in techniques of making.

The Enger cross is not large, but it emulates the great imperial crosses. The squared-off ends echo the imperial cross in Vienna, begun under the Ottonian emperor Henry II in 1024 with its squared ends and gem-encrusted panels (fig. 31). This is also the shape of the Osnabrück cross, made at the end of the eleventh century for another bishop, Benno (r. 1068–88), who—like Henry of Werl—was close to Henry IV, and was even with him during his repentance at Canossa (fig. 32).⁷⁸ Henry of Werl was in frequent contact with the bishops of Osnabrück and would have probably been aware of Benno's great cross:

an exchange of property between the abbey of Abdinghof and the church of Osnabrück, in August of 1094, was sanctioned by Bishop Henry and Benno of Osnabrück's successor Wido; the two met again in 1101 with the emperor in Cologne.⁷⁹ The emulation of these imperial crosses must have been significant in these years of the investiture crisis.

The gems on the Enger cross certainly carry past histories and evoke an imperial heritage. The bottom arm of the front of the cross is set with an ancient Roman cameo that Lasko has dated to the late Roman period, while the left arm is set with an ancient intaglio that Lasko suggests may date to the fourth century B.C.E.⁸⁰

As Ilene Forsyth has argued, the accumulation of ancient gems on a medieval object evoked the eventual triumph of Christ.⁸¹ The placement of the cameo on the Enger cross may also have echoed the ancient cameo on the late tenth-century Lothair cross, probably a gift of the Emperor Otto III to the Palatine chapel in Aachen. In the context of late eleventh- and early twelfth-century northern Germany, such political connections were heavily loaded. As Henry of Werl sought to legitimize and stabilize his office as bishop, the use of a cameo on an imperial-form cross would have been a powerful confirmation of episcopal status and imperial allegiance.

Details of the Enger cross's construction also hark back to eleventh-century examples from imperial contexts. The arms of the front of the cross are filled with a very fine gold filigree, into which are set groups of three or four pearls. Strands of pearls also frame the axial gems, the central rock crystal, and the two gems on each arm of the cross. Wide tri-lobed clasps secure the central rock crystal of the Enger cross, and smaller pointed fastenings secure the smaller stones. The rock crystals on the Gisela brooch are also attached with tri-lobed clasps, and the smaller stones with a series of filigree loops (plate 27; fig. 33).⁸² The central stone of the Theophanu cross and the four axial stones of the Essen crown are attached similarly, while the beaded pearls that frame the larger stones on the Enger cross can also be seen on the borders of the golden crown in Essen. According to Lasko, the continued use of these similar settings may be evidence of a goldsmith's technique that passed through generations, but because of the gap of time and distance it is hard to know.⁸³ However, as Hiltrud Westermann-Angerhausen

has recently argued, many of these objects were technically conservative. Because "the exemplary function of an important reliquary or gemmed cross, for instance, lay in its venerability," she writes, "this again meant that it had to be identifiable through its characteristic decoration and form. An ecclesiastical goldsmith was bound by the formal and material identity of an 'old-fashioned' but venerated model."⁸⁴ This may be the case here. The settings are similar in type but not detail: a tri-lobed clasp is used for large gems, but one is tall and one wide. Loops or rows are used for small gems, but one is curved and the other triangular. The objects may have been made in different workshops, but their resemblance attests to patterns of decorum and consistent manners of procedure. Gem settings are traces of a course of action, a manner of making, that can be taught and remembered, repeated and adjusted.

Indeed, filigree patterns on the back panels of the Enger cross are variations and embellishments of patterns discovered in older objects. The central band of filigree on the cross of Theophanu, for example, is filled with a pattern of heart shapes (fig. 34). Each heart is filled with a tri-lobe that emerges from the center meeting point of the two strands. The bottom of each heart, meanwhile, flows into double, back-to-back scrolls. This pattern is the basis for the pattern on the Enger cross, though there it is made considerably more complex. In the Enger cross, a similar heart shape is the core of the pattern. Placed in the center of the panel, it encloses a tri-lobe; surrounding it are curls and scrolls that turn and grow to fill the space and create larger heart-like forms. In both objects, too, tiny bands hold together the strands of the filigree.⁸⁵ The filigree panels on the reverse of the Enger



FIG. 33
Gisela brooch, detail: gem settings. Kunstgewerbemuseum, Berlin.

cross are more ornate than the regular pattern seen on the Theophanu cross, but they break down to a similar basic structure.

The Enger cross is modest in scale, but it is lush in its ornamentation and gracious in its execution. It shows a consistency of workshop style that cuts across the ability levels of a group of artists, and it shows a virtuosity of technique. It uses far fewer gems and gold than the great imperial crosses, but its ornamentation is nevertheless dense. Beading and filigree take ornamental motifs and schemata to a higher degree of complexity, opening up and recombining elements into new patterns. Even the

background on the niello panels on the reverse of the cross might be a display of ingenuity. Here, a row of semicircles and a row of small dots form a band, which alternates with another band filled by tall, narrow notches. This pattern may echo a pattern of gem settings on the front of the cross: Each of the rectangular panels on the arms of the cross, and the square in the middle, has four small gems filling each corner. These gems are placed in settings that comprise a row of semicircles, above a row of small beaded filigree, above narrower notches. Scholars have already noted the dynamics between the fronts and backs of such crosses; the echo of patterns



FIG. 34
Altar cross of Theophanu, ca. 1040, with later alterations. Essen
Cathedral Treasury.

in gem settings and plaque backgrounds here may be a part of this fluidity as well. Numerical significance already creates a resonance: the four small gems are placed in the corners of panels that correspond to the four Evangelist figures on the reverse. Perhaps similarity of pattern makes the connection more clear. Decorative pattern of gem settings is a tiny detail, but it brings attention to larger thematic continuities between front and back, linking the four Evangelist symbols on the back with the five sets of four gems on the front. The panel with the lamb is surrounded by a semicircular pattern perhaps made to echo metalwork or gem settings, making the entire circle a two-dimensional representation of a glowing gem like the one on the front. Even the chalice of

the lamb shows the pattern, adding another link to the chain of visual associations. Like the altar of Henry of Werl, it is as if the cross makes up in ingenuity and density what it lacks in size.

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The Modoaldus Cross and the Frankfurt Cross: Materiality and Memory

So far this chapter has shown how variety of skill and ingenuity was used to make up for fewer gems or smaller size, or to create continuities and echoes across an object. Now we shall see how two crosses, both attributed to the Helmarshausen workshop, use particular ornamental motifs to organize space and create a more ornate object without an increase in material cost. One, now in the Museum Schnütgen in Cologne, was probably made to hold the relics of Saint Modoaldus, archbishop of Trier (614/15–647/49) and advisor to King Dagobert of France, which were probably transferred from Trier to Helmarshausen in 1107 (plate 28).⁸⁶ A single copper plate that has been engraved and gilded, the cross is 42 centimeters high and 33.5 wide. It is thus slightly longer in proportion than the Imperial cross in Vienna, which at 77 × 70 cm is larger but more square, and bigger and broader than the Enger cross, which is 22.4 × 18.5 cm. The arms of the Modoaldus cross end in square panels on which the Evangelist symbols are depicted; at the center is a medallion that bears an image of the Lamb of God. The lower arm bears an image of Saint Modoaldus, and inscriptions on the cross naming the saint and referring to the Lamb of the Apocalypse confirm that this cross probably held a relic of the early medieval bishop.⁸⁷ The cross was probably once a *crux gemmata*;



FIG. 35

Cross of the abbess Mathilde and Duke Otto, 973–82, reverse. Essen Cathedral Treasury.

nail holes running around the edge of the piece suggest it was probably the reverse of a gemmed cross attached to a wooden core.

As we have seen elsewhere on the objects attributed to Roger, drapery is rendered by means of closely set lines falling in series of V-folds juxtaposed with blank spaces. These are most visible on the figure of Modoaldus, who, with his slender height, straight pose, and closely set folds of drapery is often compared to the figures of Liborius and Kilian on the altar of Henry of Werl. The division of drapery and bodies into sections, as we have seen on the eagle on the Enger cross, is most visible here on the figure of the eagle. Here the spaces of the animal's shoulders and upper chest are clearly differentiated

from the robe that drapes around his neck and falls behind the scroll he holds.

The surviving plate shows a clear organization of space. The cross is divided into nine sections: the four arms, the four end panels with Evangelist figures, and the central medallion. Borders around each section delineate the units. Small squares filled with tiny crosses frame the panels of each of the Evangelists and slightly larger rectangles with crosses line the edge of three of the arms, while a notched line runs up the panel with the image of Modoaldus. These divisions echo the spatial division on the reverse of the Enger cross, yet instead of being an assembly of separate panels, here the divisions are engraved onto a single piece of copper. The division of space clarifies the structure of the piece, framing the squared-off ends and differentiating the images of the Evangelists from the vine pattern of the arms and the figure of Modoaldus himself. If we compare it with the reverse of the cross of Otto and Mathilde at Essen, we see that the program is similar, but the concern for structure is very different (fig. 35). On the cross of Otto and Mathilde, the Evangelists are framed at the ends of the cross, and a medallion of the lamb is placed in the center. Yet here the framing of the Evangelist figures and the medallion of the lamb correspond to the shape of the cross. The ornamental frames in the Modoaldus cross also make the separation of elements more clear and, again, highlight the shape of the object. The division creates variety, where each section is framed by a different type of ornament, and order, where the variety is governed by the structure of the cross itself. These characteristics may well be evidence of the artistic skill that Theophilus urges his reader to practice.

The use of technique here might also be an adjustment for material expense. Instead of being

made of separate panels filled with gold filigree and niello, as in the Enger cross, the Modoaldus cross is less expensive gilded copper. Furthermore, it is gilt in the negative: the engraved lines are gilt, while the background is left bare. Theophilus's point about the division of silver for the chalice—where silver filed away is recovered for use elsewhere, where materials are carefully conserved and reused and nothing is wasted—might mean that the decision to gild thus on the Modoaldus cross was an ingenious way to save funds and display a different kind of technique. It is also possible to see some of the patterns as simplified versions of those used elsewhere in metalwork or in other media. The tiny horizontal bands forming a line along the edge of the Modoaldus panel, for example, evoke the borders of beaded wire on the edges of the Enger cross. Perhaps most prominent are the heart-shaped leaves filled with palmettes and marked within by a cross that together suggest the tree of life.⁸⁸ With its series of hearts connected by scrolls, this band is structurally similar to what we saw on the top of the altar of Henry of Werl, or, in a more complex arrangement, in filigree on the reverse of the Enger cross. Perhaps this heart/vine pattern was a memorized motif to be repeated and reused on different objects, by different artists. Such combinations and repetitions may be a sign of the artist's skill and know-how. Though it bears no filigree and no variety of media or technique, the Modoaldus cross nevertheless shows many of the values espoused by Theophilus, for it uses a variety of ornamental patterns and displays clear structure, a concern for economic constraints, and technical expertise.

A crucifix now in the Museum für Angewandte Kunst in Frankfurt repeats much of the same structure of the Modoaldus cross. Probably

made shortly after the Enger cross, between 1110 and 1120 (plate 29), it is smaller than the Modoaldus cross but has almost the same proportions, with measurements of 28.5 centimeters high and 22.5 centimeters wide, in comparison to the Modoaldus cross, at 42 centimeters high and 33.5 wide. Rather than being a *crux gemmata*, the Frankfurt cross bears a cast bronze figure of Christ, with niello plaques containing images of the Evangelist symbols at the end of each arm. The reverse of the cross shows four large squares, each with gems, and a central panel with the Lamb of God (fig. 36). Many of the patterns visible on the Modoaldus cross make an appearance here. The cross-in-square pattern that lines the panels of Evangelist symbols runs around the entire edge of the cross, and the image of the Lamb of God, with its body facing left and its head turned back to the right, is repeated too, as is a running heart-shaped pattern in the center band of the arms of the cross. These attributes, along with the drapery of the figure of Christ on the obverse, led Creutz to attribute the cross to the hand of Roger.⁸⁹ Beyond attribution, however, we might see in the object the continued repetition and re-working of motifs, where elements like the cross-in-square pattern are constantly varied. Around the square panels of the Modoaldus cross, for example, the border is filled with squares; in each square is a thick, equal-armed cross, bisected by a thin X, the arms of which then split into three branches as they reach the corners of the square. In short, each square is filled with an eight-armed star. The border pattern of the reverse of the Frankfurt cross is similar in structure but varies in detail. Here the square is filled with a thick equal-armed cross, but the arms do not meet at the middle, as they have been pushed out to the walls of the square. In the

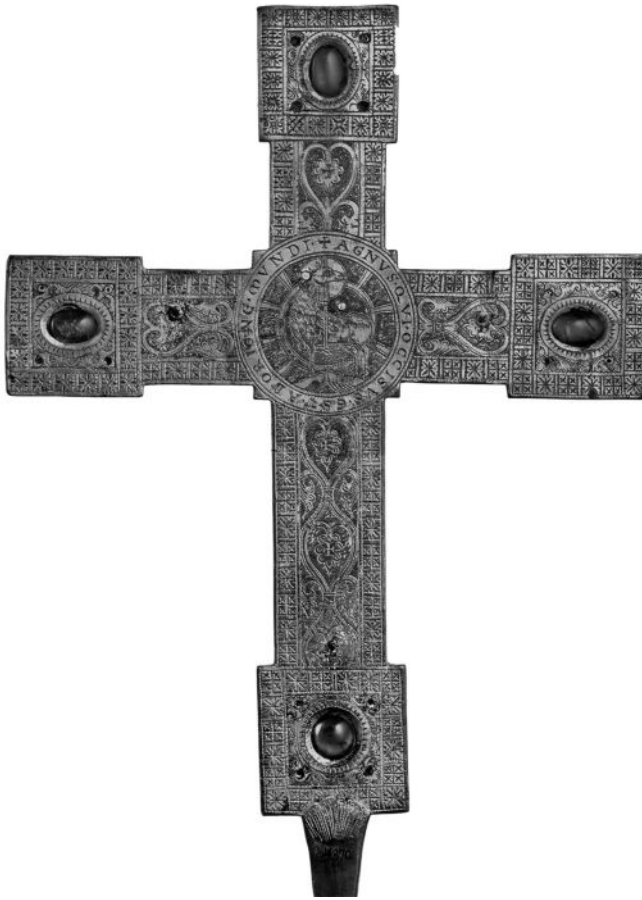


FIG. 36

Roger of Helmarshausen, crucifix, ca. 1110–20, reverse.
Museum Angewandte Kunst, Frankfurt.

middle is an X, but the arms of the X stop before the corners, and smaller lines fill the gap between thick cross and thin X. Again, the structure creates an eight-armed star, with four thick and four thin arms, but it does so in an opposite manner. The squares of the Frankfurt cross, too, alternate with another cross-in-square pattern to create further variety. Like the altar of Henry of Werl, the background areas of the Frankfurt cross are filled with punchwork, and the same double lines that separate border ornaments are used on both the Frankfurt cross and the Modoaldus cross.

The two crosses show us how patterns can be manipulated and changed to accommodate different material expectations, different types of objects, and different needs—what Theophilus might call “measure.” Ornamental structures repeat and are modified, embellished, or simplified; we might see in their repetition the practice of the workshop. Perhaps this is why Theophilus never describes the precise forms that elements should take, even though his instructions for technical processes are otherwise highly specific and detailed. It is as if he expects such details

of pattern or ornament to be determined by the workshop. In all, the two crosses suggest how economic choices might come into play. Medium and size were a matter of expense, and artistic skill could fill the gaps that material wealth could not. The repetition of motifs may even suggest they may have been a proprietary element, used by a particular workshop and handed down from artist to apprentice.

The Trier Book Cover

A book cover attributed to the workshop of Roger of Helmarshausen provides a few final threads of evidence for how motifs were probably copied and re-created. Now in the Cathedral Treasury at Trier, the book cover is usually dated somewhat later than the altar of Henry of Werl, to 1110 or 1120 (plate 30). A book of Gospels, the volume was probably used at Helmarshausen until at least the fifteenth century, for it includes feast days celebrated at Helmarshausen and a fifteenth-century inscription listing documents of the abbey.⁹⁰ The cover is adorned with four repoussé Evangelist symbols, all executed in gilded silver, that fill the spaces between the arms of a gem-studded cross. At the center of the book cover is a large gem, and the frame dividing the book cover into four is filled with more gems, filigree, and small enamels. Peter Lasko has noted that the use of filigree on the book cover, with the curving scroll ending in a pearl, can also be found on the mandorla of Christ on the altar of Henry of Werl, a technical strategy that may attest to workshop practice.⁹¹

The figures on the book cover have been noted for their fluid, confident, and smooth manner; the angel representing Matthew is

perhaps the best example of this.⁹² The angel kneels and turns his body to the viewer's left. His draped hands raise the book in a similar direction. His head, however, turns back toward the center of the cover and toward the other creatures. The images of the Lamb of God on the Modoaldus and Frankfurt crosses are arranged similarly, and the angel carved on the central gem of the Enger cross also holds a book in one direction while turning his head in the opposite. Yet far closer comparisons can be found, for the eleventh-century Gospel book kept inside the book cover contains a similar rendering of the angel of Matthew, with body and head turned in opposite directions.⁹³ The form, in fact, was repeated, for it was used again in the Gospels of Helmarshausen, a manuscript illuminated at Helmarshausen in 1120s and now at the Getty Museum.⁹⁴ In addition, the star around the central stone of the book cover mimics the star in the halo of Luke on his portrait page.⁹⁵

The enamels on the book cover are perhaps the most telling for our purposes. Presently, the cover contains six rectangular strips of enamel and six small triangular pieces. It would probably have contained twelve or possibly fourteen pieces originally. There were once four small rectangular plaques in the arms of the cross; two are missing. Four longer strips survive in the vertical bands of the frame. Seven triangular plaques surround the central gem, of which one is missing. Franz Ronig has suggested that plaques would also originally have been in the four corners. Given the large size and square shape of the pieces, this seems likely. If these were reused plaques, as Ronig has argued, they were probably made in Egbert's workshop in Trier or perhaps in Essen.⁹⁶ Yet they show patterns that were reused and recombined. The cross-in-square pattern

in the small plaques on the arms of the cross is similar in structure to the border patterns on the Modoaldus and Frankfurt crosses, while the turned-out scrolls may be a mirror version of the scroll pattern in the filigree border of the Henry altar. Even more compelling, perhaps, is the way in which these scroll patterns form the basis for filigree that fills the spaces between gems and enamels in the book cover. The scrolls in the enamel just above the central medallion may be the germinating element of the filigree scrolls that we see filling the space between the vertically set enamel and the two scrolls that flank the white stone just below, with one smaller tucked into and above the larger scroll. These comparisons may be minute, but the repetition and reconfiguration of figural and ornamental elements within a single object, and across a group of objects, alert us to modes of technical procedure.

The text composed by Theophilus and the objects made by Roger demonstrate the increasing value placed on technical skill that is evident in the creation of a work. Such display of skill is particularly visible in ornament. As we have seen in repeating patterns of filigree, some ornament may have been recognized as a series of elements that could be subject to variation or adaptation for different objects. To this end, it is the visual trace of process and of labor expended. Theophilus's concern is for how things are made, and he urges his reader to achieve a similar level of understanding. We might take a cue here from scholars like Wollheim, who, with Nelson Goodman and Richard Neer, has suggested that style does not exist outside of the object; it is something practiced, performative. As James Herbert has noted in the case of painting, it can

be "emergent."⁹⁷ Thus form is constituted in technique—be it in variety, proportion, or a broad understanding of how techniques relate to each other and to the object.

The importance of technique and skill may reflect the political and economic uncertainties of the early twelfth century, as was the case with Henry of Werl. Monastic histories and contracts such as the one that survives for Henry's altar show that the abbeys had a serious interest in establishing their position within a long chronology of institutional continuity, prestigious patronage, and Christian history. Landed property and revenues, as well as art objects, could serve in negotiations among abbeys. As we saw in the case of Henry's portable altar, the craft of the object was as much a part of its claim to authority as its figural imagery. We saw earlier in this book that the mining of metal and the production of precious objects made from these same materials were extensions of institutional powers and ambitions. The ability to acquire and manipulate such materials spoke to an abbey's influence and significance as a force in the region. Together, the charter remembering Roger and the inscription in the Vienna manuscript of *On Diverse Arts* suggest that by the mid- to late twelfth century, the artist's name could too. Finally, changing patterns of patronage may play a part. As Dominique Iogna-Prat has recently argued, in the twelfth century the buildings and objects created by an abbot for a monastery were increasingly viewed as earthly manifestations of the monastery's spiritual function.⁹⁸ Perhaps then, the understanding and recognition of artistic technique was likewise becoming a sign of prestige and sophistication.

CONCLUSION

The Practice of Seeing

*You have given them cause to praise
the Creator in the creature and pro-
claim Him wonderful in His works.
For the human eye is not able to con-
sider on what work first to fix its gaze;
if it beholds the ceilings they glow like
brocades; if it considers the walls they
are a kind of paradise; if it regards
the profusion of light from the win-
dows, it marvels at the inestimable
beauty of the glass and the infinitely
rich and various workmanship.*

THEOPHILUS,
On Diverse Arts, prologue III

On Diverse Arts is far more than a practical handbook. It reveals the modes and parameters by which art is to be made, from the learning of technique to the creation of form. Theophilus presents the learning of art as a spiritual activity, yoked to the practice of virtues as set forth by Augustine. Materials—from gold and silver to iron, pigment, and the frit for glass—are not static elements to be employed thoughtlessly or acquired greedily. From the acquisition of raw material to its transformation and use in the art object, the properties and nature of matter are paramount, and the artisan must work in harmony with that nature. It is the dynamic character of physical matter that makes the creation of the art object possible. Similarly, artistic skill is moralized. Theophilus defines each activity as the practice of virtue; patience, care, and knowledge are put into action as matter is transformed into art object. Such a moralization of labor is then extended into how form is understood, for as shown in the last chapter, Theophilus's emphasis on good technique, on understanding how things are made, suggests that visual patterns and ornamental structures are valued as evidence of a particular procedure and the trace of skill. Fine craftsmanship is a visual value. Ornament, structural forms, and display of technique are witness to a particular mode of making that is valued as a learned

subject and reflects positively on the patron, the workshop, and the abbey. The object is something that displays skill, virtue, and access to good materials; it is the trace of the good labor of the artisan and workshop. David and Solomon are exemplary viewers, and the object, too, can be witness to the learning and virtue of all those who recognize it as the trace of good work and piety, be they makers, patrons, or viewers.

Theophilus does not describe finished objects but instead how things *should be made*. A fine object is the product of accomplished skill and craft, of thoughtfulness, care, rationality, and learning. It is determined by the series of events—from the mining of metal to the careful application of filigree—that formed it. The fact that the chalice is made by dividing the metal in two equal parts, for example—one for the foot and one for the cup—determines its visual balance, a stark contrast to many Byzantine examples, with their large cups and narrow stems. Similarly, instead of prescribing iconography, Theophilus only hints at what appropriate imagery might be. One of the most extensive lists of examples of imagery is in the discussion of die-stamping:

The Lord's crucifixion is also engraved on the die, in the above way. . . . An engraving representing the Lamb of God is also made on the die, and representations of the four evangelists. . . . In the same way, the representation of Christ in Majesty is also made, and other figures of any shape and either sex. These, stamped on gold or silver or gilt-copper, add the greatest elegance to the places to which they are applied, because of their delicacy and elaborate workmanship (*propter sui subtilitatem et operositatem*).¹

The passage suggests that there is a certain decorum to figural decoration, a vocabulary or system of figural imagery to be drawn upon, but it is the technique that is valued and the “delicacy and elaborate workmanship” that makes the object elegant and fine.

From learning artistic techniques to material, labor, and the creation of visual form, the artisan's work comprises a series of processes, a trajectory, that runs in parallel to his spiritual purpose and is framed by it. Each aspect entails leaning toward God in some way. Learning follows an orderly, spiritual trajectory. Material is dynamic, and its manipulation requires knowledge; it is a transformation from raw to useful matter. Labor is a moral activity performed in accordance with monastic virtue, and it enables skill. Finally, with visual memory as a tool, skill and good procedure become a manifest value, and the object is esteemed as the trace of fine craft.

The importance of process is emphasized in one of the only self-referential passages of the entire treatise. In the first prologue, Theophilus uses a term that emphasizes procedure over finished product: “Be eager and anxious to look at this little work on the various arts (*schedula diversis artibus*),” he writes. “Read it through with a retentive memory, and cherish it with a warm affection.”² Calling his treatise a *schedula* suggests that, like the art object itself, Theophilus's text is a refashioning. According to Jean Leclercq, the word *schedula* can refer simply to the parchment upon which something is written,³ but the term is defined by Hugh of St. Victor as that which is still being added to, a collection still in progress:

A codex is composed of many books, a book is composed of one volume. And a codex is

so called, by transference, from the trunks (*codicibus*) of trees or vines, as if it were a trunk because it contains a multitude of books coming out of itself like so many branches; a volume (*volumen*) is so called from 'to roll up' (*volvere*). *Liber* is the inner rind of a tree, upon which the ancients used to write before the use of paper or parchment. For this reason they used to call writers *liberarii*, and a volume a *liber*. *Scheda* (a leaf of paper), whose diminutive form is *schedula*, is a Greek word. What is still being corrected and has not yet been bound in books is properly called a *scheda*.⁴

Lessing's initial title for Theophilus's work was the *Schedula diversarum artium*. Later publications of the text, however, most often use the title *De diversis artibus*, following the rubricator's inscription on the first folio of the Vienna manuscript and a title on the initial folio of the Cambridge manuscript.⁵ This title, however, loses some meaning in its interpretation.

Despite the medieval use of the term, the modern use of *schedula* has thus far hindered interpretation of the tract, for it implies a text more modest than "treatise" or *commentarium* suggest. Dodwell, for example, interpreted Theophilus's description of the text as a *schedula* as "probably simply the self-deprecatory kind of remark about 'this little work of mine' that any author might make of his own composition."⁶ However, it may pay to take the author at his word. Bernard Cerquiglini speaks of the way that medieval texts could be defined by their "joyful excess," their lack of finish and closure.⁷ Indeed, in the next passage Theophilus refers to the treatise as work, or *meo labore*: "When you have read through these things several times and

commended them to a retentive memory, you will recompense me for the labour of instruction (*meo labore*) if every time you make good use of my work you pray to Almighty God to have mercy on me."⁸ Thus Theophilus's reference to his *schedula* has significance after all. Moreover, if—as Thomas Haye suggests—the genre of medieval didactic literature operates according to the assumption that God is the ultimate author, then any text is a variation, a presentation of received ideas in a new form.⁹

From the writing of a treatise to the making of an art object, one is always working with pre-existing material, and the value of the project is in its transformation. This idea can also be found in Hugh of St. Victor, for whom much of the appeal of the mechanical arts is grounded in the labor of practicing them:

"Again, philosophy is the art of arts and the discipline of disciplines"—namely that toward which all arts and disciplines are oriented. Knowledge can be called an art "when it comprises the rules and precepts of an art" as it does in the study of how to write; knowledge can be called a discipline when it is said to be "full" as it is in the "instructional" science, or mathematics. Or, it is called art when it treats of matters that only resemble the true and are objects of opinion; and discipline when, by means of true arguments, it deals with matters unable to be other than they are. This last distinction between art and discipline is the one which Plato and Aristotle wished to establish. Or, that can be called an art which takes shape in some material medium and is brought out in it through manipulation of that material, as is the case in architecture; while that is called

a discipline which takes shape in thought and is brought forth in it through reasoning alone, as is the case in logic.¹⁰

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Hugh defines *ars* as that which entails the manipulation of material. In addition, he uses the verb *explicatur*: “Vel ars, dici potest, quae fit in subjecta materia et explicatur per operationem, ut architectura.” The word is significant, as it is often used in relation to verbal explanations and has connotations of pulling or drawing out. It is not a fresh creation; it is not originary, but rather, like a *schedula*, it is a remolding.

Hugh’s definition of art as a drawing out, a manipulation of material, thus corresponds with what we have seen in Theophilus’s text. The act of collecting, of ordering, of “pulling out” is not without agency; it creates effect and valuation. Here agency and effect are but the mode and manifestation of the yearning for God. In fact, in the passage quoted above, we see that Theophilus even refers to the results of his teaching not as a product or as an achievement of his own, but in terms of the change it produces in others, as *vicissitudinem*, a term often associated with spiritual transformation.¹¹

A reconsideration of the term *schedula* underscores the importance of process and transformation. Like the gathering and ordering of knowledge to compose the treatise itself, art is the virtuous manipulation of preexisting materials for a worthy end; it is driven by a yearning for God. From this point of view, we begin to read the art object not as a static entity, but rather as evidence of process, of labor, and of transformation. For the medieval viewer, the art object is a testament to craftsmanship and the virtues practiced in the course of its production. It is the trace of a series of good choices. This idea

of craftsmanship is fundamental to Theophilus’s concept of art. It may also be the English word with the closest approximation to a medieval idea of *ars*, that slippery term that denotes procedure, skill, and object all at once.

Theophilan Viewing

The notion of craftsmanship that emerges in the treatise changes how we conceive of the practice of viewing. Let us begin by looking at how Theophilus describes the act of seeing. As has been noted, Theophilus uses the verb *invenies*, to come to know, or discover, with its religious connotations of knowing God; and the use of the term *vicissitudinem*, which refers to the result of that learning.¹² Looking once more at the descriptions of techniques and countries in the first prologue, some particular values emerge: “If you will diligently examine it, you will find in it whatever kinds and blends of various colours Greece possesses: whatever Russia knows of workmanship in enamels or variety of niello: whatever Arabia adorns with repoussé or cast work, or engravings in relief: whatever gold embellishments Italy applies to various vessels or the carving of gems and ivories: whatever France esteems in her precious variety of windows: whatever skilled Germany praises in subtle work in gold, silver, copper, iron, wood and stone.”¹³ Here the Latin text is illuminating: “Quam si diligentius perscruteris, illic invenies quicquid in diversorum colorum generibus et mixturis habet Graecia, quicquid in electrorum operositate seu nigelli varietate novit Russia, quicquid ductili vel fusili seu interrasili opere distinguit Arabia, quicquid in vasorum diversitate seu gemmarum ossiumue sculptura auro

decorat Italia, quicquid in fenestrarum pretiosa varietate diligit Francia, quicquid in auri, argenti, cupri et ferri lignorum lapidumque subtilitate sollers laudat Germania.”¹⁴ Scanning the Latin again, a now-familiar vocabulary is perceptible: Theophilus praises the pigments of Greece for diversity of color (*diversorum colorum*); Russia’s enamel for its workmanship (*operositate*) and its niello for its variety (*varietate*); Arabia for its objects adorned (*distinguit*) with three kinds of techniques (*opere*); Italy for its embellished (*decorat*) ivories, gems and different kinds of vases (*vasorum diversitate*); France for its variety of windows (*fenestrarum pretiosa varietate*); and Germany for its subtle skills (*subtilitate sollers*) in various media. In short, the values of diversity, variety, subtlety, and workmanship (*diversitate, varietate, subtilitate sollers*) come to the fore. The passage thus encourages the reader to look for traces of fine execution.

In the second prologue, qualities of variety and diversity emerge. First, variety is given as a reason that art should be studied: “How sweet and delightful it is to give one’s attention to the practice of the various useful arts” (*diversarum utilitatum exercitiis operam dare*).¹⁵ Theophilus uses it again at the end of the prologue, commending the techniques and products of glass that he will describe:

I have approached the temple of holy wisdom, and beheld the sanctuary filled with a variety of all kinds of diverse colours with the usefulness and nature of each one set forth. . . . I have, like a diligent seeker, taken particular pains to discover by what ingenious techniques a building may be embellished with a variety of colours, without excluding the light of day and the rays of the sun. Having applied

myself to this task, I understand the nature of glass, and I consider that this object can be obtained simply by the correct use of the glass and its variety.¹⁶

Here the quality of glass is commended repeatedly for its variety and for its use, for even the term *ingenio*, in the comment on “ingenious techniques” (*artis ingenio*), refers to the variety of color that is possible in glass. This is a mirror of the reference to *subtilitate sollers* in the first prologue, for there, too, the ingenious aspect of *sollers*, or cleverness, refers to subtlety and, once more, to carefulness.

In the third prologue, Theophilus’s learned artist, whose heart is “filled with the Spirit of God,” is praised for creating a “variety of work.”¹⁷ In the list of attributes of the Holy Spirit, the gift of intelligence is linked to ingenuity, and the purpose of ingenuity is to create variety: “Through the spirit of understanding, you have received the capacity for skill—the order, variety and measure with which to pursue your varied work.”¹⁸ Throughout the prologues, good work is characterized primarily by variety and diversity. Ingenuity and cleverness are used much less often, but they are significant nevertheless, for they make variety and diversity possible; they are qualities based in and indicative of skill.

Such mastery of skill as a visual characteristic in contemporary objects. The variety of techniques displayed on the altar of Henry of Werl may be one of the kinds of *varietas* to which Theophilus refers in the text. Seen in light of Theophilus’s continual emphasis on craftsmanship and variety, it seems likely that such a combination of skills was particularly sought after. Technique and the execution of craft became something to be displayed.

Romanesque art has long been understood to have such a concern for variety.¹⁹ Yet as we have seen in objects like Henry's altar, variety and diversity are not limited to imagery; they are also found in ornament, structure, and technique. In the very last prologue, the importance of variety comes through even more strongly. Here, Theophilus summarizes his notion of vision and thus further defines the values related to craftsmanship and viewing. Variety and workmanship are mentioned again and again. Singled out by two hands in the margin of the Harley manuscript, one particular passage was clearly of interest to a medieval reader. It begins with a description of form: "Animated, dearest son, by these supporting virtues, you have approached the House of God with confidence, and have adorned it with so much beauty [charm] (*tanto lepore*); you have embellished the ceilings or walls with varied work in different colours, and have, in some measure, shown (*ostendisti*) to beholders the paradise of God (*speciem paradysi Dei*), glowing with varied flowers, verdant with herbs and foliage, and cherishing with crowns of varying merit the souls of the saints."²⁰ Variety plays an important role here. The adornment of the House of God with beauty and charm is the result of "varied work in different colours," which shows "in some measure" the paradise of God. Theophilus then goes on to describe effect: "You have given them cause to praise the Creator in the creature and proclaim Him wonderful in His works. For the human eye is not able to consider on what work first to fix its gaze; if it beholds the ceilings they glow like brocades; if it considers the walls they are a kind of paradise; if it regards the profusion of light from the windows, it marvels at the inestimable beauty of the glass (*inestimabilem vitri decorem*) and the

infinitely rich and various workmanship."²¹ Here again, variety and craftsmanship are paramount, for these create the effect. As viewers move their eyes, they see different things—from ceilings to walls to light—and they "marvel" at the beauty of glass and the "rich and various workmanship." Lastly, the author explains how images can cause one to yearn for God:

But if, perchance, the faithful soul observes the representation of the Lord's Passion expressed in art, it is stung with compassion (*compungitur*). If it sees how many torments the saints endured in their bodies and what rewards of eternal life they have received, it eagerly embraces the observance of a better life. If it beholds how great are the joys of heaven and how great the torments in the infernal flames, it is animated by the hope of its good deeds and is shaken with fear by reflection on its sins.²²

Here, rather than technique, material, or form, different kinds of imagery—from the Passion to torments to salvation—are necessary to bring the point home. In each section variety and workmanship play a role, creating the beauty of embellishment, the marvelous forms, and the emotional impact.

What is also significant here is that the passage describes three modes of vision and sets them in a hierarchy: the recognition of form, the effect of ornament, and the compunction for God. The first section has a Neoplatonic cast. Theophilus writes that having "adorned" the house (*decorasti*), decorating it with "varied work" and "in different colours" (*diverso opere diversisque coloribus distinguens*), you have "in some measure, shown to beholders" (*quodammodo*

aspicientibus ostendisti) the sight of the “paradise of God” (*speciem paradysi Dei*). Although this does not come through in Dodwell’s translation, the use of *ostendisti* implies a kind of revealing, a making present, while *speciem* implies the act of seeing itself. *Speciem* is a word for “sight,” not for imitation or image; a passage of Paul’s letter to the Corinthians in the Vulgate, for example, uses *speciem* as a kind of embodied, physical vision that is distinct from faith. One walks toward God, Paul says, “by faith, not by sight” (*per fidem enim ambulamus et non per speciem*).²³ It is logical, therefore, to use *speciem* in a passage about physical form and to understand the idea of sight as a process of unveiling.

The second part of the passage presents a dizzying vision: glowing brocades, paradisial walls, light from the windows. It is a description of effect, conveyed with a series of clauses that begin with “if,” or *si*. Vision, then, is divided into the physical—be it ceiling, wall, or windows—and the effect it creates: its glow, its similarity to paradise, its beauty. In the last clause the meter slows, and then Theophilus once again returns to workmanship and variety. We see here what is sometimes described as the paradox of medieval viewing: images are often understood to be transcendent, to lead to God, but at the same time they are criticized as earthly, transient objects, fabricated by man and possibly idolatrous.²⁴ This passage might be read according to this duality, with a sharp distinction between what is visible and the effect it creates. Read in the context of the entire book, however, and as the culmination of a long lesson in artistic technique and spiritual learning, it has rather different implications, for it links together making and viewing. Vision is the recognition of skill and craft.

The idea of compunction comes through in the last section. This is one of very few places where Theophilus mentions imagery per se: “But if, perchance, the faithful soul observes the representation of the Lord’s Passion expressed in art, it is stung with compassion (*compungitur*).” The passage suggests that the emotions displayed in an image—be it of the passion of Christ, the joys of heaven, or the torments of hell—can move the soul. Vision functions both physically and spiritually; like craft itself, it is an embodied, moralized experience. The passage is linked to Augustine’s division of sight into perception and understanding. Theophilus begins by describing what is seen and then claims that an image can evoke an emotional response, a yearning for God. Craft is key here; it is a vehicle. Variety and workmanship are what enable the effect to begin, and—as marveling at the beauty leads to emotional reaction—compunction to occur.

The sequence demonstrates that vision is a complex and theorized phenomenon for Theophilus. It begins with form and then moves to effect and compunction. It is not dissimilar from Hugh of St. Victor’s three modes of seeing, in which there is the eye of the physical, fleshly world; the eye of reason that is the movement of the soul; and the eye that is close to God.²⁵ As Theophilus describes it, the artist enables the viewer to “praise the Creator in the creature and proclaim Him wonderful in His works.” This attests to the workmanship of the object and makes a connection between the recognition of skilled labor in the object and the recognition of God. For example, an educated viewer seeing the raised height and textured surface of the repoussé eagle on the book cover at Trier might view the work as not simply a display of virtuosity but also the result of a prolonged procedure, the trace of a

series of repeated actions that require both sustained mental focus and physical dexterity. When such actions and technical skill are moralized, the well-made object can be read as the trace of God's presence, because the trace of virtuous behavior, rationality, and skill are evidence of the work of the Holy Spirit.

Craft, as we see it here, traverses the procedural and the visual. Theophilus teaches his reader to look for careful procedure, skill, purity of material, and calculated form; this is not only craftsmanship, but the trace of God. Vision, too, is therefore moralized, and Theophilus's reader becomes a sophisticated viewer. Theophilus's treatise seeks to teach its reader about artistic technique, and it frames this learning in spiritual terms. By extension, it makes the practice of seeing a part of participation in God. In the first prologue, Theophilus makes a clear distinction between man who is made in the image of God and man's likeness to God, and it is here that he invokes the concept of participation: "In the account of the creation of the world, we read that man was created in the image and likeness of God and was animated by the divine breath, breathed into him. By the eminence of such distinction, he was placed above other living creatures, so that, capable of reason, he acquired participation in the wisdom and skill of the Divine Intelligence, and, endowed with free will, was subject only to the will of his Creator."²⁶ The Holy Spirit makes action and learning possible, and it is man's reason and will that allow him to recognize the operation of the Holy Spirit and align his actions with the sacred work. Thus for Theophilus, action on earth leads to, and is a part of, the yearning for God.

In attributing all action to the grace of the Spirit, Theophilus is drawing directly on

Augustine, who sees the Holy Spirit as the dwelling of the whole Trinity in man, binding together God the Father and God the Son through love and enabling man's entry to the divine through Christ, the gift of that same love.²⁷ As the word made flesh, Christ enabled man to be redeemed. The possibility of redemption opens the door for man to participate in God, which means cleaving to God; this is man's fulfillment of his potentiality, or, as Mary Noreen Rita Marrocco explains it, the activation of his image-quality.²⁸ While ultimate participation can come only through death, man can turn himself to God in life. Theophilus writes that before the fall, man could participate in God's rationality, and after the fall, man's rationality is the basis of his image-quality, which enables him to turn toward God and bring himself closer to him.

For Augustine, then, the Holy Spirit is a catalyst and a vehicle for virtuous action on earth. The virtues of charity and love, for example, operate in man because of the Holy Spirit; when man acts with love, he becomes closer to God.²⁹ Augustine's *On Nature and Grace*, which refutes Pelagius's ideal of human perfection and emphasizes the role of the Holy Spirit, deals with this problem directly:

This indeed he, through grace, received from God, since he did not have it in a nature which had become corrupted and depraved,—even as we read in the Gospel, "But as many as received Him, to them gave He power to become the sons of God"; which they were not by nature, nor could at all become, unless by receiving Him they also received power through His grace. This is the power which is claimed for itself by the fortitude of that love which is only

communicated to us by the Holy Ghost bestowed upon us.³⁰

Because the action of the artist has the potential to be the practice of virtue, it has the potential, if performed well, to be the operation of the Holy Spirit in man and a vehicle for man's participation in God.³¹

The idea of participation drives Theophilus's entire concept of making, as learning and action become pursuits of grace. It expands Augustine's notion of participation by extending it to his concept of vision. The spiritual aspect of learning, where recognition of good action is recognition of the Holy Spirit at work, makes it possible for Theophilus's learned reader, who has "read through these things several times and committed them to memory," to recognize good craftsmanship in an object and value it as an operation of the Holy Spirit, thus deriving spiritual benefit from the experience. This reader also recognizes Theophilus's work in the treatise and prays for him. Similarly, Theophilus's description of seeing, and of its progression from form to effect and compunction, depends on the marveling at the variety and richness of the craft, and, especially, recognizing God in that work: "You have given them cause to praise the Creator in the creature and proclaim Him wonderful in His works."³²

For Augustine, even temporal things could lead to participation in the divine. It is when such temporal things are driven by, and framed according to, the contemplation of the eternal, that they best fulfill their highest purpose. It is perhaps precisely this contemplation of the eternal that Theophilus invokes when he describes the effect of imagery and the idea of compunction. Vision, like craft, is moralized, and the

apparent contrast between vision and effect is resolved.

It is perhaps no surprise, then, that the two main characters of the last prologue, David and Solomon, are presented as model viewers and patrons. The use of these figures as exempla was not uncommon in the twelfth century: David was a model for repentant devotion and prayer and Solomon for the building of the temple.³³ Theophilus, indeed, describes the building of the temple, but he praises David as the originator of the project. David, Theophilus writes, "applied himself with the full force of his mind to the love of his Creator," and thus "uttered these words: 'Lord I have loved the beauty of Thy House.'"³⁴ He was able to recognize beauty and appreciate it because of his devotion to and love of God. It is perhaps not surprising, then, that David is only mentioned in the last prologue. Theophilus's high praise for David suggests that vision itself is something to be learned: David had to learn to see God and beauty. Not every viewer is equal in his ability to see God, nor is every viewer equal in his ability to recognize beauty.

Theophilus's passage on beauty has become famous because it seems to be a justification of art-making. Theophilus praises the embellishment of the Temple of Solomon, the artists of the tabernacle, and the quality of their work. These artists, Beseleel and his assistant Ooliab, are "filled . . . with the spirit of wisdom and understanding and knowledge in all learning for contriving and making works in gold and silver, bronze, gems, wood and in art of every kind."³⁵ It is only in the highest stages of the quest for wisdom, when the artist has achieved the gift of understanding and acquired a variety of virtuous skills, that Beseleel can be invoked as the model artist. Similarly, it is in the last prologue, in the

highest stages of the journey toward wisdom, that Theophilus invokes David and Solomon as exemplary viewers.

According to the theory of participation, not every being is equal, even though every made thing eventually contains God. Just as man participates in the intelligence of God while beasts participate only in the being-ness of God, so too objects are hierarchized and evaluated. Some are the trace of participation in God and some are not, depending on the presence of skill to which they attest. Accordingly, some viewers can recognize them as such, some cannot.

Theory, Practice, and Craft

On Diverse Arts has long been read as a practical treatise, a religious text, or a third voice in the debates between Bernard of Clairvaux and Abbot Suger of St. Denis. While these interpretations are true, in part, they are but pieces of the story. Physical evidence from the manuscripts shows us how *On Diverse Arts* was read, what interested its readers, and, in some cases, for whom it was written, revealing patterns of readership that bring to light the broader cultural values at stake in the book. Discrepancies exist—the Vienna manuscript seems to have been kept in a studio, and the Wolfenbüttel one in a library, for example—but a broad view of manuscript patterns asks us to reconsider whether these are discrepancies in the first place. Similarly, traces of readership and evidence from manuscripts can help us understand the literary realm in which the tract was composed and read. The Wolfenbüttel copy was read alongside, or even following, Vitruvius; the Vienna copy was kept alone as a pamphlet and perhaps as a book

of memory, containing scattered drawings of ornamental motifs and identifying Theophilus as the figure Roger; and the Harley manuscript preserves markings in the prologues that single out passages linking the practice of art to a spiritual agenda. These copies of *On Diverse Arts* suggest that it was not just another in a long line of technical treatises but a highly learned tract, probably owned by a practicing artist. Examined in light of the exegetical and theological sources that permeated the Benedictine realm, and with which the treatise shows familiarity, *On Diverse Arts* emerges as an articulation of the Christian, and particularly monastic, values embedded in artistic practice. This presumed technical treatise, therefore, reveals the discourses and values of its time.

C. Stephen Jaeger has argued that the eleventh and twelfth centuries saw a shift in primary modes of teaching and learning. Schools of the eleventh century were dependent on charismatic teachers and taught mores and manners, while twelfth-century learning, led by figures like Peter Abelard, made debate and dialogue a primary pedagogic mode.³⁶ *On Diverse Arts* is situated at this crucial moment of transition, for while Theophilus extols the virtues of actions, he presents his ideas through the medium of writing. The notion that outer action forms the inner soul is a fundamental assumption of the treatise, yet the primacy of the physical is, through the medium of text, transformed into an orderly body of knowledge and a hierarchical system of activities. It is thus perhaps anachronistic to read any technical treatise as a straightforward description of artistic activity or a nostalgic transcription of past events for posterity. *On Diverse Arts* belies an interest in the text itself: its structure, its vocabulary, and its descriptions. It

also shows what a text can do: how it might be read, circulated, and memorized. Theophilus is keenly aware that writing is an activity of significance. Manner and a charismatic persona shape our idea of the humble, pseudonymic, monastic author in the treatise, yet they are used as the basis for an elaborate written theory of art.

In the tumultuous first half of the twelfth century—as bishops and abbots jockeyed for position, as the mercantile and lay economies grew ever more active and the number of lay artisans grew, as monastic circles debated and tolerated a series of movements for reform—Theophilus’s systematizing of the knowledge of art might be seen as an attempt to control a practice that was increasingly outside Benedictine influence. In the region of the Rhine-Meuse in particular, the growth of trade, exchange of metals, and establishment of “burgher” cities like Huy seem to have also given rise to a new class of artists, for it is precisely from this region that the names Godefried and Reiner, and slightly later, Eilbertus of Cologne, are known. By the end of the twelfth century, the situation for artists seems rather different: Hugo d’Oignies and Nicholas of Verdun are the best examples of this, for their names are better remembered and a body of works are more closely associated with them. These two goldsmiths have since become personas in their own right, elevated as figures whose virtuosity and fame foreshadow the artistic activity of the Renaissance.

On Diverse Arts defines artistic practice as a learned activity that operated according to a monastic Christian system of values. By expanding the traditional “recipe” book into a treatise covering a number of different media, by moralizing skill and the acquisition of material, and by tying technical instructions not only to liturgical

objects but also to spiritual ladders of learning, Theophilus defines all aspects of artistic practice as a distinctly Benedictine activity. Thus we begin to see that the question of whether or not the treatise was used for practical purposes or theoretical purposes is a modern one. Conceptual divisions can, and do, remain, but the very purpose of *On Diverse Arts* is to valorize practice, to show how artistic practice makes monastic values manifest. In the end, we see that for the skilled artist, for the educated viewer, and for the reader of *On Diverse Arts*, the object is more than its material substance and more than the images it portrays. As a finely made product of labor, of skill, of learning, of devotion, it is an eloquent testament to man’s participation in the divine.

NOTES

INTRODUCTION

1. Belting, *Likeness and Presence*, 9; Arnold, *What Is Medieval History?*, 20–22.
2. See, for example, Williams and Overbey, *Transparent Things*, i–iii, and Hahn, *Strange Beauty*, 3–8.
3. For example, Kessler, *Neither God nor Man*, 26–38.
4. Theophilus, *De diversis artibus* (ed. Dodwell), 1. All quotations and references, unless otherwise specified, are taken from Dodwell's edition of the text, abbreviated hereafter as *DDA*.
5. Lessing, *Vom Alter der Oelmalerey aus dem Theophilus Presbyter*, 159–97, and “Theophili Presbyteri Diversarum Artium Schemata,” 291–424.
6. Leventhal, *Disciplines of Interpretation*, 10–12 and 71–92; Vasari, “Of Divers Flemings,” in *Lives of the Most Eminent Painters*, 9:34.
7. Raspe, *Critical Essay on Oil Painting* (1781); Eastlake, *Methods and Materials* (1847); and Merrifield, *Original Treatises* (1849).
8. Theophilus, “Essai sur divers arts” (ed. Bourassé), *PL*, cols. 729–1014 (1863); *Libri 3 seu Diversarum artium schemata* (ed. De L'Escalopier; 1843); *Diversarum artium schemata liber secundus Theophilus* (ed. Bontemps; 1876).
9. Theophilus, *Schemata Diversarum Artium* (ed. Ilg; 1874) and Eitelberger and Ilg, *Quellenschriften für Kunstgeschichte* (1871–82), reviewed by A. L. F. Jr. in *The American Journal of Archaeology and of the History of the Fine Arts* 5, no. 1 (1889): 63–64; Von Schlosser, *Die Kunstliteratur*, 22–25.
10. See Von Schlosser, *Die Kunstliteratur*, 20–27; “*Mappae Clavicula*” (ed. Smith and Hawthorne), 3–128. For the manuscript tradition of the text, see also Johnson, “Notes on Some Manuscripts,” 72–81; and “Some Continental Manuscripts,” 84–103.
11. Heraclius, *Von den Farben und Künsten der Römer*, translated into English in Merrifield, *Original Treatises*, 2:166–257; see also Merrifield's introduction to her translation, 1:166–80, and Von Schlosser, *Die Kunstliteratur*, 21–22. On the *Lumen animae*, see *Light of the Soul*, 17; Raspe, *Critical Essay on Oil Painting*, 145; and Rouse and Rouse, “Texts Called *Lumen Anime*,” 5–113.
12. For example, Villela-Petit, “Copies, Reworkings and Renewals,” 167–81; Tosatti, *Trattati medievali di tecniche artistiche*, 61–96; Scholtka, “Theophilus

- Presbyter," 1–53; and Bischoff, "Die Überlieferung der technischen Literatur," 277–97.
13. For example, Buckton, "Theophilus and Enamel," in Buckton and Heslop, *Studies in Medieval Art and Architecture*, 1–13; and Raub, "How to Coat Objects with Gold," 101–10.
 14. Cerquiglini, *In Praise of the Variant*, 21–22; Nichols and Wenzel, *Whole Book*, 1–6.
 15. For example, Speer and Westermann-Angerhausen, "Ein Handbuch mittelalterlicher Kunst?" in Stiegemann and Westermann-Angerhausen, *Schatzkunst am Aufgang der Romanik*, 249–58; Virginia C. Raguin, "The Reception of Theophilus' *De diversis artibus*," in Boulanger and Hérol, *Le Vitrail et les traités du Moyen Âge à nos jours*, 11–28; and in the same volume, Brigitte Kurmann-Schwarz, "[...] quicquid discere, intelligere vel excogitare possis artium [...]": Le traité *De diversis artibus* de Théophile, état de la recherche et questions," 29–44.
 16. For example, Bruno Reudenbach, "Werkkünste und Künstlerkonzept in der *Schedula* des Theophilus," in Stiegemann and Westermann-Angerhausen, *Schatzkunst am Aufgang der Romanik*, 243–48, esp. 244–45; Sears, "Afterlife of Scribes," 75–96; and Curtius, *European Literature in the Latin Middle Ages*, 407–13, 515–18.
 17. Hourihane, introduction to *From Minor to Major*, xvii–xxiii, and in the same volume, Paul Binski, "London, Paris, Assisi, Rome Around 1300: Questioning Art Hierarchies," 3–21, here, 3–4.
 18. See, most recently, Hamburger, "Hand of God and the Hand of the Scribe," 55–80; Nees, "Originality of the Early Medieval Artists," 77–109; and Hanning, "*Ut enim faber . . . sic creator*," 95–149.
 19. Kittler, "History of Communication Media," 66–81. My thanks to Ken Rogers for this reference. See also Kwakkel, "The Cultural Dynamics of Medieval Book Production," 243–52; and Rouse and Rouse, *Authentic Witnesses*, 1–12.
 20. Panofsky, *Idea: A Concept in Art Theory*, 50.
 21. For example, Hamburger, "The Place of Theology in Medieval Art History: Problems, Positions, Possibilities," in Hamburger and Bouché, *The Mind's Eye*, 11–12; and Burns, "Cennino Cennini's *Il Libro dell'Arte*," 1–13.
 22. Hugh of St. Victor, book II, chap. 1 of *Didascalicon* (trans. Taylor), 62.
 23. Binski, "London, Paris, Assisi, Rome," 3–4.
 24. Speer and Westermann-Angerhausen, "Ein Handbuch mittelalterlicher Kunst?" in Stiegemann and Westermann-Angerhausen, *Schatzkunst am Aufgang der Romanik*, 249–58; Dodwell, "Introduction," *DDA*, xviii–xliv.
 25. Wolfenbüttel, HAB cod. Guelph Gudianus lat. 2°69.
 26. Vienna, ÖNB Cod. 2527.
 27. London, British Library, Harley MS 3915.
 28. Brussels, BR/KB MS 10147–58.
 29. Cambridge, University Library MS Ee. 6.39, and London, BL Egerton MS 840a.
 30. Leipzig, Universitätsbibliothek MS 1157 (formerly Karl Marx Universitätsbibliothek MS 1144; Saxony, fourteenth century); Paris, BnF MS nouv. acq. lat. 1422 (Germany, perhaps the Rhineland, fifteenth century); Amiens, Bibliothèque municipale MS fonds Lescalopier 46 (Italy, fifteenth century).
 31. Partial copies include: Oxford, Magdalene College MS 173; Florence, Biblioteca Nazionale Centrale MS Palat. 951; Paris, BnF MS lat. 6741; Wolfenbüttel, HAB Guelph Helmst. 1127 (Wolf. MS 1234); Klosterneuburg, Augustiner-Chorherrenstift cod. 331. Manuscripts containing excerpts from the text are: Munich, Bayerische Staatsbibliothek Clm. 444; London, BL Harley MS 273; London, BL Sloane MS 1754; Wrocław, Biblioteka Uniwersytecka, MS IV 8°9; Montpellier, École de Médecine MS lat. 277; Oxford, Corpus Christi College 125; Paris, BnF MS lat. 11212; BnF MS lat. 6830F. The excerpts contained in the Corpus Christi manuscript and these two latter Paris manuscripts contain chapters that are not exclusively attributable to Theophilus. Two manuscripts in Rome, Biblioteca Apostolica Vaticana, Reg. 2079 and Urbin. Lat. 293, have been said to contain Theophilus but do not.
 32. These are Venice, Biblioteca Nazionale Marciana MS lat. VI, 199 (3597) and Vienna, ÖNB Cod. 11236.
 33. London, BL Sloane MS 781; Cambridge, University Library MS Ee. 6.39.
 34. De L'Escalopier, *Théophile: Essai sur divers arts*. The manuscript copies now in Amiens are Bibliothèque municipale, L'Escalopier MSS 47 and 117.

35. Speer and Westermann-Angerhausen, "The *Schedula diversarum artium*—A Digital Critical Edition" (<http://schedula.uni-koeln.de>) and the essays in Speer, with Mauriège and Westermann-Angerhausen, *Zwischen Kunsthandwerk und Kunst*; see also Gearhart, "Theophilus' *On Diverse Arts*."
36. Vienna, ÖNB Cod. 2527, fol. 1r.
37. For recent work questioning other assumptions—such as gender—in medieval artistic practice, see also Therese Martin, "The Margin to Act: A Framework of Investigation for Women's (and Men's) Medieval Art-Making," *Journal of Medieval History* 42, no. 1 (2016): 1–25, and Therese Martin, ed., *Reassessing the Roles of Women as "Makers" of Medieval Art and Architecture*, Visualising the Middle Ages 7 (Leiden: Brill, 2012).
38. Snijders, *Manuscript Communication*, especially "Contextualisation and Monastic Networks," 241–81, here, 242–44.
39. *Ex miraculis sancti Modualdi auctore Stephano*, MGH SS 8:223–26; *Translatio sancti Modoaldi, Miracula s. Modoaldi*, MGH SS 12:284–315.
40. Freise, "Roger von Helmarshausen in seiner monastischen Umwelt," 180–293; "Roger von Helmarshausen ein Maaslandischer Künstler und Mönch in Westfalen," 287–307; and "Zur Person des Theophilus" in Legner, *Ornamenta Ecclesiae*, 1:357–62.
41. Blumenthal, *The Investiture Controversy*, 43.
42. Van Engen, "Rupert of Deutz (ca. 1075–1129)," 42.
43. The charter will be discussed in further detail in chapter 4. The document is now conserved as Paderborn, Erzbistumsarchiv, Generalvikariat, Urkunde 11.
44. Leyser, "The Crisis of Medieval Germany," 21–50.
45. See Constable, *The Reformation of the Twelfth Century*, 46–48; Gerhart B. Ladner, "Terms and Ideas of Renewal" in Benson and Constable, *Renaissance and Renewal in the Twelfth Century*, 1–33; for a counterargument, see Van Engen, "'Crisis of Cenobitism' Reconsidered," 269–304.
46. Van Engen, "'Crisis of Cenobitism' Reconsidered," 269–304.
47. Constable, *The Reformation of the Twelfth Century*, 46–47; Berman, "Were There Twelfth-Century Cistercian Nuns?," 836; and Constable and Smith, introduction to *Libellus de diversis ordinibus*, xx–xxi.
48. *Libellus de diversis ordinibus*.
49. White, "Theophilus Redivivus," 224–33.
50. Rudolph, *The "Things of Greater Importance,"* 64; see also Rudolph, "Bernard of Clairvaux's *Apolo-gia*," 125–32.
51. Verhulst, *The Rise of Cities*, 56.
52. *Ibid.*, 71–72.
53. *Ibid.*, 24–33; see also 68–75, 133–35.
54. Klemm, "Die Anfänge der romanischen Buchmalerei," 465–81; Von Euw, "Zur Problematik stilverwandter Phänomene," 37–46; Lasko, "Roger of Helmarshausen, Author and Craftsman," 180–201.
55. Castelnovo, introduction to *Artifex bonus*, v–xxxv.
56. Alexandre, "Nécrologe de l'abbaye de Neufmoustier," V:280–303; VI:101–43; and esp. VI:135.
57. Van Engen, "Rupert of Deutz (ca. 1075–1129)," 39–47.
58. Jaeger, *Envy of Angels*, 49, 62–79.
59. For example, Kurz, "Zur handschriftlichen Überlieferung der Werke Hugos von St. Viktor," 469–82; and Esmeijer, "The Open Door and the Heavenly Vision," 43–56.
60. Hamburger, "The Place of Theology in Medieval Art History," 25.

CHAPTER 1

1. Wolfenbüttel, HAB cod. Guelph Gudianus lat. 2°69. The gutter margin is 1.5 cm throughout.
2. Bischoff, "Die Überlieferung des Theophilus-Rugerus," 176.
3. Wolfenbüttel, HAB cod. Guelph Gudianus lat. 2°69, fol. 1r.
4. Buzas, *Deutsche Bibliotheksgeschichte des Mittelalters*, 26. See also Pütz, *Die Bibliothek des Klosters St. Pantaleon in Köln*.
5. London, BL Harley MS 2767; Brussels, BR/KB MS 5253; Vatican, Biblioteca Apostolica Vaticana, Urb. Lat. 293 (ex-292); and Wolfenbüttel, HAB cod. Guelph Gudianus lat. 2°69. See Krinsky, "Seventy-Eight Vitruvius Manuscripts," 36–70; for London, BL Harley MS 2767:51–52; Brussels, BR/

- KB MS 5253, 44–45; Vatican, Biblioteca Apostolica Vaticana, Urb. Lat. 293 (ex-292), 65; and Wolfenbüttel, HAB cod. Guelph Gudianus lat. 2°69, 48. Jones, “The Provenance of the London Vitruvius,” 64–70; Frank Granger’s introduction to Vitruvius, *On Architecture* (ed. and trans. Granger), xvii–xix. Granger’s introduction remains valuable, though he argues the London manuscript originated at Jarrow, not Cologne, as is now generally believed.
6. *Godera(m)nus p(ro)posit(us)*, London, BL Harley MS 3915, fol. 145v; Krinsky, “Seventy-Eight Vitruvius Manuscripts,” 51–52; Jones, “The Provenance of the London Vitruvius,” 64–70, esp. 65n2; and Granger, introduction to *On Architecture*, xvi–xviii.
 7. Jones, “The Provenance of the London Vitruvius,” 65; and Granger, introduction to *On Architecture*, xvii–xix. For evidence of readership of Vitruvius in the Middle Ages, see Wirth, “Bemerkungen zum Nachleben Vitruvs,” 281–91.
 8. For example: Isidore of Seville, “*De argento*,” *De lapidis et metallis* in *Etymologies*, PL 82:585–86, and in English translation, Barney, et al., *The Etymologies of Isidore of Seville*, 330; Bede, chapter 27 (*Ordo ventorum*) in *De natura rerum*, PL 90:249, and in *Opera didascalica*, CCSL 123A:173–234; Hugh of St. Victor, book III, chapter 2 (*De auctoribus artium*) of *Didascalicon*, PL 176, col. 766, and in Hugh of St. Victor, *Didascalicon* (ed. and trans. Jerome Taylor), 84; Hucbald of St. Amand, on the winds (*De harmonica institutione*), PL 132:946; and William of Malmesbury, book II of *Gesta regum anglorum*, PL 179, col. 1140d. See also Krinsky, “Seventy-Eight Vitruvius Manuscripts,” 36–37.
 9. Fols. 27v, 98v, 144v, and 159r. Jones, “The Provenance of the London Vitruvius,” 65; and Krinsky, “Seventy-Eight Vitruvius Manuscripts,” 51. The marginal notes have not been published.
 10. Wolfenbüttel, HAB cod. Guelph Gudianus lat. 2°69, fol. 6v, 59r.
 11. “*Ut Vitruvius dixit in libro de Architectura, secundum physicos non plus sunt, quam octo venti: principales quatuor, et subjecti quatuor.*” Hucbald of St. Amand, *De harmonica institutione*. See also Schuler, *Vitruv im Mittelalter*, 49–51.
 12. London, BL Harley MS 2767, fol. 16v.
 13. Schuler, *Vitruv im Mittelalter*, 4–6.
 14. My thanks to Helmut Puff for bringing this to my attention. Minnis, Scott, and Wallace, *Medieval Literary Theory and Criticism*, 37–39 and 65–71. See also Quain, “The Medieval Accessus ad Auctores,” 215–64. Hildegard of Bingen’s *Physica* is a rare exception to this rule.
 15. Wolfenbüttel, HAB Guelph Gudianus lat. 2°69, fol. 85v.
 16. See Minnis, *Medieval Theory of Authorship*, 18–25.
 17. Vitruvius, *De architectura*, 21.
 18. For example, Alcuin and Einhard. Schuler, *Vitruv im Mittelalter*, 48–49.
 19. Bischoff, *Manuscripts and Libraries in the Age of Charlemagne*, 148.
 20. Dolbeau, “Un nouveau catalogue des manuscrits de Lobbes” (1979), 191–248, esp. 13 and 30–31; Rouse and Rouse, “*Florilegia* and the Latin Classical Authors of Twelfth- and Thirteenth-Century Orléans,” 131–60; Taylor, “Medieval Proverb Collections,” 19–35.
 21. Avranches Bibliothèque municipale MS 235, dating to the twelfth century; Paris BnF 6830F and Paris BnF 11212, both twelfth- or thirteenth-century.
 22. John of Salisbury, prologue to the *Metalogicon*, 5–6.
 23. Colish, *Medieval Foundations of the Western Intellectual Tradition*, 178.
 24. Reynolds and Wilson, *Scribes and Scholars*, 110–15. A manuscript now in Berlin and copied at Corvey in the twelfth century may be Wibald’s manuscript (Berlin Lat. 20 252).
 25. Hugh of St. Victor, *Didascalicon*, especially book VI (135–51 in Taylor). See also Colish, *Medieval Foundations of the Western Intellectual Tradition*, 175–82, esp. 176–77.
 26. See, for example, Rouse and Rouse, “*Florilegia* and the Latin Classical Authors,” 131–60.
 27. *DDA*, 62–63.
 28. For example, McCague, “Le don des métiers,” 47–66; Van Engen, “Theophilus Presbyter and Rupert of Deutz,” 147–63; Reudenbach, “Werk-künste und Künstlerkonzept,” 243–48; Mariaux, “La ‘double’ formation de l’artiste,” 42–45; and Roland Sanfaçon, “Le sacré et le vitrail dans les traités du Moyen Âge” in Boulanger and Hérol, *Le Vitrail et les traités du Moyen Âge à nos jours*, 133–48.

29. Isaiah 11:1–3.
30. Ecclesiasticus 1:16.
31. Augustine, book I, chap. 3 of *The Lord's Sermon on the Mount* (trans. Jepsen), 17.
32. Ibid.
33. Rupert of Deutz, book IV of *De gloria et honore filii hominis super Mattheum*, CCCM 29.
34. DDA, 62: "Quapropter, fili dilectissime, non cuncteris, sed plena fide crede spiritum Dei cor tuum implesse, cum eius ornasti domum tanto decore tantaque operum varietate. Et ne forte diffidas, pandam evidenti ratione, quicquid discere intelligere vel excogitare possis artium, septiformis Spiritus gratiam tibi ministrare."
35. Ibid. 4.
36. Ibid. 1.
37. Compare Abbot Suger's *Ordinatio*, which begins: "Sugerus, beati Dionysii qualiscumque abbas . . .," quoted in Erwin Panofsky, *Abbot Suger*, 122.
38. Augustine, book I, chap. 1 of *De trinitate*, CCSL 50; Augustine, *On the Trinity* (trans. McKenna), 3.
39. DDA, 1.
40. Ibid.
41. Ibid. 2.
42. Augustine, book I, chap. 3 of *The Lord's Sermon on the Mount*, 17.
43. Ibid.
44. Proverbs 9:10.
45. Augustine, book I, chap. 4 of *The Lord's Sermon on the Mount*, 11.
46. DDA, 2.
47. Augustine, book I, chap. 15 of *De doctrina christiana*, CCCL 32; see also Augustine, *The Confessions, The City of God, On Christian Doctrine* (ed. Hutchins), 628.
48. Hugh of St. Victor, book I, chap. 1 of *Didascalicon*, 47.
49. DDA, 2.
50. Ibid., 3.
51. Augustine, book I, chap. 3 of *The Lord's Sermon on the Mount*, 17.
52. DDA, 4.
53. For example, Bruno Reudenbach, "Ornatus materialis domus Dei" in Beck and Hengevoss-Durküp, *Studien zur Geschichte der Europäischen Skulptur*, 1–17; Oltrogge, "Cum sesto et rigula," 67–99; and Schuler, "Campum artium perscrutari," 45–55.
54. White, "Theophilus Redivivus," 224–33, esp. 229.
55. London, BL Add. MS 28106, vol. 1, fol. 161r. The attribution has been largely accepted. See Lasko, *Ars Sacra*, 165.
56. Minnis, *Medieval Theory of Authorship*, 18–25. Examples include Conrad of Hirsau and Bernard D'Utrecht's *Accessus ad Auctores*. See also Nikolaus M. Häring, "Commentary and Hermeneutics," in Benson and Constable, *Renaissance and Renewal in the Twelfth Century*, 173–200; Reudenbach, "Werkkünste und Künstlerkonzept," 243–48; and Quain, "The Medieval Accessus ad Auctores," 215–64.
57. For example, Anselm of Canterbury, meditatio 3 of *Orationes sive meditationes*, in *Opera Omnia*, III:91.
58. DDA, 4.
59. Augustine, book I, chap. 3 of *The Lord's Sermon on the Mount*, 17.
60. Ibid.
61. DDA, 4.
62. Ibid., 36. The quotation is from Marcus Porcius Cato's book of proverbs, the *Disticha Catonis*, IV:29: "Non pudeat quae nescieris te velle doceri; Scire aliquid laus est, culpa est nihil discere velle." See Cato, *Disticha uel dicta Catonis* (ed. Baehrens), III:233.
63. DDA, 36.
64. Augustine, book I, chap. 3 of *The Lord's Sermon on the Mount*, 17.
65. Ibid., 17–20.
66. Benedict of Nursia, chapter 48 ("The Daily Manual Labor") in *RB 1980*, 249–50. See also Van Engen, "Theophilus Presbyter and Rupert of Deutz," 155–56.
67. DDA, 36.
68. London, BL Harley MS 3915 fol. 20r.
69. DDA, 37.
70. Cahn, *Masterpieces*, 7–11.
71. Augustine, book I, chap. 4 of *The Lord's Sermon on the Mount*, 20.
72. DDA, 37.
73. Matthew 5:8.
74. Augustine, book I, chap. 3 of *The Lord's Sermon on the Mount*, 18.
75. For example, Mariaux, "La 'double' formation de l'artiste," 42–45; and Reudenbach, "Ornatus

- materialis domus Dei,” 1–17; see also Sanfaçon, “Le sacré et le vitrail dans les traités du Moyen Âge,” 133–48.
76. DDA, 61.
 77. Ibid.
 78. Augustine, book I, chap. 3 of *The Lord's Sermon on the Mount*, 18.
 79. DDA, 61–62.
 80. Ibid., 62.
 81. Augustine, book I, chap. 4 of *The Lord's Sermon on the Mount*, 20.
 82. DDA, 62.
 83. Ibid., 63.
 84. Ibid., 64.
 85. For example, Walter Cahn, “Solomonic Elements in Romanesque art,” in Gutman, *The Temple of Solomon*, 45–72.
 86. See Frank, “Abbot Suger and the Temple in Jerusalem,” 109–29; and Speer, “L'abbé Suger et le trésor de Saint-Denis,” 59–82.
 87. Augustine, book I, chap. 1 of *The Lord's Sermon on the Mount*, 11–12.
 88. Schandorff's study is helpful in seeing these broad patterns: in the eighth century, they are referred to by Alcuin (d. 804) and Charlemagne (d. 814); in the ninth, Drogo of Metz, Godescalcus (d. ca. 867), Hincmar of Reims (d. 882), Ratramnus of Corbie (d. 868), and Smaragdus (fl. 809–19); in the tenth, Aelfric of Eynshaw (d. ca. 1020); and in the eleventh, Anselm of Canterbury (d. 1109), Bruno of Segni (d. 1123), Guidmundi of Aversoni, Humbertus, Cardinal (d. 1061?), Pope Leo IX (d. 1054), and Peter Damian (d. 1072). In the late eleventh and early twelfth centuries, the list increases significantly and includes, for the late eleventh century, Roscellinus (d. 1125), Anselm of Laon (d. 1117), Hildebert of Tours (d. 1134), Odo of Cambrai (d. 1113), Otto of Bamberg (d. 1139), and Rupert of Deutz (d. 1135); and for the early twelfth century, Peter Abelard (d. 1142), Anselm of Havelberg (d. 1158), Bernard of Clairvaux (d. 1153), Ernardus Bonaevalis, Gauthier de Castillon (d. 1200), Gilbert de la Porrée (d. 1154), Hugh of St. Victor (d. 1141), Joachim of Fiore (d. 1202), Peter of Blois (fl. 1190), Petrus Chrysolanus (fl. 1102–12), Peter Lombard (d. 1169), Reiner of St. Laurent, Liège (d. 1182), and Richard of St. Victor (d. 1173). For the late twelfth and early thirteenth centuries, the list is again smaller, and includes Alexander of Hales (d. 1245) and William of Auxerre (1150–1231). See Schandorff, *The Doctrine of the Holy Spirit* II:850–53. See also Dodwell, “Introduction,” DDA, xx–xxiii.
 89. The story is recounted in Numbers 20:11. Isidore of Seville, chap. 24 (*In Exodum*) of *Quaestiones in uetus testamentum*, CCSL 108B; also Isidore of Seville, *Quaestiones in veterum testamentum*, PL 83, col. 299.
 90. Haymo of Halberstadt, homily LXXVI (*feria quarta paschae*) in *Homiliarum sive concionum ad plebem in evangelia de tempore et sanctis*, PL 118, col. 474; Remigius of Auxerre, *Enarrationes in Psalmos, Psalmus Primus*; PL 131, col. 149.
 91. Rabanus Maurus, chap. 14 (*De candelabro et lucernis ejus*) of *Enarrationum in librum numerorum libri quatuor*, PL 108, col. 634.
 92. Hugh of St. Victor, chap. 1 (*Quaenam sint quinque septena in sacra scriptura contenta*) and chap. 5 (*De septem donis spiritus sancti, seorsum*) in *De quinque septenis seu septenariis opusculum*, PL 175 cols. 405b and 410c–d.
 93. Rupert of Deutz, book VIII, chap. 5–17 of *Liber de divinis officiis*, CCCM 7:272–99, and book XXVIII, chap. 6 (*In Isaiam II*) of *De sancta trinitate et operibus eius*, CCCM 23:1514–16.
 94. Rupert of Deutz, book VII, chap. 11 of *Liber de divinis officiis*, CCCM 7:2051; and book IV of *De gloria et honore filii hominis super Mattheum*, CCCM 29, line 38. See also Van Engen, *Rupert of Deutz* (1983), 304–6.
 95. Jaeger, *Envy of Angels*, especially “*Cultus Virtutum*,” 76–117.
 96. *Libellus de Willigisi consuetudinibus*, MGH SS 15.2, section 70, pp. 743 and 745; the latter translated and quoted in Jaeger, *Envy of Angels*, 78.
 97. Leo IX, *Epistolae et decreta pontifica XXII*, PL 143, col. 623c–d; translated and quoted in Jaeger, *Envy of Angels*, 107.
 98. Susanne Wittekind, “*Caput et corpus*,” in Reudenbach and Toussaint, *Reliquiare im Mittelalter*, 107–35, and “Das Alexanderreliquiare im liturgischen Festkreis von Stablo” in *Altar—Reliquiar—Retabel*, 173–224.
 99. Wittekind, “*Caput et corpus*,” 114; Wittekind has also connected the ascension of virtue to

- Augustine's *Ennarationes in psalmus* and *De timore Dei* in *Altar-Reliquiar-Retabel*, 179.
100. Cynthia Hahn, "Body Part Reliquaries," in *Strange Beauty*, 129–31, and "The Spectacle of the Charismatic Body," in Bagnoli, *Treasures of Heaven*, 163–72.
 101. *DDA*, 62.
 102. *Ibid.*
 103. Augustine, *Quaestiones in Heptateuchum*, PL 34:cxviii.
 104. Leclercq, *Love of Learning*.
 105. Hugh of St. Victor, book I, chap. 8 ("In What Man is like unto God") of *Didascalicon*, 54–55; see also Rudolph, "Inventing the Exegetical Stained-Glass Window," 399–402.
 106. Hugh of St. Victor, book I, chap. 1 of *Didascalicon*, 47; see also Taylor's introduction in the same volume, 18–19.
 107. See, for example, Grover Zinn Jr., "Minding Matter," in Nederman, Van Deusen, and Matter, *Mind Matters*, 47–67.
 108. *DDA*, 1.
 109. See, for example, Eileen C. Sweeney, "Hugh of St. Victor: The Augustinian Tradition," in English, *Reading and Wisdom*, 61–83; and in the same volume, Grover Zinn Jr., "The Influence of Augustine's *De Doctrina Christiana*," 48–60.
 110. Book I, chap. 1, "The Mixing of Colours for Nude Bodies"; chap. 2, "The Colour, Green Earth"; chap. 3, "The First Shadow Colour for Flesh"; chap. 11, "The Beards of Youths." *DDA*, 5–6, 8.
 111. Book I, chap. 17, "The Panels of Altars and Doors and Casein Glue"; chap. 18, "Glue Made from Hide and Hartshorn"; chap. 19, "Whitening Hide and Wood with Gesso." *DDA*, 16–18.
 112. Bruno Reudenbach, "Praxisorientierung und Theologie," in Baumgärtner, *Helmarshausen*, 199–218.
 113. Book II, chap. 1, "The Construction of the Kiln for Working Glass"; chap. 2, "The Annealing Kiln"; chap. 3, "The Kiln for Spreading the Glass and the Tools for the Work"; chap. 4, "The Frit." *DDA*, 37–39.
 114. Book II, chap. 17, "How to Construct Windows"; chap. 18, "Cutting the Glass"; chap. 19, "The Colour with which the Glass Is Painted"; chap. 20, "Three Shades of Colour for Highlights in the Glass"; chap. 21, "The Embellishment of a Painting on Glass"; chap. 22, "The Kiln in which Glass Is Fired"; chap. 23, "How Glass Is Fired"; chap. 24, "The Iron Moulds"; chap. 25, "Casting Comes"; chap. 26, "The Wooden Mould"; chap. 27, "Assembling and Soldering Windows"; chap. 28, "Setting Gems in Painted Glass." *DDA*, 47–58.
 115. Book II, chap. 29, "Simple Windows"; chap. 30, "How to Mend a Broken Glass Vessel"; chap. 31, "Rings." *DDA*, 58–60.
 116. Book III, chap. 1, "The Construction of the Workshop"; chap. 2, "Seating Accommodation for the Workmen"; chap. 3, "The Work Furnace"; chap. 4, "Bellows"; chap. 5, "Anvils"; chap. 6, "Hammers"; chap. 24, "Dividing up the Silver for the Work"; chap. 25, "Casting Silver"; chap. 26, "Making the Smaller Chalice"; chap. 27, "The Large Chalice and Its Mold"; chap. 32, "Refining Gold"; chap. 34, "The Same as Above"; chap. 25, "Milling the Gold"; chap. 36, "The Same in Another Way"; chap. 37, "The Same as Above"; chap. 38, "Coating and Gilding the Handles." *DDA*, 64–67, 75–90.
 117. Book III, chap. 26, "Making the Smaller Chalice"; chap. 30, "Casting Handles for the Chalice"; chap. 28, "Niello"; chap. 60, "The Repoussé Censer"; chap. 61, "The Cast Censer." *DDA*, 76–79, 80–83, 111–19.
 118. Book III, chap. 66, "Making Brass"; chap. 81, "Organs"; chap. 85, "Founding Bells." *DDA*, 124–25, 142–44, 150–58.
 119. Hugh of St. Victor, book VI, chap. 3 ("Concerning History") of *Didascalicon*, 138. See also Rudolph, *The Mystic Ark*, esp. 45–46.
 120. Hugh of St. Victor, book VI, chapter 4 ("Concerning Allegory") of *Didascalicon*, 140. See also Nicolai, *Hermeneutical Principles*, 169–70, and Zinn, "Hugh of St. Victor and the Ark of Noah," 261–72.
 121. Donatus, *Ars Grammatica* in *The Ars Minor of Donatus* (trans. and ed. Chase); Purcell, *Ars Poetica*, 22–31; Murphy, *Rhetoric in the Middle Ages*, 140–42.
 122. Leclercq, *Love of Learning*, 6–7. See also Fleischman, "Philology, Linguistics, and the Discourse of the Medieval Text," 19–37.
 123. See Curtius, *European Literature and the Latin Middle Ages*, 76–77; and, still fundamental for

the subject, McGarry, "Educational Theory in the Metalogicon of John of Salisbury," 659–75.

124. John of Salisbury, *Metalogicon*, 10–11.
125. Rupert of Deutz, book I, chap. 15 (*De annulo*) of *Liber de divinis officiis*, CCCM 7:20–21.
126. Van Engen, "Theophilus Presbyter and Rupert of Deutz," 147–63.
127. Esmeijer, "The Open Door and the Heavenly Vision," 43–56; Von Euw, "Rupert von Deutz und das Lütticher Taufbecken," 217–44.
128. John of Salisbury, book III, chap. 4 of *Metalogicon*, CCCM 98, translated and quoted in Colish, *Medieval Foundations of the Western Intellectual Tradition*, 178.
129. Jaeger, "Pessimism in the Twelfth-Century 'Renaissance,'" 1151–83.

CHAPTER 2

1. Manuscripts dating before 1300 include Wolfenbüttel, HAB Guelph Gudianus lat 2°69; Paris, BnF MS 6830; Paris, BnF MS 11212; London, BL Egerton MS 840a; London, BL Harley MS 3915; and Cambridge, University Library MS Ee. 6.39. The fourteenth-century copies are Leipzig, Universitätsbibliothek MS 1157, Montpellier, École de Médecine MS lat. 277, and Munich, Bayerische Staatsbibliothek CLM. 444.
2. Dodwell, "Introduction," *DDA*, lxvi.
3. Bos and Menschung, "Macer Floridus: A Middle Hebrew Fragment with Romance Elements," 17–18, 17.
4. Cambridge, University Library MS Ee. 6.39. Fols. 1r–2r are blank; 2v includes the table of contents; and the text of Palladius begins on 3r. The first quire includes folios 1–8.
5. Cambridge, University Library MS Ee. 6.39, fol. 2v; see also *A Catalogue of the Manuscripts Preserved in the Library of the University of Cambridge* 2:276–77.
6. See for example, the serifs and tails of letters such as *h*, in Macer fol. 1v and Palladius fol. 31r.
7. Dodwell, "Introduction," *DDA*, lxvi.
8. "Prologus primi libri Theophili monachi et presbyteri, de diversis artibus in primis de coloribus," Cambridge, University Library MS Ee. 6.39, table of contents.
9. *Mappae Clavicula* (ed. Phillipps), 187; "Mappae Clavicula" (ed. Smith and Hawthorne), 26; and Johnson, "The Manuscripts of the Schemata" 87.
10. For example, see Mary A. Rouse and Richard H. Rouse, "Ordinatio and Compilatio Revisited," in Jordan and Emery, *Ad litteram*, 113–34.
11. Reynolds and Marshall, *Texts and Transmission*, 287.
12. Hunt, "English Learning in the Late Twelfth Century," 19–42, esp. 20–22, and Charland, "Artes praedicandi," *contribution à l'histoire de la rhétorique au moyen âge*, 23–24.
13. Platearius, "The Rationalization of Pharmacy," in Grant, *A Source Book in Medieval Science*, 787; see also Kristeller, "The School of Salerno," 513.
14. Montpellier, École de Médecine MS lat. 277.
15. For example, Bede, book III, chap. 37, section 21, verse 19 (*Fundamenta muri civitatis omni lapide pretioso ornate*) in *Explanatio Apocalypsis*, CCSL 121A, 531. Marbode of Rennes, *Marbode of Rennes' "De Lapidibus"* (ed. Riddle), 7–14, 131–32, and Giliberto, "Stone Lore in Miscellany Manuscripts," 258, 272–75.
16. London, BL Egerton MS 840a, formerly Cambridge, Trinity College MS R 15 5. See Watson, "A St. Augustine's Abbey, Canterbury, Manuscript" in *Medieval Manuscripts in Post-medieval England*, 211–17. See also Dodwell, "Introduction," *DDA*, lxvii; and Johnson, "The Manuscripts of the Schemata," 94.
17. Schuler, *Vitruv im Mittelalter*, 4–5.
18. Ibid.
19. Hucbald of St-Amand, *De harmonica institutione*, PL 132:945–46.
20. Ibid.
21. Schuler, *Vitruv im Mittelalter*, 50.
22. Tübingen, University Library manuscript on deposit from Berlin Staatsbibliothek, where it was cod. lat. fol. 601. Krinsky, "Seventy-Eight Vitruvius Manuscripts," 48. I have been unable to locate this manuscript.
23. William of Malmesbury, *Gesta regum Anglorum*, PL 179:1140d–1141d. See also Bittermann, "The Organ in the Early Middle Ages," 390–410.

24. London, British Museum Add. 38818. Krinsky, "Seventy-Eight Vitruvius Manuscripts," 49; and J. P. Gilson, *Catalogue of Additions to the Manuscripts in the British Museum*, 251–55.
25. See Hugh of St. Victor, book III, chap. 2 ("Concerning the Authors of the Arts") of *Didascalicon*, 84.
26. Ibid. The original Latin is instructive: "*Mechanica diversos habuit auctores. Hesiodus Ascræus primus apud Graecos in describendis rebus rusticis studuit. Deinde Democritus; Mago quoque Carthaginensis XXVII, voluminibus studium agriculturæ conscripsit; apud Latinos primus Cato opus De agricultura instituit, quod deinde Marcus Terentius Varro expolivit. Virgilius quoque Georgica fecit. Deinde Cornelius et Julius Atticus Aemilianus, sive Columella insignis orator, qui totum corpus disciplinae hujus complexus est. Palladius quoque De agricultura scripsit; Vitruvius autem De architectura.*" PL 176:766c–766d.
27. Cambridge, University Library MS Ee. 6.39.
28. Cambridge, Corpus Christi MS 297, fols. 23v–24r. See René Martin's introduction to Palladius, *Traité d'agriculture*, xxxvii.
29. Paris, BnF MS lat. 11212, and Paris, BnF MS lat. 6830 F.
30. Gesner, *De vitro & speculis* in book XIII (*de mechanicis et aliis illiteratis artibus*) of *Pandectarum*, 28.
31. Simmler, *Epitome, bibliothecae Conradi Gesneri*, 173r–v.
32. Agricola, *Bermannus*, 95; see also William McPeak, review of Halleux and Yans's edition of *Agricola's Bermannus (le mineur)*, 124–25; and Baxandall, "Rudolph Agricola and the Visual Arts," 409–19.
33. Agricola, book I of *De re metallica*, 1–24; see also Hannaway, "Georgius Agricola as Humanist," 553–60.
34. Haye, *Das Lateinische Lehrgedicht*, 169–70.
35. See also Leipzig, Universitätsbibliothek MS 1157, Munich, Bayerische Staatsbibliothek Clm. 444; and Montpellier, École de Médecine MS lat. 277.
36. Macer, *De viribus herbarum*, lines 161–78, quoted and translated in Bos and Mensching, "Macer Floridus," 43.
37. Murdoch, "Charms, Recipes, and Prayers," in *German Literature of the Early Middle Ages*, 52–77. See also Peter Murray Jones, "Image, Word, and Medicine in the Middle Ages," and Jean A. Givens, "Reading and Writing the Illustrated *Tractatus de herbis*, 1280–1526," both in Givens, Reeds, and Touwaide, *Visualizing Medieval Medicine and Natural History*, 1–24 and 115–45, respectively.
38. *Mappae Clavicula* (ed. Phillipps), chap. 97; Smith and Hawthorne, "*Mappae Clavicula*," 41.
39. Ibid.
40. Murdoch, "Charms, Recipes, and Prayers," 52–77.
41. These are chapters 285–87 of *Mappae Clavicula* (ed. Phillipps), 241; Smith and Hawthorne, "*Mappae Clavicula*," 71.
42. Hildegard of Bingen, book IX of *Physica* (trans. Throop), 238.
43. Ibid.
44. *DDA*, 33.
45. Lambert of St. Omer, *Liber Floridus*, Ghent, University Library MS 92, fol. 228v.
46. Faith Wallis, "Medicine in Medieval Calendar Manuscripts" in Schleissner, *Manuscript Sources of Medieval Medicine*, 125.
47. London, BL Egerton MS 840a.
48. Dodwell, "Introduction," *DDA*, lxxvii.
49. Adelard of Bath, *Quaestiones naturales* 1 in *Adelard of Bath, Conversations with His Nephew*, 92–93.
50. *DDA*, 4.
51. Marilyn Jenkins, cat. no. 43, "Box," in *The Art of Medieval Spain, A.D. 500–1200*, 97; George, "Un reliquaire, 'souvenir' du pèlerinage des Liégeois à Compostelle en 1056?"; Pirenne-Hulin, "Textiles du Moyen âge de l'ancien diocèse de Liège," 15–26.
52. For example, Hoffman, "Translating Image and Text," 584–623.
53. The titles appear in Vienna, ÖNB Cod. 11236, fol. 1r; and Venice, Biblioteca Nazionale Marciana, MS lat. VI, 199 (3597), unfoliated title page: "I. *de temperamentis colorum*, II. *de arte vitriana*, III. *De arte fusili*." For modern editions, see Theophilus, *On Divers Arts* (ed. and trans. Hawthorne and Smith), 10, 45, 75; the earliest text to use the titles in this way is Raspe, *A Critical Essay on Oil Painting*, 45–46.
54. Wolfenbüttel, HAB Cod. Guelph Gudianus lat. 2°69, fols. 86r–v, 92v–93r, 97v–98v; Vienna, ÖNB Cod. 2527, fol. 8v–9r.

55. Wolfenbüttel, HAB Cod. Guelph Gudianus lat. 2°69, fol. 85v.
56. London, BL Egerton MS 840a, fol. 6r: "Hic incipit tractatus Lumbardicus: Qualiter temperantur colores ad depindendum."
57. Cambridge, University Library MS Ee. 6.39, quire xvii, fol. 1r.
58. *DDA*, 4.
59. Theophilus treats ivory-carving in book III, chap. 93; on gems, book III, chaps. 95–96; and their settings, book III, chap. 53; on the organ, book III, chaps. 81–84. *DDA* 166–67, 168–71, 104–5, 142–50.
60. Reudenbach, "Praxisorientierung und Theologie," 199–218.
61. Book I, chap. 1, "The Mixing of Colours for Nude Bodies." *DDA*, 5.
62. Book I, chap. 2, "The Colour, Green Earth." *DDA*, 5.
63. Book I, chap. 1, "The Mixing of Colours for Nude Bodies"; chap. 3, "The First Shadow Color for Flesh"; chap. 10, "The Hair of Boys, Youths, and Young Men." *DDA*, 5–6, 8.
64. Book I, chap. 3, "The First Shadow Colour for Flesh," and chap. 4, "The First Rose Colour." *DDA*, 5–6.
65. Book I, chap. 7, "The Second Shadow Colour for Flesh." *DDA*, 7.
66. Book I, chap. 9, "The Second Highlight." *DDA*, 8.
67. Saint Luke, Stavelot Bible, London, BL Add. MS 28107, fol. 161v. See also Thomas E. Dale, "Transcending the Major/Minor Divide," in Hourihane, *From Minor to Major*, 23–42, esp. 33–34.
68. Book II, chap. 5, "The Vessels for the Work and the Fusing of White Glass" and chap. 6, "How Sheet Glass Is Made." *DDA*, 40–41. Cf. Strobl, *Glasechnik des Mittelalters*, 40–59; and Oidtmann, *Die rheinischen Glasmalereien*, 5–14.
69. Reudenbach, "Werkkünste und Künstlerkonzept," 243–48.
70. Book II, chap. 7, "Yellow Glass." *DDA*, 41–42.
71. These are book II, chap. 10, "How Glass Vessels aAe Made"; chap. 11, "Flasks with a Long Neck"; chap. 12, "The Various Colours of Opaque Glass"; chap. 13, "Glass Goblets which the Greeks Embellish with Gold and Silver"; chap. 14, "The Same"; chap. 15, "The Greek Glass which Decorates Mosaic Work"; and chap. 16, "Earthenware Vessels Glazed in Various Colours." *DDA*, 43–47. Four lost chapters, listed in tables of contents of the eldest copies of the book—the Vienna, Wolfenbüttel, Harley, and Leipzig manuscripts—would have been here, between chapters 10 and 11. They address the coloring of glass from copper, lead and salt: *De coloribus qui fiunt ex cupro et plumbo et sale; De viridi vitro; De vitro saphireo; De vitro quod vocatur gallien* (The Colors which Are Made from Copper and Lead and Salt; The Green Glass; The Blue Glass, The Glass which Is Called Gaulian). See *DDA*, xvi–xviii and 44; also Wolfenbüttel, HAB Guelph Gudianus lat. 2°69, fol. 92v; Vienna, ÖNB Cod. 2527, fol. 33r; London, BL Harley MS 3915, fol. 21r; and Leipzig Universitätsbibliothek MS 1157, fol. 20r.
72. Book II, chap. 12, "The Various Colours of Opaque Glass." *DDA*, 44–45.
73. Book II, chap. 13, "Glass Goblets which the Greeks Embellish with Gold and Silver." *DDA*, 45.
74. Book II, chap. 16, "Earthenware Vessels Glazed in Various Colours." *DDA*, 47.
75. Book III, chap. 1, "The Construction of the Workshop," *DDA*, 64, and following, chap. 2, "Seating Accommodation for the Workmen" and chap. 3, "The Work Furnace." *DDA*, 64–65.
76. Book III, chap. 17, "Files"; chap. 19, "The Same as Above"; and chap. 20, "Tempering Iron." *DDA*, 72–73.
77. Book III, chap. 22, "Crucibles for Melting Gold and Silver," and following, chap. 23, "The Refining of Silver"; chap. 24, "Dividing up the Silver for the Work"; chap. 25, "Casting Silver"; and chap. 26, "Making the Smaller Chalice." *DDA*, 74–76.
78. Book III, chap. 27, "The Large Chalice and Its Mould"; chap. 28, "Niello"; chap. 29, "Applying the Niello." *DDA*, 79–82.
79. Book III, chap. 31, "Soldering the Silver"; chap. 33, "Refining Gold"; chap. 38, "Coating and Gilding the Handles"; chap. 51, "The Flux for Soldering the Gold"; chap. 53, "Setting the Gems and Pearls"; chap. 54, "Enamel"; chap. 63, "Copper"; and chap. 66, "Making Brass." *DDA*, 83–85, 89, 99, 104–7, 120–21, 124–25.
80. Paul Binski, "Questioning Art Hierarchies" in Hourihane, *From Minor to Major*, 6.
81. *DDA*, 63.
82. *Ibid.*, 37.

83. Caviness, "Stained Glass," 548–54, esp. 548–49; see also Kessler, "They preach not by speaking out loud but by signifying," 55–70, and Rudolph, "Inventing the Exegetical Stained-Glass Window," esp. 104.
84. *DDA*, 64.
85. Book II, chap. 10, "How Glass Vessels Are Made," and chap. 11, "Flasks with a Long Neck." *DDA*, 43–44.
86. Book III, chap. 58, "The Cruet" (*de ampulla*), and chap. 88, "Tin Jugs" (*de ampullis stagneis*). *DDA*, 109, 160, and 160n1.
87. Book III, chap. 58, "The Cruet" (*de ampulla*). *DDA*, 109.
88. Dodwell translates the silver vessel as a cruet and the tin vessel as a jug, reasoning that "it is hardly likely that [one] would make a cruet out of tin." See Book III, chap. 58, "The Cruet" (*de ampulla*), and chap. 88, "Tin Jugs" (*de ampullis stagneis*). *DDA*, 109, 160, and 160n1.
89. Book III, chap. 91, "Iron"; chap. 10, "Files Hollowed out Underneath"; chap. 11, "Chisels"; and chap. 12, "Rasps." *DDA*, 162–64, 69–70.
90. Book III, chap. 24, "Making the Smaller Chalice"; chap. 27, "The Large Chalice and Its Mould"; chap. 28, "Niello"; chap. 38, "Coating and Gilding the Handles"; chap. 60, "The Gold Chalice." *DDA*, 76–81, 89–90, 98–99. Hiltrud Westermann-Angerhausen, "The Two Censers in the *Schedula diversarum artium* of Theophilus and Their Place in the Liturgy," in *Les cinq sens au Moyen Age*, ed. E. Palazzo (Paris: Éditions du Cerf, 2016), 189–212. My thanks to Dr. Westermann-Angerhausen for sharing this with me prior to its publication.
91. Book I, chap. 2, "The Colour, Green Earth": "*Qui prasinus est quasi confectio quaedam habens similitudinem viridis coloris et nigri, cuius natura talis est, quod non teritur super lapidem sed missus in aquam resoluitur et per pannum diligenter colatur.*" *DDA*, 5.
92. Pliny, book XXXVI, chap. 26 of *The Historie of the World* (trans. Holland); Isidore of Seville, book XVI, chap. 16 ("Glass") of *Etymologies*, PL 82, col. 583c. Here translated by Stephen A. Barney in *The Etymologies of Isidore of Seville*, 328. Heraclius also quotes the story; see Raspe, *A Critical Essay on Oil Painting*, 3, 111.
93. Book II, chap. 1, "The Construction of the Kiln for Working Glass." *DDA*, 37. For an analysis of Theophilus's description of a kiln, see also Strobl, *Glastechnik des Mittelalters*, 45–46.
94. *Mappae Clavicula* (ed. Philipps), chap. 224, and "*Mappae Clavicula*" (ed. Smith and Hawthorne), 62; *Mappae Clavicula* (ed. Philipps), chaps. 158 and 159, and "*Mappae Clavicula*" (ed. Smith and Hawthorne), 50.
95. Book II, chap. 8, "Purple Glass." *DDA*, 42.
96. Book III, chap. 46, "Gold of the Land of Havilah." *DDA*, 96.
97. Book III, chap. 47, "Arabian Gold"; chap. 48, "Spanish Gold"; chap. 49, "Sand Gold." *DDA*, 96–98.
98. Book III, chap. 43, "Copper." *DDA*, 120–21.
99. Book III, chap. 91, "Iron." *DDA*, 162.
100. Theophilus, *On Divers Arts* (ed. and trans. Hawthorne and Smith), 139–40 and note.
101. Book III, chap. 43, "Copper." *DDA*, 120–21.
102. Berghaus, "Die Münzprägung westfälischer Stifte und Klöster," in Jászai, Schulze, and Luckhardt, *Monastisches Westfalen*, 55–57.
103. Verhulst, *The Rise of Cities*, esp. 133–34.
104. Maurice Lombard, "La route de la Meuse" in Francastel, *L'Art Mosan*, 9–28, and in the same volume, Germaine Faider-Faytmans, "Les arts du métal," 29–37.
105. Verhulst, *The Rise of Cities*, 24–25, 32.
106. Ammann, *Une importante contribution suisse à l'histoire hutoise*, 8; Joris, *La Ville de Huy*, 88–89.
107. Ammann, *Une importante contribution suisse à l'histoire hutoise*, 8–11; Rousseau, *La Meuse et le Pays Mosan*, 71, 85–89. Rousseau's study has been criticized in recent years for its grand depiction of the role of the Meuse region in European culture; this criticism aside, the text contains much valuable information about the trade of metals in the period.
108. Hansotte, "Abbey of Neufmoustier, Huy," 283–99.
109. Rousseau, *La Meuse et le Pays Mosan*, 105.
110. *Ibid.*, 106–7; Höhlbaum, *Hansisches Urkundenbuch* 3:386–87 and 1:3. See also Bormans, *Cartulaire de la commune de Dinant*, 1:4.
111. Ammann, *Une importante contribution suisse à l'histoire hutoise*, 12–13.
112. Verhulst, *The Rise of Cities*, 73.
113. Rousseau, *La Meuse et le Pays Mosan*, 90 and 104.

114. Ibid., 106–7; Höhlbaum, *Hansisches Urkundenbuch*, 3:386–87 and 1:3; Bormans, *Cartulaire de la commune de Dinant*, 1:4.
115. Rousseau, *La Meuse et le Pays Mosan*, 83 and 104–5; Höhlbaum, *Hansisches Urkundenbuch*, 3:387; Bormans, *Cartulaire du Dinant*, 1:20.
116. Rousseau, *La Meuse et le Pays Mosan*, 100–101, 117–18; Berlière, *Documents inédits*, 1:30–31; Barbier, *Histoire de l'abbaye de Floreffe*, 2:28, no. 43.
117. Conradus II Germaniae, *Diploma* XXIX, PL 189, cols. 1496d–1497a.
118. Wyckoff, “Albertus Magnus on Ore Deposits,” 112.
119. Verhulst, *The Rise of Cities*, 124.
120. Ibid.
121. Gerhard Weilandt, “Krise der Königshofes—Krise der Kunst?” in Jarnut and Wemhoff, *Vom Umbruch zur Erneuerung?*, 453–67.
122. *Herimannus comes Wincenburgensis Eugenio III papae Wibaldum commendat*. (ca. December 1147), no. 71 in *Notae Stabulenses de Wibaldo in Monumenta Corbeiensia* (ed. Jaffé), 147.
123. *DDA*, 3.
124. Herbert L. Kessler, “Image and Object,” in Davis and McCormick, *The Long Morning of Medieval Europe*, 291–319, and “Matter,” in *Seeing Medieval Art*, 19–43.
125. Abbot Suger, *De Administratione*, xxxiv; Panofsky, *Abbot Suger*, 73–77.
126. Hugh of St. Victor, book II, chap. 1 of *Didascalicon*, PL 176, col. 751c (p. 61 in Taylor).
127. Hugh of St. Victor, book II, chap. 21 (*Hae mechanicae appellantur, id est adulterinae: quia de opere artificis agunt quod a natura formam mutuatur*) in *Didascalicon* PL 176, col. 760b (p. 75 in Taylor, who identifies it as book II, chap. 21).
128. Ambrose, book I, chap. 48 of *Les devoirs*, 207–211.
129. For example, Gramaccini, “Zur Ikonologie der Bronze in Mittelalter,” 147–70; and Brigitte Buettner, “From Bones to Stones,” in Reudenbach and Toussaint, *Reliquiare im Mittelalter*, 43–59.
130. Honorius of Autun, book I, chap. 141 (*De corona*) of *Gemma Animae*, PL 172:588.
131. Droth, introduction to Droth and Curtis, *Bronze: The Power of Life and Death*, 14; and Cole, “The Demonic Arts and the Origin of the Medium,” 621–40.

CHAPTER 3

1. London, BL Harley MS 3915, fol. 20r.
2. *DDA*, 36.
3. Dodwell, “Introduction,” *DDA*, lxiii–lxiv; *A Catalogue of the Harleian Manuscripts*, 3:96; Singer, *Catalogue of Latin and Vernacular Alchemical Manuscripts*, II:594–96 (nos. 873 and 875). The older copies are Vienna, ÖNB Cod. 2527 and Wolfenbüttel, HAB Cod. Guelph Gudianus lat. 2°69.
4. Dodwell, “Introduction,” *DDA*, lxiv–lxv, also Bischoff, “Die Überlieferung des Theophilus-Rugerus,” 179; Johnson, “The Manuscripts of the *Schedula*,” 90.
5. London, BL Harley MS 3915, fol. 149v. Although part of the inscription is lost due to a wormhole, it reads: “*Emi ego N. hunc librum munster [. . .] 1444, in die sancti lamberti in dieta inter dominum Eugenium papam et antipapam felicem.*”
6. See, for example, the entry for BL Harley MS 3915 in the online catalogue to Harleian manuscripts (www.bl.uk/manuscripts/FullDisplay.aspx?ref=Harley_MS_3915, accessed January 6, 2016); Hallauer, “Cod. Harl. 3631 and Cod. Harl. 3915,” 94–103, esp. 99–103.
7. Meuthen and Hallauer, *Acta Cusana*, vol. 1, part 2, p. 469.
8. Snijders, *Manuscript Communication*, 241–81.
9. Böker, *Die Marktpfarrkirche St. Lamberti zu Münster*, 15–16.
10. Joachim W. Stieber, *Pope Eugenius IV, the Council of Basel and the Secular and Ecclesiastical Authorities in the Empire: The Conflict Over Supreme Authority and Power in the Church* (Brill, 1978), 214–15; and Alois Schröer, *Die Kirche in Westfalen vor der Reformation*, II, 110–111.
11. Coulton, “The Last Generations of Mediaeval Monachism,” 448–50.
12. Paris, BnF ms lat. 6741. See Ouy, “Jean de Montreuil,” 53–6; Bidez, *Catalogue des manuscrits alchimiques latins*, I:39–48.
13. *DDA*, 1.
14. Genesis 1:26–7.
15. *DDA*, 1.
16. Reudenbach: “Werkkünste und Künstlerkonzept,” 243–48; “Praxisorientierung und Theologie,”

- 199–218; and, primarily, “Ornatus materialis domus Dei,” 1–17. See also Marco Collareta, “Teofilo, ‘qui et Rogerus,’” in Castelnovo, *Artifex Bonus*, 50–55.
17. Van Engen, “Theophilus Presbyter and Rupert of Deutz,” 149–51.
 18. Rupert of Deutz, book VII of *Liber de diuinis officiis*, CCCM 7:225.
 19. Augustine, book XXXVI of *De vera religione*, CCSL 32:231, and Quaestio 51 in *De diuersis quaestionibus octoginta tribus*, CCSL 44A:74; and Ladner, “St. Augustine’s Conception of the Reformation of Man,” 595–608.
 20. Rupert of Deutz, book VI of *Liber de diuinis officiis*, CCCM 7:187–88, esp. 187. See also Rupert’s *Commentaria in evangelium Sancti Iohannis*, CCCM 9:14.
 21. Rupert of Deutz, book I, section I, chapter 4 of *Commentaria in Evangelium Sancti Iohannis*, CCCM 9:17; and book IX, section 8, chapter 49, CCCM 9:481. See also Constable, “The Ideal of the Imitation of Christ,” 186.
 22. 1 Corinthians 13:12.
 23. Hugh of St. Victor, book I, part 3 (*De Cognitione diuinitatis*), chap. 4 of *De sacramentis*, PL 176 col. 219A. See also Hugh of St. Victor, book I, part 3, chap. 3 (“By what ways the knowledge of God comes to Man”) and chap. 6 (“On that kind of Knowledge by which the rational mind can see God in itself”) of *On the Sacraments of the Christian Faith* (trans. Ferrari), 42–43. See also Hugh of St. Victor, book V of *Expositio in Hierarchiam Coelestem S. Dionysii*, PL 175, col. 1007b.
 24. Hugh of St. Victor, book 1, chap. 1 of *Didascalicon*, 46.
 25. Hugh of St. Victor, book I, chapters 1–3 and 8 of *Didascalicon*; see also Taylor’s introduction in the same volume, 18–19.
 26. Bell, *The Image and Likeness*, 33.
 27. DDA, 1. The passage, quoted in English above, describes the deception of man by the devil: “*Qui astu diabolico misere deceptus, licet propter inobedientiae culpam privilegium immortalitatis amiserit, tamen scientiae et intellegentiae dignitatem adeo in posteritatis propaginem transtulit, ut quicumque curam sollicitudinemque addiderit, totius artis ingeniique capacitatem quasi hereditario iure adipisci possit.*”
 28. Augustine, book V, chap. 10 of *De civitate dei*, in *The City of God* (trans. Zema and Walsh), 264–65.
 29. Van’t Spijker, *Fictions of the Inner Life*, 3.
 30. DDA, 36.
 31. See for example, Constable, “The Ideal of the Imitation of Christ,” esp. 166–67, 186; Jaeger, “Introduction,” in *Magnificence and the Sublime*, 1–16; and in the same volume, Martino Rossi Monti, “‘Opus es magnificum’: The Image of God and the Aesthetics of Grace,” 17–34.
 32. Book III, chap. 24, “Dividing up the Silver for the Work.” DDA, 75.
 33. The first two chapters of the *Mappae clavicula*, for example, begin thus: “*De Vermiculo. Si vis facere Vermiculum, accipe ampullam vitream et lini deforis de luto; and De Lazorio. Si vis facere Lazorium optimum, ollam novam que nunquam fuit in opus. . . .*” *Mappae Clavicula* (ed. Phillipps), 187.
 34. “*Quomodo deauratur oes, vel auricalcum, vel argentum. Accipe fuliginem et purum sal; teres fortiter, accipiesque glaream ovi et distemperabis. . . .*” Heraclius, book III, chap. 15 of *De coloribus romanorum*, in Merrifield, *Original Treatises*, I:222–23.
 35. “*Si vero thuribula ductili opere componere volveris in auro vel argento sive cupro, primum purificabis ordine que supra, atque funde in infusoriis ferris duas marcas vel tres sive quatuor, secundum quantitatem quam vis habere superiore partem thuribuli.*” Book III, chap. 60, “The Repoussé Censer.” DDA, 111.
 36. Leipzig, Universitätsbibliothek MS 1157.
 37. DDA, 64.
 38. Leipzig, Universitätsbibliothek MS 1157, fol. 22r.
 39. See fols. 22r–v. The unnumbered chapters are as follows. In fol. 22r: chap. 17, “Files,” chap. 18, “The Tempering of Files,” chap. 19, “The Same as Above,” and chap. 22, “Crucibles for Melting Gold and Silver,” DDA, 72–74. In fol. 22v: chap. 28, “Niello,” chap. 29, “Applying the Niello,” chap. 32, “Applying the Niello,” and chap. 41, “Polishing the Niello,” DDA, 81, 84, and 92.
 40. Christopher Mackert, of the Leipzig University Library, has also noted that the diamond X pattern on the leather binding, which probably dates to

the sixteenth century, is typical for the Dominican monastery of St. Paul in Leipzig, from which many books of the Leipzig collection came. My thanks to Christopher Mackert for this information. See also Kadner, "Klostervisitation und Reformation im albertinischen Sachsen 1539/40," 120–26.

41. See, for example, Talbot, "The Cistercian Attitude Towards Art," 56–64; and Rudolph, "The 'Principal Founders' and the Early Artistic Legislation of Cîteaux," 1–45. For the monastery of Altle, see Löffler, "Handschriftenmigration unter den Zisterzienserklöstern Georgenthal und Altle im Mittelalter," 1–12.
42. DDA, 1.
43. See, for example, Hugh of St. Victor, book III, chap. 11 of *De arca Noe*, CCCM 176:76; Bernard of Clairvaux, sermo 65, part 1 of *Sermones de diuersis in St. Bernardi opera*, vol. 6, 1:73–406, here page 298; Andrew of St. Victor, *Expositio historica in parabolis in Expositiones historicae in libros Salomonis*, CCCM 53B, line 190.
44. DDA, 2.
45. Hugh of St. Victor, book I, chap. 5 of *Didascalicon*, 52.
46. Carruthers, *Craft of Thought*, 14–16.
47. Hugh of St. Victor, book II, chap. 1 of *Didascalicon*, 61.
48. *Ibid.*, 62.
49. *Ibid.*
50. London, BL Harley MS 3915, fol. 20v.
51. DDA, 36.
52. Hugh of St. Victor, book II, chap. 1 of *Didascalicon*, 62.
53. DDA, 36; see also Leclercq, *Love of Learning*, 256.
54. DDA, 36.
55. Carruthers, *Craft of Thought*, 80–81, 83.
56. Book I, chap. 2, "The Colour, Green Earth," DDA, 5: "sed missus in aquam resoluitur et per pannum diligenter colatur; cuius usus in recenti muro pro viridi colore satis utilis habetur"; book II, chap. 4, "The Frit," DDA, 39: "deinde tollens duas partes cinerum de quibus supra diximus, et tertiam sabuli diligenter de terra et lapidibus purgati, quod de aqua tuleris, commisce in loco mundo"; book III, chap. 54, "The Enamel," DDA, 105: "sufflansque diligenter considerabis si aequaliter liquefiant."
57. "Quam si diligentius prescruteris, illic inuenies quicquid in diversorum colorum generibus et mixturis habet Graecia." DDA, 4.
58. "Ex quibus illud intellegitur, ut ad officii formam revertamur, appetitus omnes contrahendos sedandosque esse excitandamque animadversionem et diligentiam, ut nequid temere ac fortuito, inconsiderate neglegenterque agamus." Cicero, book 1, chap. 29, section 103 of *De Officiis*, 104–5. For *De officiis* in the twelfth century, see Patrick Gerard Walsh's introduction to Cicero's *On Obligations*, xxxvii–xxxix, and Reynolds and Wilson, *Scribes and Scholars*, 109–15.
59. Perry, "Notes on *diligentia*," 445–58. Perry also notes a passage of Pliny: "memorabili exemplo adhibendi et curae modum, . . . gratiam omnem diligentia abstulerit" (Pliny, book 34, chap. 19 of *Historia Naturalis*, quoted in Perry, 452).
60. Perry, "Notes on *diligentia*," 452–55.
61. Benedict of Nursia, chap. IV of *Regula* (*Quae sint instrumenta bonorum operum*), PL 66, col. 0298b-d.
62. Cicero, book I, chap. 39, sections 138–39 of *De officiis* (p. 140 in Miller); see also Leen, "Cicero and the Rhetoric of Art," 229–45, esp. 237.
63. See, for example, Curtius, *European Literature and the Latin Middle Ages*, esp. 49–50; also Reynolds and Wilson, *Scribes and Scholars*, 109–15.
64. "et commiscens haec tria simul, ita ut sit tertia pars pulvis et tertia viride tertia saphirum, teres pariter super ipsum lapidem cum vino vel urina diligentissime, et mittens in vas ferreum sive plumbeum, pinget vitrum cum omni cautela secundum tractus qui sunt in tabula." Book II, chap. 19, "The Colour with which Glass Is Painted." DDA, 49.
65. "Quo facto pertrahe imagines quot volueris, imprimis plumbo vel stagno sicque rubeo colore sive nigro, faciens omnes tractus studiose, quia necessarium erit cum vitrum pinxeris, ut secundum tabulam coniungas umbras et lumina." Book II, chap. 17, "How to Construct Windows." DDA, 47.
66. "Quam studiose foveri, quam diligenter haberi et teneri debeat, vestra melius novit eruditio." Letter 27 (Wibaldum abbatem iis concessum commendat; ca. 1146–47), *Monumenta Corbeiensia*, 105.

67. “et non sculpantur profunde nimis, sed mediocriter ac studiose.” Book III, chap. 75, “Die Stamping.” DDA, 135.
68. “Age ergo nunc, vir prudens, felix apud Deum et homines in hac vita, felicius in futura, cuius labore et studio Deo tot exhibentur holocausta, ampliori deinceps accendere sollertia.” DDA, 64.
69. The Wolfenbüttel manuscript, notably, uses *vir bone* here, as does Amiens, Bibliothèque municipale ms fonds Lescalopier 46.
70. This object will be discussed further in chapter 4.
71. “ordinabis particulas subtiliter et diligenter, unamquamque in suo loco, atque confirmabis humidia farina super carbones. Cumque impleveris unam partem solidabis eam cum maxima cautela, ne opus gracile et aurum subtile disiungatur aut liquefiat.” Book III, chap. 53, “Setting the Gems and Pearls.” DDA, 105.
72. “et pictura perfecta et siccata delato opere ad solem, diligenter linies illud glutine vernition, et cum defluere coeperit a calore, leniter manu fricabis.” Book I, chap. 26, “How Many Times the Same Colour May Be Applied.” DDA, 25.
73. “Moxque eiciens appone ori tuo et suffla modicum, statimque removens ab ore, tene iuxta maxillam, ne forte, si retraxeris anhelitum, trahas flammam in os tuum.” Book II, chap. 6, “How Sheet Glass Is Made.” DDA, 40.
74. “Quod aurum vel argentum, cum primo fuderis, diligenter circumradendo vel fodiendo inspice, ne forte aliqua vesica siva fissura in eo si, quae saepe contingunt ex incuria sive negligentia vel ignorantia aut inscitia fundentis, cum aut nimis calidum aut nimis frigidum, aut nimis festinato aut nimis productim effunditur. Cumque considerate et caute fuderis, si huiusmodi vitium in eo deprehenderis, cum ferro ad hoc apto diligenter effodies, si possis.” Book III, chap. 74, “Repoussé Work.” DDA, 131–32.
75. “Quod si contigerit per negligentiam glutinis non bene cocti,” book I, chap. 29, “How [Powdered] Gold and Silver Are Applied in Books,” DDA, 28; “Quod si contigerit per negligentiam, ut non satis nigrum sit incaustum, accipe ferrum grossitudine unius digiti, et ponens in ignem sine candescere, moxque in incaustum procie,” book I, chap. 38, “Ink,” DDA, 35; “Et si per aliquam negligentiam contigerit, ut argentum fustum non sit sanum, iterum funde, donec sanum fiat,” book III, chap. 25, “Casting Silver,” DDA, 76; “Nisi contingat ex magna negligentia, semer est sanum ad operandum in eo quodcumque voveris,” book III, chap. 27, “The Large Chalice and Its Mould,” DDA, 79; “Si vero ex negligentia contigerit, ut aliqua macula appareat in argento, ubi aurum tenue sit in inaequaliter positum, cum cupro superpone, et cum siccis setis, aequa, donec per omnia aequale sit,” book III, chap. 38, “Coating and Gilding the Handles [of the Chalice],” DDA, 90; “Et sic interius cum minoribus curvis ferris elevabis leniter et diligenter, summo-pere cavens ut non rumpatur opus aut perforetur. Quod si ex ignorantia vel negligentia contigerit, hoc modo solidari debet,” book III, chap. 74, “Repoussé Work,” DDA, 133.
76. Book III, chap. 31, “Soldering the Silver.” DDA, 83. Emphasis added.
77. Conversation with the author, August 23, 2014.
78. Book III, chap. 74, “Repoussé Work,” DDA, 132–33: “Quo facto et tabula per se refrigerata, iterum in inferiori parte cum curvo ferro fricabis leniter et diligenter fossam capitis interius, convertensque tabulam in superiori parte denuo cum aequali ferro fricabis, et depones campum et monticulus capitis elevetur, rursumque cum malleo mediocri circa ipsum leniter percutiens, appositis carbonibus recoques. Sic saepe facies diligenter elevando interius et exterius et crebro percutiendo, totiensque recoquendo, donec monticulus ille ducatur ad altitudinem trium digitorum aut quatuor, sive plus vel minus secundum quantitatem imaginum.”
79. This object will be discussed further in chapter 4.
80. Wibald, epistles 127 and 167 in *Monumenta Corbeiensia* (nos. 205 and 286), quoted and translated by Constable in “The Ideal of the Imitation of Christ,” 186.
81. C. Stephen Jaeger uses the term “virtue made visible.” See Jaeger, *Envy of Angels*, 111.
82. Forsyth, “The Vita Apostolica,” 75–82.
83. Hahn, “Seeing and Believing,” 1079–1106.
84. Van’t Spijker, *Fictions of the Inner Life*, 8–11, and “Model Reading,” 135–56, esp. 136, 140, and 144.
85. Van’t Spijker, *Fictions of the Inner Life*, 10–11, 14.
86. Jaeger, *The Envy of Angels*, 5–14, 240–241.

87. Book III, chap. 27, "The Large Chalice and Its Mould," *DDA*, 80; and chap. 58, "The Cruet," *DDA*, 109.
88. Book III, chap. 31. "Soldering the Silver." *DDA*, 83.
89. Augustine, book V, chap. 11 of *City of God* (trans. Zema and Walsh), 265; and *De civitate dei*, CCSL 47.
90. Augustine, book V, chap. 11 of *City of God* (trans. Zema and Walsh), 264–65.
91. Augustine, *On the Trinity*, 509–10, and book XV, chap. 23 of *De trinitate*, CCSL 48.
92. "Qui altari deserviunt, cum altario participantur." Rupert of Deutz, book III (*de officio altaris*), chap. 6 of *Super quaedam capitula regulae divi Benedicti abbatis*, PL 170, col. 514.
93. *DDA*, 62. In this chapter and elsewhere I follow the spelling of the Vulgate, Douay-Rheims translation for Beseleel.
94. McCague, "Le don des métiers," 47–66; Castellnuovo, "Dedalo e Beseleel," 65–68; Dodwell, "Medieval Attitudes to the Artist," in *Aspects of Art*, 153–71.
95. Exodus 31:1–5.
96. Theodulf of Orleans, book I, chap. 16 of *Opus Caroli regis contra synodum*, MGH, Concilia 2, Supplementum 1, p. 176, line 17; and Walafrid of Strabo, *Carmine (De Einhardo magno)*, MGH *Poetae latini aevi carolini*, vol. 2, Carmine 5, *De diversis rebus*, section 23, page 377, verse 221. See also the translations of Alcuin's letters in Dutton, "Some Reflections on Einhard" in *Charlemagne's Courtier: The Complete Einhard*, nos. 1–2.
97. "Conditus ecce iacet tumulo vir nobilis isto, / Einhardus nomen cui genitor dederat. / Ingenio hic prudens, probus actu atque ore facundus / Extitit, ac multis arte fuit utilis." Hrabanus Maurus, *Carmine*, MGH *Poetae latini aevi carolini*, vol. 2, Carmine 85, page 238, verse 3.
98. Walafrid of Strabo, *Glossa Ordinaria*, PL 113, col. 286.
99. Peter Damian, Letter 83, to the Senator Peter, in *The Letters of Peter Damian*, 242. See also Peter Damian, *Epistola V, ad petrum senatorum urbis*, PL 144, col. 470–71.
100. *DDA*, 69.
101. Augustine, book 2, quaestio 138 in *Quaestiones in Heptateuchum*, PL 34, col. 0643.
102. Rupert of Deutz, book XIII, part 4 (*In Exodum*) of *De Sancta Trinitate*, CCCM 22:802.
103. Schroder, "Die Rekonstruktion des Salomonischen Tempels," 157–65.
104. Rupert of Deutz, book XIII, part 4 (*In Exodum*) of *De Sancta Trinitate*, CCCM 22:802.
105. Smalley, "An Early Twelfth-Century Commentator on the Literal Sense of Leviticus," 27–48.
106. Benedict, chap. 48 ("The Daily Manual Labor") of the *Regula*, *RB* 1980, 69.
107. Idung of Prüfening, "Dialogus duorum monachorum," in *Le moine Idung et ses deux ouvrages*, 91–186, and "A Dialogue between a Cluniac and a Cistercian," 19–141, in *Cistercians and Cluniacs: The Case for Cîteaux* (trans. O'Sullivan). See especially, for example, 91–94, on manual labor.
108. Le Goff, "Labor, Techniques, and Craftsmen," 71–86.
109. Van Engen, "The Crisis of Cenobitism Reconsidered," 292–94.
110. *Herimannus comes Wincenburgensis Eugenio III papae Wibaldum commendat*, letter 71 (ca. December 1147) in *Notae Stabulenses de Wibaldo, Monumenta Corbeiensia*, 147. This letter is also published as *Epistola 59 of Epistolae Wibaldi*, PL 189, cols. 1166c–1167a.
111. Téxier, *Dictionnaire d'orfèvrerie*, 1047–48.
112. Rupert of Deutz, *Super quaedam capitula regulae divi Benedicti abbatis*, PL 170, col. 513–14.
113. From St-Laurent in Liège, a twelfth-century manuscript, possibly the second half, except fol. 159r, which is fifteenth-century, now Brussels, BR/KB MS 11144, fols. 1v–159r; from Gembloux, a twelfth-century manuscript, now Brussels, BR/KB MS 5194–96, fols. 1v–100r; from Great St. Martin, Cologne, a tenth-century manuscript, with corrections from the fifteenth, now Berlin, Staatsbibliothek, Theol. lat. fol. 339, fols. 1r–89v and from Stavelot, a ninth-century text with a twelfth- to thirteenth-century chapter index, now London, BL Add. MS 16961, fols. 3r–182v. There is also a fifteenth-century copy from Great St. Martin, which is now Brussels, BR/KB MS 670–73, fols. 17r–95r.
114. Benedict of Nursia, chap. IV of *Regula (Quae sint instrumenta bonorum operum)*, PL 66, col. 0298b–d; Smaragdi Abbatis, chap. 43 (*De opere manuum*

cotidiano) of *Exposito in Regulam S. Benedicti* (ed. Spannagel and Engelbert), 271–72. Translation my own.

115. Joris, *La Ville de Huy*, 103ff, and “A propos de ‘burgus’ à Huy et à Namur,” 192–99.
116. Joris, *La Ville de Huy*, 121.
117. *DDA*, 36, 63.

CHAPTER 4

1. Vienna, ÖNB Cod. 2527, visible between folios 70v and 71r; the stub is continuous with folio 75.
2. Book III, chap. 36, “The Same in Another Way (Milling the Gold),” *DDA*, 87.
3. Dodwell, “Introduction,” *DDA*, lix.
4. Vienna, ÖNB Cod. 2527, folio 32r (book I, chap. 36 [“Spanish Green”], *DDA*, 33); and folio 57r (book III, chaps. 9 [“The Implement which Is Called the ‘Organarium’”] and 10 [“Files Hollowed Underneath”], *DDA*, 68–69).
5. Book I, chap. 17. “The Panels of Altars and Doors and Casein Glue.” *DDA*, 16.
6. London, BL Harley MS 3915, fols. 20r, 20v, 36r, and 36v.
7. Dodwell, “Introduction,” *DDA*, lix–lxii.
8. “*Facies quoque anulum singulariter, qui stare debet inter nodum et vas superius, eadem quantitate et specie sicut est ille, quem ductili ferro formasti sub nodo, et accipiens ferrum obtusum fricabis super cotem aequalem.*” Book III, chap. 26, “Making the Smaller Chalice.” *DDA*, 78.
9. “*Si vero thuribula ductili opere componere volveris in auro vel argento sive cupro primum purificabis ordine quo supra, atque funde in infusoriis ferreis duas marcas vel tres sive quatuor, secundum quantitatem quam vis habere superiorem partem thuribuli.*” Book III, chap. 60, “The Repoussé Censer.” *DDA*, 111.
10. Vienna, ÖNB Cod. 2527, fol. 5r. Translated in *DDA*, 37.
11. See Holcomb, “Strokes of Genius” in *Pen and Parchment*, 28.
12. Dodwell, “Introduction,” *DDA*, lxii.
13. *Ibid.*, lix–lxii.
14. Vienna, ÖNB Cod. 2527, fol. 1r.
15. Lessing, *Vom Alter der Oelmalerey*, rpt. in *Sämtliche Schriften* 12, 159–97.
16. Albert Ilg, “Introduction” in Theophilus, *Schedula Diversarum Artium* (1874), xliii.
17. See Erzbistumsarchiv, Paderborn, Generalvikariat, doc. 11.
18. Most recently, Clemens M. M. Bayer, “Der Paderborner Dom-Tragaltar,” in Stiegemann and Westermann-Angerhausen, *Schatzkunst am Aufgang der Romanik*, 65–77, and Eckhard Freise, “Adelstiftung, Reichsabtei, Bischofskloster” in Baumgärtner, *Helmarshausen*, 12–27, esp. 23.
19. Von Falke, “Das Kunstgewerbe im Mittelalter,” 190–420, esp. 240–43; Von Falke and Frauberger, *Deutsche Schmelzarbeiten des Mittelalters*, 13–17.
20. Creutz, *Kunstgeschichte der edlen Metalle*, 154–55, and “Aus der Werkstatt des Rogerus,” 357–70, esp. 359–61; Hermann Fillitz, “Rogerus von Helmarshausen” in Gosebruch, *Helmarshausen und das Evangeliar Heinrichs des Löwen*, 43–62, esp. 58; *Kunst und Kultur in Weserraum*, 2:559–562.
21. Freise, “Roger von Helmarshausen in seiner monastischen Umwelt,” 180–293, esp. 249–51, and “Zur Person des Theophilus,” 357–61.
22. *Liber vitae* of the abbey of Corvey, Münster, Staatsarchiv, Msc. I Nr. 133, p. 26.
23. Freise, “Zur Person des Theophilus,” 360.
24. For example, Degering, “Theophilus Presbiter, qui et Rogerus,” in *Westfälische Studien*, 248–62.
25. Hoffman, “Pathways of Portability,” 17–50.
26. Freise, “Zur Person des Theophilus,” 359; Lasko, “Anthropomorphic Evangelist Symbols,” *England and the Continent*, 18.
27. Bayer, “Der Paderborn Dom-Tragaltar und die zu 1100 gefälschte Urkunde,” in Stiegemann and Westermann-Angerhausen, *Schatzkunst am Aufgang der Romanik*, 65–77, esp. 65–66; see also Freise, “Adelstiftung, Reichsabtei, Bischofskloster,” 32–33.
28. Eldevik, *Episcopal Power*, 10–15 and 215–16.
29. Chap. 123 of *Vita Meinweri*, MGH SS 59, p. 61, line 22–24.
30. Eldevik, *Episcopal Power*, 221–22.
31. Bischoff, “Die Überlieferung des Theophilus-Rugerus,” 175–82.
32. Lectionary, Stuttgart Hauptstaatsarchiv, J 522 B Xa 588; Hartmut Hoffmann, “Die Paderborner

- Schreibschule," in Stiegemann and Wemhoff, *Canossa 1077*, 2:449–64.
33. The charter is now preserved in Marburg Staatsarchiv as Kloster Helmarshausen 1017 Juli 11; rpt. MGH DH II: item number 371, pp. 474–76. See also Hoffmann, "Die Paderborner Schreibschule," 450; Freise, "Adelstiftung, Reichsabtei, Bischofskloster," 12–27; and John W. Bernhardt, "King Henry II of Germany: Royal Self-Representation and Historical Memory," in Althoff, Fried, and Geary, *Medieval Concepts of the Past*, 39–70, esp. 61–63.
 34. Jaeger, "The Courtier Bishop," 291–325.
 35. Geary, "Archival Memory and the Destruction of the Past," in *Phantoms of Remembrance*, 81–114; and Nees, "Reading Aldred's Colophon," 333–77.
 36. Parkes, "Archaizing Hands," 101–44.
 37. See, for example, Sears, "The Afterlife of Scribes," 75–96.
 38. Lasko, *Ars Sacra*, 164–65, and Karl Hermann Usener, "Vorgotische Goldschmiedekunst," *Kunst und Kultur in Weserraum, 800–1600*, 1:560.
 39. Lasko, *Ars Sacra*, 151.
 40. Cohen, "Making Memories," 135–52.
 41. *Ibid.*, 142.
 42. Wollheim, "Style in Painting," 43.
 43. *Ibid.*
 44. Holcomb, "Strokes of Genius," in *Pen and Parchment*, 26.
 45. Vienna, ÖNB Cod. 2527, fol. 52v.
 46. Carruthers, *The Craft of Thought*; see, for example, 150–55.
 47. "Quae cum saepe relegeris et tenaci memoriae commendaueris." *DDA*, 4.
 48. For example, see Hahn, "A Case Study: Wibald of Stavelot," in *Strange Beauty*, 209–21.
 49. Book III, chap. 3, "The Work Furnace." *DDA*, 65–66.
 50. *DDA*, 62.
 51. *Ibid.*, 63.
 52. Book III, chaps. 27–43, *DDA*, 79–94.
 53. Book III, chap. 42, "Embellishing the Bowl of the Chalice." *DDA*, 93.
 54. Book III, chap. 24, "Dividing up the Silver for the Work." *DDA*, 75–76.
 55. Book III, chap. 60, "The Repoussé Censer." *DDA*, 111.
 56. Book III, chaps. 66–69, "Gold of the Land of Havilah," "Arabian Gold," "Spanish Gold," and "Sand Gold." *DDA*, 96–98.
 57. Book III, chap. 60, "The Repoussé Censer." *DDA*, 111. See also Westermann-Angerhausen, "The Two Censers." My thanks to Dr. Westermann-Angerhausen for sharing this with me prior to its publication.
 58. Book III, chap. 27, "The Large Chalice and Its Mould." *DDA*, 79.
 59. Lasko, *Ars Sacra*, 164; Dodwell, "Introduction," *DDA*, xlii–xliii; Von Falke and Frauberger, *Deutsche Schmelzarbeiten des Mittelalters*, 14.
 60. Book III, chap. 73, "Punched Work." *DDA*, 130–31.
 61. Book III, chap. 74, "Repoussé Work." *DDA*, 131–135, esp. 132–33.
 62. Book III, chap. 72, "Openwork." *DDA*, 130.
 62. "Gertrvdis xpo [christo] felix/vt vivat in ipso/obtulit hvnc lapidem/gemmis avroq[ue] nitentem." See cat. no. 42 in Bagnoli, *Treasures of Heaven*, 86.
 64. For a contrary view, see Clemens M. M. Bayer, "Die gefälschte Urkunde Bischof Heinrichs II," in Stiegemann and Westermann-Angerhausen, *Schatzkunst am Aufgang der Romanik*, 69.
 65. Budde, *Altare Portatile*, Textband, pp. 17–18, 27–30; Katalogband I, cat. no. 13, pp. 90–98; cat. no. 32, pp. 207–27.
 66. The altar of Henry of Werl measures 34.5 × 21.2 × 16 cm. The altar of Gertrude is 27 × 21 × 10.5 cm.
 64. *Annalista Saxo* A. 1105, MGH SS VI:740. See also Thomas Vogtherr, "Handlungsspielräume Bishöflicher Parteinahme" in Jarnut and Wemhoff, *Vom Umbruch zur Erneuerung?*, 417–25, here 422.
 68. Garrison, *Ottonian Imperial Art and Portraiture*.
 69. See cat. no. 33 in Budde, *Altare Portatile*, 208.
 70. Book III, chap. 53, "Setting the Gems and Pearls." *DDA*, 104. See also Barbara Drake Boehm, "A Brilliant Resurrection," in Bagnoli, *Treasures of Heaven*, 149–61, esp. 151.
 71. For example, cat. no. 32 in Budde, *Altare Portatile*, 194–206; Jürgen Werinhard Einhorn, "Zu Provenienz und Werkgeschichte des Abdinghofer Tragaltars im 19. und 20. Jahrhundert," in Stiegemann and Westermann-Angerhausen, *Schatzkunst am Aufgang der Romanik*, 184–96; cat. no. 65 in Bagnoli, *Treasures of Heaven*, 126.

72. Neer, "Connoisseurship and the Stakes of Style," *Critical Inquiry* 32, no. 1 (Autumn 2005): 1–26, esp. 15–23.
73. For example, Michael Peter, "Neue Fragen und alte Probleme" in Stiegemann and Westermann-Angerhausen, *Schatzkunst am Aufgang der Romanik*, 80–96.
74. Sauerländer, "Antiqui et Moderni at Reims," 19–37.
75. The cross of Abbess Mathilde and Otto, Duke of Swabia, now in Essen, is 44.5 cm high; the Lothair cross is 50 cm high; the now-lost Borghorst cross is 41.1 cm high; and the Gisela cross is 44.5 cm high. See Schnitzler, "Das sogenannte grosse Bernwardkreuz," 382–94, and Lasko, "The Enger Cross," 197–236.
76. Von Falke and Frauberger, *Deutsche Schmelzarbeiten*, 16–17.
77. Lasko, "The Enger Cross," 208–9.
78. Johnson, "Bishop Benno II," 399.
79. See entries 1266 and 1297 in *Regesta Historiae Westphaliae, accedit Codex Diplomaticus: Die Quellen der Geschichte Westfalens* 1:208 and 212.
80. Lasko, "The Enger Cross," 202.
81. Forsyth, "Art with History," 153–62.
82. My thanks to Genevra Kornbluth and Lothar Lombacher for their observations about this object.
83. Lasko, "The Enger Cross," 204–5.
84. Westermann-Angerhausen, "Did Theophano Leave Her Mark on Ottonian Sumptuary Arts?" 254.
85. Lasko has found parallels for these bands on the hilt of a sword from Essen. Lasko, "The Enger Cross," 204.
86. Creutz, "Aus der Werkstatt des Rogerus," 359–61; *Translatio S. Modualdi*, MGH SS 12:289–310.
87. The reference is to Revelation 5:12; Legner, *Ornamenta Ecclesiae* 462–66.
88. This motif is also visible in the Hezilo cross, for example. See cat. no 13 in Barnet, Brandt, and Lutz, *Medieval Treasures from Hildesheim*, 55.
89. Creutz, "Aus der Werkstatt des Rogerus," 355–70.
90. Franz J. Ronig, "Der Prachteinband" in Stiegemann and Westermann-Angerhausen, *Schatzkunst am Aufgang der Romanik*, 123–33; see especially 123 and 131–32; see also Jörg Baumgarten, "Der Buchdeckel des Roger von Helmarshausen, Veruch einer ikonographischen Bestimmung," in Ronig, *Schatzkunst Trier*, 35–44.

91. Lasko, *Ars Sacra*, 166–67.
92. Ronig, "Der Prachteinband," 129–31; Ursula Mende, "Goldschmiedekunst in Helmarshausen," *Helmarshausen* (ed. Baumgärtner), 163–98, esp. 177–80.
93. Trier Cathedral Treasury MS 139, fol. 14v; see also Lasko, "The Enger Cross," 202–3.
94. Getty Museum MS Ludwig II 3, fol. 7v.
95. Trier Cathedral Treasury MS 139, fol. 81v.
96. Ronig, "Der Prachteinband," 126–28.
97. Herbert, "Courbet, Incommensurate, and Emergent," 339–81; Wollheim, "Style in Painting," 40–41; Neer, "Connoisseurship and the Stakes of Style," 12.
98. Iogna-Prat, *La Maison Dieu*, esp. 345–51. My thanks to Thomas Dale for this reference.

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CONCLUSION

1. Book III, chap. 75, "Die Stamping." *DDA*, 136–37.
2. "*nec defatigati discendi desiderio, intolerabili tamen acquirunt labore, hanc diversarum artium scedulam avidis obtutibus concupisce, tenaci memoria perlege, ardenti amore complectere.*" *DDA*, 4.
3. Leclercq, *Love of Learning*, 177.
4. Hugh of St. Victor, book IV, chap. 16 ("Some Etymologies of Things Pertaining to Reading") of *Didascalicon*, 118–19.
5. Vienna, ÖNB Cod. 2527, fol. 1r reads "*Incipit prologus libri primi Theophilus qui et Rogerus de diversis artibus.*" Cambridge, University Library MS Ee. 6.39, part III, fol. 1r introduces *On Diverse Arts* with a title that reads "*prologus primum liber Theophili monachi et presbyteri de diversis artibus in primis de coloribus.*"
6. Dodwell, "Introduction," *DDA*, lxxiii.
7. Cerquiglini, *In Praise of the Variant*, 33.
8. "*Quae cum saepe relegeris et tenaci memoriae commendaveris, hanc vissitudinem institutori tuo recompensabis ut, quotiens labore meo bene usus fueris, ores pro me apud misericordiam Dei omnipotentis.*" *DDA*, 4.
9. Haye, *Das lateinische Lehrgedicht*, 91.
10. Hugh of St. Victor, book II, chap. 1 ("Concerning the Distinguishing of the Arts") of *Didascalicon*, 61–62. The Latin of this last phrase, as given in

- Migne's edition for the *Patrologia Latina*, uses the word *operatio*: "*De discretione artium. Quam differentiam Plato et Aristoteles esse voluerunt inter artem et disciplinam. Vel ars dici potest, quae fit in subjecta materia et explicatur per operationem, ut architectura.*" PL 176:751d–52b.
11. "*Quae cum saepe relegeris et tenaci memoriae commendaveris, hanc vicissitudinem institutori tuo recompensabis ut, quotiens labore me bene usus fueris.*" DDA, 4.
 12. See chapter 1 in this volume.
 13. DDA, 4.
 14. Ibid.
 15. "*Quamque dulce ac delectabile diversarum utilitatum exercitiis operam dare.*" Ibid., 36.
 16. "*Apprehendi atrium agiae Sophiae conspicioque cellulam diversorum colorum omnimoda varietate refertam et monstrantem singulorum utilitatem ac naturam. . . . quasi curiosus explorator omnimodis elaboravi cognoscere, quo artis ingenio et colorum varietas opus decoraret, et lucem diei solisque radios non repelleret. Huic exercitio dans operam vitri naturam comprehendo, eiusque solius usu et varietate id effici posse considero.*" Ibid., 37.
 17. "*Cum eius ornasti domum tanto decore tantaque operum varietate.*" Ibid., 62.
 18. "*Per spiritum intellectus cepisti capacitatem ingenii, quo ordine qua varietate qua mensura valeas insistere diverso operi tuo.*" Ibid., 62.
 19. Binski, "London, Paris, Assisi, Rome Around 1300," 6–7, and—famously—Schapiro, "On the Aesthetic Attitude Toward Romanesque Art," 1–27.
 20. DDA, 63.
 21. Ibid.
 22. Ibid., 63–64.
 23. 2 Corinthians 5:5–9: "Now he that maketh us for this very thing, is God, who hath given us the pledge of the Spirit. Therefore having always confidence, knowing that, while we are in the body, we are absent from the Lord. (For we walk by faith, and not by sight.) But we are confident, and have a good will to be absent rather from the body, and to be present with the Lord. And therefore we labour, whether absent or present, to please him."
 24. Hamburger, "The Medieval Work of Art: Wherein the 'Work'? Where in the 'Art'?", in Hamburger and Bouché, *The Mind's Eye*, 374–412, and in the same volume, Kessler, "Turning a Blind Eye," 413–39. See also Hamburger, "Images and the *Imago Dei*" in *St. John the Divine*, 185–202.
 25. Hugh of St. Victor, book I, part 10, chap. 2 ("Quid sit fides") of *De sacramentis Christiane fidei*, PL 176, col. 329c–d. See also Dale, "Transcending the Major/Minor Divide," 35–36.
 26. DDA, 1.
 27. Augustine, book VI, chap. 10 of *De trinitate* (*On the Trinity*; trans. McKenna), 212–15.
 28. Marrocco, *Participation in the Divine Life*, 89.
 29. Augustine, book 15, chap. 18 of *De trinitate* (*On the Trinity*; trans. McKenna), 496–97.
 30. Augustine, chap. 64, par. 77 of *De natura et gratia* in *Corpus Scriptorum Ecclesiasticorum Latinorum* 60:233–99, 291.
 31. Bell, *The Image and Likeness*, 58–62; Augustine, chap. 64, part 77 of *De natura et Gratia* in *Corpus Scriptorum Ecclesiasticorum Latinorum* 60:233–99, 291; book 15, chap. 17 (29–31) of *De trinitate*, in *On the Trinity*, 493–96; on the renewal of man, book 14, chaps. 17, 15, and 8–11 of *De trinitate*, in *On the Trinity*, 444–45 and 469–80.
 32. DDA, 63.
 33. Cynthia Hahn, "Production, Prestige, and Patronage of Medieval Enamels" in Hourihane, *From Minor to Major*, 152–68, here 163; and Palazzo, *L'Evêque et son image*, 56.
 34. DDA, 61.
 35. Ibid., 62.
 36. Jaeger, *Envy of Angels*, esp. 4–7.

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